

UMC 0.11 um Logic/Mixed_Mode AI Advanced Enhancement LVT 5 V MOSFET SPICE Model Document

**Version 0.4 Phase1
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1 Introduction

1.1 Revision History

Table 1-1 Revision history

Version	Date	Author	Remark
0.2_p1	2012/05/18	Eason Cheng Yan Zung Wang	1. Original release.
0.3_p1	2012/06/01	Eason Cheng	1. Modify parameter SCREF in model card, and table, plot of WPE chapter in the document. 2. Modify parameter SAREF in model card, and table, plot of LOD chapter in the document. 3. Update RO tables, plots, dimension ranges and some typo in the document. 4. Update Device Performance Reference Tables.
0.4_p1	2016/09/13	Eason Cheng	1. Set source terminal off current aligned to drain terminal by adding jtssws=jtsswd, jtsswgs=jtsswgd, and xtsswgs=xtsswgd.

1.2 General Information

This document serves as a reference to the design of products that will be manufactured by UMC using the following process.

0.11 um Logic and Mixed-Mode 1P8M Metal Metal Capacitor AI Advanced Enhancement Process

For more information about this process and technology, please see the electrical design rule listed below.

G-02-LOGIC/MIXED_MODE11-1P8M-MMC/AL/L110AE-EDR

The creation, verification, and usage of the compact model of the following devices are presented in this document.

5 V n-ch MOSFET
5 V p-ch MOSFET

The following compact model is chosen for the modeling of these devices.

BSIM3V3.3.24

For detailed descriptions of these compact model equations and parameters, please refer to this material.

BSIM3V3 MOSFET Model User Manual

In BSIM3V3.3.24, there are two types of models that can be used. One is a binning model, which means the whole device dimension range is chopped into several "bins" and different model parameter sets are provided for each bin. The other one is a global model, which means one scalable parameter set is provided and it can cover the whole device geometry range. Each of these two approaches has advantages over the other.

UMC provides global models for their many benefits and the model accuracy is guaranteed by a quantitative report of model fitting error against the silicon data, which can be found in the following sections.

1.3 Model Usage

The compact model is provided in various formats. It is guaranteed that the simulation results of this model from different simulators show differences less than 0.5% and 1% for DC and AC simulations respectively. It should be noted that the model has been verified on different simulators of the specific versions listed below and the simulation results from other versions of the simulators might be different.

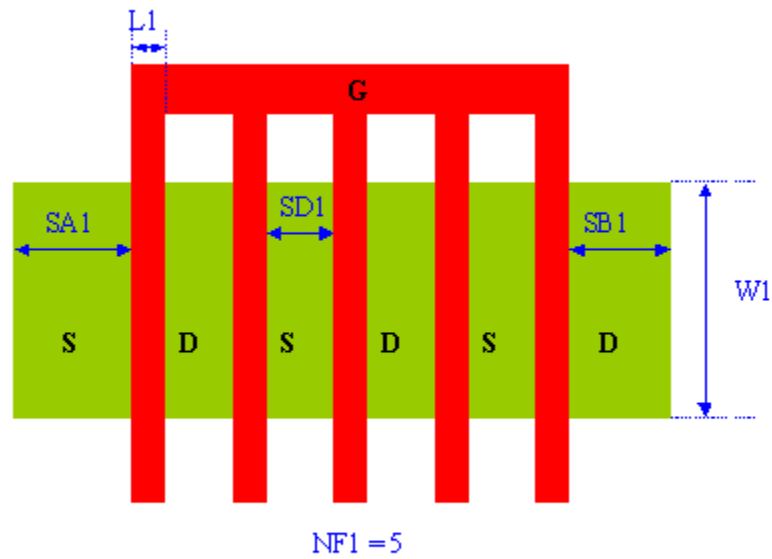
Synopsys HSPICE release 2014.9SP2
Cadence SPECTRE version 15.1.0.257 64bit
Mentor Graphics ELDO version 15.4

To cope with the impact of process variations on circuit performance, worst-case models are provided along with the typical case and they are listed below. Please note that 'fast' here means lower threshold voltage, higher leakage and driving current, and 'slow' means higher threshold voltage, lower leakage and driving current.

TT: Typical n-ch MOSFET and Typical p-ch MOSFET
FF: Fast n-ch MOSFET and Fast p-ch MOSFET
SS: Slow n-ch MOSFET and Slow p-ch MOSFET
FNFP: Fast n-ch MOSFET and Slow p-ch MOSFET
SNFP: Slow n-ch MOSFET and Fast p-ch MOSFET

Follow these steps to invoke the model file during simulation.

(a) Example for Odd Finger Number



(b) Example for Even Finger Number

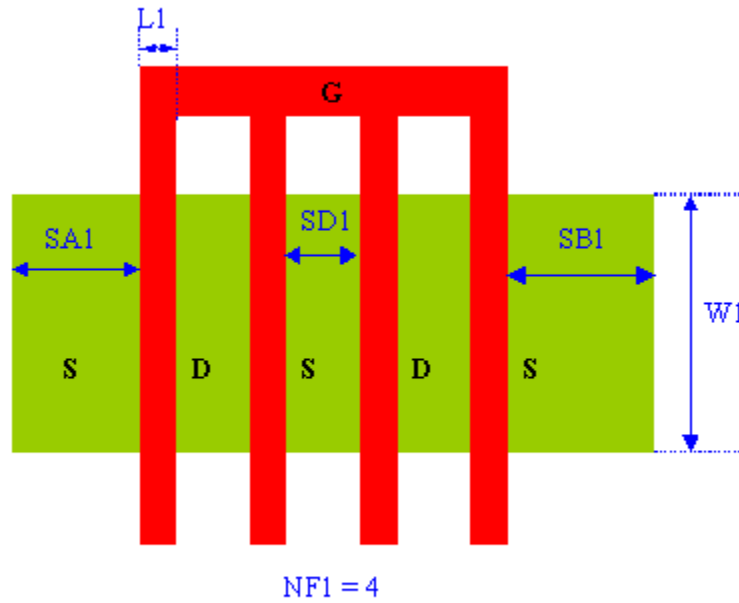


Figure 1-1 Layout example for LOD effect

For Synopsys HSPICE

- Copy ModelFileName.mdl and ModelFileName.lib into your simulation directory.

.lib "/ModelFileName.lib" CaseName

, where CaseName can be TT, FF, SS, FNFP, or SNFP.

- Define device condition by 'Mxx' statement;

Example 1: For odd finger Number device (Refer to Figure 1-1 (a))

```
M1 Vdd Vgg Vss Vbb DeviceName W='W1*NF1' L=L1 NF=NF1 SA=SA1
+ SB=SB1 SD=SD1
+ PS='2*(SA1+W1)+(SD1+W1)*(NF1-1)'
+ PD='2*(SB1+W1)+(SD1+W1)*(NF1-1)'
+ AS='SA1*W1+SD1*W1*(NF1-1)/2'
+ AD='SB1*W1+SD1*W1*(NF1-1)/2'
```

Example 2: For even finger number device (Refer to Figure 1-1(b))

```
M1 Vdd Vgg Vss Vbb DeviceName W='W1*NF1' L=L1 NF=NF1 SA=SA1
+ SB=SB1 SD=SD1
+ PS='2*(SA1+W1)+2*(SD1+W1)*((NF1)/2-1)+2*(SB1+W1)'
+ PD='(SD1+W1)*NF1'
+ AS='SA1*W1+SD1*W1*((NF1)/2-1)+SB1*W1'
+ AD='SD1*W1*(NF1)/2'
```

For Cadence Spectre

- Copy ModelFileName.mdl.scs and ModelFileName.lib.scs into your simulation directory.

include "/ModelFileName.lib.scs" section=CaseName

, where CaseName can be TT, FF, SS, FNFP, or SNFP.

- Define device condition by 'Mxx' statement;

Example 1: For odd finger number device (Refer to Figure 1-1 (a))

```
M1 ( Vdd Vgg Vss Vbb) DeviceName w=W1*NF1 l=L1 nf=NF1 sa=SA1
+ sb=SB1 sd=SD1 sca=SCA1 scb=SCB1 scc=SCC1
+ ps=2*(SA1+W1)+(SD1+W1)*(NF1-1)
+ pd=2*(SB1+W1)+(SD1+W1)*(NF1-1)
+ as=SA1*W1+SD1*W1*(NF1-1)/2
+ ad=SB1*W1+SD1*W1*(NF1-1)/2
```

Example 2: For even finger number device (Refer to Figure 1-1(b))

```
M1 (Vdd Vgg Vss Vbb) DeviceName w=W1*NF1 l=L1 nf=NF1 sa=SA1
+ sb=SB1 sd=SD1
+ ps=2*(SA1+W1)+2*(SD1+W1)*((NF1)/2-1)+2*(SB1+W1)
+ pd=(SD1+W1)*NF1
+ as=SA1*W1+SD1*W1*((NF1)/2-1)+SB1*W1
+ ad=SD1*W1*(NF1)/2
```

For Mentor Graphics Eldo

- Copy ModelFileName.mdl.eldo and ModelFileName.lib.eldo into your simulation directory.
- .lib "/ModelFileName.lib.eldo" CaseName
- , where CaseName can be TT, FF, SS, FNFP, or SNFP.
- Define device condition by 'Mxx' statement;

Example 1: For odd finger number device (Refer to Figure 1-1 (a))

```
M1 Vdd Vgg Vss Vbb DeviceName W='W1*NF1' L=L1 NF=NF1 SA=SA1
+ SB=SB1 SD=SD1 + PS='2*(SA1+W1)+(SD1+W1)*(NF1-1)'
+ PD='2*(SB1+W1)+(SD1+W1)*(NF1-1)'
+ AS='SA1*W1+SD1*W1*(NF1-1)/2'
+ AD='SB1*W1+SD1*W1*(NF1-1)/2'
```

Example 2: For even finger number device (Refer to Figure 1-1(b))

```
M1 Vdd Vgg Vss Vbb DeviceName W='W1*NF1' L=L1 NF=NF1 SA=SA1
+ SB=SB1 SD=SD1
+ PS='2*(SA1+W1)+2*(SD1+W1)*((NF1)/2-1)+2*(SB1+W1)'
+ PD='(SD1+W1)*NF1'
+ AS='SA1*W1+SD1*W1*((NF1)/2-1)+SB1*W1'
+ AD='SD1*W1*(NF1)/2'
```

ModelFileName for 5 V MOSFET: l110ae_lvt50_v041

DeviceName for 5 V n-ch MOSFET: n_50_lvtl110ae

DeviceName for 5 V p-ch MOSFET: p_50_lvtl110ae

As stated above, one global model is provided and this model is valid within the following geometry, voltage, and temperature ranges. Device behavior beyond these ranges may not be well described by the model and is not guaranteed.

Note: Three choices of thick oxide IO devices are available, 2.5V, 3.3V and HG 3.3V, 5V, which cannot coexist on the same chip. If any design requirement is necessary, please discuss in advance with UMC through CE department.

Geometry range (After 90% shrink):

For 5 V n-ch MOSFET:

$$\begin{aligned}0.488 \text{ um} &\leq L_{DES} \leq 10 \text{ um} \\0.35 \text{ um} &\leq W_{DES} \leq 10 \text{ um} \\0.324 \text{ um} &\leq SA, SB\end{aligned}$$

For 5 V P-ch MOSFET:

$$\begin{aligned}0.488 \text{ um} &\leq L_{DES} \leq 10 \text{ um} \\0.35 \text{ um} &\leq W_{DES} \leq 10 \text{ um} \\0.324 \text{ um} &\leq SA, SB\end{aligned}$$

Note: It is recommended to adopt multi-finger structures if $W_{DES} > 10 \text{ um}$.

Adopt multi-finger structures if $W_{DES} > 10 \text{ um}$.
 $0.324 \text{ um} \leq SA, SB \leq 5 \text{ um}$

Note: $SAREF = SBREF = 3.6 \text{ um}$ is the reference dimension of the STI-stress (LOD) effect.

Voltage range:

For 5 V n-ch MOSFET:

$$\begin{aligned}0 \text{ V} &\leq V_{GS} \leq 5 \text{ V}(*1.1) \\0 \text{ V} &\leq V_{DS} \leq 5 \text{ V}(*1.1) \\-3 \text{ V} &\leq V_{BS} \leq 0 \text{ V}\end{aligned}$$

For 5 V p-ch MOSFET:

$$\begin{aligned}0 \text{ V} &\geq V_{GS} \geq -5 \text{ V}(*1.1) \\0 \text{ V} &\geq V_{DS} \geq -5 \text{ V}(*1.1) \\3 \text{ V} &\geq V_{BS} \geq 0 \text{ V}\end{aligned}$$

Temperature range:

$$-40 \text{ C} \sim +125 \text{ C}$$

Feature\ GDSII	Drawing in shrinkable L130AE GDS database (unit : um)	L110AE GDSII =0.9*(L130AE GDS) Dimension (unit : um)	L110AE SPICE Dimension (unit : um)
Core (W/L)	10/0.12	9/0.108	HS(L + 8 nm): 9/0.116 LL/SP(L +12 nm): 9/0.12
2.5V I/O (W/L)	10/0.24	9/0.216	(L +12 nm): 9/0.228
3.3V I/O (W/L)	10/0.34	9/0.306	(L +12 nm): 9/0.318
5V I/O (W/L)	10/0.542	9/0.488	(L +12 nm): 9/0.5
HG 3.3V I/O (W/L)	HG IO-PMOS : 10/0.32 HG IO-NMOS : 10/0.34	HG IO-PMOS : 9/0.228 HG IO-NMOS : 9/0.306	HG IO-PMOS (L +12 nm): 9/0.3 HG IO-NMOS (L +12 nm): 9/0.318
Salicide and Non-salicide resistor	(Width/Length)	0.9*(Width/Length)	0.9*(Width/Length)
Mixed-Mode Device	(Width/Length)	0.9*(Width/Length)	0.9*(Width/Length)
RF Device	(Width/Length)	0.9*(Width/Length)	0.9*(Width/Length)
Contact size	0.16*0.16	0.144*0.144	0.144*0.144
Via size(1X;2X)	0.2*0.2; 0.4*0.4	0.18*0.18; 0.36*0.36	MVA1(W -0.02um) : 0.16*0.2 Other 1X MVA : 0.18*0.18 2X MVA (W -0.04um): 0.28*0.44
Metal size	Width/Space	0.9*(Width/Space)	0.9*(Width/Space)
Sloting inserting	Metal line width>5.556um	Metal line width>5um	

Note:

1. Because process bias make drawn and physical dimension are different, please refer to interconnect capacitance model (G-04 document in individual DSM) for precise interconnection-capacitance extraction.
2. Please refer to SPICE model (G-05 document in individual DSM) for precise device characteristics.

2 5 V N-ch MOSFET

2.1 Silicon Wafer for Modeling

The information about the silicon wafer used for characterization and model parameter extraction is summarized in the table below.

Table 2-1 Test wafer information for NMOS

	MOSFET DC model	MOSFET AC Model	Junction DC Model	Junction AC Model
Mask ID	BSW1PA	BSW1PA	BSW1PA	BSW1PA
Lot ID	D6FW8	D6FW8	D6FW8	D6FW8
Wafer Number	#2	#2	#2	#2

2.2 Electrical Parameters

The important electrical parameters for the characterization of this device are summarized in the table below.

Linear and saturation threshold voltage is extracted at
 $I_D = 300\text{nA} \cdot W_{DES} \cdot 0.9 / (L_{DES} \cdot 0.9 + 12\text{nm})$ for NMOS and
 $I_D = -70\text{nA} \cdot W_{DES} \cdot 0.9 / (L_{DES} \cdot 0.9 + 12\text{nm})$ for PMOS, respectively.

All parameters are given for $T=25\text{ C}$, $V_{BS}=0\text{ V}$ unless specified otherwise.

Table 2-2 Electrical parameters for NMOS

All parameters are given for $T=25\text{ }^\circ\text{C}$, $V_{BS}=0\text{ V}$ unless specified otherwise.

Target & Centered Value are given for $SA=3.6\mu$ $SB=3.6\mu$

Measured & Extracted Value are given for SA=3.6u SB=3.6u

Parameter	Condition	Unit	Target	Measured Value	Extracted Value	Centered Value
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Long channel device (W/L= 10 μm / 10 μm)

V_{TLIN}	$V_{DS} = 0.1 \text{ V}$	V	0.562	0.564	0.565	0.562
I_{ON}	$V_{DS} = 5 \text{ V}$	$\mu\text{A}/\mu\text{m}$	71.16	71.34	71.00	70.81

Wide and short channel device (W/L= 10 μm / 0.5 μm)

V_{TLIN}	$V_{DS} = 0.1 \text{ V}$	V	0.552	0.552	0.550	0.552
V_{TSAT}	$V_{DS} = 5 \text{ V}$	V	0.440	0.441	0.449	0.441
Body Effect V_T shift	$V_{DS} = 5 \text{ V}$ $V_{BS} = 0 \sim -3 \text{ V}$	V	--	0.407	0.505	0.504
DIBL	$V_{DS} = 0.1 \sim 5 \text{ V}$	V	0.112	0.111	0.100	0.111
I_{ON}	$V_{DS} = V_{GS} = 5 \text{ V}$	$\mu\text{A}/\mu\text{m}$	612	613.46	615.49	612.00
I_{OFF}	$V_{DS} = 5 \text{ V}, V_{GS} = 0 \text{ V}$	A/ μm	7.0E-12	1.1E-11	7.7E-12	7.7E-12

Narrow and long channel device (W/L= 0.35 μm / 10 μm)

V_{TLIN}	$V_{DS} = 0.1 \text{ V}$	V	0.558	0.559	0.559	0.558
V_{TSAT}	$V_{DS} = 5 \text{ V}$	V	0.545	0.548	0.541	0.540
I_{ON}	$V_{DS} = V_{GS} = 5 \text{ V}$	$\mu\text{A}/\mu\text{m}$	62.86	64.08	63.68	62.86

Narrow and short channel device (W/L= 0.35 μm / 0.5 μm)

V_{TLIN}	$V_{DS} = 0.1 \text{ V}$	V	0.521	0.529	0.514	0.521
V_{TSAT}	$V_{DS} = 5 \text{ V}$	V	0.422	0.424	0.405	0.423
I_{ON}	$V_{DS} = V_{GS} = 5 \text{ V}$	$\mu\text{A}/\mu\text{m}$	702.86	708.29	712.43	702.86
I_{OFF}	$V_{DS} = 5 \text{ V}, V_{GS} = 0 \text{ V}$	A/ μm	1.36E-11	9.1E-11	3.0E-11	1.4E-11

Please set gmindc=4.0543e-12 in SPECTRE simulator for achieving the consistent I_{OFF} result with HSPICE's

2.3 MOSFET DC Model

2.3.1 Test devices

The geometry of devices that were used for MOSFET DC model parameter extraction is summarized below.

Table 2-3 Geometry of NMOS devices measured for DC parameter extraction

W/L	10	5	1	0.8	0.6	0.5
10	X	X	X	X	X	X
5	X		X			X
1	X	X	X	X	X	X
0.8	X		X			X
0.6	X	X	X	X	X	X
0.43	X		X		X	X
0.35	X	X	X	X	X	X

2.3.2 Measurement conditions

The drain current was measured at various bias conditions summarized in the table below:

Table 2-4 Test conditions for IV curves of NMOS

I_D vs. V_{GS}				I_D vs. V_{DS}			
	Start	Stop	Step		Start	Stop	Step
V_{DS} (V)	0.1	5	4.9	V_{DS} (V)	0	5.5	0.05
V_{GS} (V)	-0.5	5.5	0.05	V_{GS} (V)	0.9	5	0.41
V_{BS} (V)	0	-3	-0.6	V_{BS} (V)	0	-3	-3

2.3.3 Model verification plots

The simulation results obtained from the extracted model were plotted together with the measurement results for comparison. The L and W in the plots refer to physical dimension (LDES *0.9+12nm and WDES *0.9) respectively.

I_D vs. V_{DS} plots

The following plots show measured and simulated output I-V curves for selected devices at 25 C. (The dimensions are after 90% shrink and 12nm sizing channel length back.)

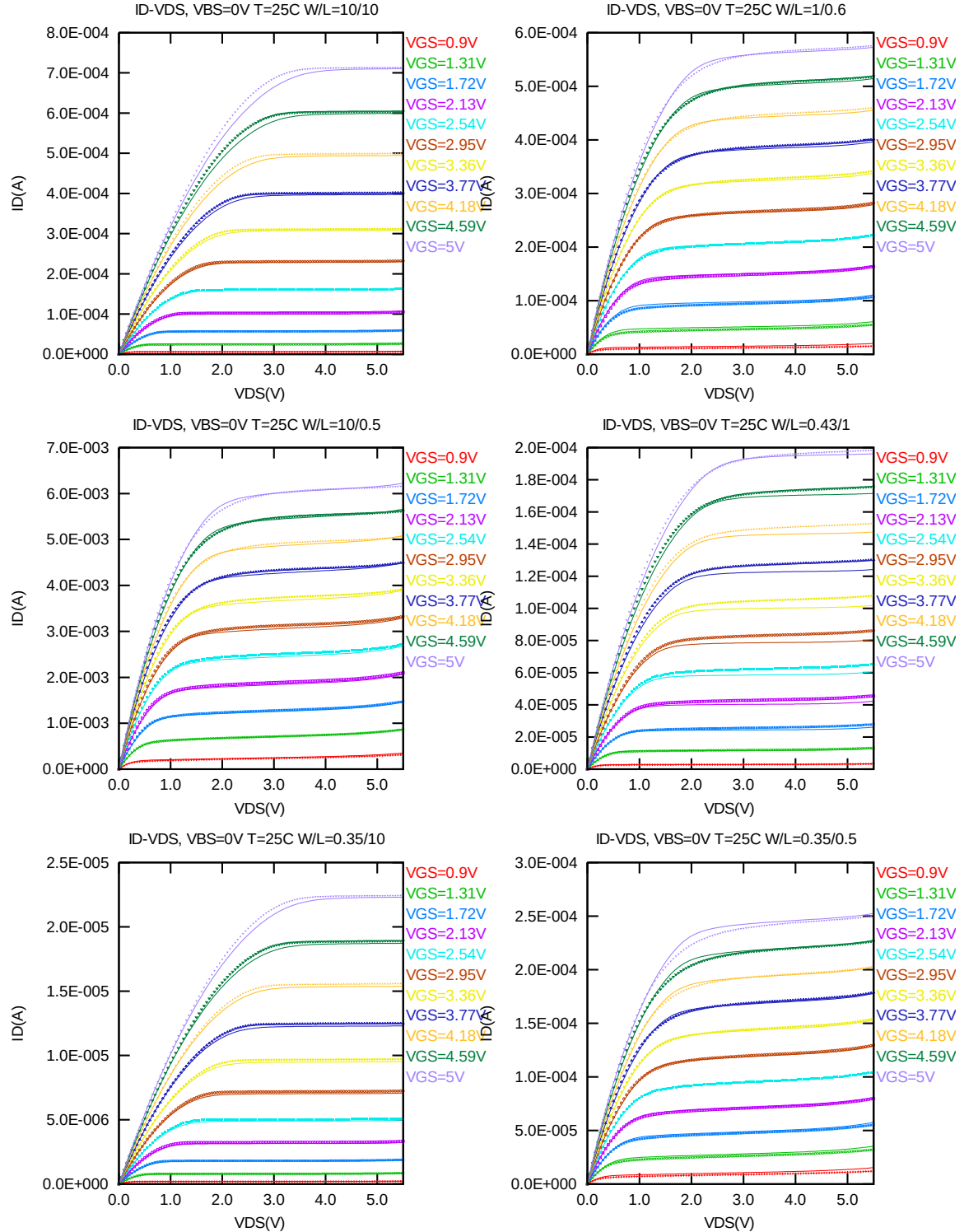


Figure 2-1 I_D vs. V_{DS} at $V_{BS}=0V$ for NMOS

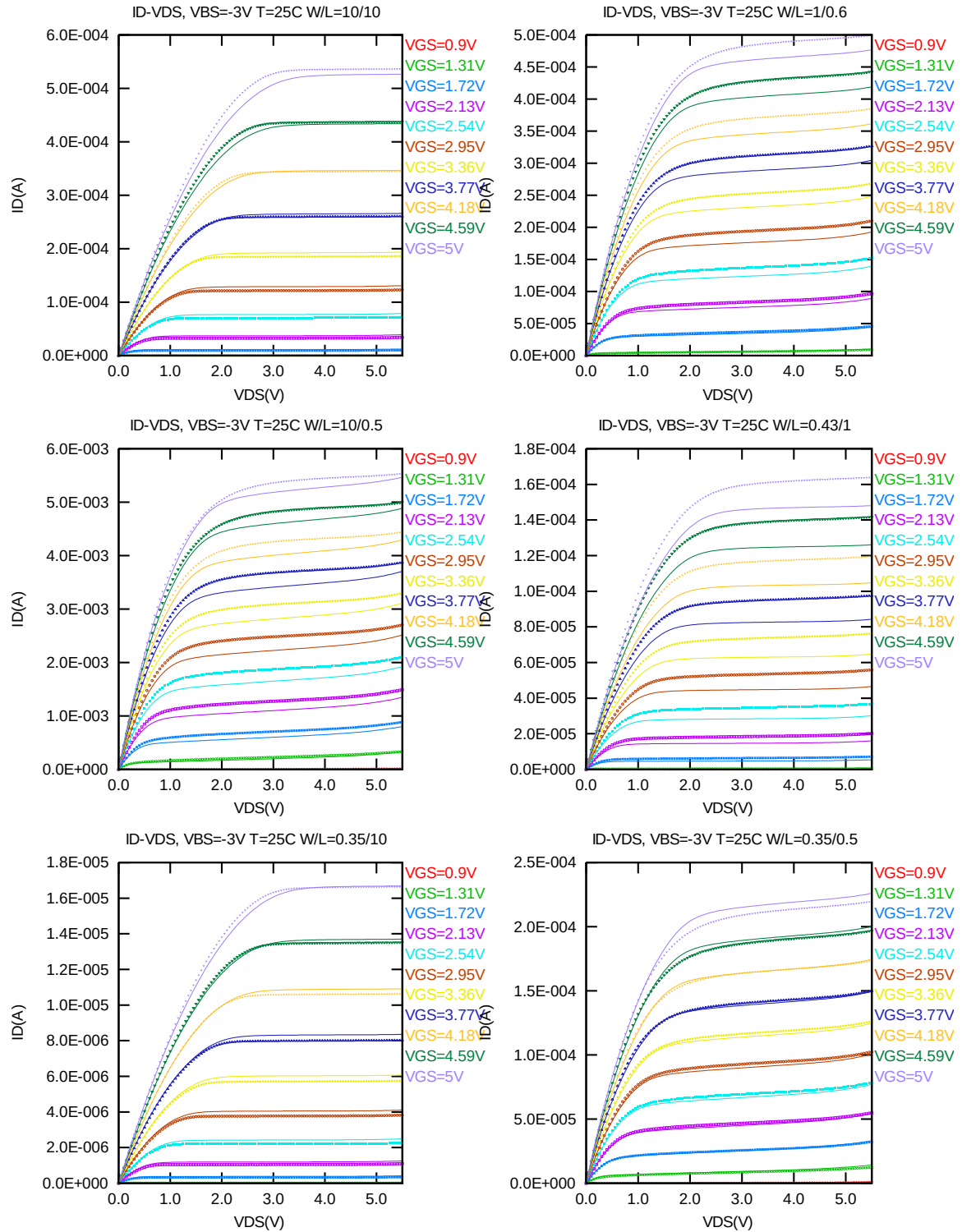


Figure 2-2 I_D vs. V_{DS} at $V_{BS} = -3V$ for NMOS

G_{DS} vs. V_{DS} plots

The following plots show measured and simulated output I-V curves for selected devices at 25 C. (The dimensions are after 90% shrink and 12nm sizing channel length back.)

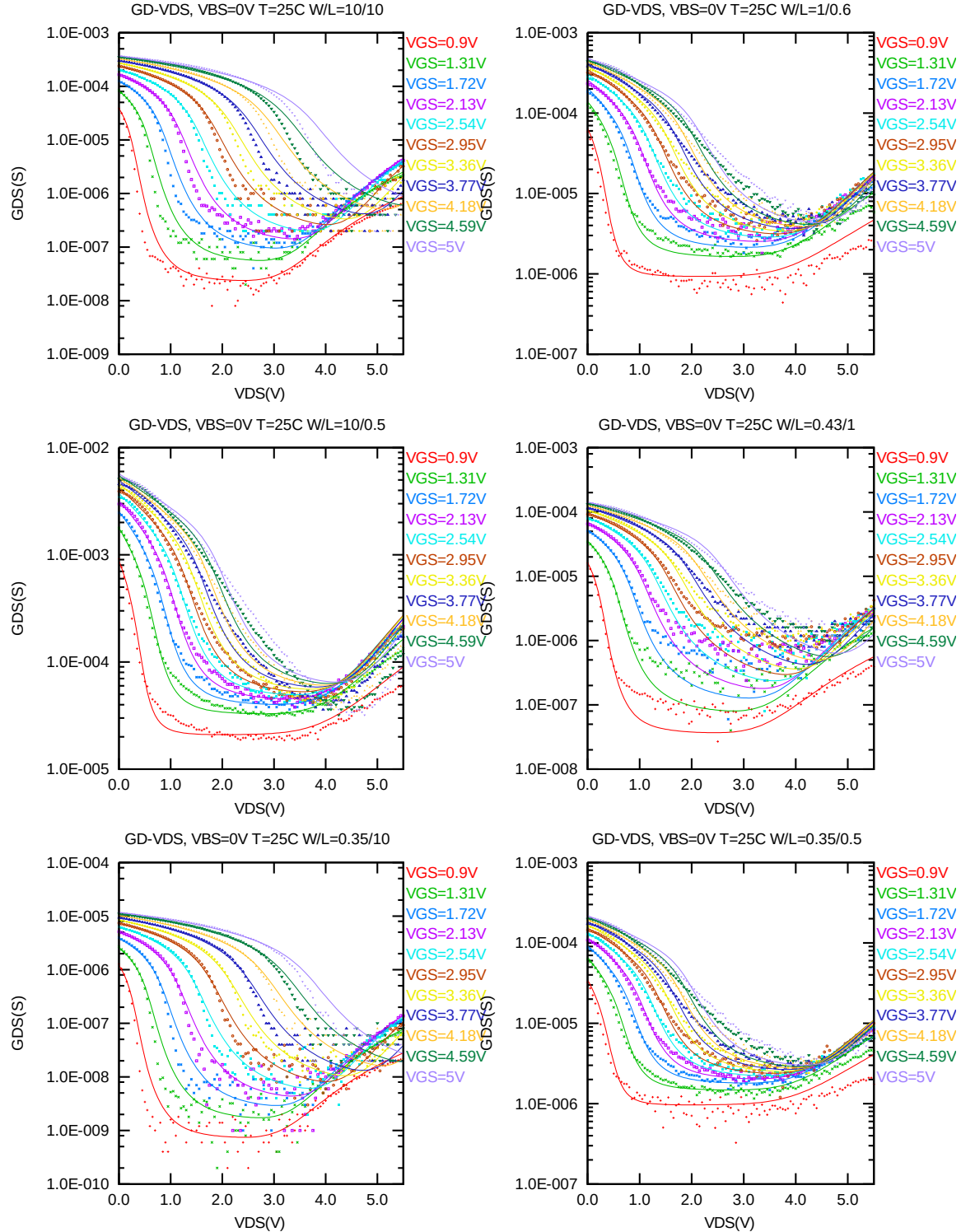


Figure 2-3 G_{DS} vs. V_{DS} at $V_{BS}=0V$ for NMOS

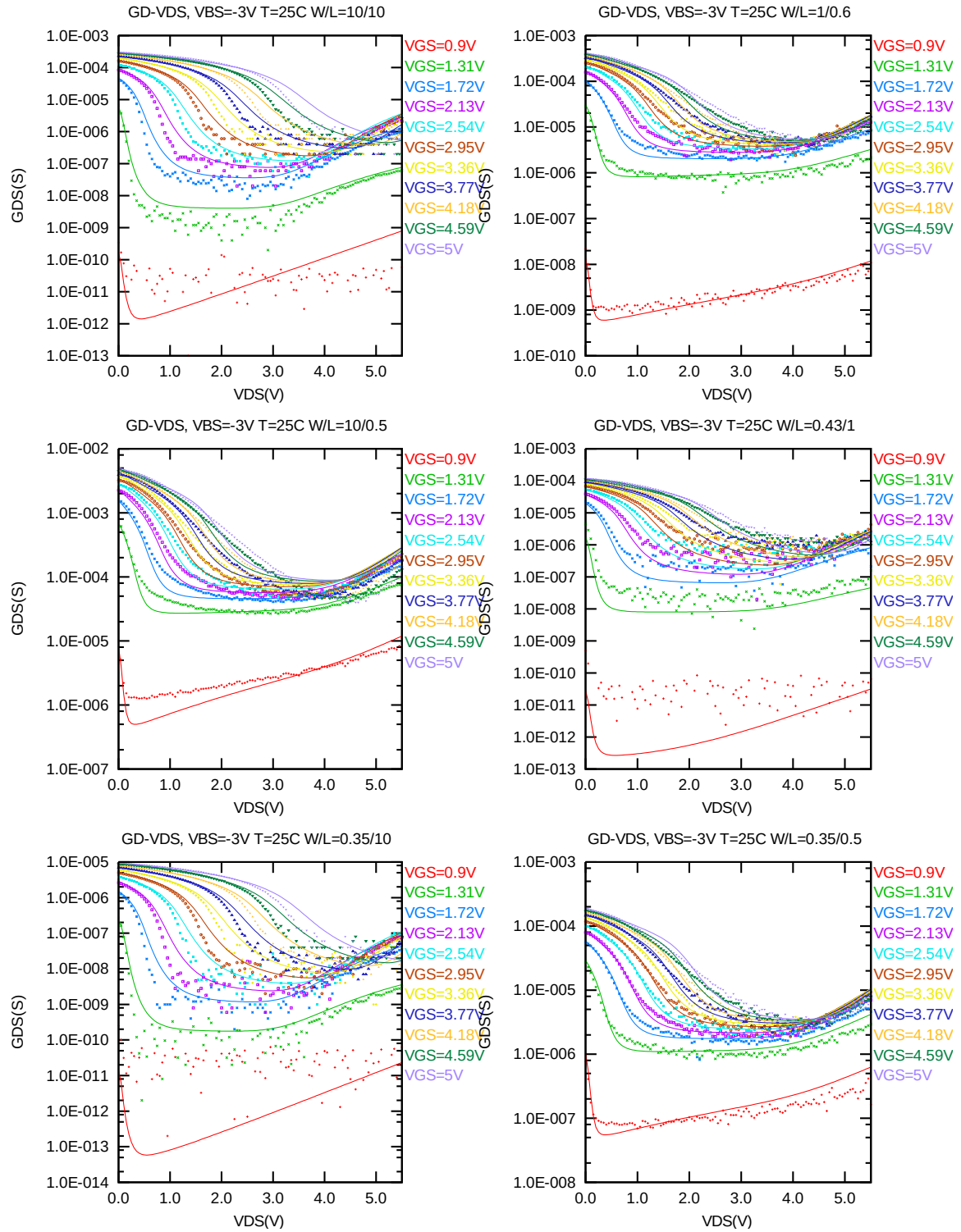


Figure 2-4 G_{DS} vs. V_{DS} at $V_{BS} = -3V$ for NMOS

I_D vs. V_{GS} plots (linear scale)

The following plots show measured and simulated output I-V curves for selected devices at 25 C. (The dimensions are after 90% shrink and 12nm sizing channel length back.)

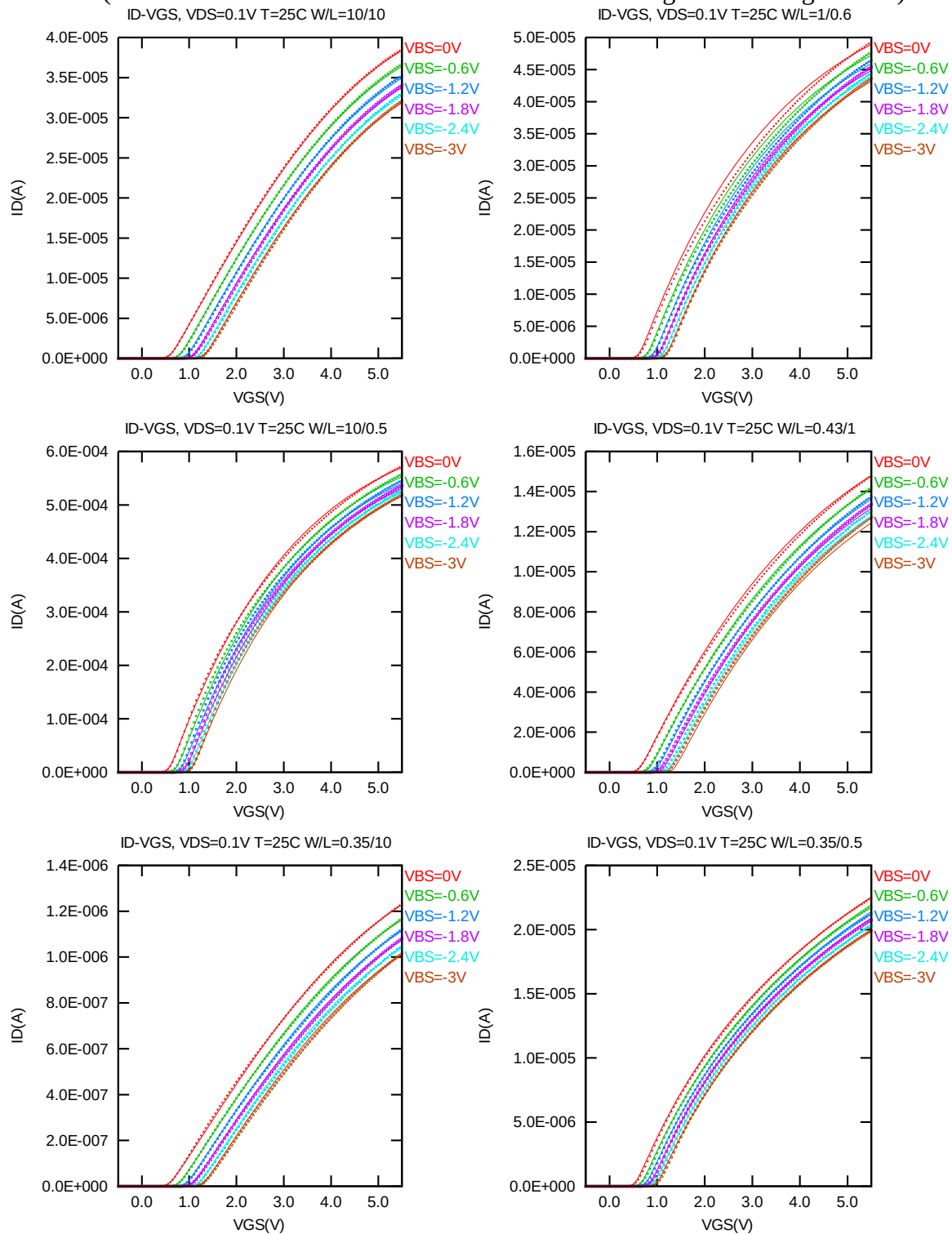


Figure 2-5 I_D vs. V_{GS} at $V_{DS}=0.1V$ for NMOS

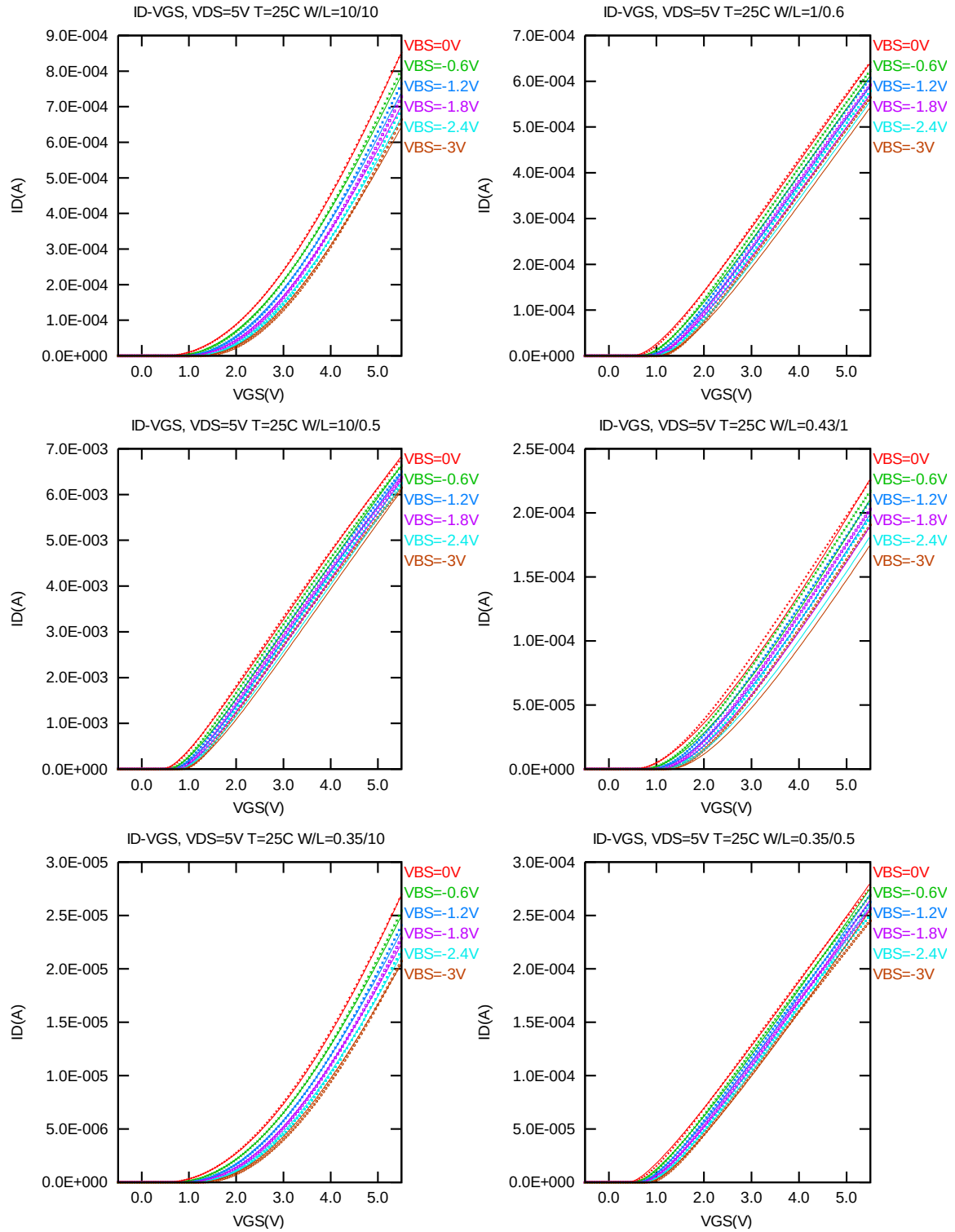


Figure 2-6 I_D vs. V_{GS} at $V_{DS}=5V$ for NMOS

I_D vs. V_{GS} plots (logarithmic scale)

The following plots show measured and simulated output I-V curves for selected devices at 25 C. (The dimensions are after 90% shrink and 12nm sizing channel length back.)

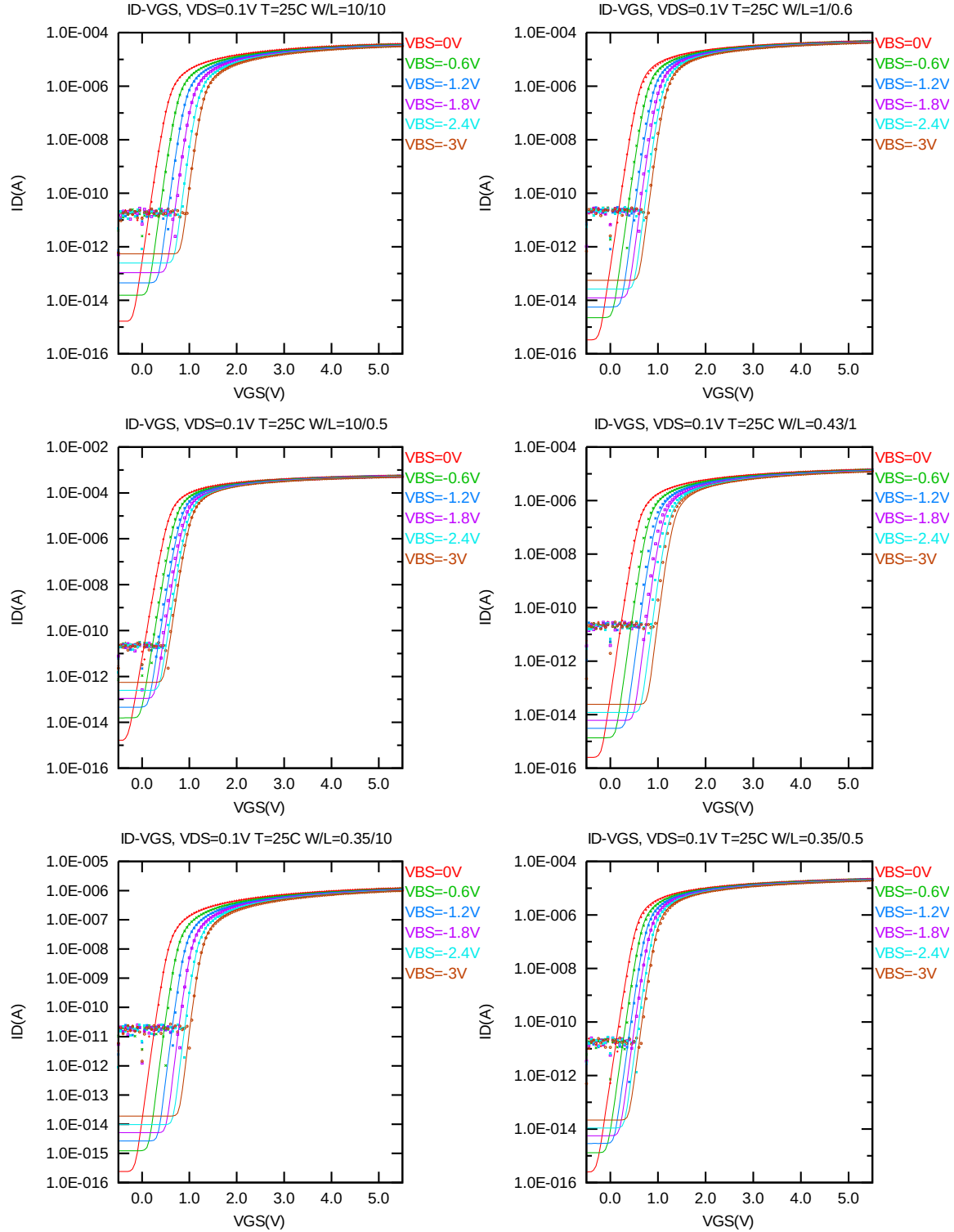


Figure 2-7 Log scaled I_D vs. V_{GS} at $V_{DS}=0.1$ V for NMOS

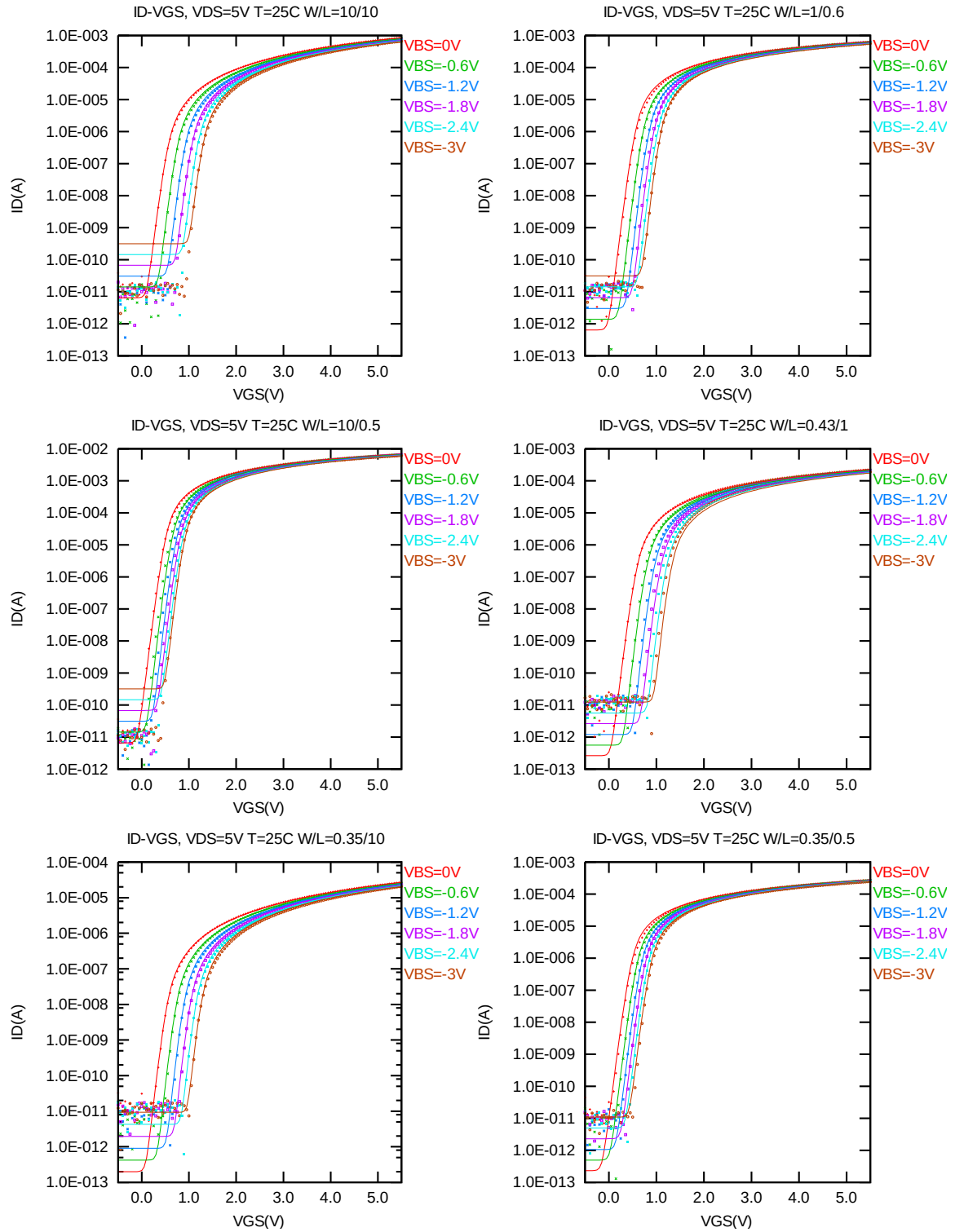


Figure 2-8 Log scaled I_D vs. V_{GS} at $V_{DS}=5V$ for NMOS

G_M vs. V_{GS} plots

The following plots show measured and simulated output I-V curves for selected devices at 25 C. (The dimensions are after 90% shrink and 12nm sizing channel length back.)

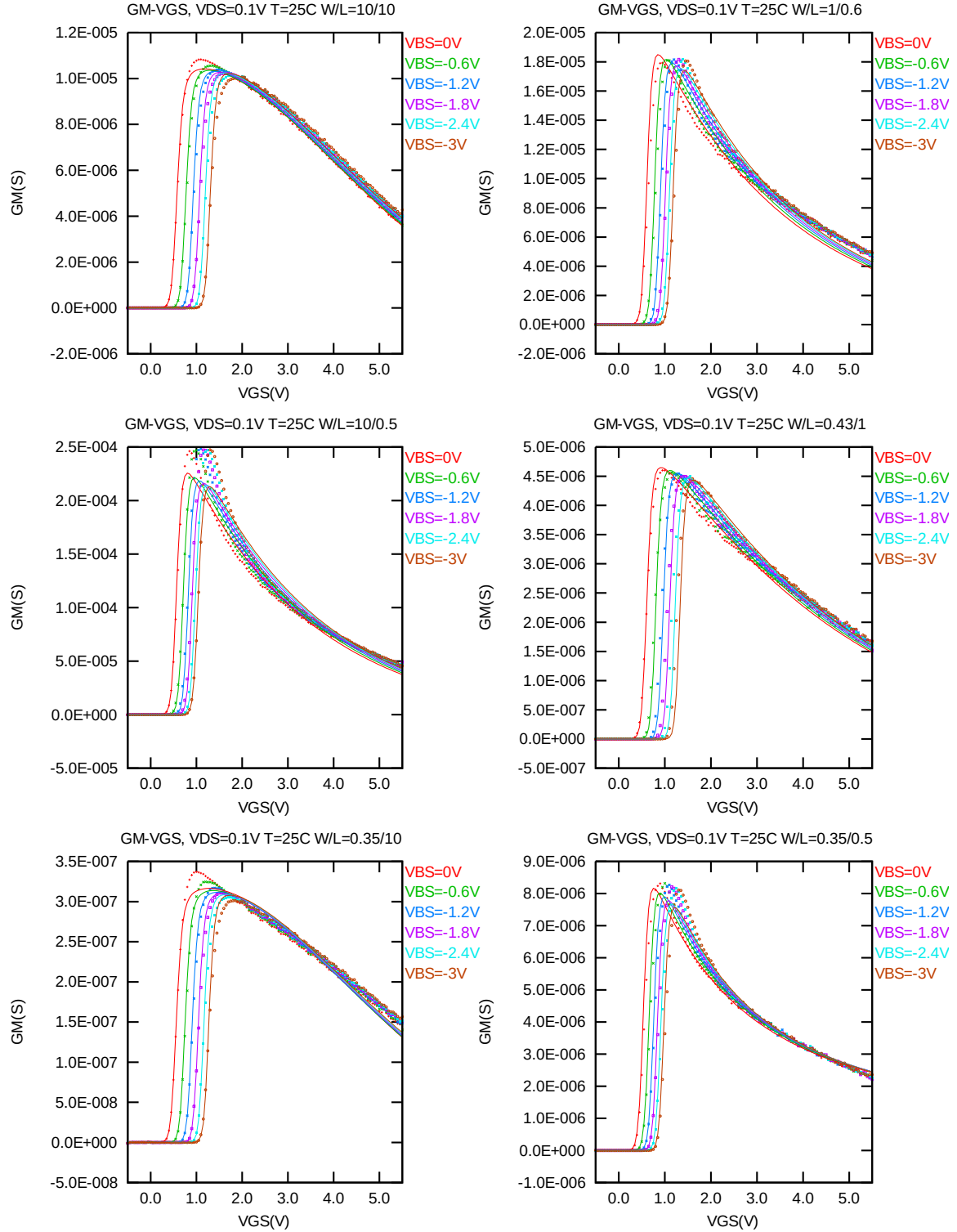


Figure 2-9 G_M vs. V_{GS} at $V_{DS}=0.1V$ for NMOS

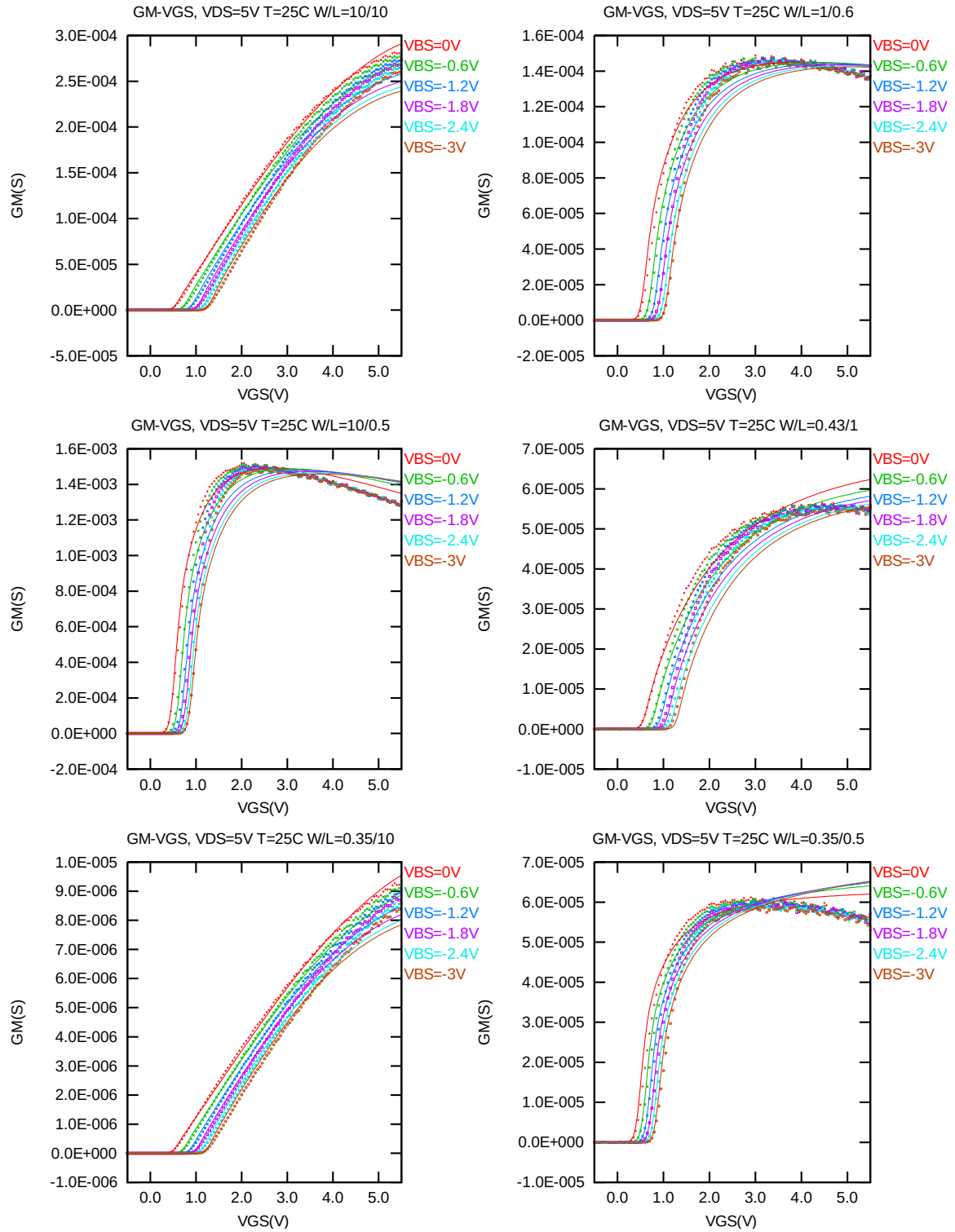


Figure 2-10 G_M vs. V_{GS} at $V_{DS}=5V$ for NMOS

The L and W in the plots refer to L_{DES} and W_{DES} respectively.

The following figures show the channel length and width dependence of linear and saturation threshold voltage extracted at $I_D = 300\text{nA} \cdot W_{DES} \cdot 0.9 / (L_{DES} \cdot 0.9 + 12\text{nm})$ at 25°C . The plots show the physical dimension.

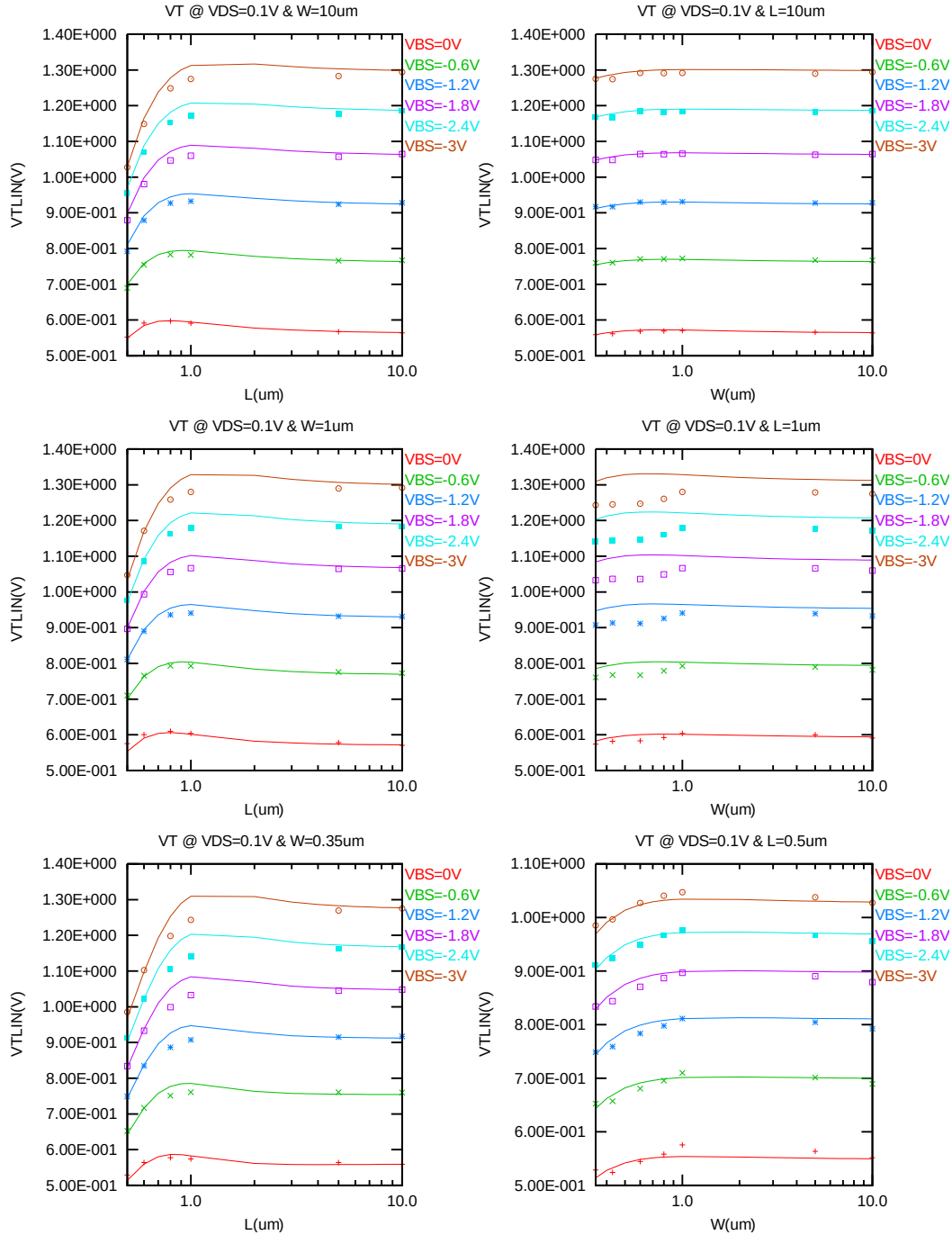


Figure 2-11 V_T vs. channel length & channel width for low drain bias ($V_{DS}=0.1\text{V}$) for NMOS

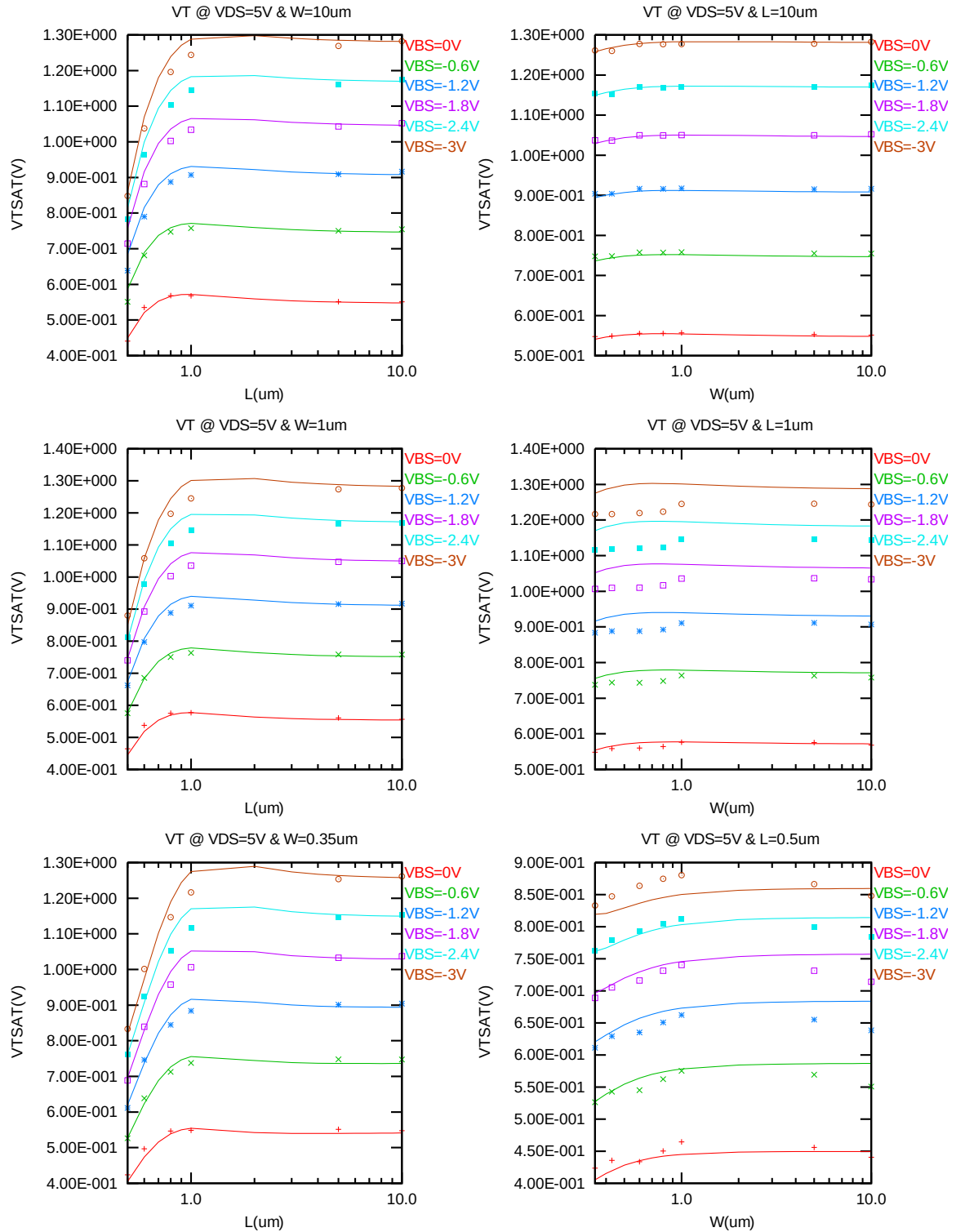


Figure 2-12 V_T vs. channel length & channel width for high drain bias ($V_{DS}=5V$) for NMOS

The L and W in the plots refer to L_{DES} and W_{DES} respectively.

The following plots show the channel length and width dependence of I_{ON} extracted at $V_{DS}=V_{GS}=5\text{ V}$ for $V_{BS}=0\text{ V}$ and $V_{BS}=-3\text{ V}$ at 25 C. (The dimensions are after 90% shrink and 12nm sizing channel length back.)

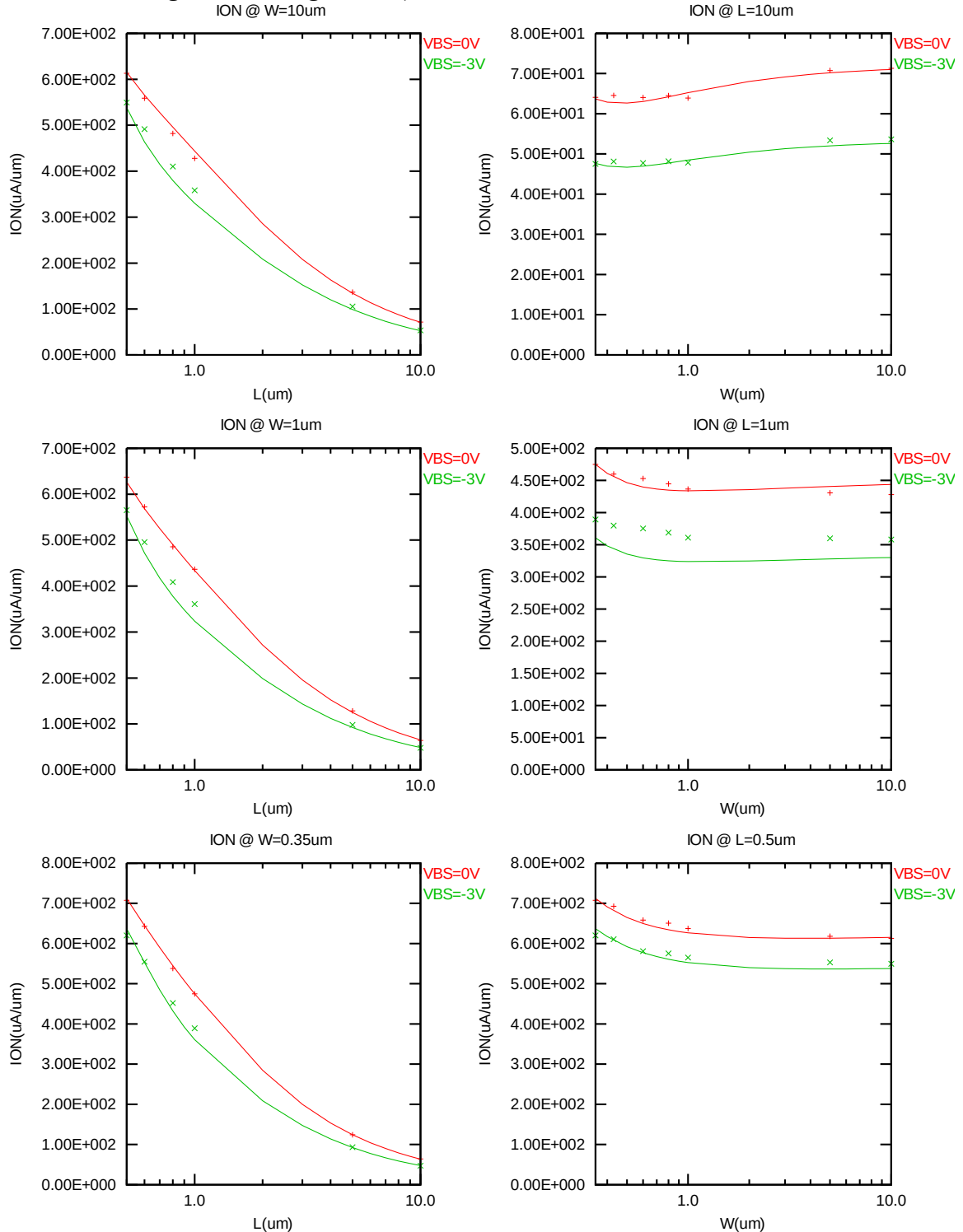


Figure 2-13 Saturation current I_{ON} vs. channel length and channel width for NMOS

The L and W in the plots refer to LDES and WDES respectively.

The following plots show the channel length and width dependence of I_{OFF} extracted at $V_{DS}=5\text{ V}$ and $V_{GS}=0\text{ V}$ for $V_{BS}=0\text{ V}$ to -3 V at 25 C . A large discrepancy of I_{OFF} can be seen in the graphs, which is due to the presence of junction leakage current.

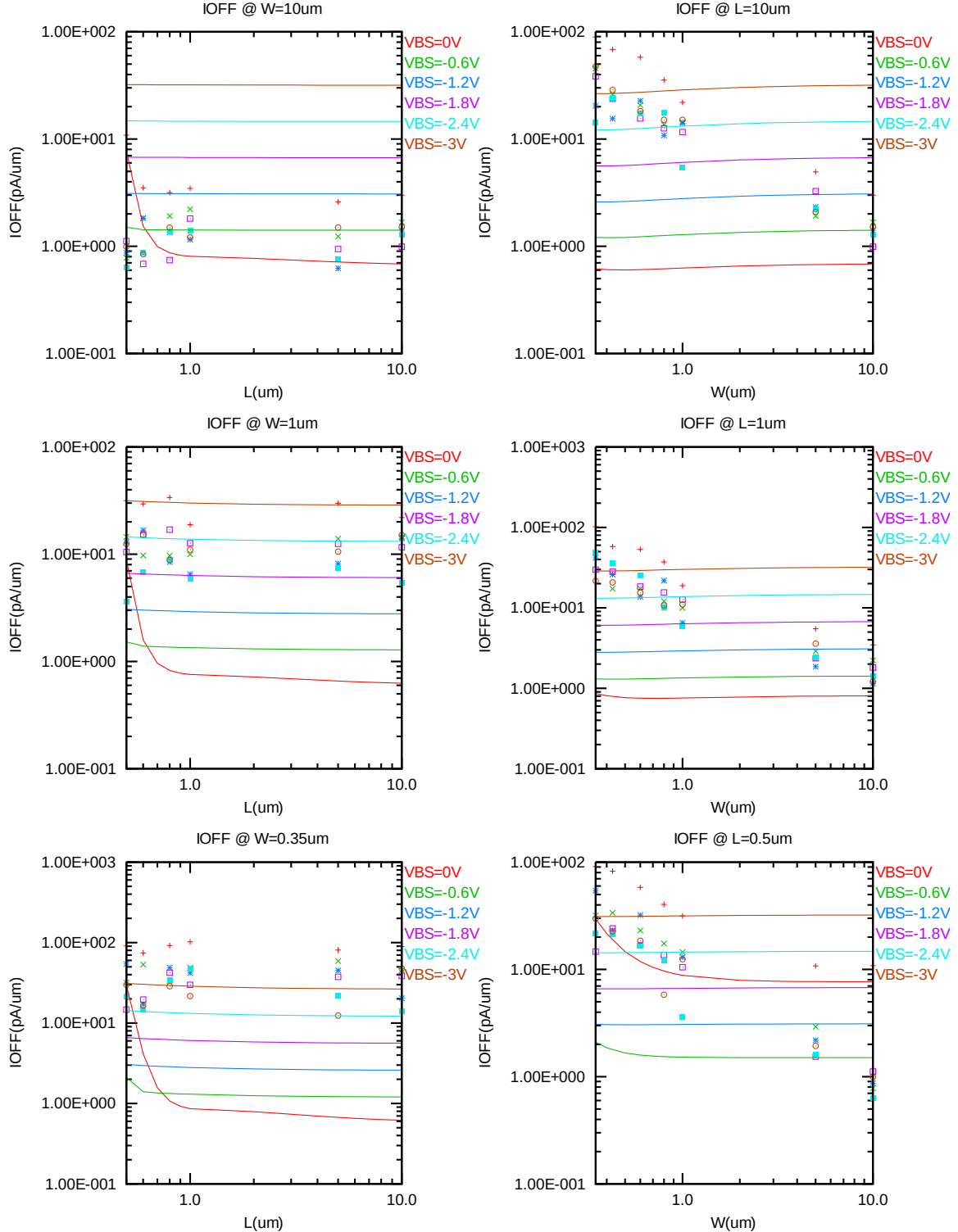


Figure 2-14 I_{OFF} vs. channel length and channel width for NMOS

V_{TLIN} vs. temperature

The following plots show the temperature dependence of the linear and saturation threshold voltage extracted at $I_D=300nA \cdot (W_{DES} \cdot 0.9) / (L_{DES} \cdot 0.9 + 0.012 \mu m)$. (The dimensions are after 90% shrink and 12nm sizing channel length back.)

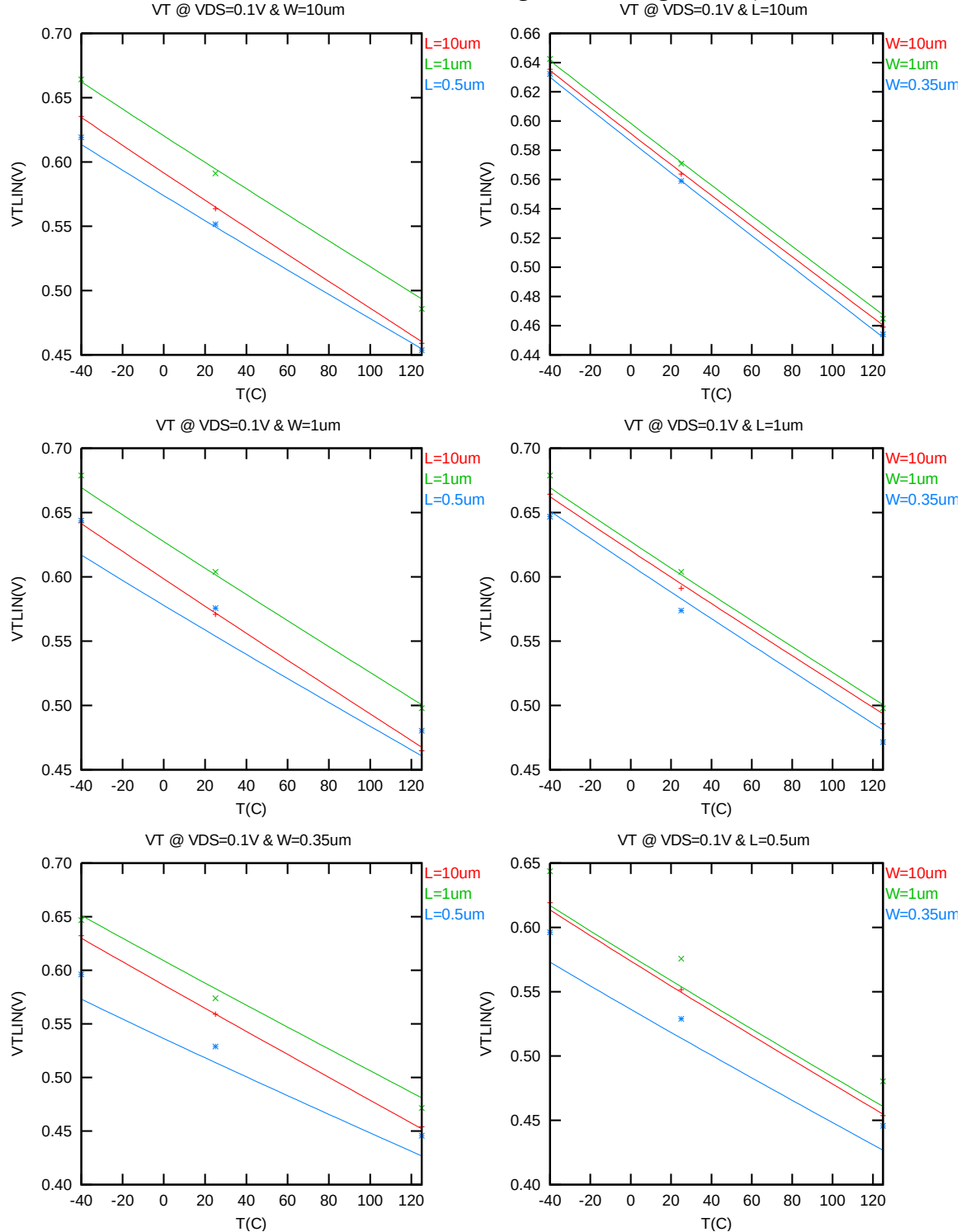


Figure 2-15 V_T vs. temperature for NMOS for low drain bias ($V_{DS}=0.1V$)

V_{TSAT} vs. temperature

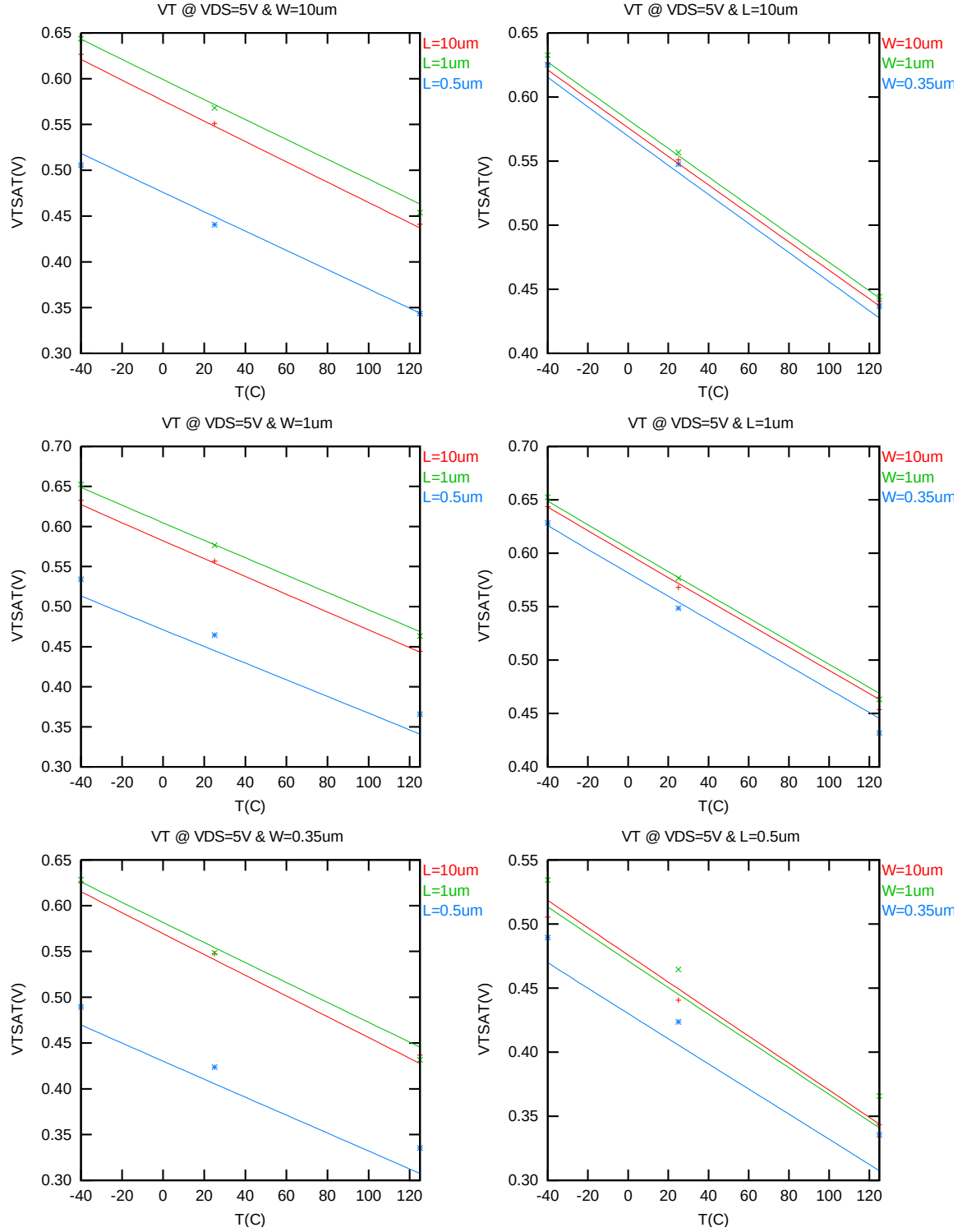


Figure 2-16 V_T vs. temperature for NMOS for high drain bias ($V_{DS}=5V$)

I_{ON} vs. temperature

The following plots show the temperature dependence of I_{ON} extracted at $V_{GS}=V_{DS}=5$ V, $V_{BS}=0$ V. (The dimensions are after 90% shrink and 12nm sizing channel length back.)

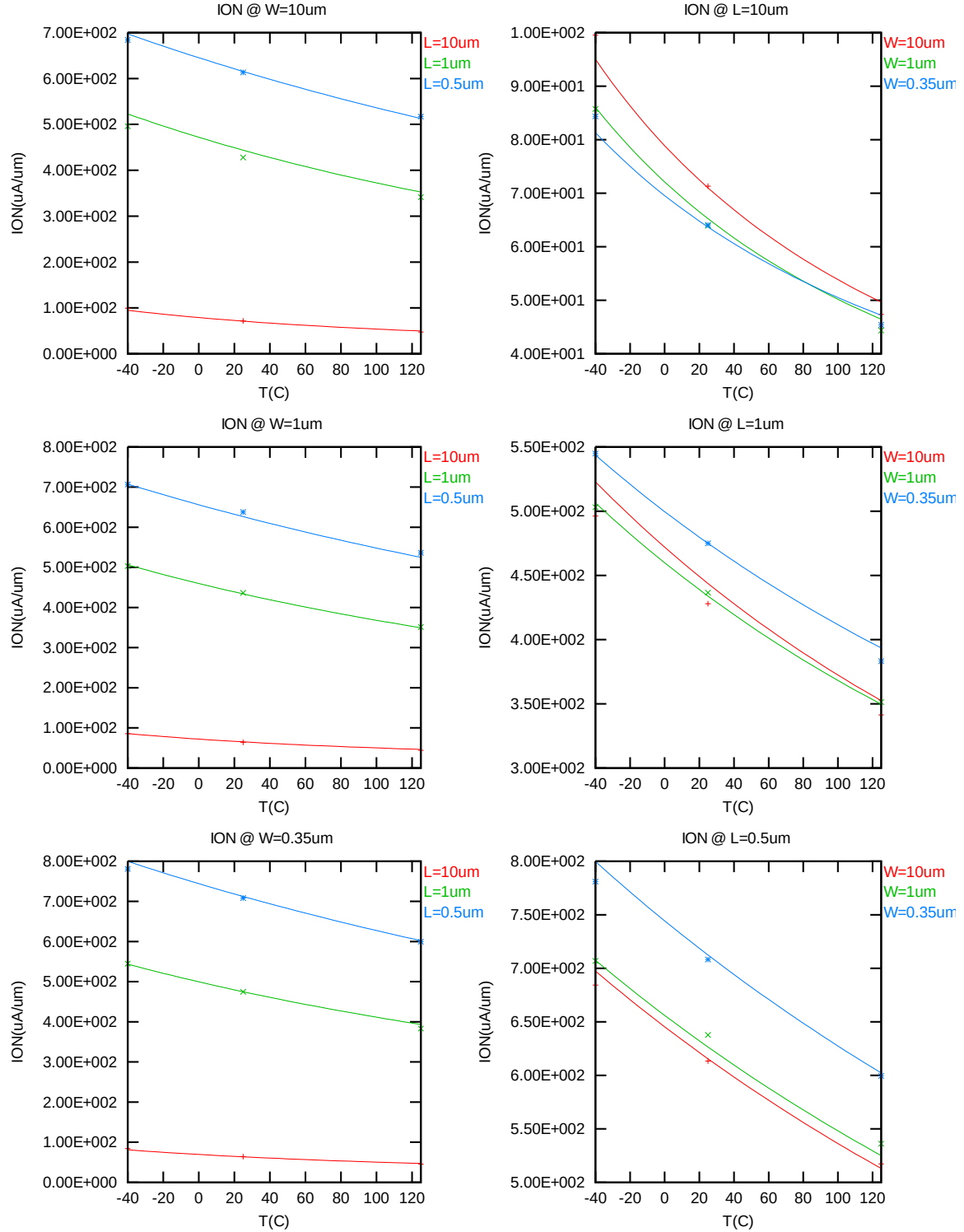


Figure 2-17 Saturation current vs. temperature for NMOS

I_{OFF} vs. temperature

The following plots show the temperature dependence of I_{OFF} extracted at $V_{DS}=5$ V, $V_{GS}=V_{BS}=0$ V. (The dimensions are after 90% shrink and 12nm sizing channel length back.)

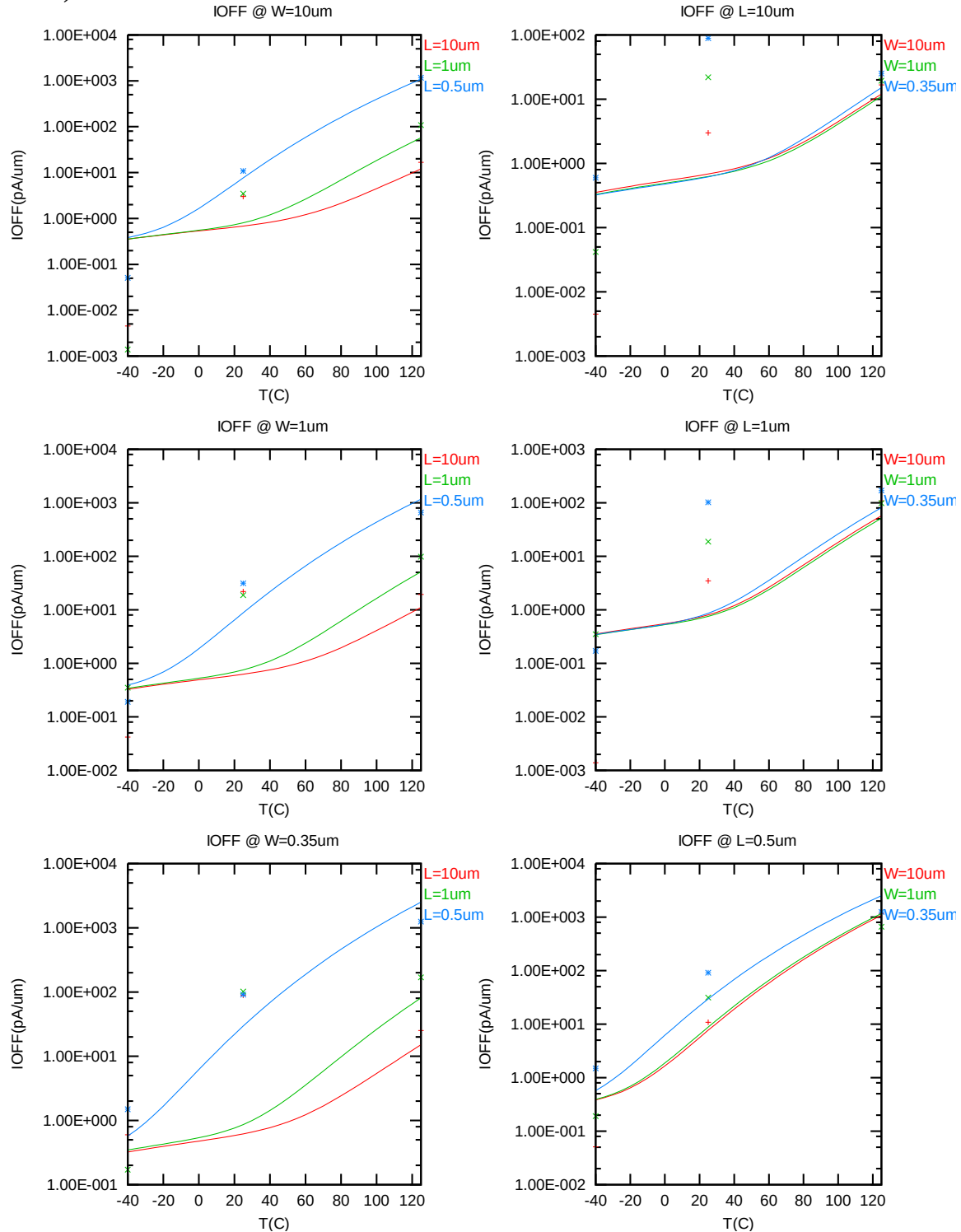


Figure 2-18 I_{OFF} vs. temperature for NMOS

2.3.4 Model fitting accuracy

The model fitting accuracy of parameters I_D , G_M and G_{DS} is quantified using the following equation:

$$\text{error \%} = 100 * | \text{measurement} - \text{simulation} | / [(\text{measurement} + \text{simulation}) / 2]$$

Different error criteria are defined for different parameters (I_D , G_M and G_{DS}) as well as for different operation regions. These criteria are explained below.

1) I_D accuracy criteria

- a. I_D in sub-threshold region, i.e. $V_{GS} < V_T$
- b. I_D in moderate inversion region, i.e. $V_T \leq V_{GS} < V_T + V_X$
- c. I_D in strong inversion region, i.e. $V_T + V_X \leq V_{GS} \leq V_{CC}$

, where $V_X = (V_{CC} - V_T) / 2$

2) G_M and G_{DS} accuracy criteria

- d. G_M and G_{DS} in sub-threshold region, i.e. $V_{GS} < V_T$
- e. G_M and G_{DS} in inversion region, i.e. $V_T \leq V_{GS} \leq V_{CC}$

Table 2-5 Requirements for model fit quality

Target of Quality Check	$V_{GS} < V_T$ (sub-threshold)	$V_T \leq V_{GS} < V_T + V_X$ (moderate inversion)	$V_T + V_X \leq V_{GS} \leq V_{CC}$ (strong inversion)
I_D	$\leq 75\%$	$\leq 25\%$	$\leq 7\%$
G_M	$\leq 50\%$	$\leq 20\%$	$\leq 20\%$
G_{DS}	$\leq 100\%$	$\leq 50\%$	$\leq 50\%$

I_D accuracy

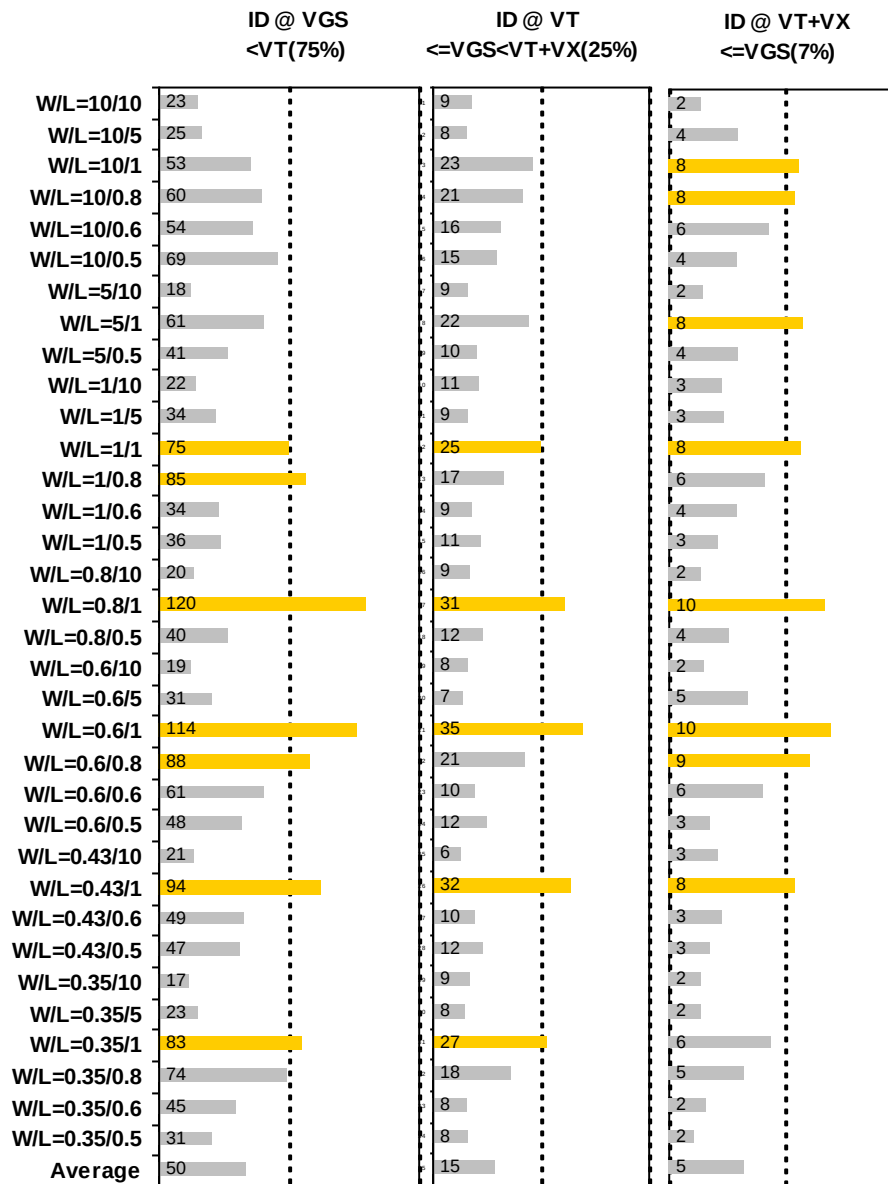


Figure 2-19 I_D accuracy at 25 C for NMOS

G_M and G_{DS} accuracy

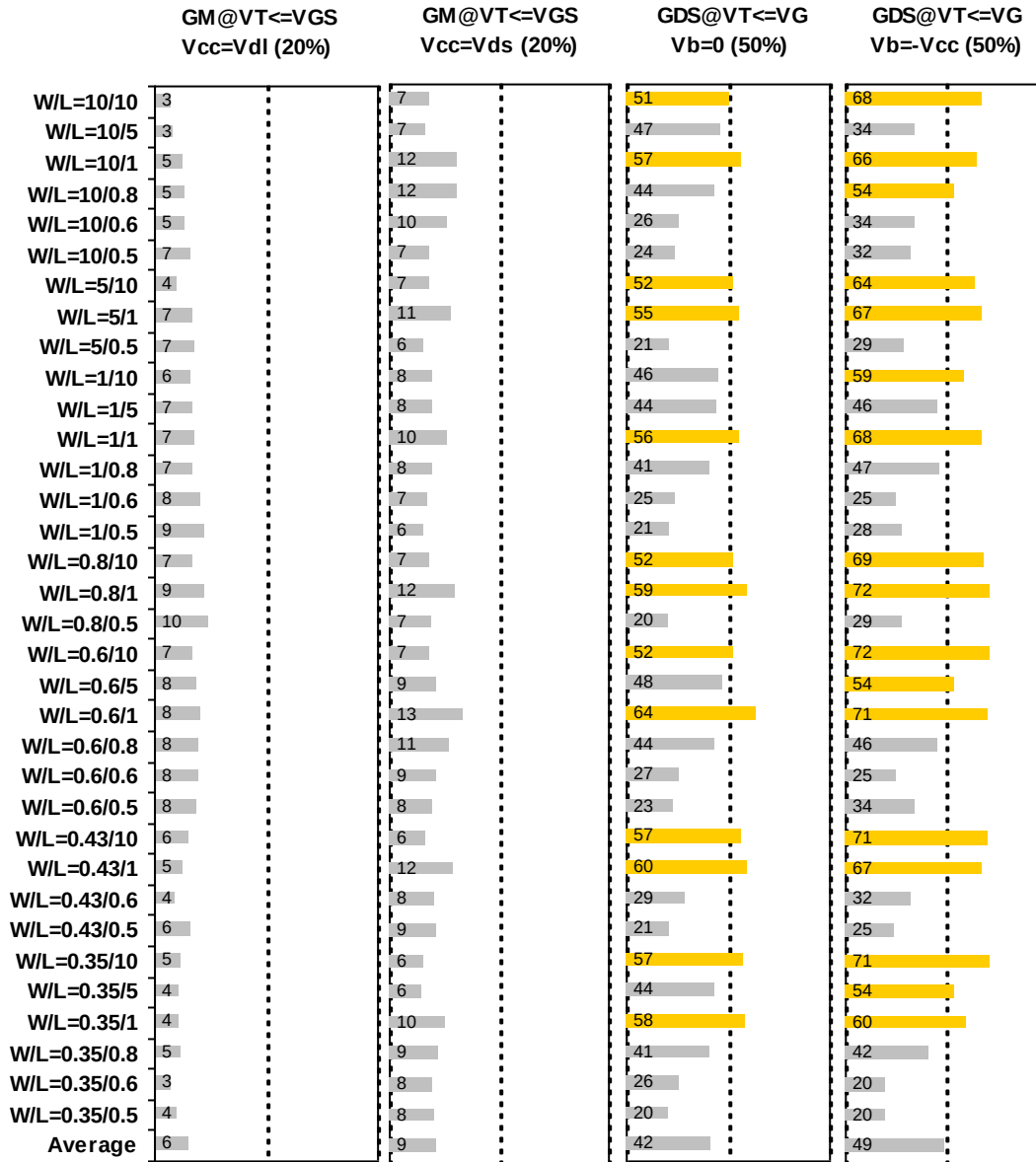


Figure 2-20 G_M and G_{DS} accuracy at 25 C for NMOS

2.4 MOSFET Capacitance Model

2.4.1 Test devices

The following test structures were used for MOSFET capacitance model parameter extraction.

Table 2-6 Measurement structure for NMOS C_{gg} capacitance model

Structure	L[μm]	W[μm]
Gate capacitance components, C _{GDS}	80	80

Table 2-7 Large area measurement structure for NMOS C_{gc} capacitance model

Structure	L[μm]	W[μm]
Gate capacitance components	.5	100
	2	100
	1	100
	2	2.08
	.5	2.08
	2	1.72
	.5	1.72
	2	1.36
	.5	1.36
	2	1
	.5	1
	2	.64
	.5	.64
	2	.35

	.5	.35
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2.4.2 Measurement conditions

The measurement conditions for MOSFET capacitance are summarized in Table 2-8 below.

Table 2-8 Measurement conditions for NMOS capacitance model

V _{BIAS} [V]	From -5.5 to 5.5 V in 0.1 V step
Frequency [kHz]	100
V _{OSC} [mV]	50
Temp [°C]	25

2.4.3 T_{ox} extraction

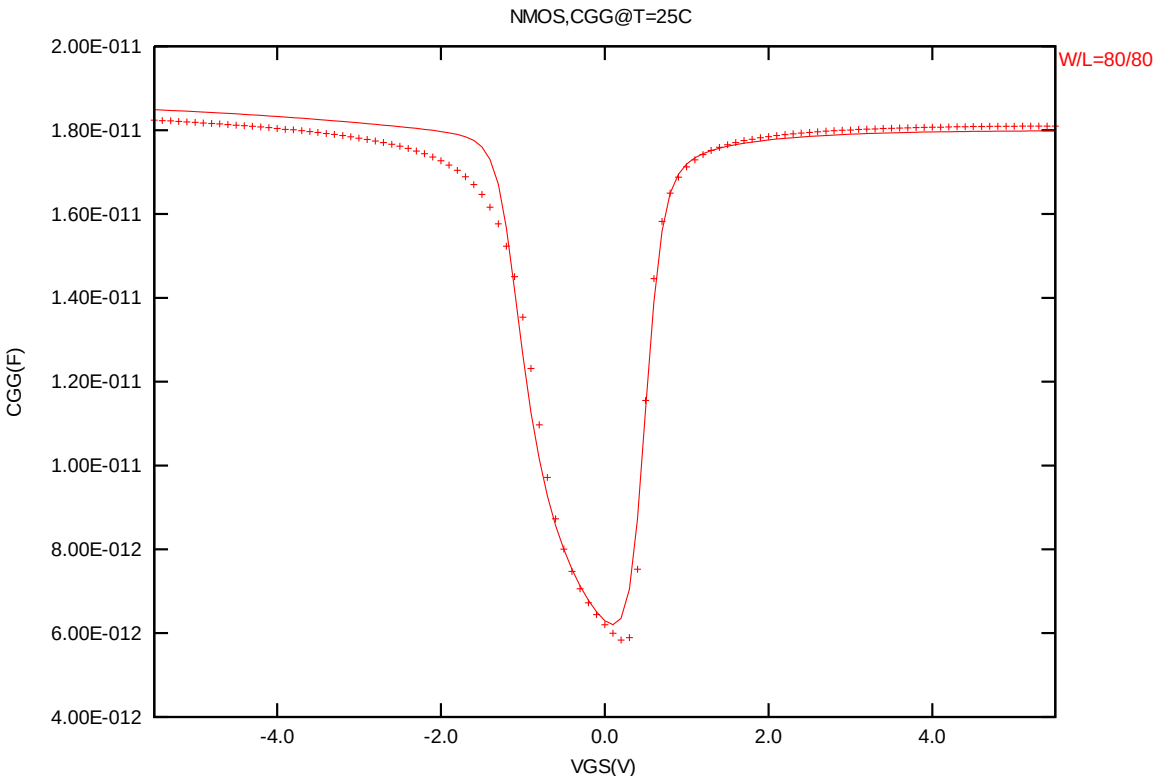
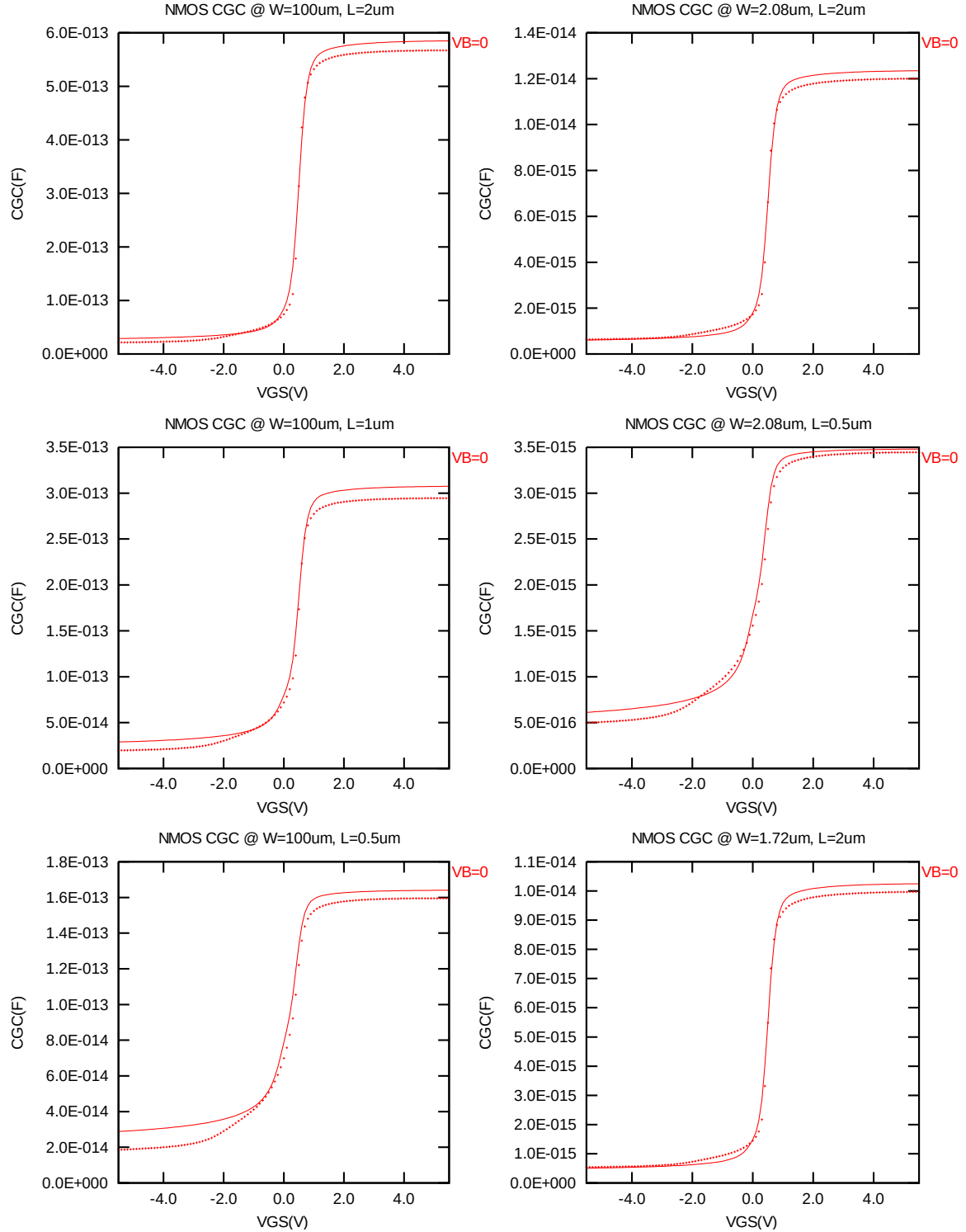
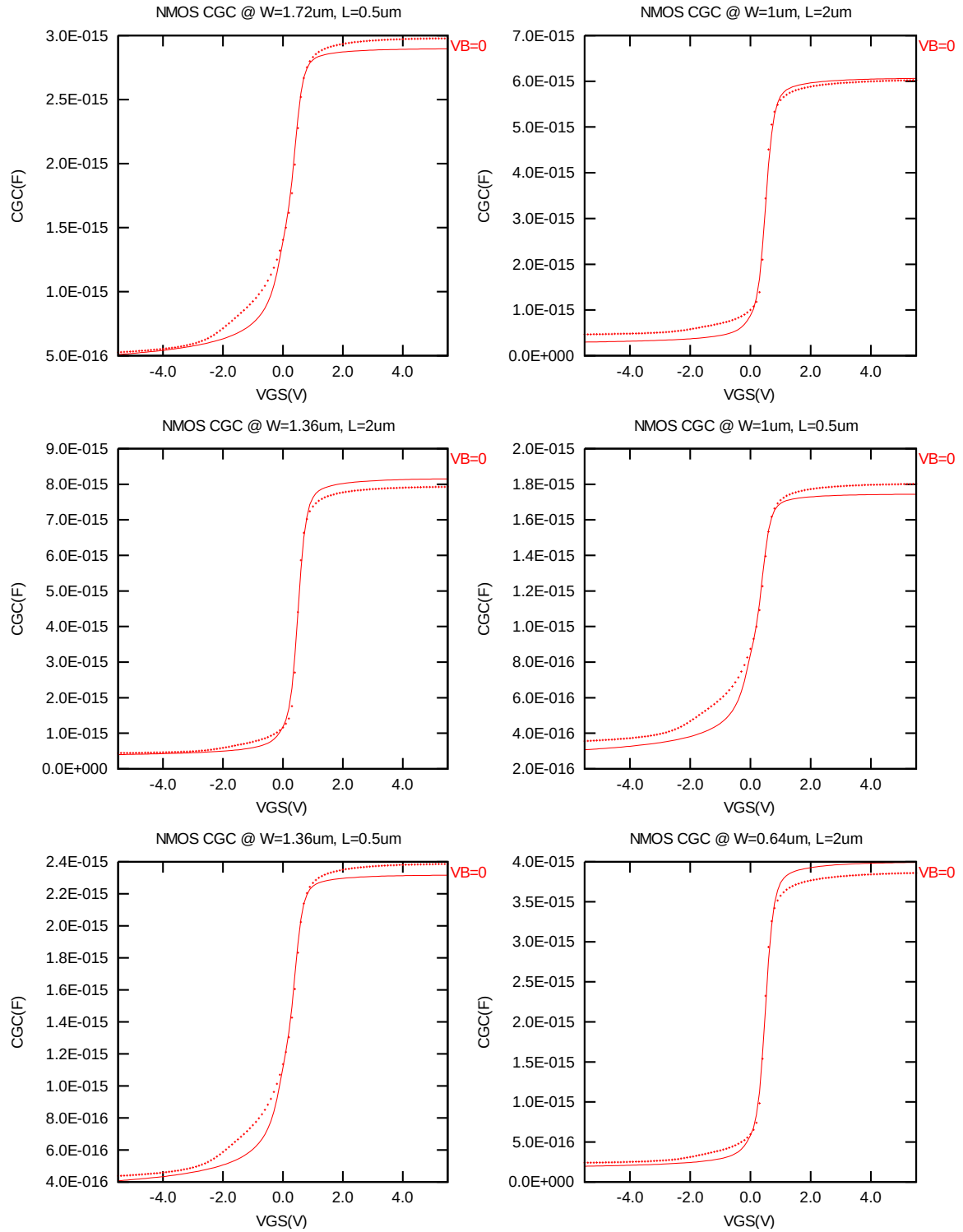


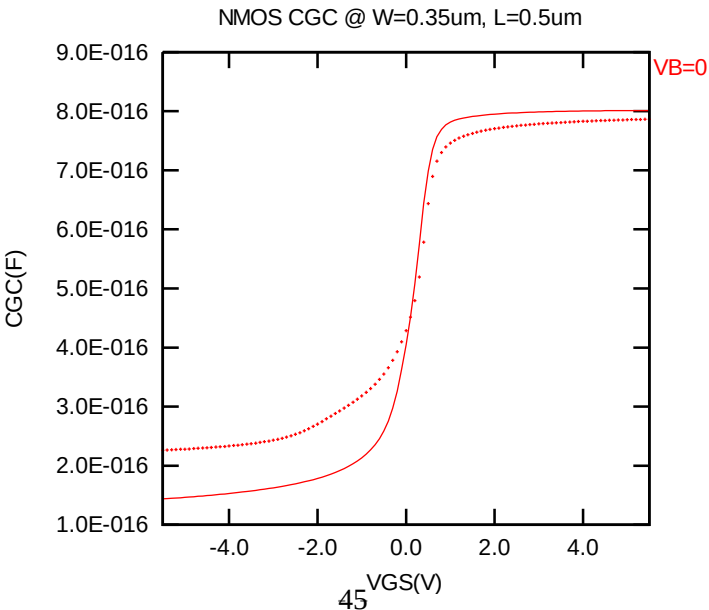
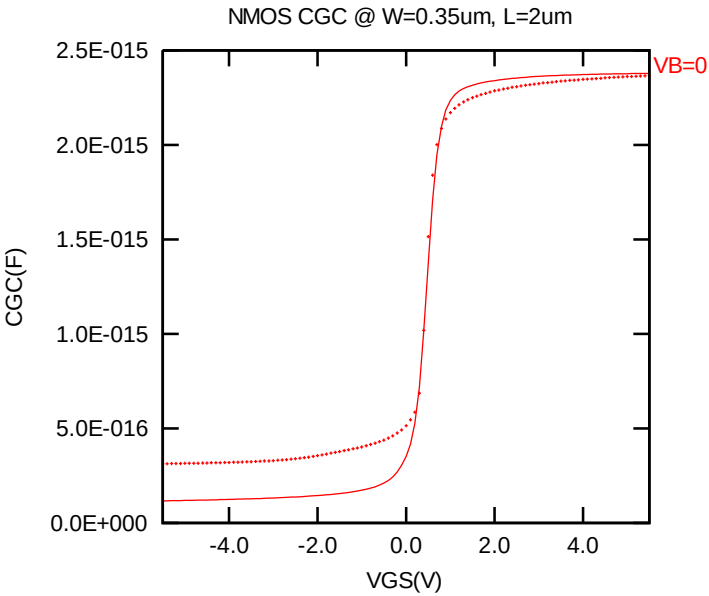
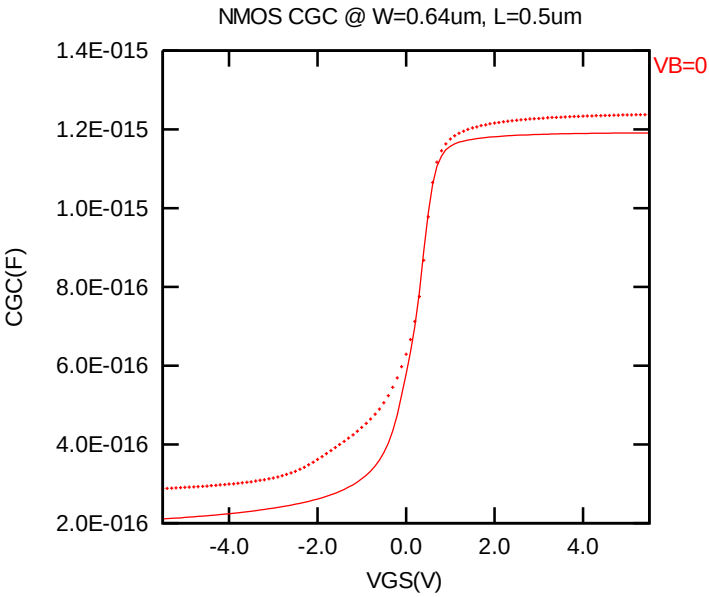
Figure 2-21 Measured and simulated C_{GG} curves

2.4.4 Width and length AC model accuracy

Width and length AC model accuracy







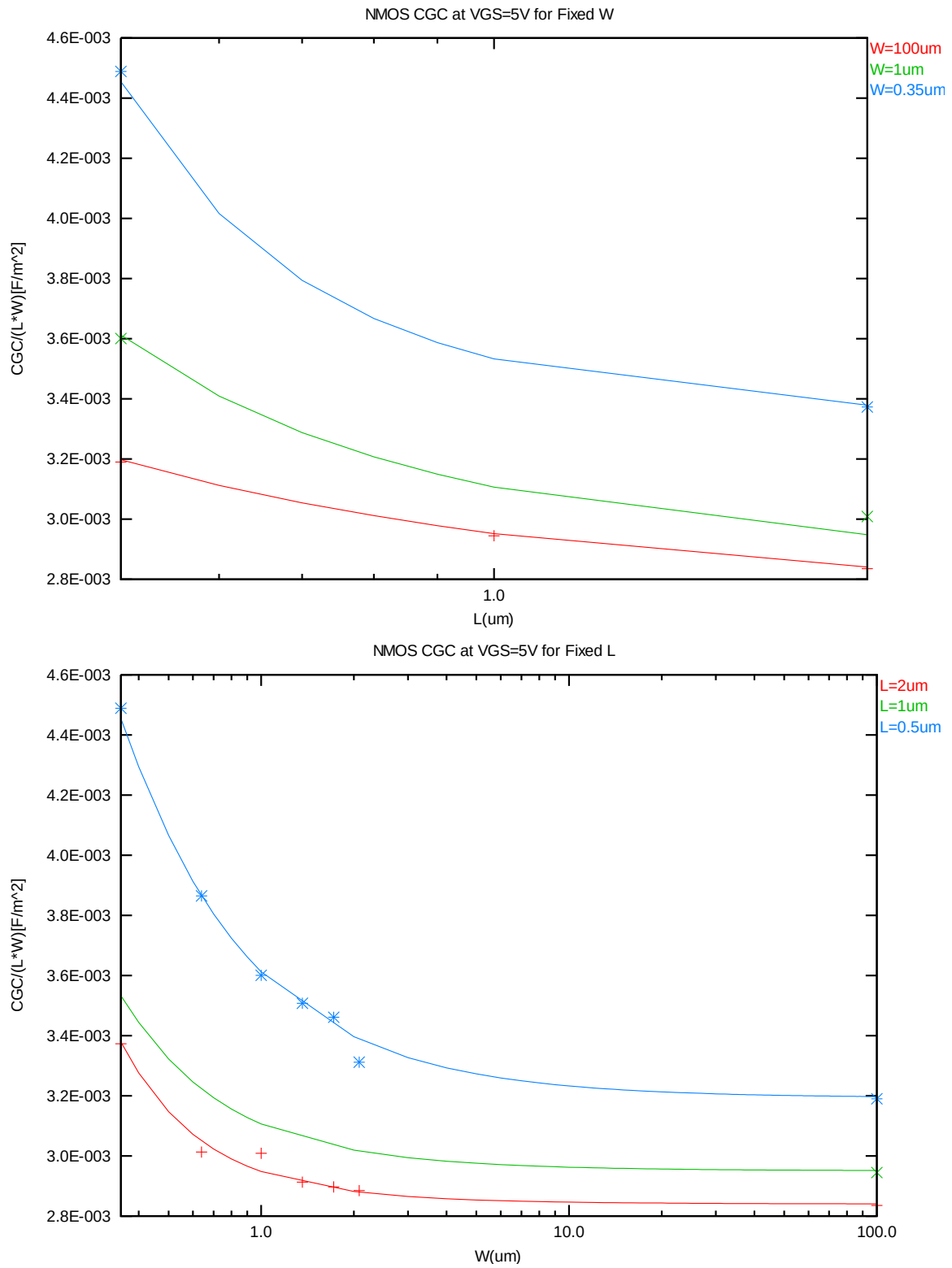


Figure 2-22 C_{GC} vs. channel length & channel width for NMOS

2.5 Parasitic Source / Drain Junction Capacitance Model

2.5.1 Test devices

The following test structures were used for parameter extraction.

Table 2-9 Junction capacitor structures for NMOS

Structures	Area[m ²]	Perimeter[m]	Description
Bottom area junction capacitance	5.00E-08	2.00E-03	Plate junction capacitor
STI bounded sidewall junction capacitance	3.60E-08	3.64E-02	Finger diode
Inner gate sidewall junction	2.76E-08	1.82E-02	Poly finger structure

2.5.2 Measurement conditions

All junction capacitance was measured under voltage sweep conditions summarized in the following table.

Table 2-10 Junction capacitor measurement conditions for NMOS

V _{BIAS} [V]	VH: pwell & VL: n+ From 0 to 5.5 V in 0.05 V step
Frequency [kHz]	100
V _{OSC} [mV]	50
Temp [°C]	-40,25,150

2.5.3 Extracted model parameters and verification plots

CJ, CJSW vs. Vbias plots

The following plots show the variation of bottom area junction and STI bounded sidewall at different bias voltages and temperatures.

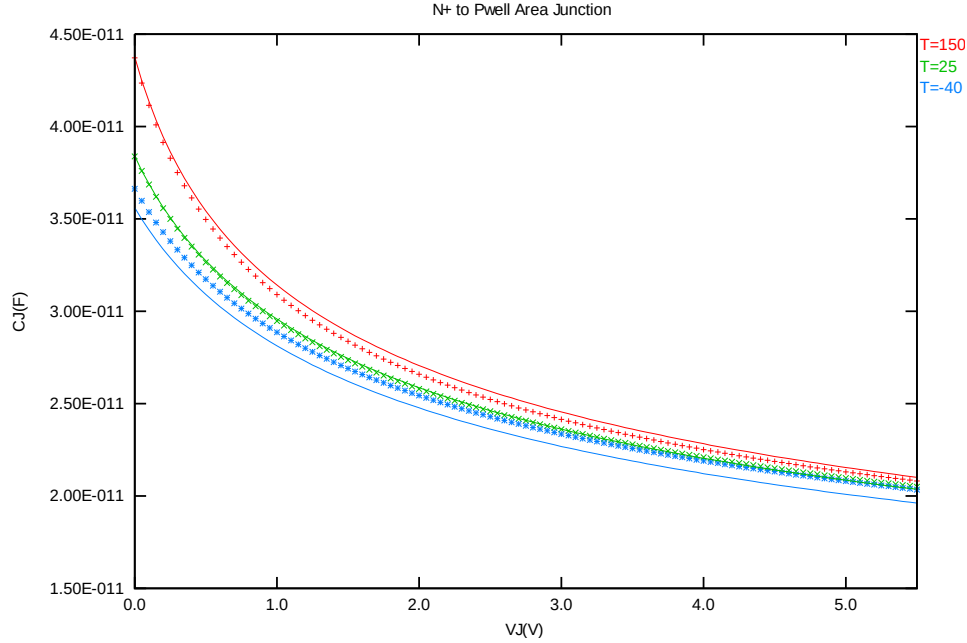


Figure 2-23 Bottom area junction capacitance at -40 C, 25 C and 150 C for NMOS

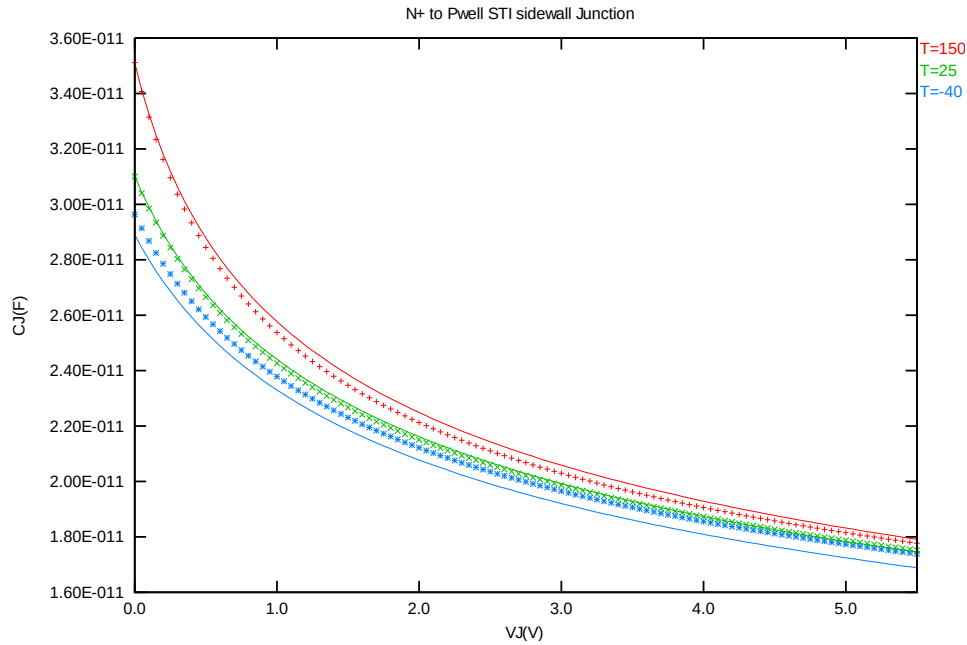


Figure 2-24 STI bounded sidewall junction capacitance at -40 C, 25 C and 150 C for NMOS

3 5 V P-ch MOSFET

3.1 Silicon Wafer for Modeling

The information about the silicon wafer used for characterization and model parameter extraction is summarized in the table below.

Table 3-1 Test wafer information for PMOS

	MOSFET DC model	MOSFET AC Model	Junction DC Model	Junction AC Model
Mask ID	BSW1PA	BSW1PA	BSW1PA	BSW1PA
Lot ID	D6FW8	D6FW8	D6FW8	D6FW8
Wafer Number	#2	#2	#2	#2

3.2 Electrical Parameters

The important electrical parameters for the characterization of this device are summarized in the table below.

Linear and saturation threshold voltage is extracted at
 $I_D = 300\text{nA} \cdot W_{DES} \cdot 0.9 / (L_{DES} \cdot 0.9 + 12\text{nm})$ for NMOS and
 $I_D = -70\text{nA} \cdot W_{DES} \cdot 0.9 / (L_{DES} \cdot 0.9 + 12\text{nm})$ for PMOS, respectively.

All parameters are given for $T=25\text{ C}$, $V_{BS}=0\text{ V}$ unless specified otherwise.

Table 3-2 Electrical parameters for PMOS

All parameters are given for $T=25\text{ }^{\circ}\text{C}$, $V_{BS}=0\text{ V}$ unless specified otherwise.

Target & Centered Value are given for $SA=3.6\mu$ $SB=3.6\mu$

Measured & Extracted Value are given for SA=3.6u SB=3.6u

Parameter	Condition	Unit	Target	Measured Value	Extracted Value	Centered Value
-----------	-----------	------	--------	----------------	-----------------	----------------

Long channel device (W/L= 10 μm / 10 μm)

V_{TLIN}	$V_{\text{DS}} = -0.1 \text{ V}$	V	-0.652	-0.652	-0.645	-0.651
I_{ON}	$V_{\text{DS}} = -5 \text{ V}$	$\mu\text{A} / \mu\text{m}$	-15.91	-15.97	-15.83	-15.91

Wide and short channel device (W/L= 10 μm / 0.5 μm)

V_{TLIN}	$V_{\text{DS}} = -0.1 \text{ V}$	V	-0.610	-0.610	-0.612	-0.610
V_{TSAT}	$V_{\text{DS}} = -5 \text{ V}$	V	-0.560	-0.558	-0.546	-0.560
Body Effect V_{T} shift	$V_{\text{DS}} = -5 \text{ V}$ $V_{\text{BS}} = 0 \sim 3 \text{ V}$	V	--	-0.596	-0.703	-0.685
DIBL	$V_{\text{DS}} = -0.1 \sim -5 \text{ V}$	V	-0.050	-0.052	-0.065	-0.050
I_{ON}	$V_{\text{DS}} = V_{\text{GS}} = -5 \text{ V}$	$\mu\text{A} / \mu\text{m}$	-263	-263.66	-262.89	-262.99
I_{OFF}	$V_{\text{DS}} = -5 \text{ V}, V_{\text{GS}} = 0 \text{ V}$	$\text{A} / \mu\text{m}$	-2.44E-12	-4.9E-12	-3.3E-12	-2.5E-12

Narrow and long channel device (W/L= 0.35 μm / 10 μm)

V_{TLIN}	$V_{\text{DS}} = -0.1 \text{ V}$	V	-0.675	-0.672	-0.674	-0.675
V_{TSAT}	$V_{\text{DS}} = -5 \text{ V}$	V	-0.670	-0.665	-0.669	-0.670
I_{ON}	$V_{\text{DS}} = V_{\text{GS}} = -5 \text{ V}$	$\mu\text{A} / \mu\text{m}$	-15.96	-16.12	-16.17	-15.96

Narrow and short channel device (W/L= 0.35 μm / 0.5 μm)

V_{TLIN}	$V_{\text{DS}} = -0.1 \text{ V}$	V	-0.619	-0.621	-0.635	-0.619
V_{TSAT}	$V_{\text{DS}} = -5 \text{ V}$	V	-0.561	-0.571	-0.565	-0.560
I_{ON}	$V_{\text{DS}} = V_{\text{GS}} = -5 \text{ V}$	$\mu\text{A} / \mu\text{m}$	-255.26	-258.26	-263.52	-255.25
I_{OFF}	$V_{\text{DS}} = -5 \text{ V}, V_{\text{GS}} = 0 \text{ V}$	$\text{A} / \mu\text{m}$	-3.59E-12	-1.5E-11	-2.7E-12	-2.2E-12

Please set gmindc=4.7099e-12 in SPECTRE simulator for achieving the consistent I_{OFF} result with HSPICE's

3.3 MOSFET DC Model

3.3.1 Test devices

The geometry of devices that were used for MOSFET DC model parameter extraction is summarized below.

Table 3-3 Geometry of PMOS devices measured for DC parameter extraction

W/L	10	5	1	0.8	0.6	0.5
10	X	X	X	X	X	X
5	X		X			X
1	X	X	X	X	X	X
0.8	X		X			X
0.6	X	X	X	X	X	X
0.43	X		X		X	X
0.35	X	X	X	X	X	X

3.3.2 Measurement conditions

The drain current was measured at various bias conditions summarized in the table below:

Table 3-4 Test conditions for IV curves of PMOS

I_D vs. V_{GS}				I_D vs. V_{DS}			
	Start	Stop	Step		Start	Stop	Step
V_{DS} (V)	-0.1	-5	-4.9	V_{DS} (V)	0	-5.5	-0.05
V_{GS} (V)	0.5	-5.5	-0.05	V_{GS} (V)	-0.9	-5	-0.41
V_{BS} (V)	0	3	0.6	V_{BS} (V)	0	3	3

3.3.3 Model verification plots

The simulation results obtained from the extracted model were plotted together with the measurement results for comparison. The L and W in the plots refer to physical dimension (LDES *0.9+12nm and WDES *0.9) respectively.

I_D vs. V_{DS} plots

The following plots show measured and simulated output I-V curves for selected devices at 25 C. (The dimensions are after 90% shrink and 12nm sizing channel length back.)

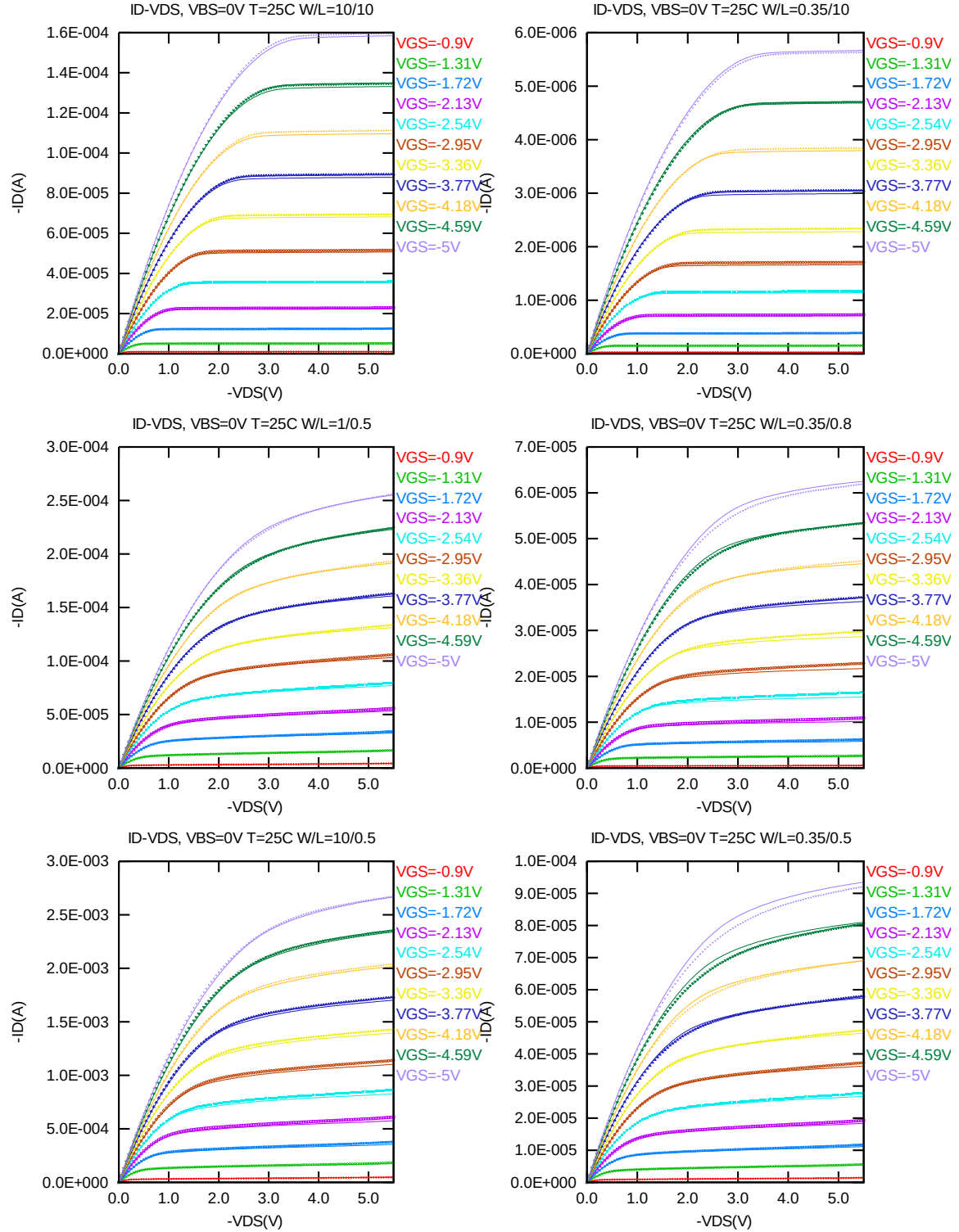


Figure 3-1 I_D vs. V_{DS} at $V_{BS}=0V$ for PMOS

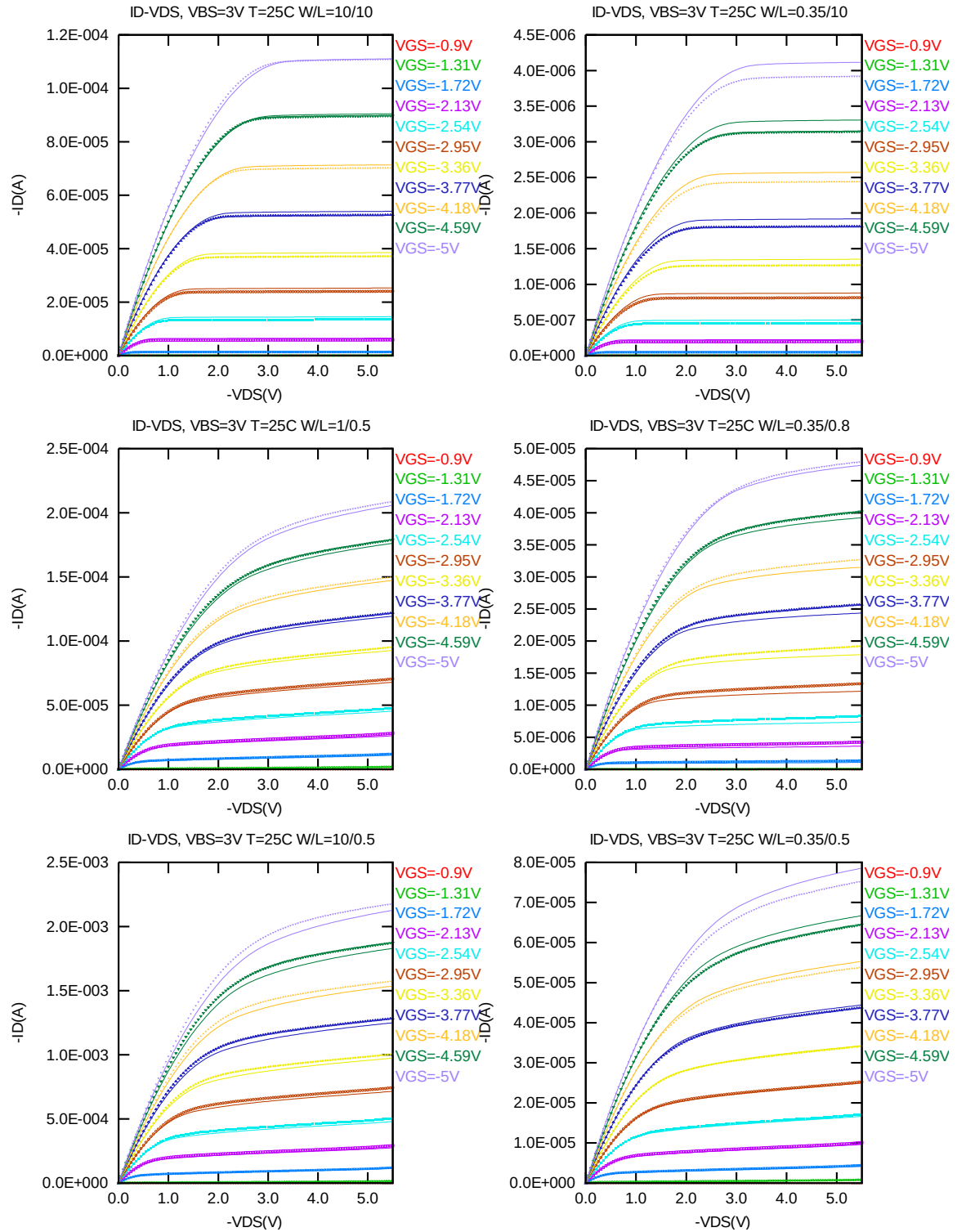


Figure 3-2 I_D vs. V_{DS} at $V_{BS}=3$ V for PMOS

G_{DS} vs. V_{DS} plots

The following plots show measured and simulated output I-V curves for selected devices at 25 C. (The dimensions are after 90% shrink and 12nm sizing channel length back.)

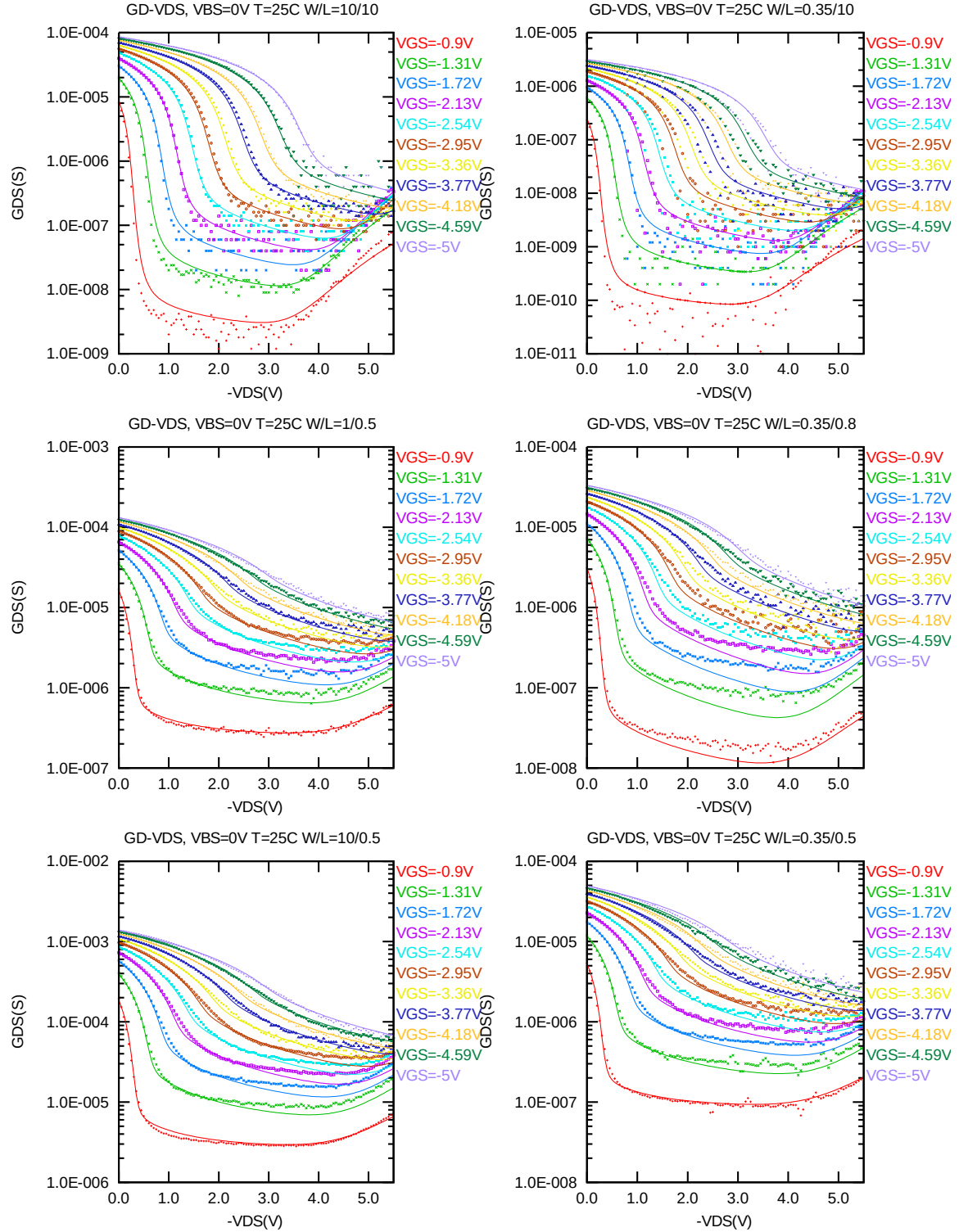


Figure 3-3 G_{DS} vs. V_{DS} at $V_{BS}=0V$ for PMOS

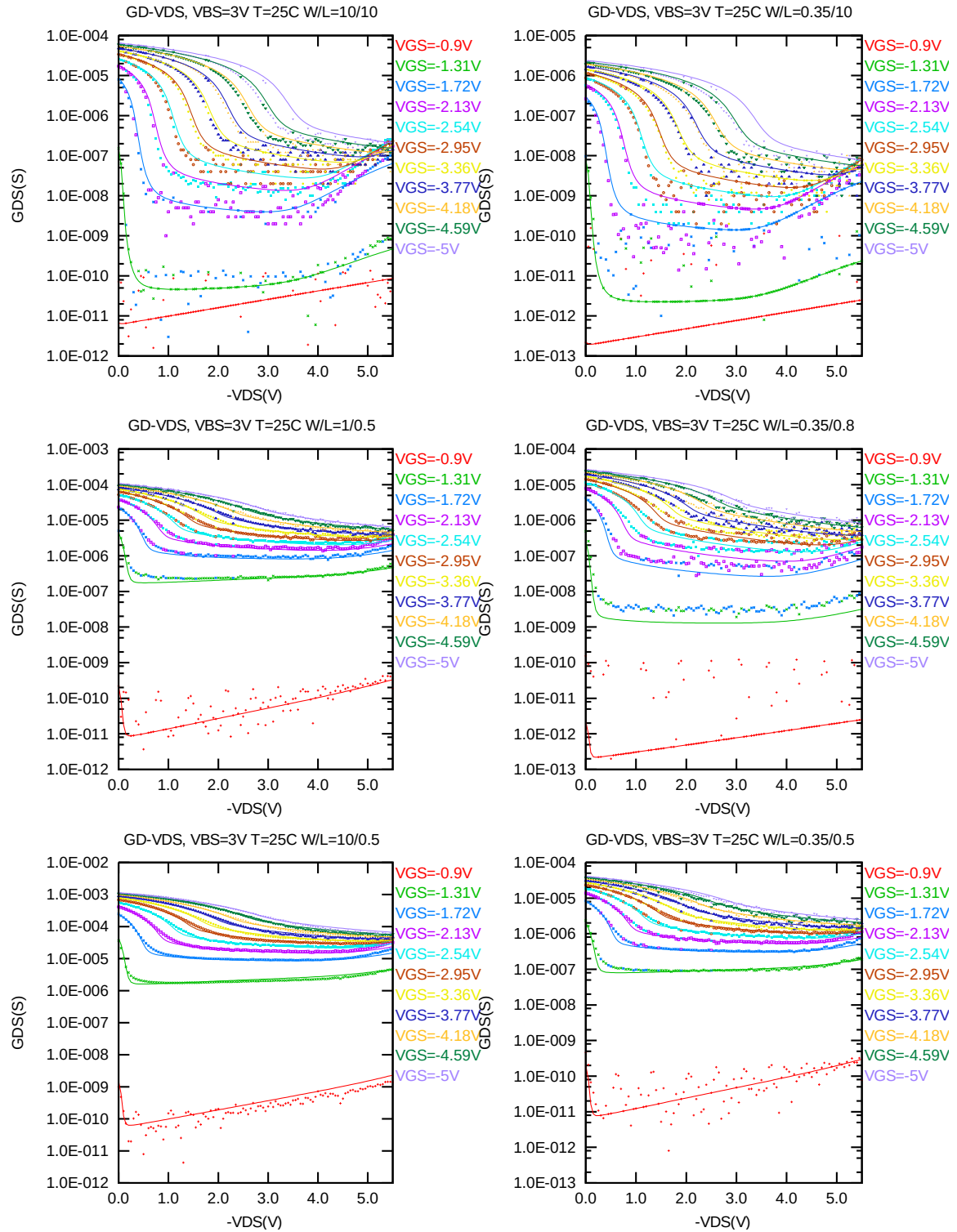


Figure 3-4 G_{DS} vs. V_{DS} at $V_{BS}=3$ V for PMOS

I_D vs. V_{GS} plots (linear scale)

The following plots show measured and simulated output I-V curves for selected devices at 25 C. (The dimensions are after 90% shrink and 12nm sizing channel length back.)

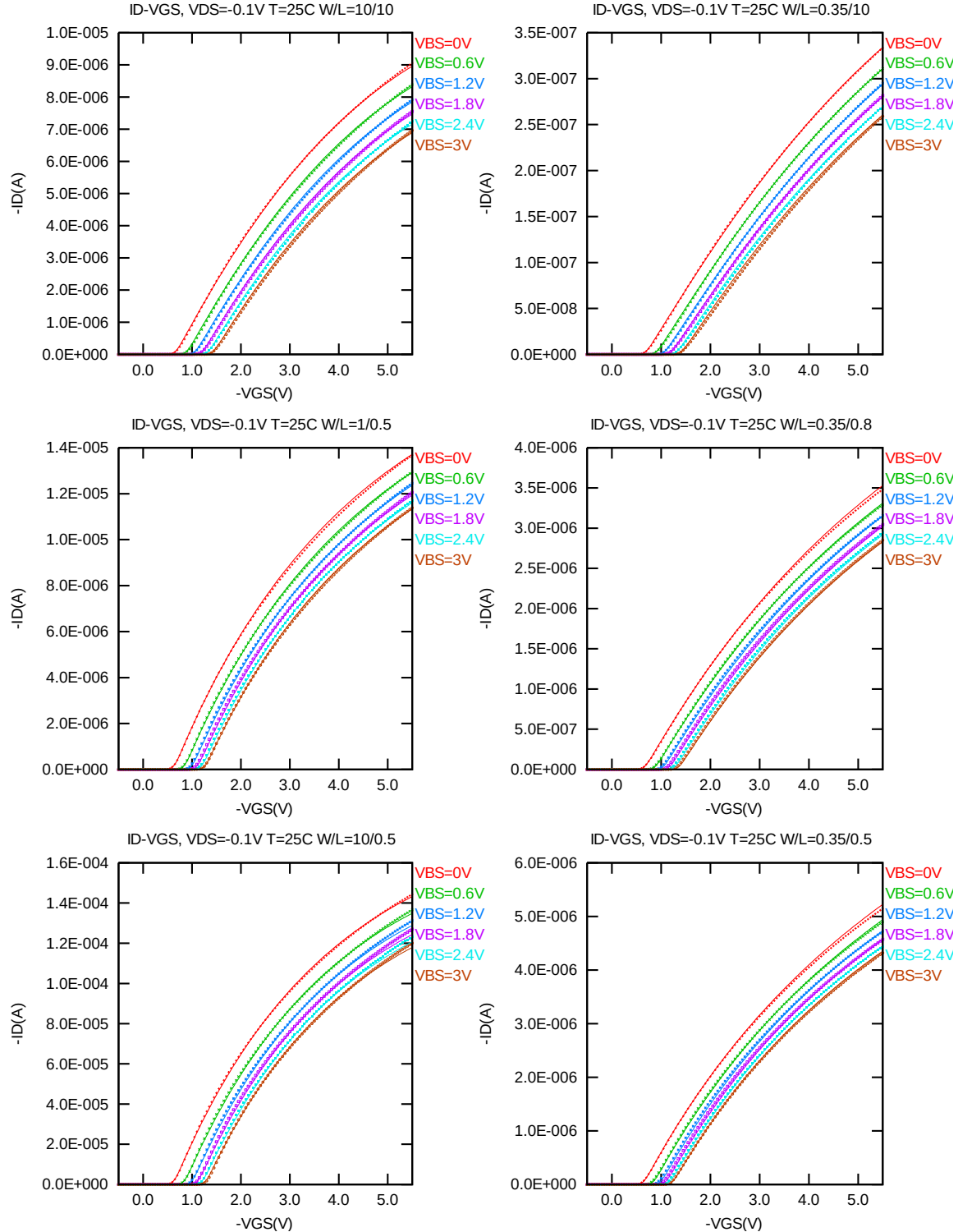


Figure 3-5 I_D vs. V_{GS} at $V_{DS} = -0.1V$ for PMOS

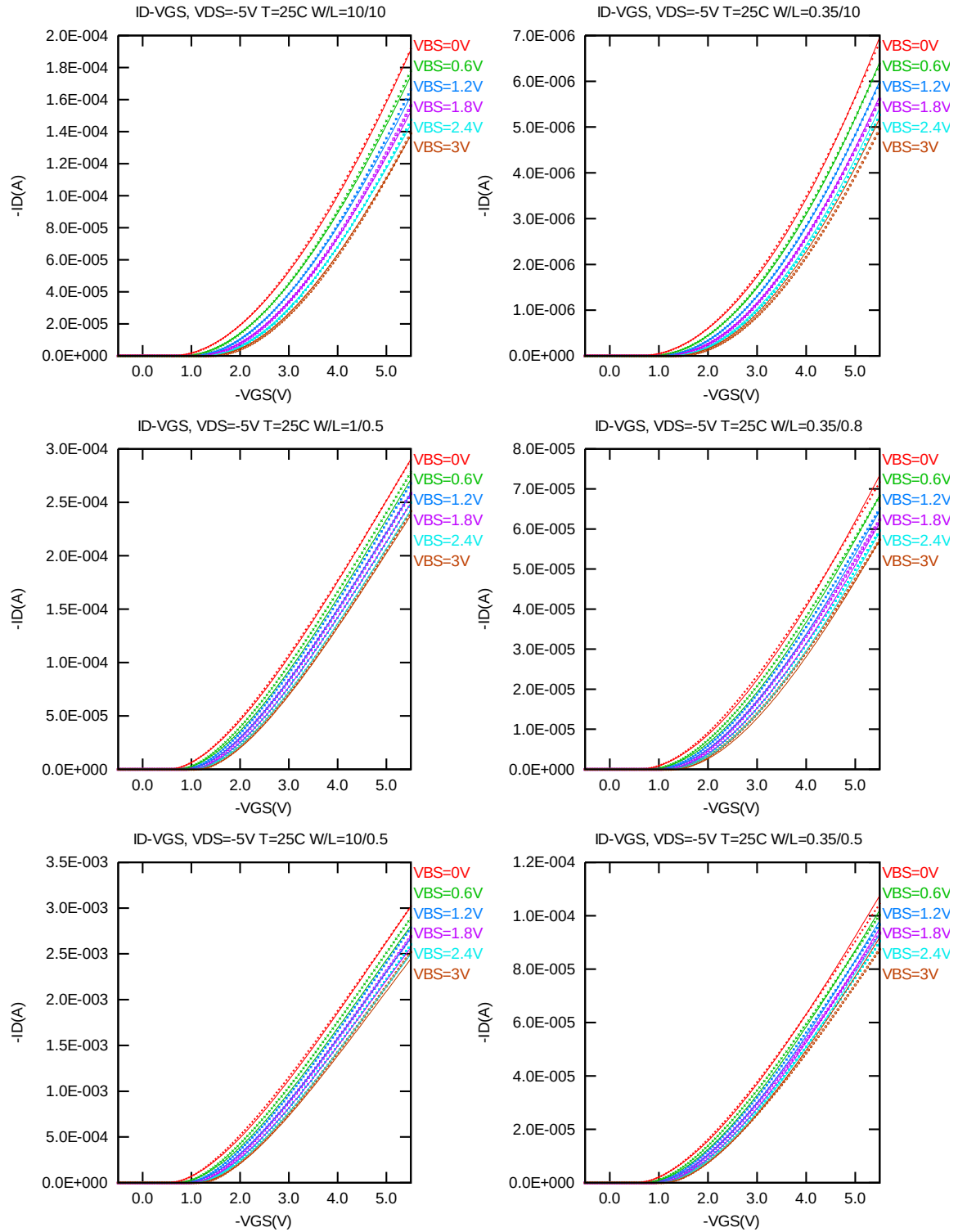


Figure 3-6 I_D vs. V_{GS} at $V_{DS} = -5V$ for PMOS

I_D vs. V_{GS} plots (logarithmic scale)

The following plots show measured and simulated output I-V curves for selected devices at 25 C. (The dimensions are after 90% shrink and 12nm sizing channel length back.)

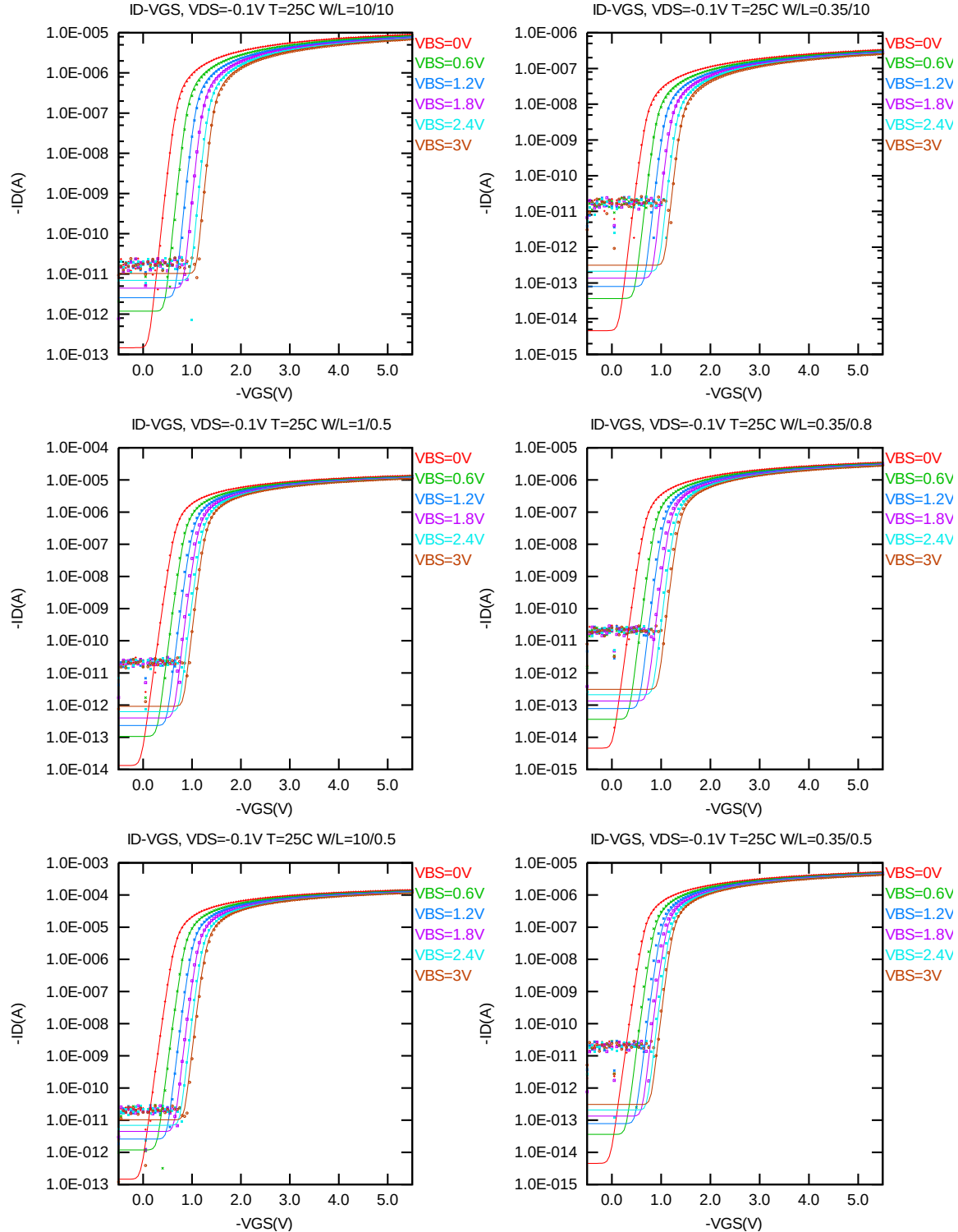


Figure 3-7 Log scaled I_D vs. V_{GS} at $V_{DS} = -0.1V$ for PMOS

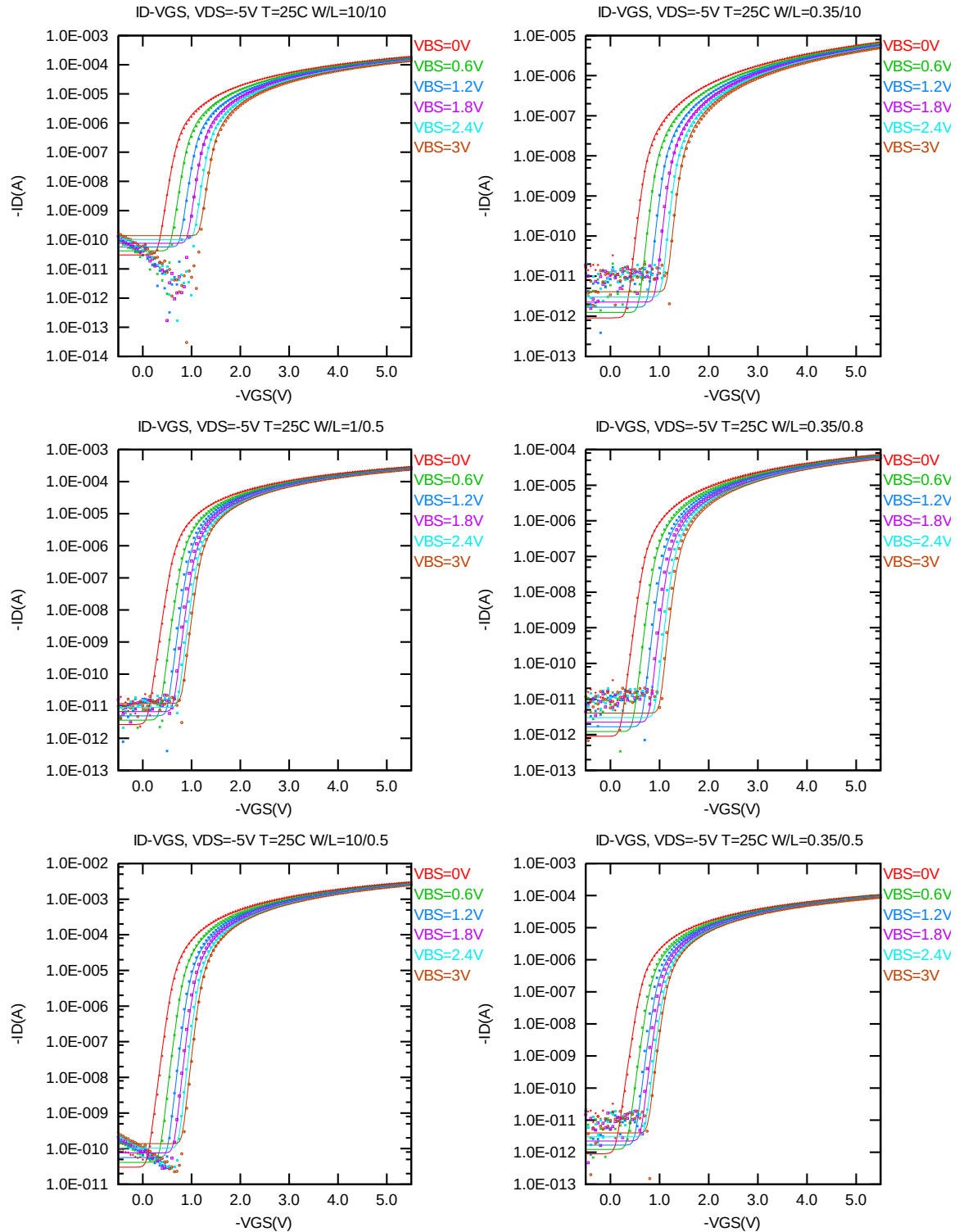


Figure 3-8 Log scaled I_D vs. V_{GS} at $V_{DS} = -5$ V for PMOS

G_M vs. V_{GS} plots

The following plots show measured and simulated output I-V curves for selected devices at 25 C. (The dimensions are after 90% shrink and 12nm sizing channel length back.)

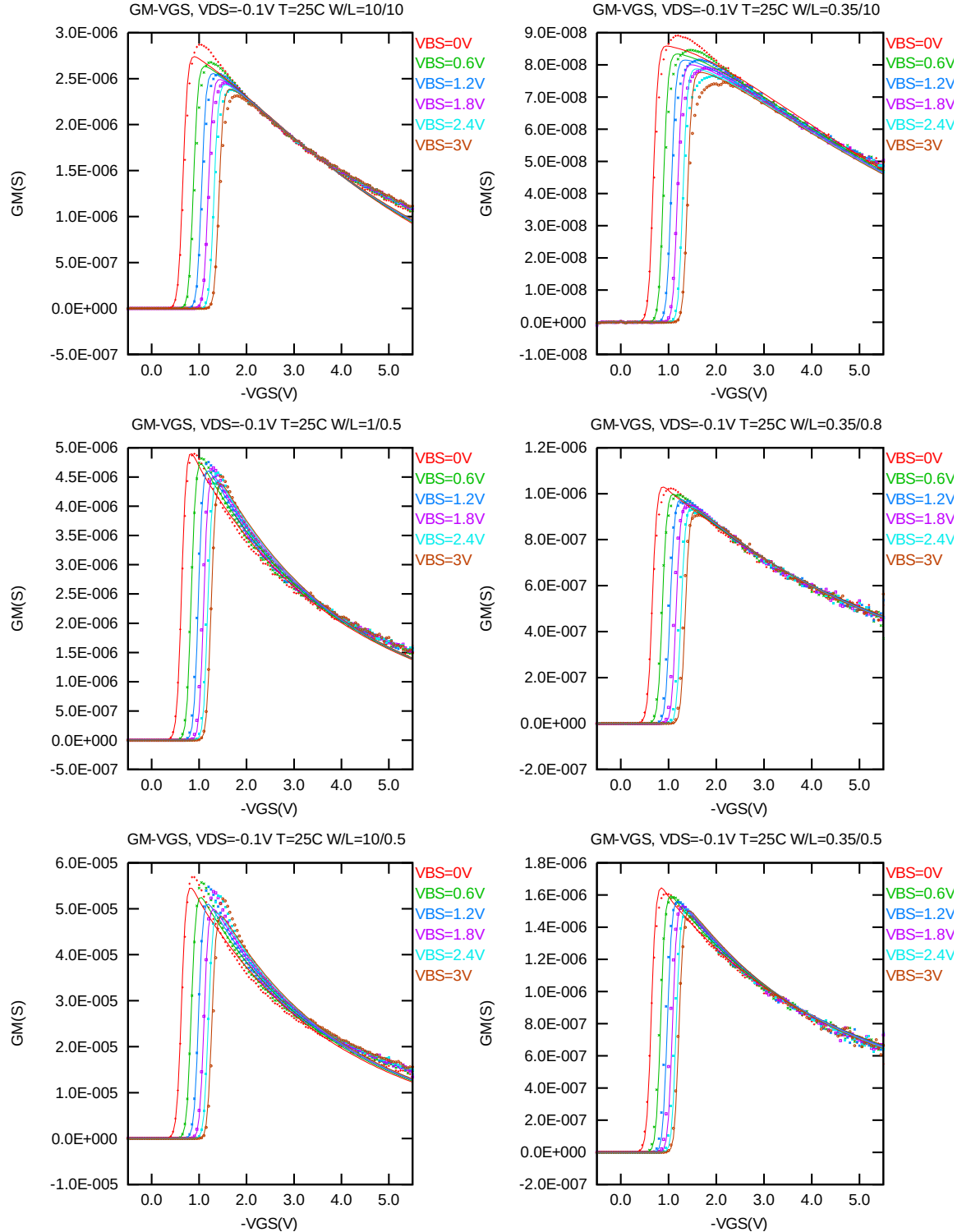


Figure 3-9 G_M vs. V_{GS} at $V_{DS}=-0.1V$ for PMOS

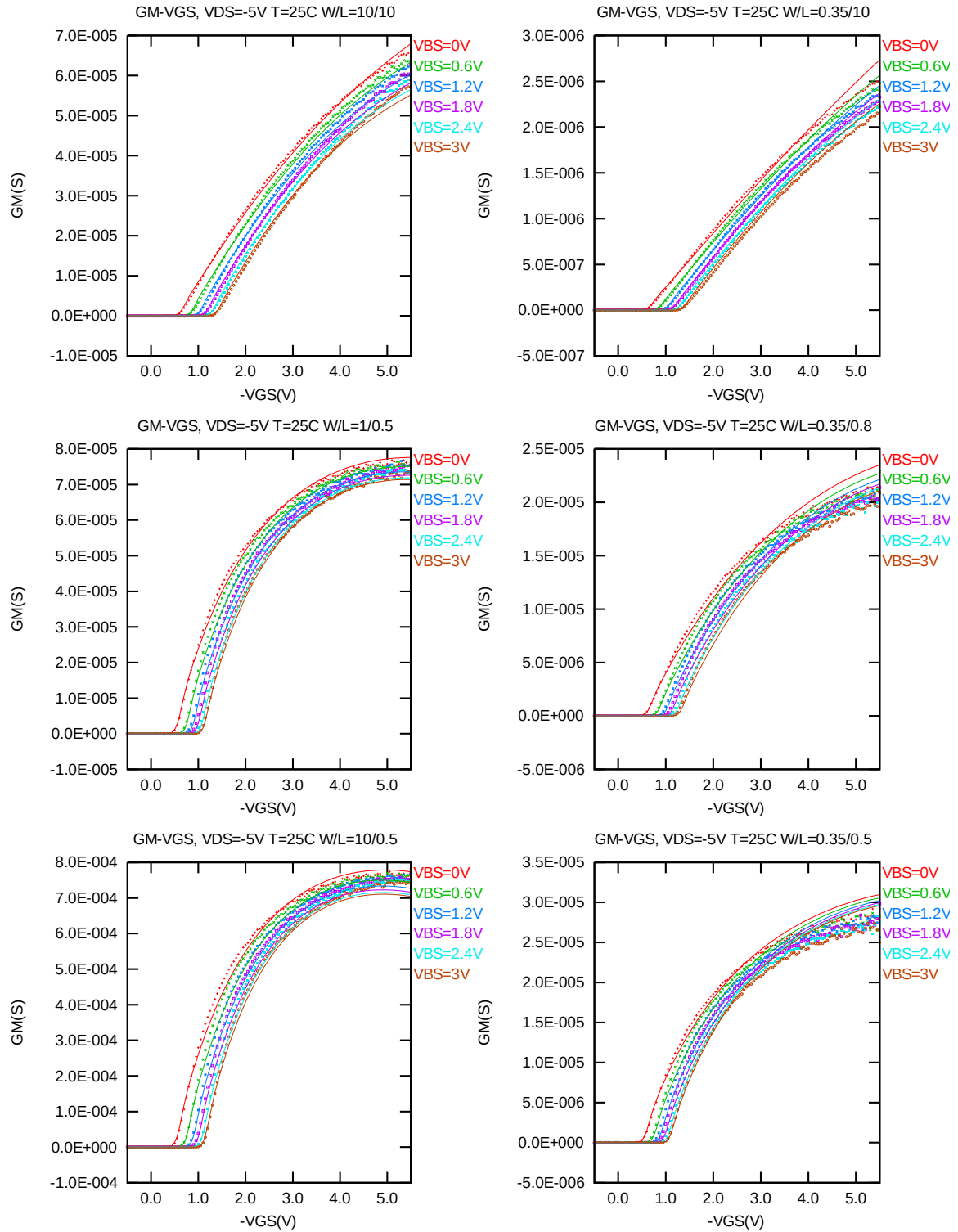


Figure 3-10 G_M vs. V_{GS} at $V_{DS} = -5V$ for PMOS

The L and W in the plots refer to L_{DES} and W_{DES} respectively.

The following figures show the channel length and width dependence of linear and saturation threshold voltage extracted at $I_D = -70\text{nA} \cdot W_{DES} \cdot 0.9 / (L_{DES} \cdot 0.9 + 12\text{nm})$ at 25°C . The plots show the physical dimension.

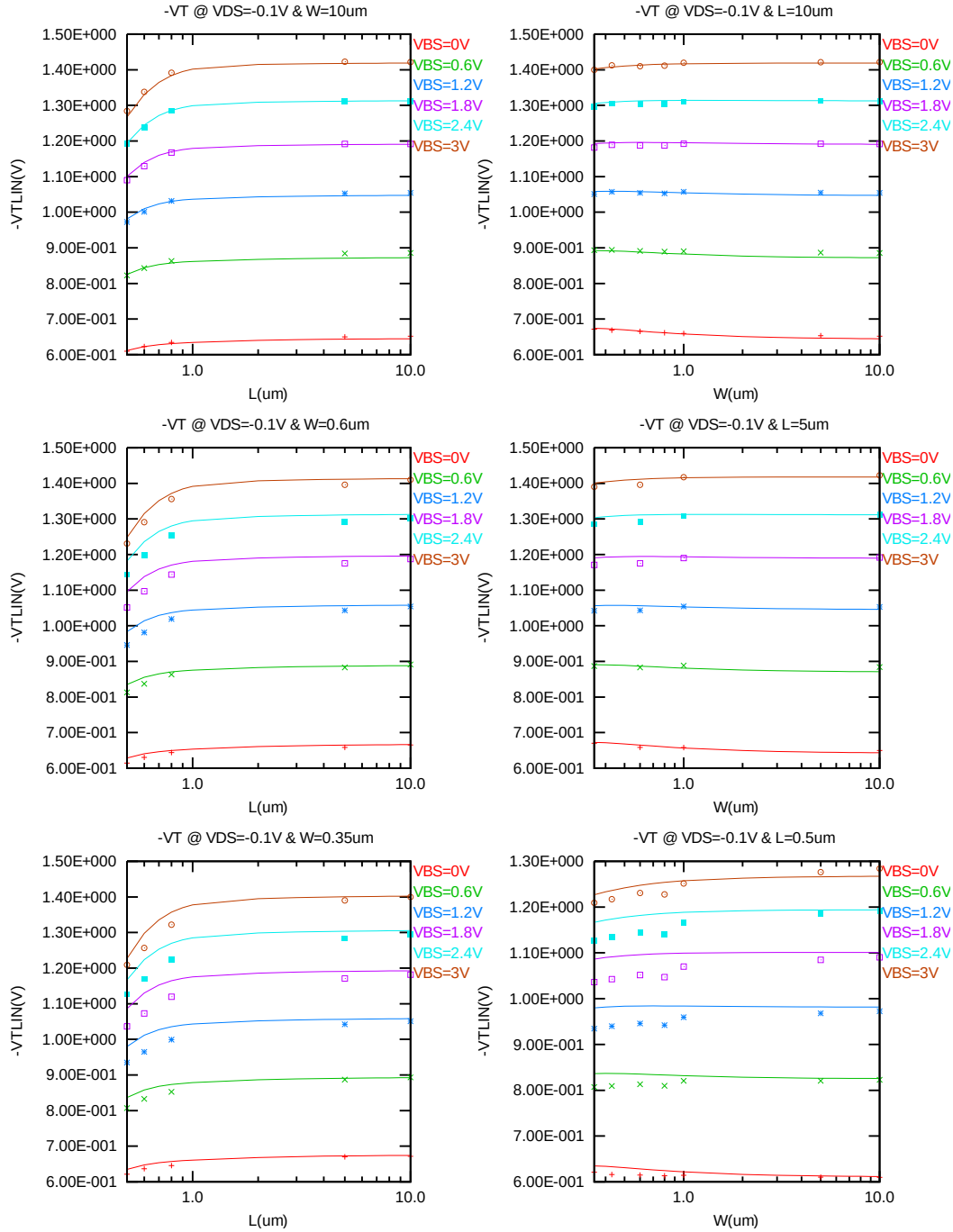


Figure 3-11 V_T vs. channel length & channel width for low drain bias ($V_{DS} = -0.1\text{V}$) for PMOS

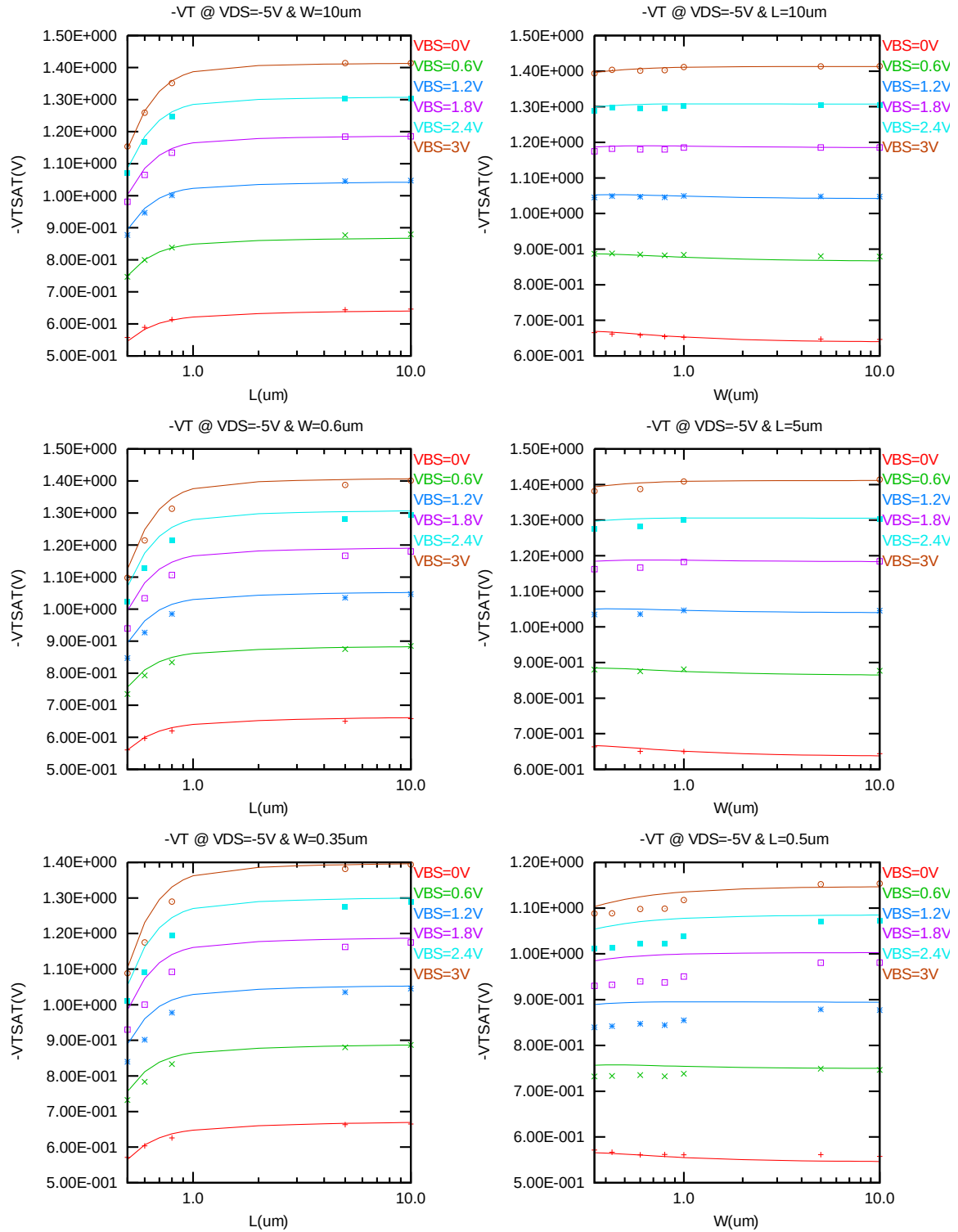


Figure 3-12 V_T vs. channel length & channel width for high drain bias ($V_{DS} = -5V$) for PMOS

The L and W in the plots refer to L_{DES} and W_{DES} respectively.

The following plots show the channel length and width dependence of I_{ON} extracted at $V_{DS}=V_{GS}=-5$ V for $V_{BS}=0$ V and $V_{BS}=3$ V at 25 C. (The dimensions are after 90% shrink and 12nm sizing channel length back.)

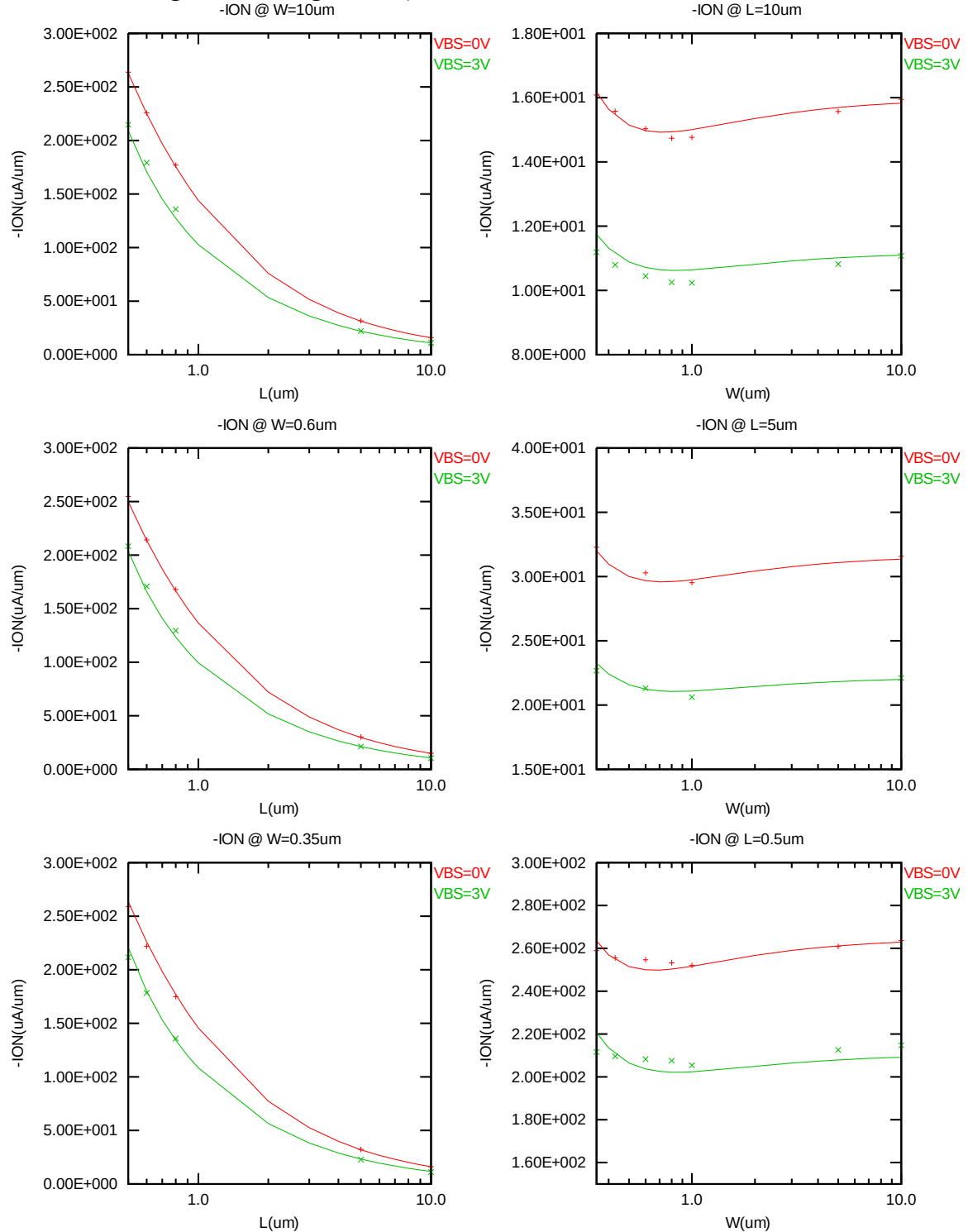


Figure 3-13 Saturation current I_{ON} vs. channel length and channel width for PMOS

The L and W in the plots refer to L_{DES} and W_{DES} respectively.

The following plots show the channel length and width dependence of I_{OFF} extracted at $V_{DS}=5\text{ V}$ and $V_{GS}=0\text{ V}$ for $V_{BS}=0\text{ V}$ to 3 V at 25 C . A large discrepancy of I_{OFF} can be seen in the graphs, which is due to the presence of junction leakage current.

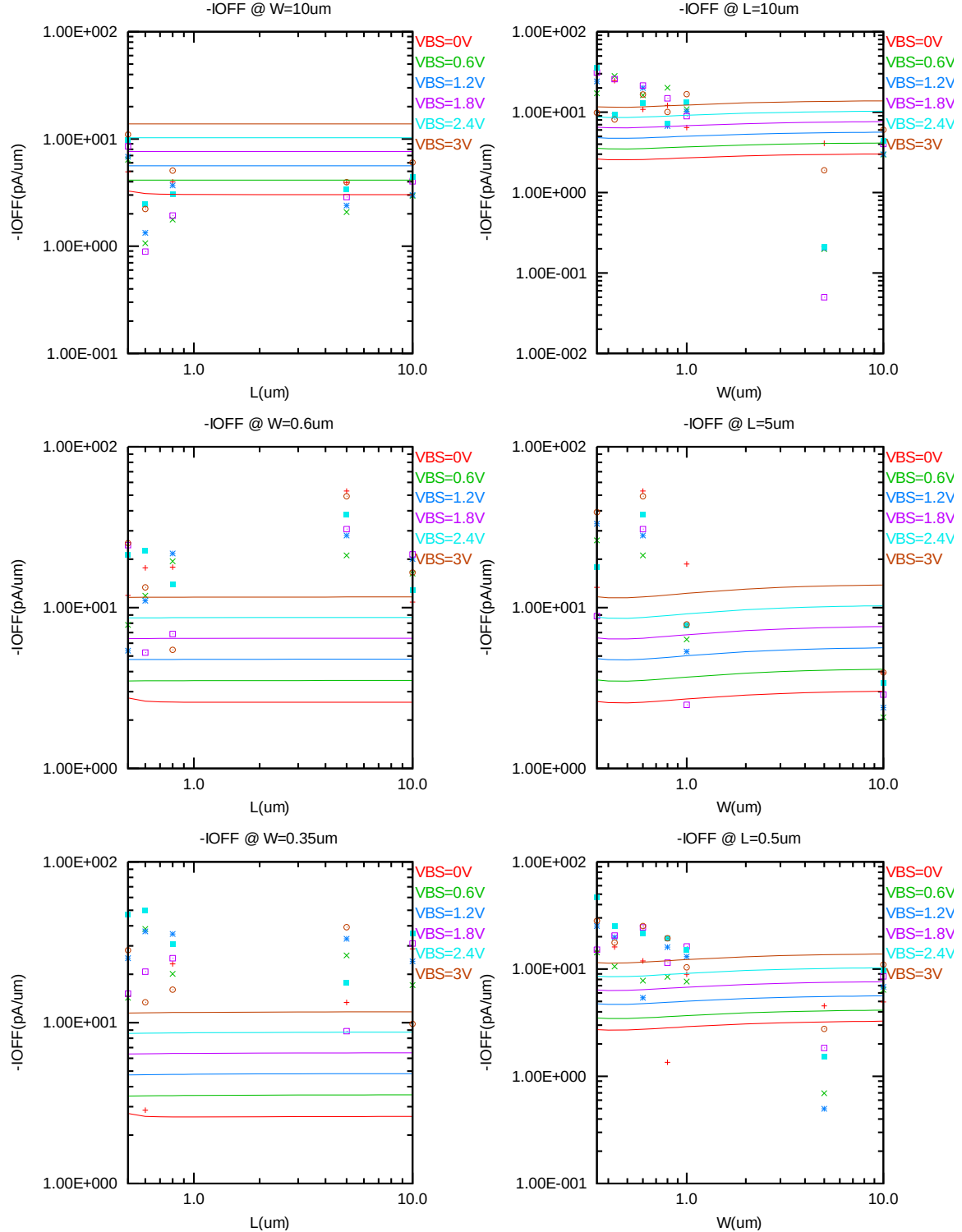


Figure 3-14 I_{OFF} vs. channel length and channel width for PMOS

V_{TLIN} vs. temperature

The following plots show the temperature dependence of the linear and saturation threshold voltage extracted at $I_D = -70nA * W_{DES} * 0.9 / (L_{DES} * 0.9 + 12nm)$. (The dimensions are after 90% shrink and 12nm sizing channel length back.)

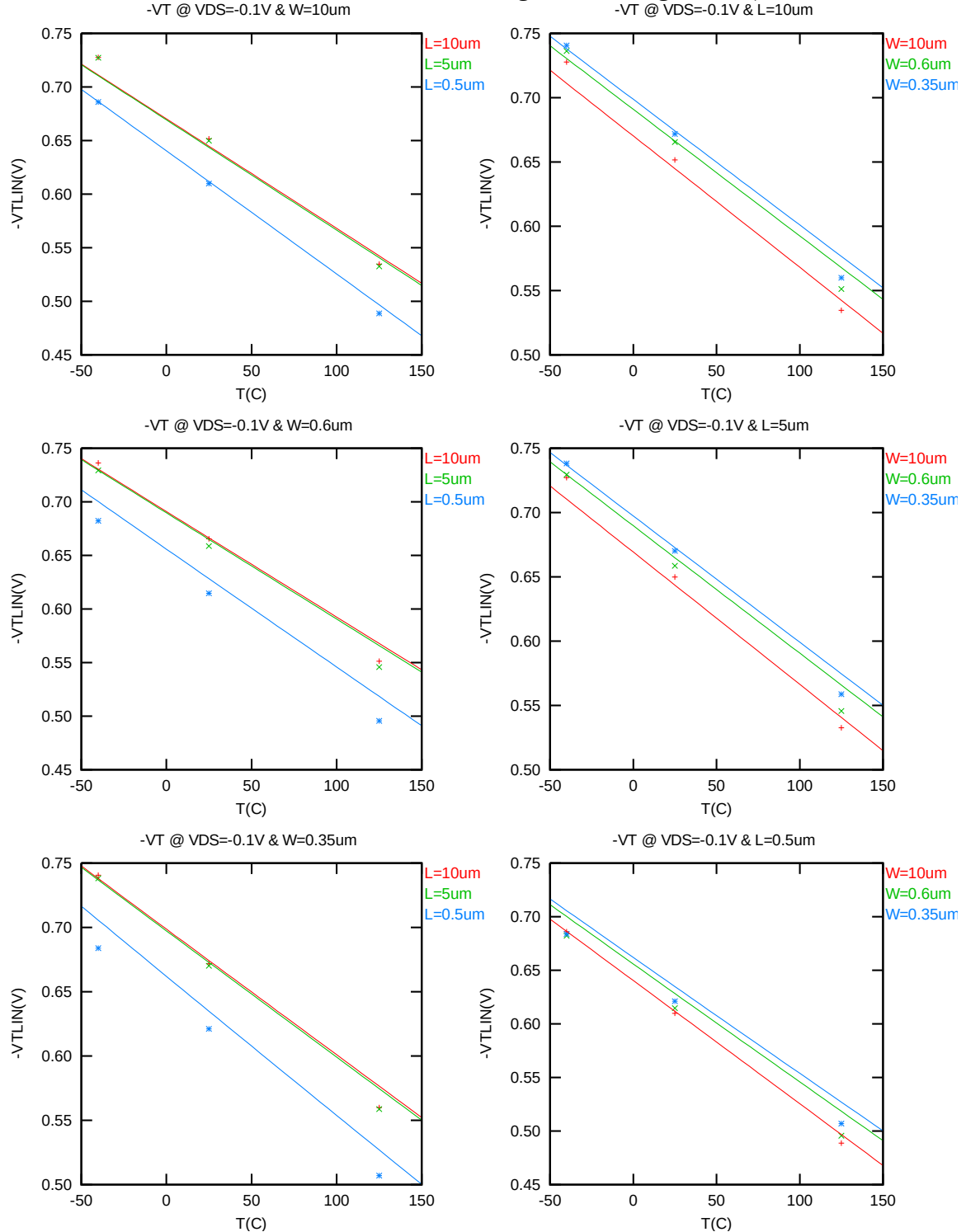


Figure 3-15 V_T vs. temperature for PMOS for low drain bias ($V_{DS} = -0.1V$)

V_{TSAT} vs. temperature

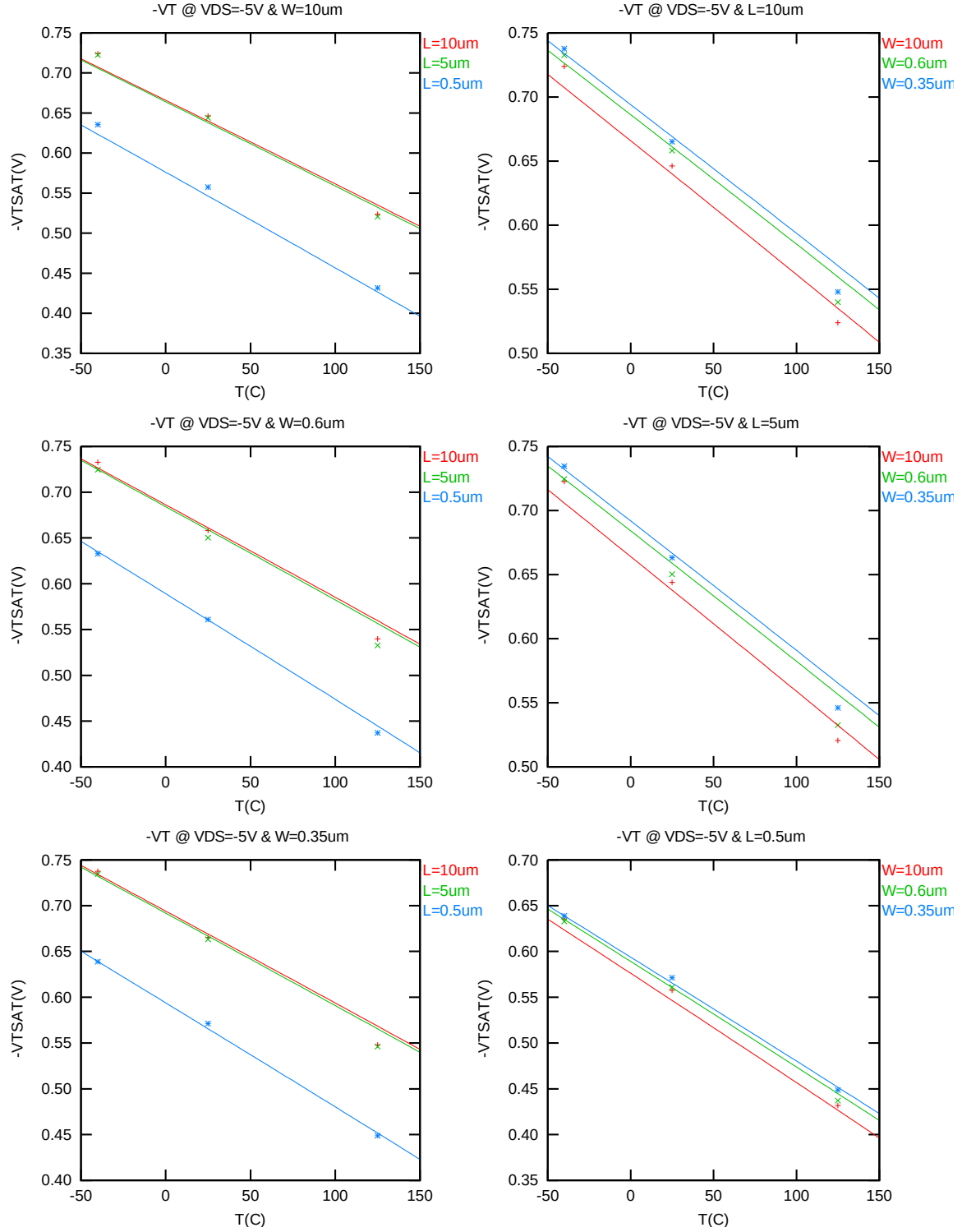


Figure 3-16 V_T vs. temperature for PMOS for high drain bias ($V_{DS} = -5V$)

I_{ON} vs. temperature

The following plots show the temperature dependence of I_{ON} extracted at $V_{GS}=V_{DS}=5$ V, $V_{BS}=0$ V. (The dimensions are after 90% shrink and 12nm sizing channel length back.)

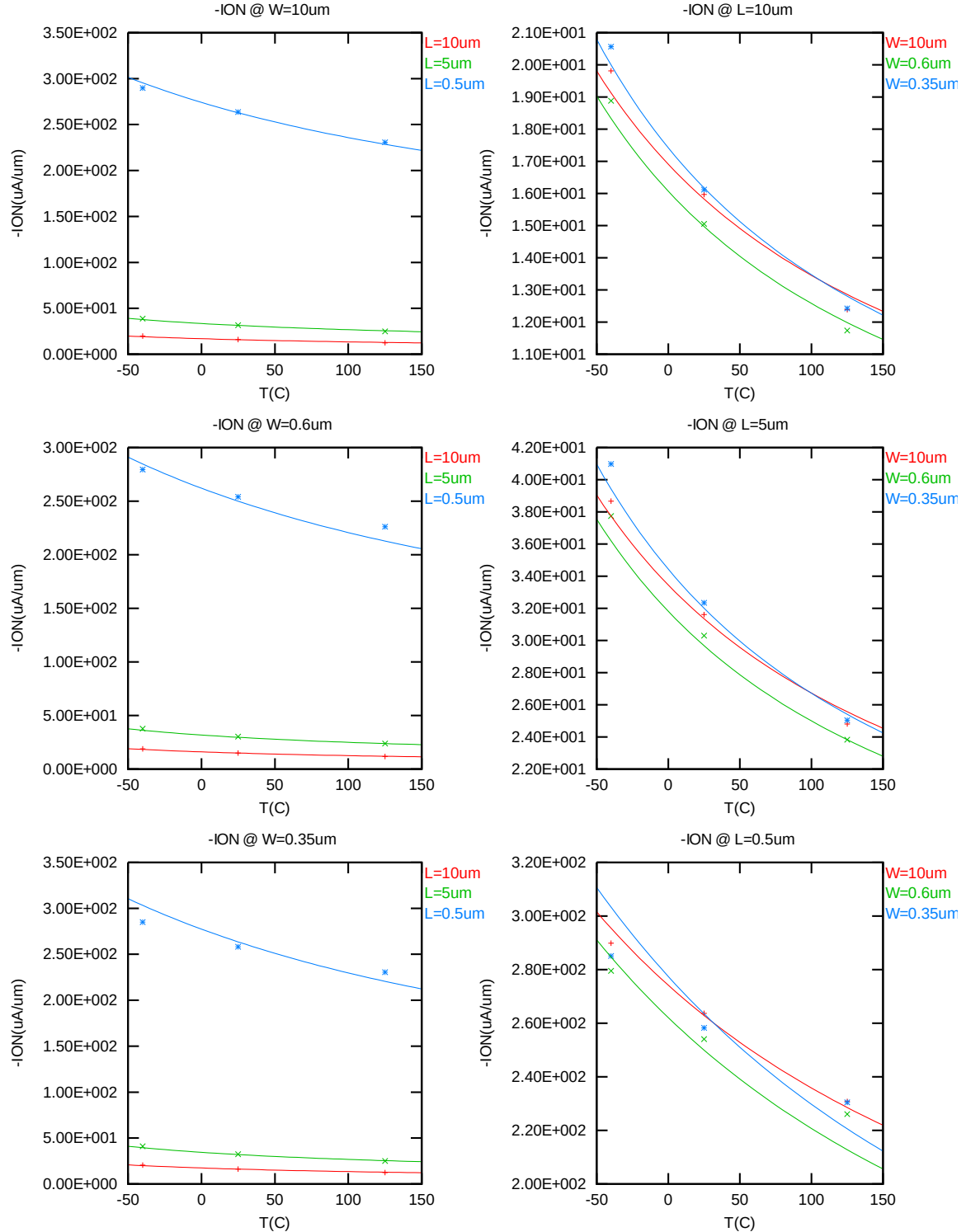


Figure 3-17 Saturation current vs. temperature for PMOS

I_{OFF} vs. temperature

The following plots show the temperature dependence of I_{OFF} extracted at $V_{\text{DS}}=5\text{ V}$, $V_{\text{GS}}=V_{\text{BS}}=0\text{ V}$. (The dimensions are after 90% shrink and 12nm sizing channel length back.)

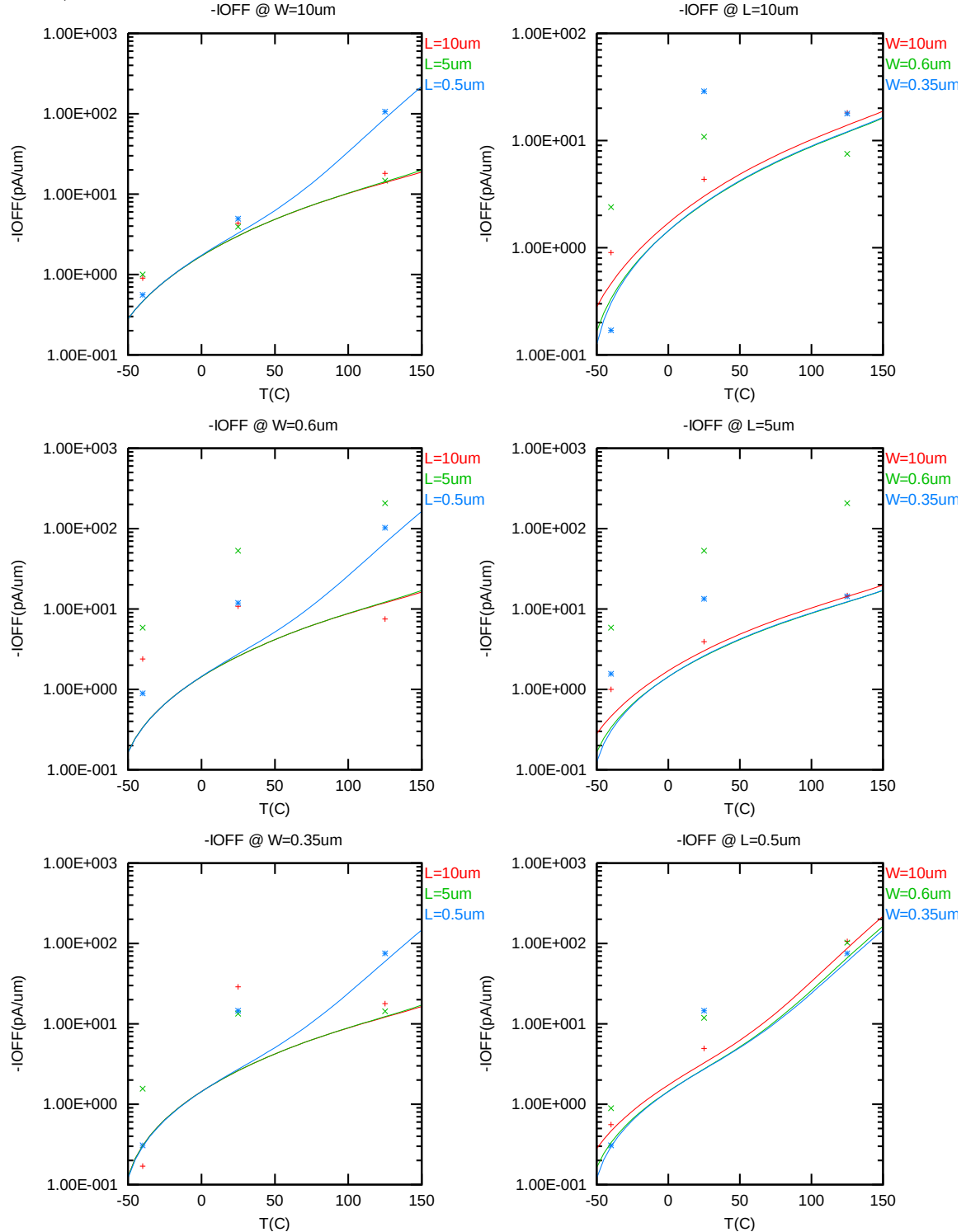


Figure 3-18 I_{OFF} vs. temperature for PMOS

3.3.4 Model fitting accuracy

The model fitting accuracy of parameters I_D , G_M and G_{DS} is quantified using the following equation:

$$\text{error \%} = 100 * | \text{measurement} - \text{simulation} | / [(\text{measurement} + \text{simulation}) / 2]$$

Different error criteria are defined for different parameters (I_D , G_M and G_{DS}) as well as for different operation regions. These criteria are explained below.

1) I_D accuracy criteria

- f. I_D in sub-threshold region, i.e. $V_{GS} < V_T$
- g. I_D in moderate inversion region, i.e. $V_T \leq V_{GS} < V_T + V_X$
- h. I_D in strong inversion region, i.e. $V_T + V_X \leq V_{GS} \leq V_{CC}$

, where $V_X = (V_{CC} - V_T) / 2$

2) G_M and G_{DS} accuracy criteria

- i. G_M and G_{DS} in sub-threshold region, i.e. $V_{GS} < V_T$
- j. G_M and G_{DS} in inversion region, i.e. $V_T \leq V_{GS} \leq V_{CC}$

Table 3-5 Requirements for model fit quality

Target of Quality Check	$V_{GS} < V_T$ (sub-threshold)	$V_T \leq V_{GS} < V_T + V_X$ (moderate inversion)	$V_T + V_X \leq V_{GS} \leq V_{CC}$ (strong inversion)
I_D	$\leq 75\%$	$\leq 25\%$	$\leq 7\%$
G_M	$\leq 50\%$	$\leq 20\%$	$\leq 20\%$
G_{DS}	$\leq 100\%$	$\leq 50\%$	$\leq 50\%$

I_D accuracy

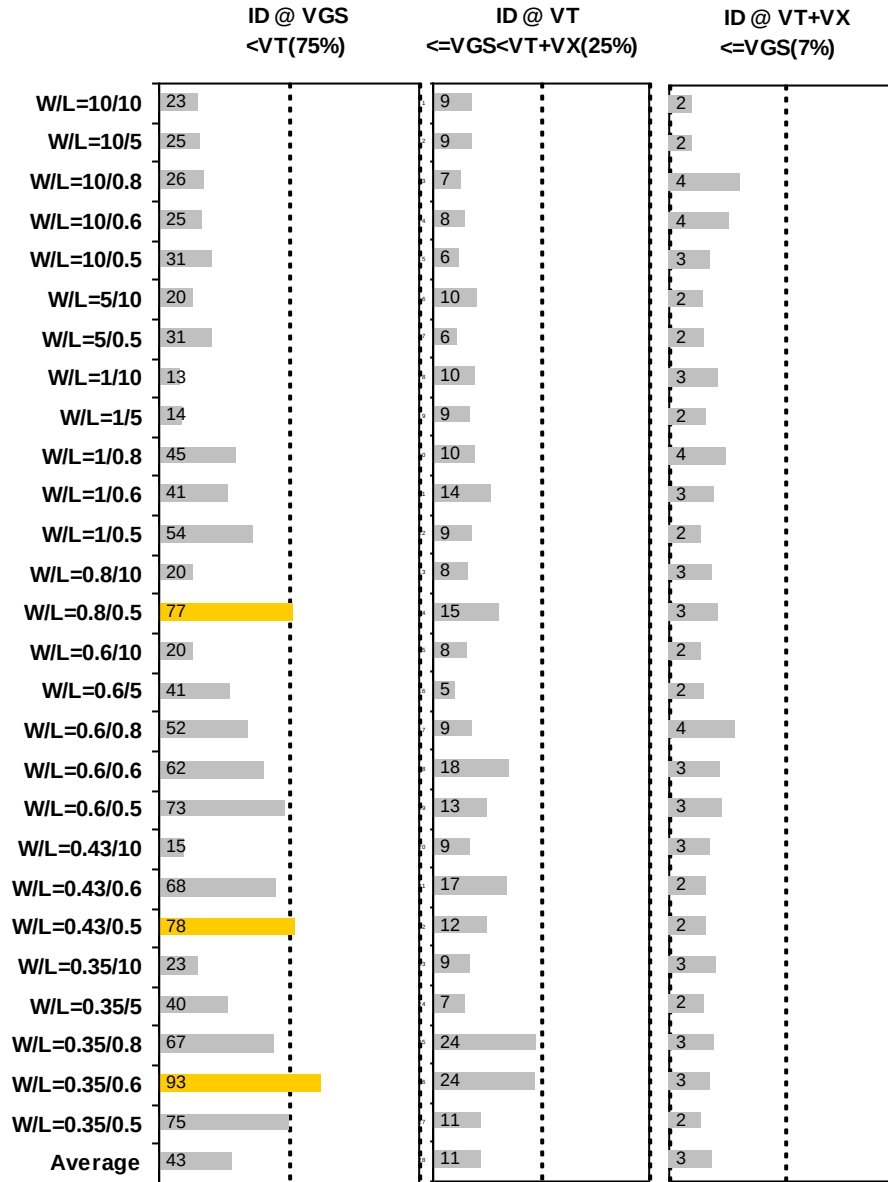


Figure 3-19 I_D accuracy at 25 C for PMOS

G_M and G_{DS} accuracy

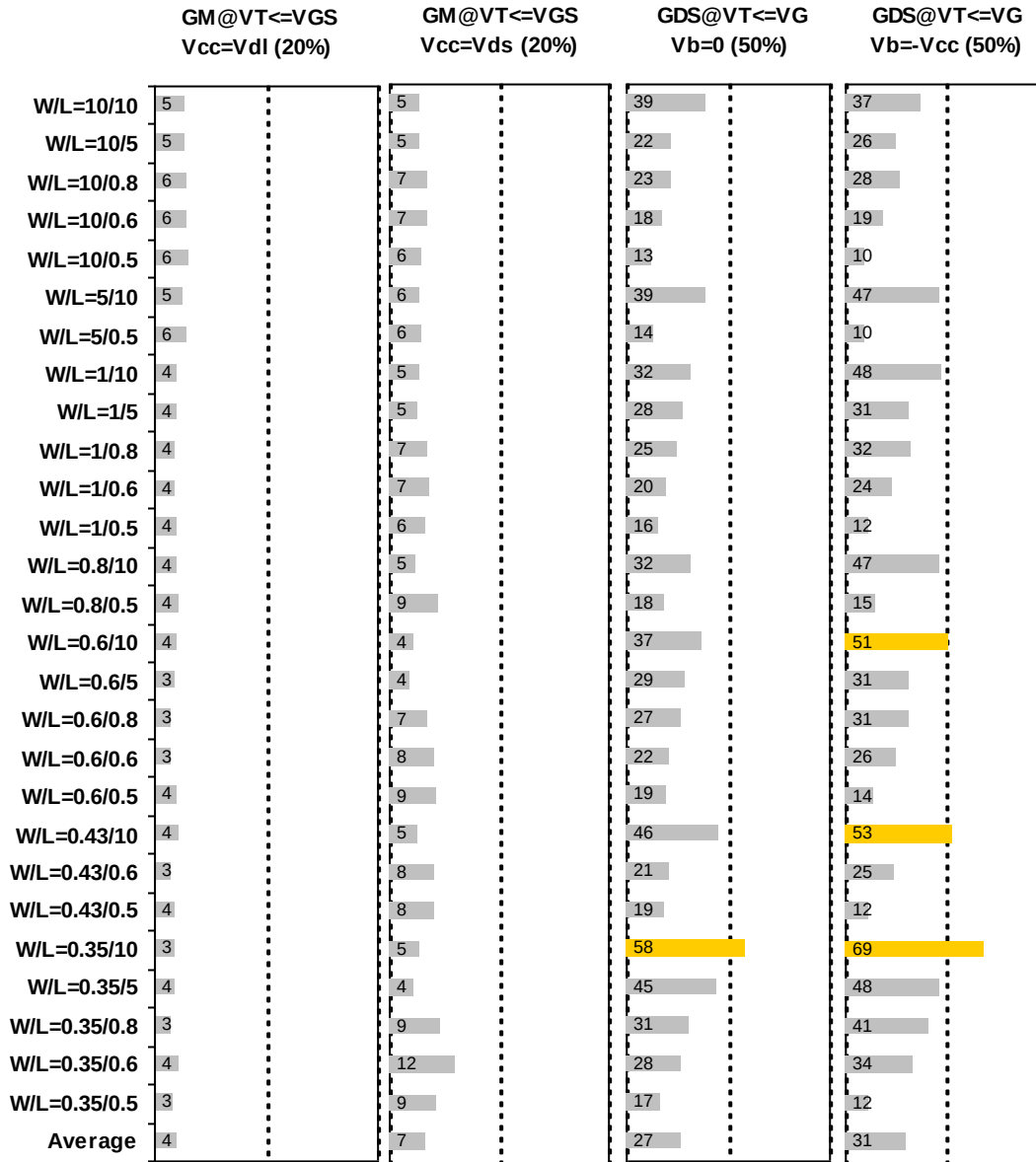


Figure 3-20 G_M and G_{DS} accuracy at 25 C for PMOS

3.4 MOSFET Capacitance Model

3.4.1 Test devices

The following test structures were used for MOSFET capacitance model parameter extraction.

Table 3-6 Measurement structure for PMOS C_{gg} capacitance model

Structure	L[μm]	W[μm]
Gate capacitance components, C _{GDS}	80	80

Table 3-7 Large area measurement structure for PMOS C_{gc} capacitance model

Structure	L[μm]	W[μm]
Gate capacitance components	1	100
	.5	100
	2	100
	2	2.08
	.5	2.08
	2	1.72
	.5	1.72
	2	1.36
	.5	1.36
	2	1
	.5	1
	2	.64
	.5	.64
	2	.35

	.5	.35
--	----	-----

3.4.2 Measurement conditions

The measurement conditions for MOSFET capacitance are summarized in Table 2-8 below.

Table 3-8 Measurement conditions for PMOS capacitance model

V _{BIAS} [V]	From -5.5 to 5.5 V in 0.1 V step
Frequency [kHz]	100
V _{OSC} [mV]	50
Temp [°C]	25

3.4.3 T_{ox} extraction

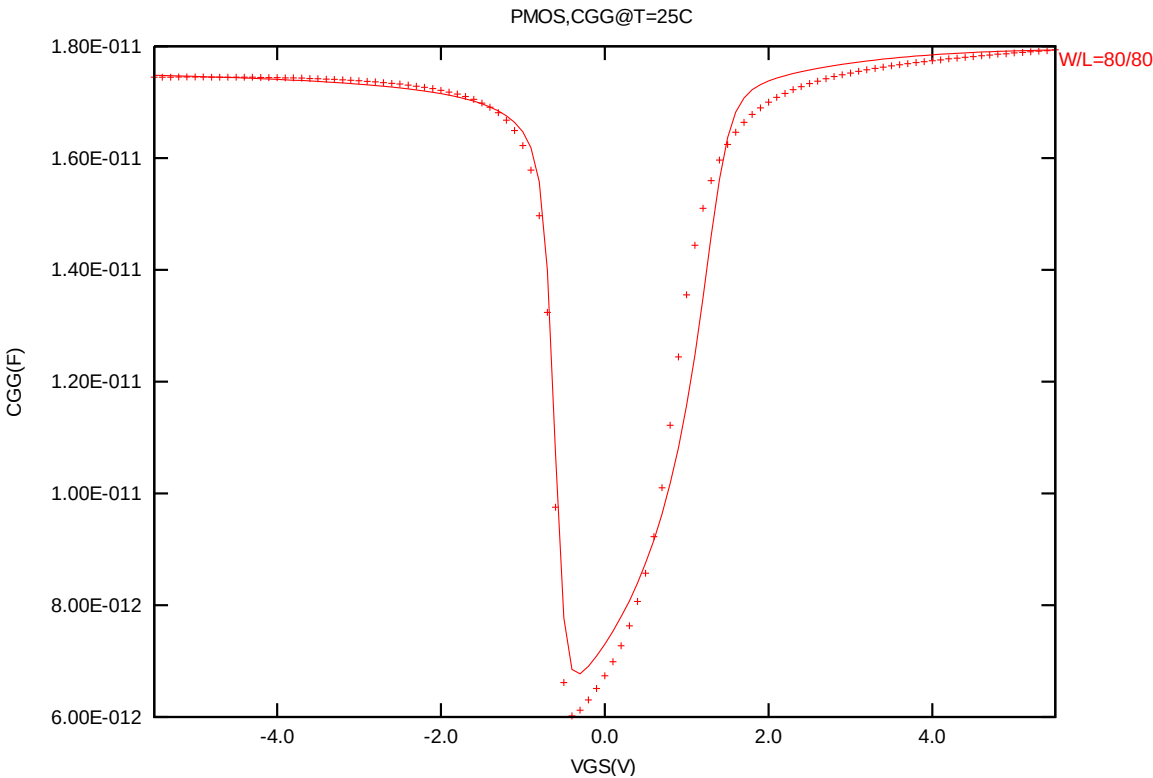
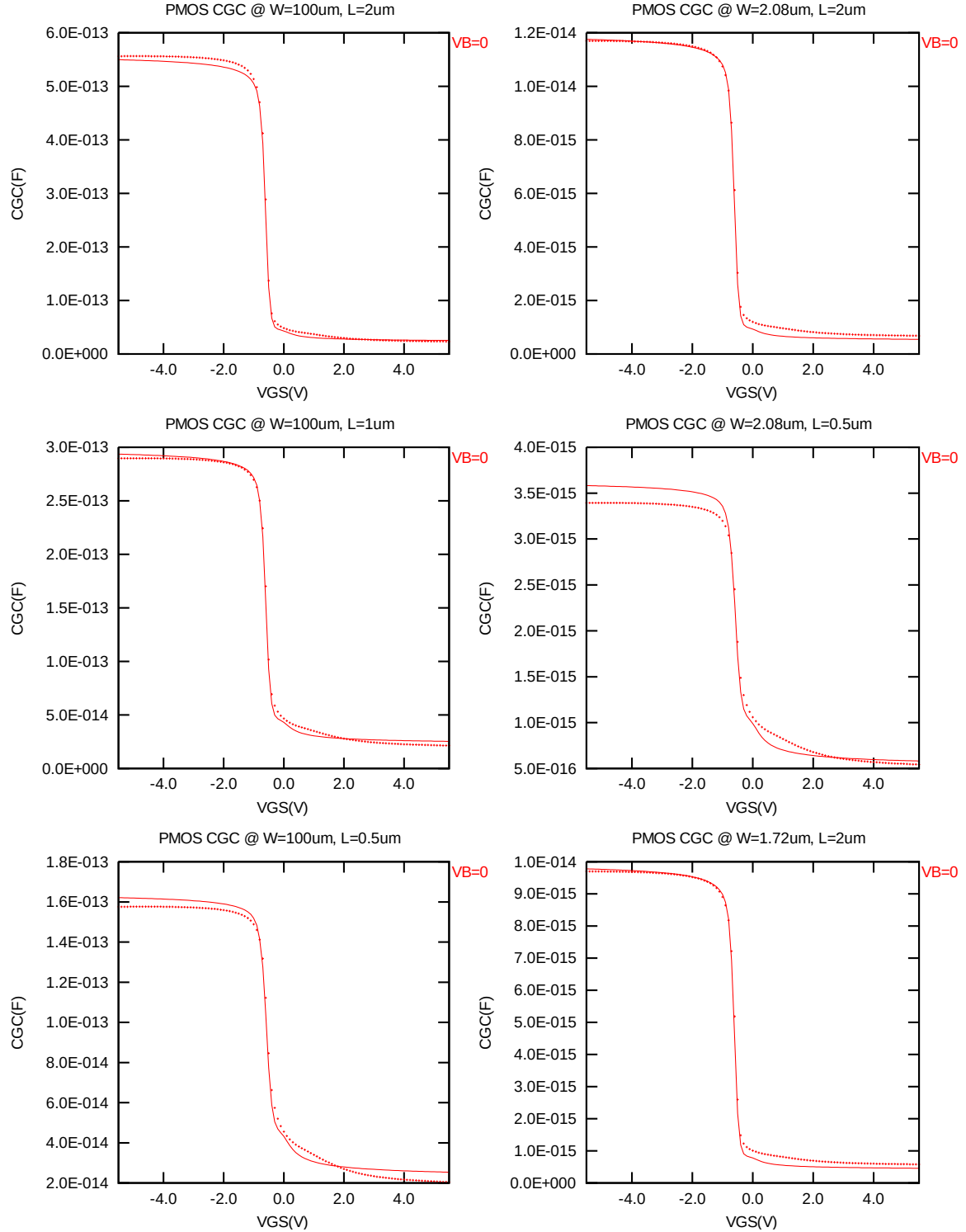
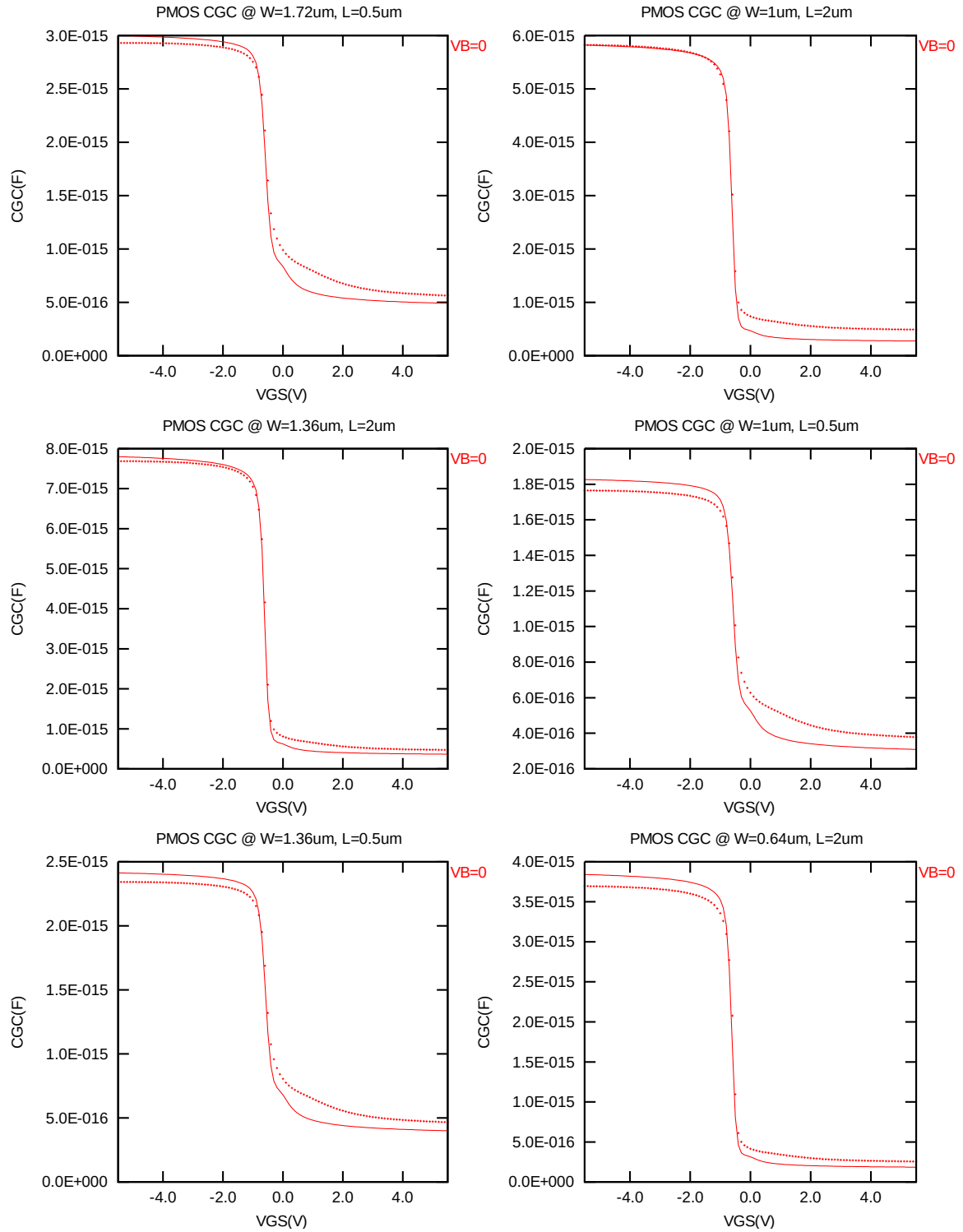


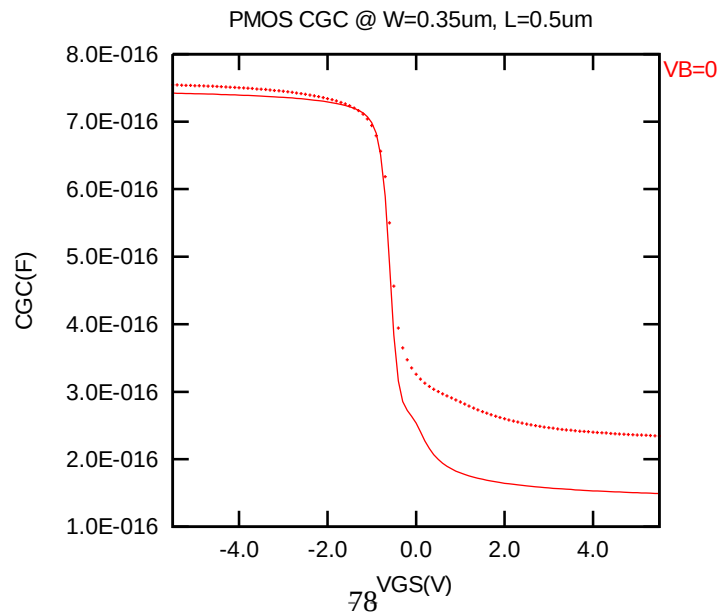
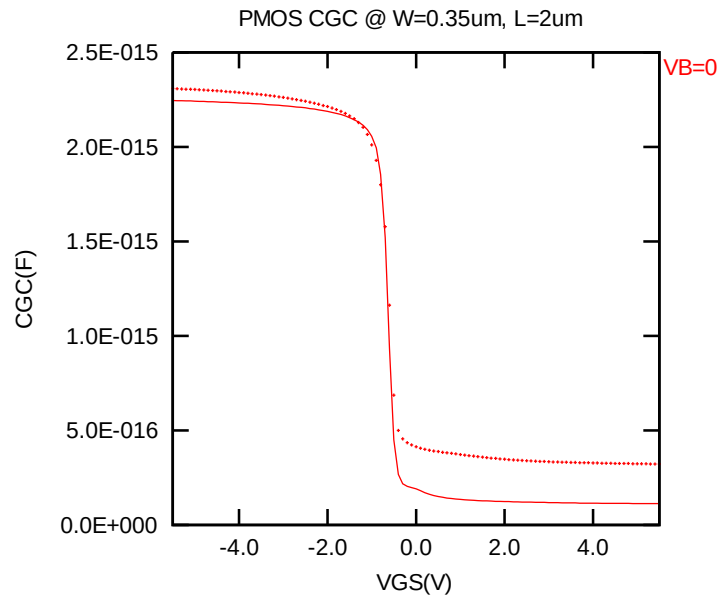
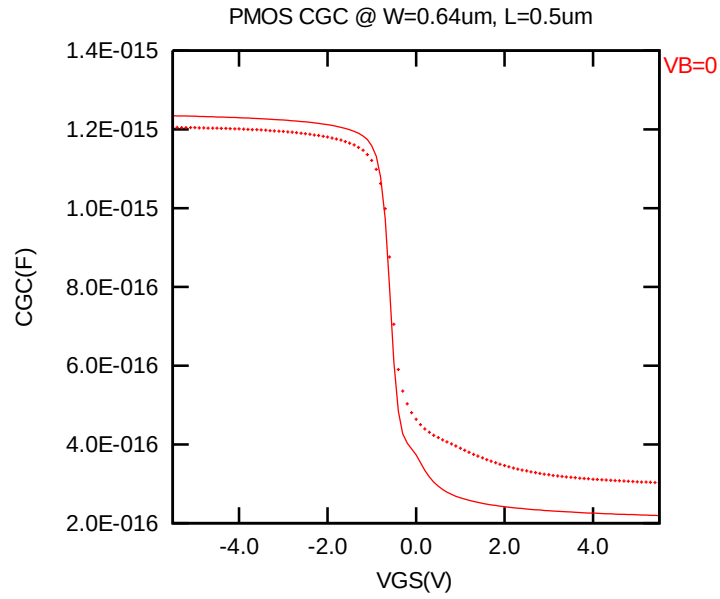
Figure 3-21 Measured and simulated C_{GG} curves

3.4.4 Width and length AC model accuracy

Width and length AC model accuracy







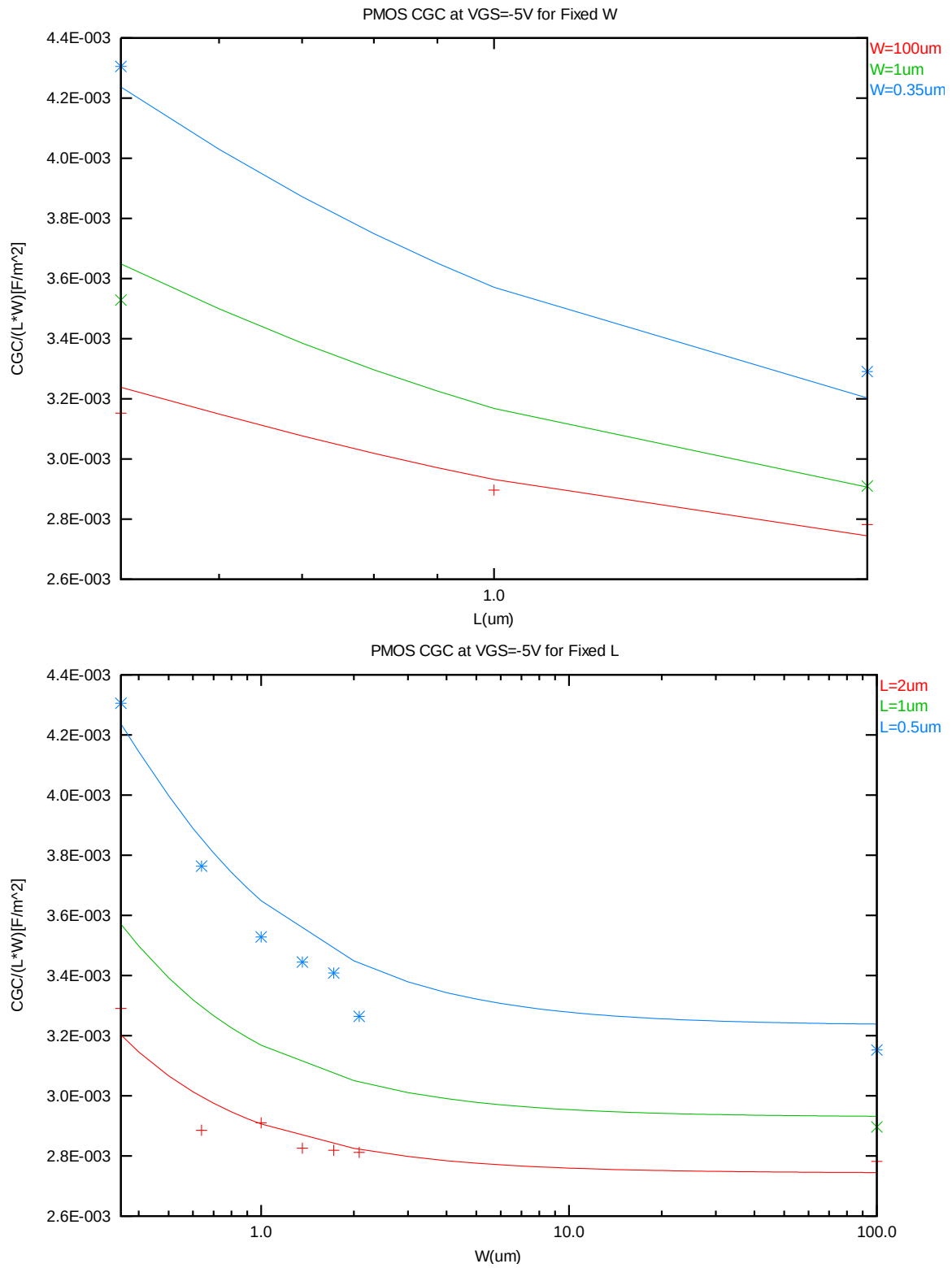


Figure 3-22 C_{GC} vs. channel length & channel width for PMOS

3.5 Parasitic Source / Drain Junction Capacitance Model

3.5.1 Test devices

The following test structures were used for parameter extraction.

Table 3-9 Junction capacitor structures for PMOS

Structures	Area[m ²]	Perimeter[m]	Description
Bottom area junction capacitance	5.00E-08	2.00E-03	Plate junction capacitor
STI bounded sidewall junction capacitance	3.60E-08	3.64E-02	Finger diode
Inner gate sidewall junction	2.76E-08	1.82E-02	Poly finger structure

3.5.2 Measurement conditions

All junction capacitance was measured under voltage sweep conditions summarized in the following table.

Table 3-10 Junction capacitor measurement conditions for PMOS

V _{BIAS} [V]	VH: nwell & VL: p+ From 0 to 5.5 V in 0.05 V step
Frequency [kHz]	100
V _{OSC} [mV]	50
Temp [°C]	-40,25,150

3.5.3 Extracted model parameters and verification plots

CJ, CJSW vs. Vbias plots

The following plots show the variation of bottom area junction and STI bounded sidewall at different bias voltages and temperatures.

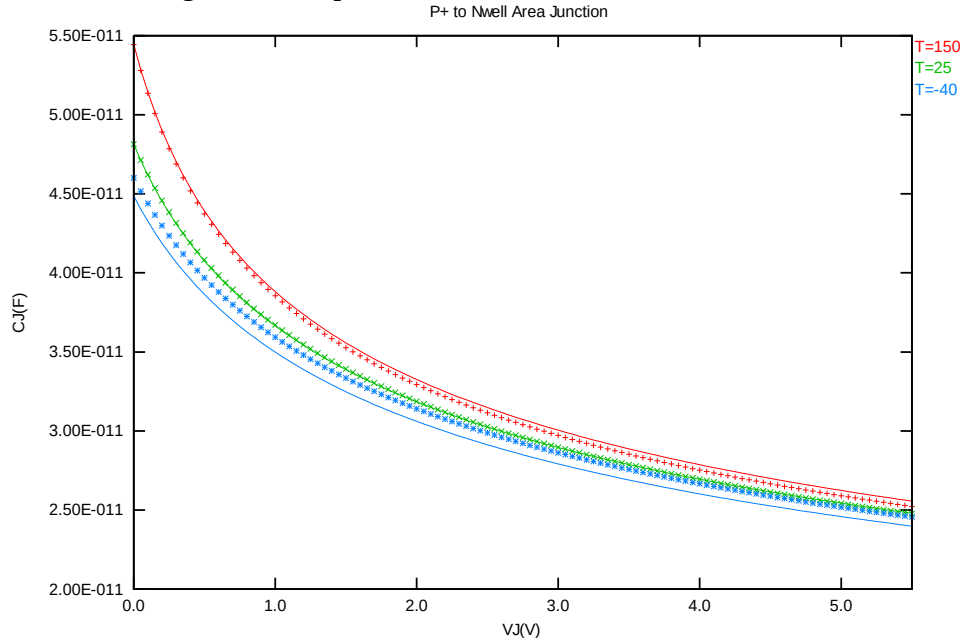


Figure 3-23 Bottom area junction capacitance at -40 C, 25 C and 150 C for PMOS

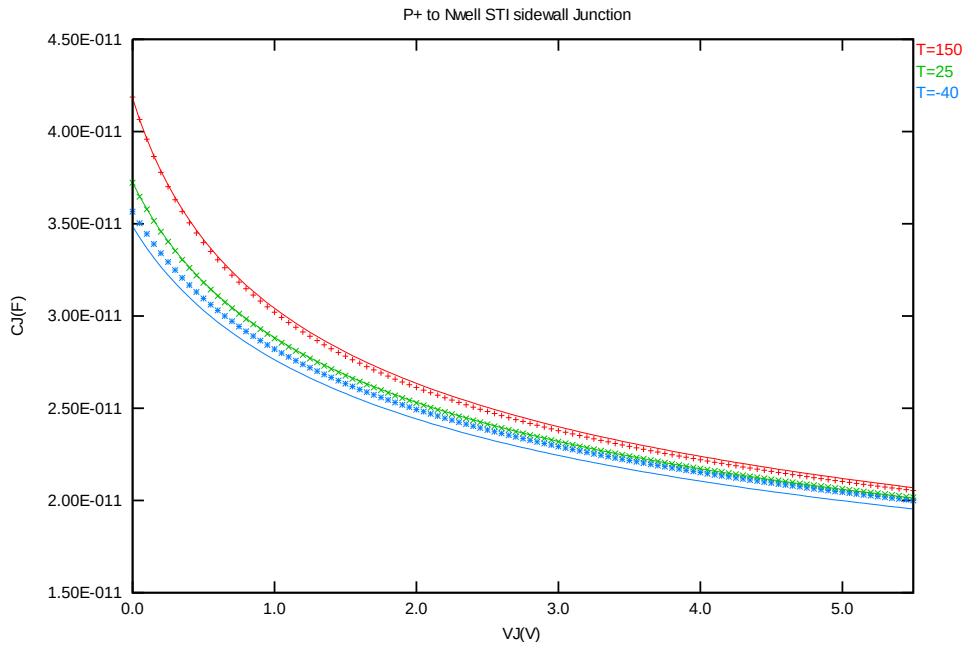


Figure 3-24 STI bounded sidewall junction capacitance at -40 C, 25 C and 150 C for PMOS

4 Worst Case Model

4.1 Wafer Data vs. Corner Plots

Threshold voltage in linear region V_{TLIN}

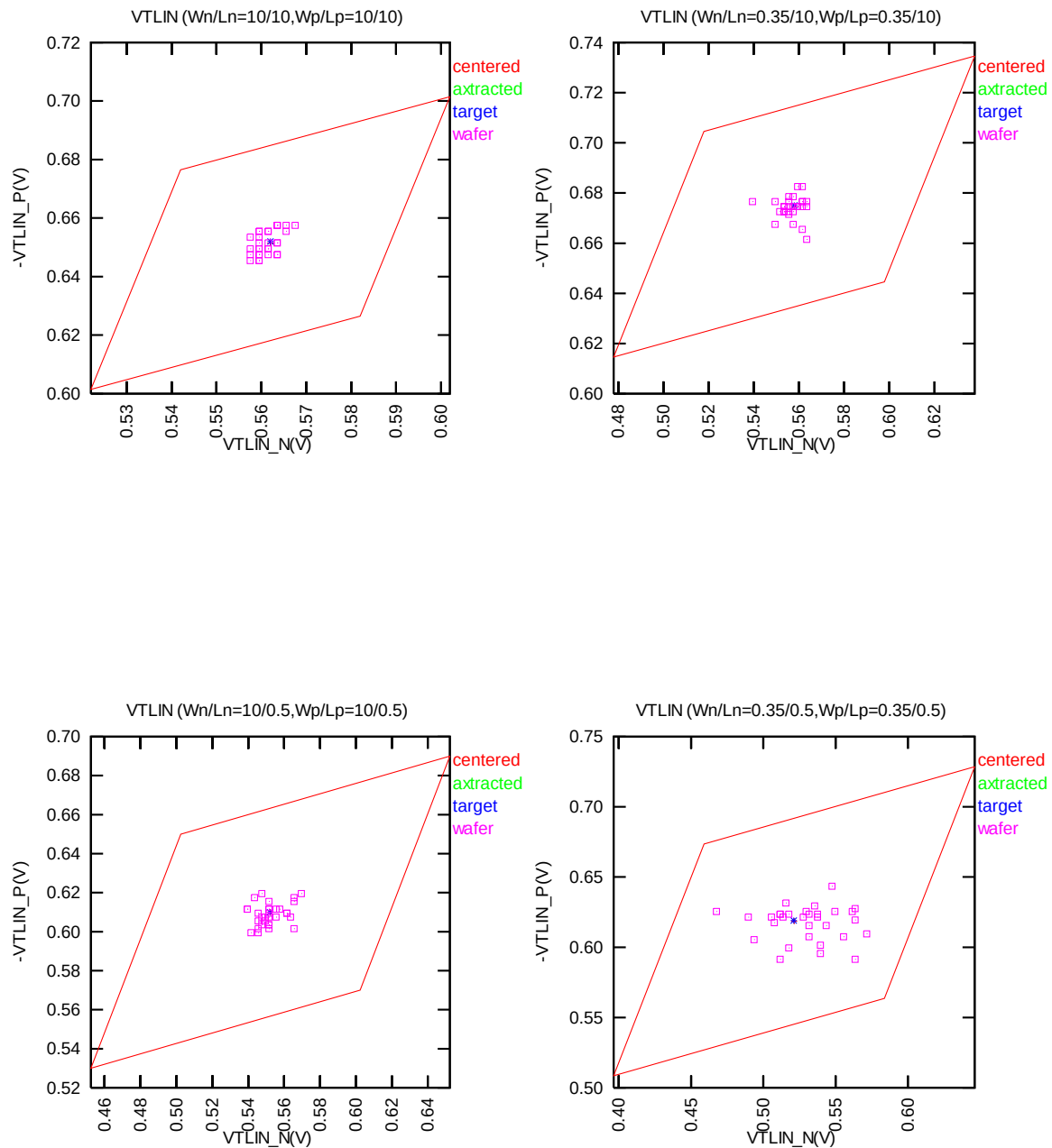


Figure 4-1 Corner plots of threshold voltage at low drain bias

Threshold voltage in saturation region V_{TSAT}

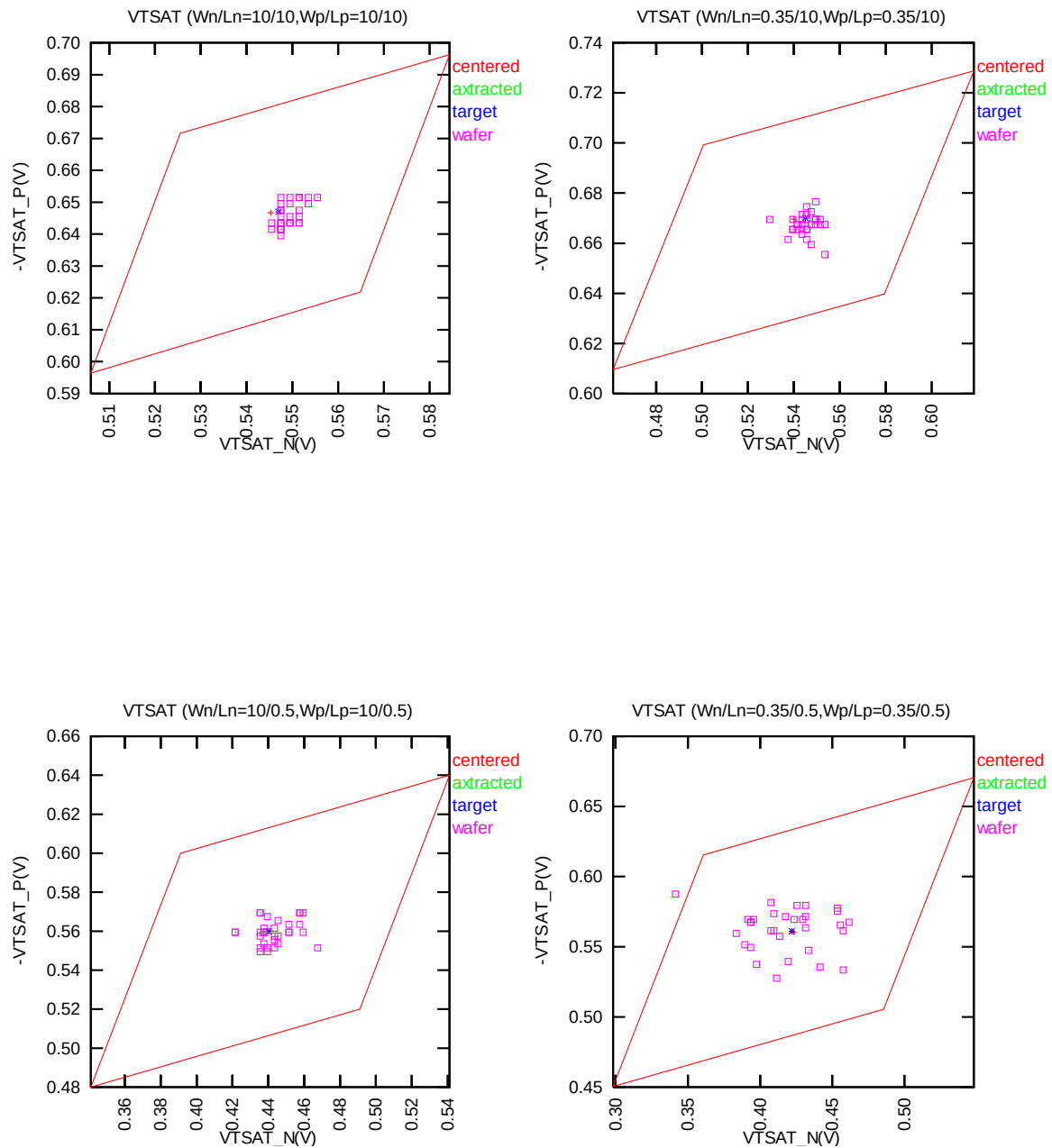


Figure 4-2 Corner plots of threshold voltage at high drain bias

Saturation current I_{ON}

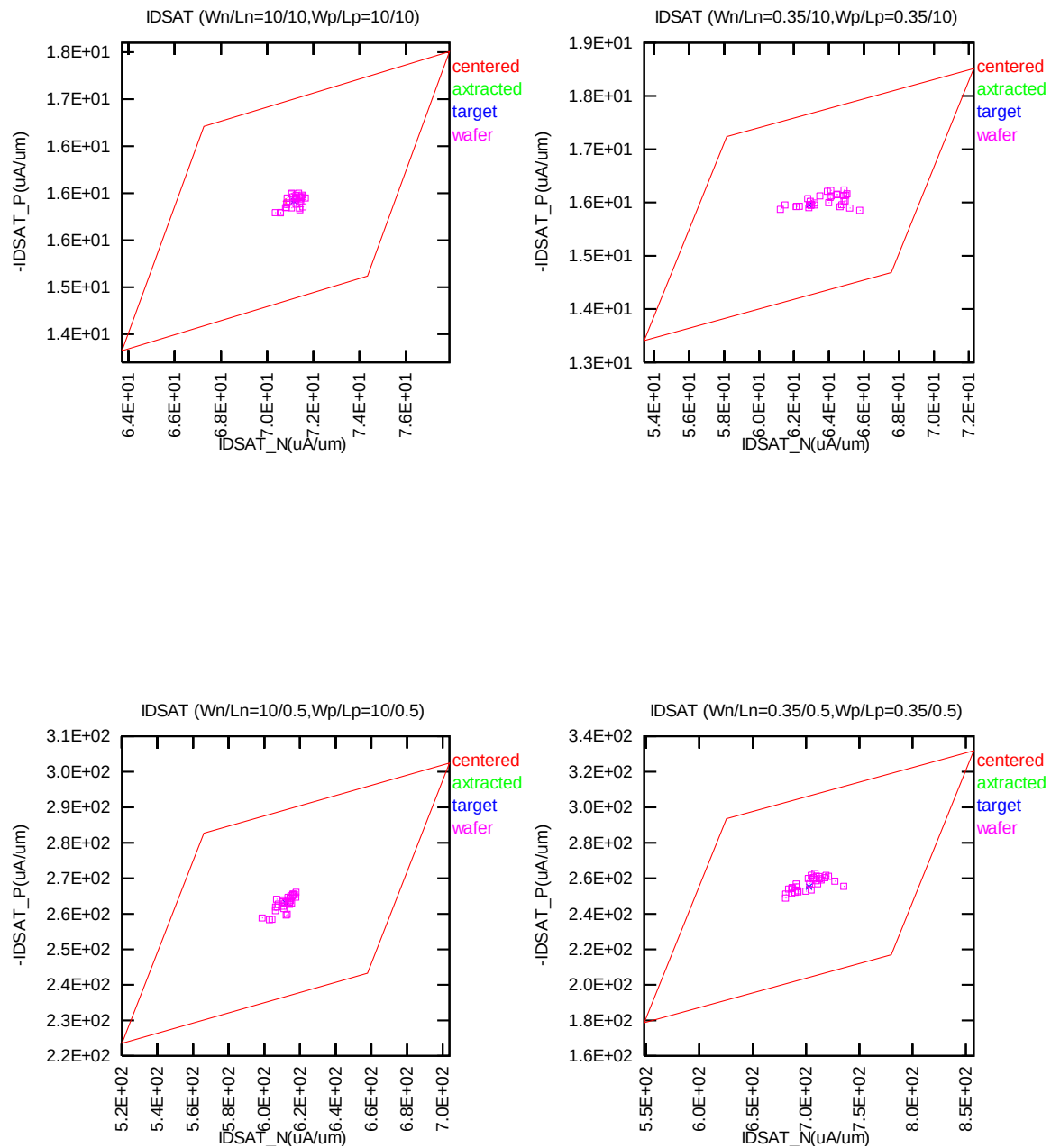


Figure 4-3 Corner plots of on-current

Off state current I_{OFF}

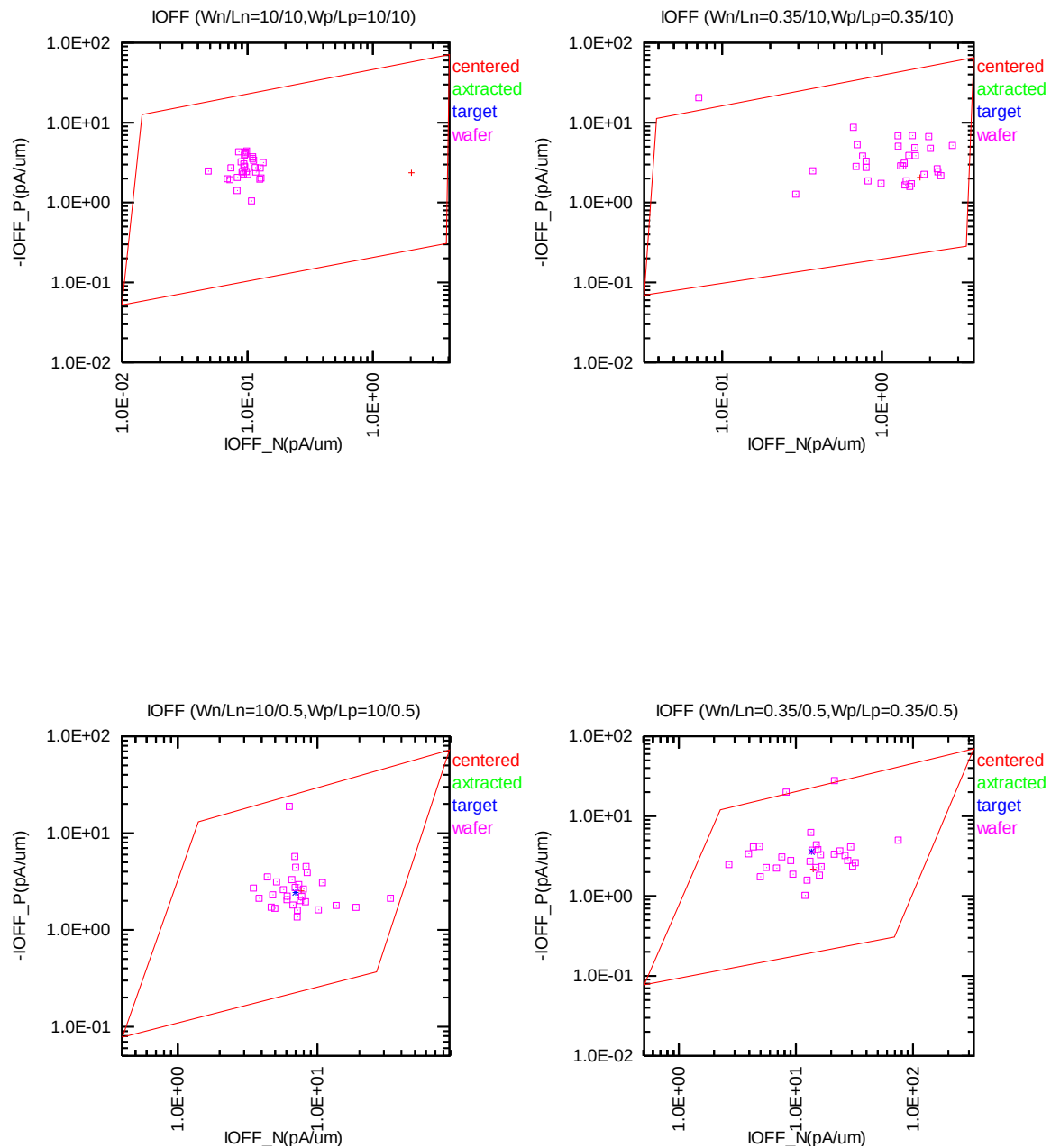


Figure 4-4 Corner plots of off-current

Note: Due to the GIDL effect, the corner graph of off-current is slightly strange.

4.2 Device Performance Reference Tables

Table 4-1 NMOS ID current at $V_{GS}=5\text{ V}$, $V_{DS}=2.5\text{ V}$ and $V_{GS}=5\text{ V}$, $V_{DS}=5\text{ V}$.

NMOS	Temp. (°C)	-40 °C		25 °C		125 °C	
IDSAT (μA)	VDS	2.5V	5V	2.5V	5V	2.5V	5V
TT	9/9	200.44	936.26	152.53	701.88	111.49	492.34
SS	9/9	175.08	842.72	133.57	631.06	97.95	442.00
FF	9/9	227.39	1029.00	172.62	772.65	125.79	543.13
SNFP	9/9	188.77	888.06	143.76	666.49	105.18	468.07
FNFP	9/9	212.32	984.52	161.44	737.26	117.90	516.59
TT	9/0.488	2383.30	6228.80	2102.00	5504.20	1763.20	4596.40
SS	9/0.488	1918.30	5282.90	1690.90	4678.30	1418.90	3915.20
FF	9/0.488	2874.20	7189.40	2532.60	6333.40	2121.10	5272.40
SNFP	9/0.488	2153.40	5754.20	1899.30	5092.10	1593.80	4259.30
FNFP	9/0.488	2620.60	6709.80	2310.40	5919.50	1936.80	4933.90
TT	0.35/9	6.46	31.03	5.16	24.36	4.00	18.09
SS	0.35/9	5.20	26.30	4.18	20.69	3.26	15.40
FF	0.35/9	7.83	35.75	6.22	28.02	4.79	20.78
SNFP	0.35/9	5.85	28.61	4.69	22.52	3.64	16.78
FNFP	0.35/9	7.08	33.45	5.65	26.19	4.36	19.39
TT	0.35/0.488	101.36	276.16	89.23	246.00	74.69	208.01
SS	0.35/0.488	73.99	214.07	65.34	191.92	54.95	163.56
FF	0.35/0.488	130.70	339.46	114.55	300.21	95.35	251.79
SNFP	0.35/0.488	87.47	244.96	77.14	218.99	64.70	186.00
FNFP	0.35/0.488	115.80	307.78	101.75	273.13	84.96	229.91

Table 4-2 NMOS ID current density at $V_{GS}=5\text{ V}$, $V_{DS}=2.5\text{ V}$ and $V_{GS}=5\text{ V}$, $V_{DS}=5\text{ V}$.

NMOS	Temp. (°C)	-40 °C		25 °C		125 °C	
IDSAT ($\mu\text{A}/\mu\text{m}$)	VDS	2.5V	5V	2.5V	5V	2.5V	5V
TT	9/9	22.27	104.03	16.95	77.99	12.39	54.70
SS	9/9	19.45	93.64	14.84	70.12	10.88	49.11
FF	9/9	25.27	114.33	19.18	85.85	13.98	60.35
SNFP	9/9	20.97	98.67	15.97	74.05	11.69	52.01
FNSP	9/9	23.59	109.39	17.94	81.92	13.10	57.40
TT	9/0.488	264.81	692.09	233.56	611.58	195.91	510.71
SS	9/0.488	213.14	586.99	187.88	519.81	157.66	435.02
FF	9/0.488	319.36	798.82	281.40	703.71	235.68	585.82
SNFP	9/0.488	239.27	639.36	211.03	565.79	177.09	473.26
FNSP	9/0.488	291.18	745.53	256.71	657.72	215.20	548.21
TT	0.35/9	18.45	88.66	14.74	69.59	11.42	51.67
SS	0.35/9	14.86	75.13	11.94	59.10	9.33	44.01
FF	0.35/9	22.37	102.14	17.77	80.07	13.68	59.36
SNFP	0.35/9	16.72	81.75	13.39	64.35	10.41	47.94
FNSP	0.35/9	20.24	95.58	16.13	74.83	12.47	55.40
TT	0.35/0.488	289.60	789.03	254.96	702.86	213.39	594.31
SS	0.35/0.488	211.39	611.63	186.70	548.34	157.01	467.31
FF	0.35/0.488	373.43	969.89	327.29	857.74	272.42	719.40
SNFP	0.35/0.488	249.92	699.89	220.39	625.69	184.87	531.43
FNSP	0.35/0.488	330.86	879.37	290.71	780.37	242.73	656.89

Table 4-3 NMOS V_{TLIN} and V_{TSAT} .

NMOS	Temp. (°C)	-40 °C		25 °C		125 °C	
VTH (V)	VDS	0.1V	5V	0.1V	5V	0.1V	5V
TT	9/9	0.633	0.619	0.562	0.546	0.457	0.434
SS	9/9	0.672	0.658	0.603	0.585	0.499	0.474
FF	9/9	0.593	0.580	0.522	0.506	0.415	0.393
SNFP	9/9	0.652	0.639	0.583	0.565	0.478	0.454
FNSP	9/9	0.613	0.600	0.542	0.526	0.436	0.413
TT	9/0.488	0.617	0.511	0.552	0.441	0.457	0.335
SS	9/0.488	0.715	0.609	0.652	0.540	0.559	0.437
FF	9/0.488	0.517	0.411	0.451	0.340	0.354	0.232
SNFP	9/0.488	0.666	0.559	0.602	0.490	0.507	0.385
FNSP	9/0.488	0.567	0.460	0.502	0.390	0.405	0.283
TT	0.35/9	0.629	0.615	0.558	0.540	0.450	0.426
SS	0.35/9	0.708	0.692	0.638	0.618	0.533	0.506
FF	0.35/9	0.550	0.537	0.478	0.461	0.367	0.345
SNFP	0.35/9	0.668	0.653	0.598	0.579	0.491	0.466
FNSP	0.35/9	0.590	0.576	0.518	0.500	0.409	0.385
TT	0.35/0.488	0.581	0.488	0.521	0.423	0.433	0.325
SS	0.35/0.488	0.703	0.610	0.645	0.547	0.562	0.453
FF	0.35/0.488	0.457	0.363	0.395	0.297	0.304	0.196
SNFP	0.35/0.488	0.641	0.548	0.583	0.485	0.497	0.389
FNSP	0.35/0.488	0.519	0.425	0.458	0.360	0.368	0.260

Table 4-4 NMOS ID current at $V_{GS}=0$ and $V_{DS}=0.1$ V or $V_{DS}=5$ V (off current).

NMOS	Temp. (°C)	-40 °C		25 °C		125 °C	
IOFF (A)	VDS	0.1V	5V	0.1V	5V	0.1V	5V
TT	9/9	1.72E-14	2.75E-12	3.27E-13	1.83E-11	7.13E-11	1.80E-10
SS	9/9	2.22E-16	1.07E-14	6.97E-14	1.03E-13	3.38E-11	4.20E-11
FF	9/9	3.45E-14	5.50E-12	7.78E-13	3.67E-11	1.52E-10	3.73E-10
SNFP	9/9	2.78E-16	1.07E-14	1.05E-13	1.42E-13	4.67E-11	5.81E-11
FNSP	9/9	3.25E-14	5.22E-12	5.92E-13	3.47E-11	1.08E-10	3.09E-10
TT	9/0.488	2.23E-14	2.95E-12	3.22E-12	6.89E-11	9.89E-10	9.17E-09
SS	9/0.488	4.44E-16	1.78E-14	2.28E-13	3.69E-12	1.38E-10	1.24E-09
FF	9/0.488	1.97E-13	1.07E-11	4.73E-11	8.18E-10	7.72E-09	7.06E-08
SNFP	9/0.488	1.11E-15	3.55E-14	7.82E-13	1.29E-11	3.53E-10	3.25E-09
FNSP	9/0.488	6.41E-14	6.26E-12	1.31E-11	2.44E-10	2.88E-09	2.64E-08
TT	0.35/9	7.56E-16	9.90E-14	1.45E-14	6.08E-13	3.53E-12	7.40E-12
SS	0.35/9	1.98E-16	9.99E-15	1.44E-15	1.14E-14	7.85E-13	9.80E-13
FF	0.35/9	1.53E-15	1.95E-13	8.94E-14	1.33E-12	1.69E-11	2.73E-11
SNFP	0.35/9	2.05E-16	9.99E-15	3.40E-15	1.40E-14	1.57E-12	1.96E-12
FNSP	0.35/9	1.31E-15	1.83E-13	3.85E-14	1.18E-12	7.99E-12	1.58E-11
TT	0.35/0.488	1.68E-15	1.36E-13	3.20E-13	4.92E-12	6.57E-11	5.01E-10
SS	0.35/0.488	2.15E-16	1.03E-14	1.27E-14	1.82E-13	5.67E-12	4.33E-11
FF	0.35/0.488	6.35E-14	1.64E-12	9.38E-12	1.19E-10	8.38E-10	5.99E-09
SNFP	0.35/0.488	3.05E-16	1.25E-14	5.88E-14	8.14E-13	1.84E-11	1.42E-10
FNSP	0.35/0.488	9.24E-15	4.00E-13	1.82E-12	2.49E-11	2.45E-10	1.81E-09

*GMINDC=1.0E-15

Table 4-5 PMOS ID current at $V_{GS}=-5$ V, $V_{DS}=-2.5$ V and $V_{GS}=-5$ V, $V_{DS}=-5$ V.

PMOS	Temp. (°C)	-40 °C		25 °C		125 °C	
-IDSAT (μA)	-VDS	2.5V	5V	2.5V	5V	2.5V	5V
TT	9/9	40.81	191.57	33.71	158.60	27.35	128.96
SS	9/9	35.22	172.61	29.14	142.63	23.70	115.59
FF	9/9	46.82	210.37	38.62	174.56	31.27	142.48
SNFP	9/9	43.59	201.43	35.95	166.58	29.12	135.24
FNSP	9/9	38.10	181.73	31.52	150.61	25.63	122.66
TT	9/0.488	745.11	2661.40	645.22	2365.00	545.73	2053.90
SS	9/0.488	610.60	2247.00	532.59	2009.40	454.03	1754.80
FF	9/0.488	891.47	3073.00	768.08	2720.60	645.95	2356.50
SNFP	9/0.488	814.42	2872.00	702.66	2542.80	591.88	2198.80
FNSP	9/0.488	678.24	2451.00	589.79	2187.20	501.19	1908.80
TT	0.35/9	1.47	7.66	1.19	6.19	0.94	4.90
SS	0.35/9	1.17	6.45	0.95	5.20	0.75	4.09
FF	0.35/9	1.79	8.86	1.45	7.19	1.14	5.72
SNFP	0.35/9	1.62	8.28	1.31	6.69	1.03	5.29
FNSP	0.35/9	1.32	7.04	1.07	5.69	0.85	4.51
TT	0.35/0.488	27.07	103.13	22.54	89.34	18.08	74.53
SS	0.35/0.488	18.01	72.31	14.93	62.54	11.89	51.85
FF	0.35/0.488	37.07	134.07	30.89	116.14	24.81	97.15
SNFP	0.35/0.488	31.88	118.79	26.54	102.74	21.29	85.59
FNSP	0.35/0.488	22.46	87.56	18.69	75.94	14.98	63.38

Table 4-6 PMOS ID current density at $V_{GS}=-5$ V, $V_{DS}=-2.5$ V and $V_{GS}=-5$ V, $V_{DS}=-5$ V.

PMOS	Temp. (°C)	-40 °C		25 °C		125 °C	
-IDSAT (μ A/ μ m)	-VDS	2.5V	5V	2.5V	5V	2.5V	5V
TT	9/9	4.53	21.29	3.75	17.62	3.04	14.33
SS	9/9	3.91	19.18	3.24	15.85	2.63	12.84
FF	9/9	5.20	23.37	4.29	19.40	3.47	15.83
SNFP	9/9	4.84	22.38	4.00	18.51	3.24	15.03
FNSP	9/9	4.23	20.19	3.50	16.73	2.85	13.63
TT	9/0.488	82.79	295.71	71.69	262.78	60.64	228.21
SS	9/0.488	67.84	249.67	59.18	223.27	50.45	194.98
FF	9/0.488	99.05	341.44	85.34	302.29	71.77	261.83
SNFP	9/0.488	90.49	319.11	78.07	282.53	65.76	244.31
FNSP	9/0.488	75.36	272.33	65.53	243.02	55.69	212.09
TT	0.35/9	4.20	21.88	3.39	17.69	2.68	14.00
SS	0.35/9	3.36	18.42	2.71	14.84	2.14	11.69
FF	0.35/9	5.13	25.31	4.14	20.54	3.26	16.34
SNFP	0.35/9	4.64	23.65	3.74	19.11	2.95	15.12
FNSP	0.35/9	3.78	20.11	3.06	16.27	2.42	12.88
TT	0.35/0.488	77.33	294.66	64.40	255.25	51.66	212.93
SS	0.35/0.488	51.46	206.59	42.66	178.68	33.98	148.14
FF	0.35/0.488	105.91	383.06	88.25	331.83	70.89	277.57
SNFP	0.35/0.488	91.09	339.40	75.83	293.54	60.83	244.55
FNSP	0.35/0.488	64.17	250.17	53.41	216.97	42.79	181.09

Table 4-7 PMOS V_{TLIN} and V_{TSAT} .

PMOS	Temp. (°C)	-40 °C		25 °C		125 °C	
-VTH (V)	-VDS	0.1V	5V	0.1V	5V	0.1V	5V
TT	9/9	0.718	0.714	0.651	0.646	0.549	0.542
SS	9/9	0.767	0.763	0.702	0.696	0.601	0.593
FF	9/9	0.668	0.665	0.601	0.596	0.497	0.488
SNFP	9/9	0.693	0.690	0.626	0.622	0.523	0.516
FNSP	9/9	0.743	0.739	0.677	0.672	0.575	0.567
TT	9/0.488	0.684	0.637	0.610	0.560	0.495	0.440
SS	9/0.488	0.763	0.716	0.690	0.640	0.577	0.522
FF	9/0.488	0.605	0.557	0.530	0.480	0.413	0.359
SNFP	9/0.488	0.645	0.597	0.570	0.520	0.454	0.400
FNSP	9/0.488	0.724	0.676	0.650	0.600	0.536	0.481
TT	0.35/9	0.738	0.734	0.674	0.669	0.577	0.569
SS	0.35/9	0.796	0.792	0.734	0.729	0.641	0.631
FF	0.35/9	0.679	0.675	0.614	0.609	0.514	0.504
SNFP	0.35/9	0.709	0.705	0.644	0.639	0.545	0.537
FNSP	0.35/9	0.767	0.763	0.704	0.699	0.609	0.600
TT	0.35/0.488	0.689	0.633	0.619	0.560	0.512	0.447
SS	0.35/0.488	0.794	0.740	0.729	0.670	0.630	0.564
FF	0.35/0.488	0.582	0.526	0.509	0.450	0.396	0.333
SNFP	0.35/0.488	0.635	0.580	0.564	0.505	0.454	0.390
FNSP	0.35/0.488	0.742	0.687	0.674	0.615	0.570	0.505

Table 4-8 PMOS ID current at $V_{GS}=0$ and $V_{DS}=-0.1$ V or $V_{DS}=-5$ V (off current).

PMOS	Temp. (°C)	-40 °C		25 °C		125 °C	
-IOFF (A)	-VDS	0.1V	5V	0.1V	5V	0.1V	5V
TT	9/9	1.19E-14	3.89E-12	1.06E-13	2.12E-11	3.67E-12	9.66E-11
SS	9/9	4.44E-16	8.88E-14	3.89E-15	4.65E-13	1.44E-12	3.55E-12
FF	9/9	3.51E-13	1.17E-10	3.08E-12	6.39E-10	2.72E-11	2.83E-09
SNFP	9/9	6.24E-14	2.08E-11	5.53E-13	1.13E-10	8.31E-12	5.05E-10
FNSP	9/9	1.78E-15	5.15E-13	1.59E-14	2.76E-12	2.03E-12	1.42E-11
TT	9/0.488	1.24E-14	3.89E-12	5.81E-13	2.27E-11	2.05E-10	6.03E-10
SS	9/0.488	4.44E-16	9.59E-14	7.75E-14	7.03E-13	4.92E-11	1.23E-10
FF	9/0.488	3.58E-13	1.17E-10	6.27E-12	6.50E-10	8.92E-10	5.02E-09
SNFP	9/0.488	6.46E-14	2.08E-11	1.87E-12	1.18E-10	4.43E-10	1.60E-09
FNSP	9/0.488	1.89E-15	5.15E-13	1.87E-13	3.32E-12	9.49E-11	2.48E-10
TT	0.35/9	5.93E-16	1.41E-13	3.72E-15	7.22E-13	1.11E-13	3.25E-12
SS	0.35/9	2.08E-16	1.27E-14	2.95E-16	2.43E-14	4.48E-14	1.20E-13
FF	0.35/9	1.27E-14	4.20E-12	1.10E-13	2.28E-11	9.40E-13	1.01E-10
SNFP	0.35/9	2.37E-15	7.32E-13	1.93E-14	3.94E-12	2.66E-13	1.75E-11
FNSP	0.35/9	2.50E-16	2.62E-14	6.80E-16	9.95E-14	6.13E-14	4.66E-13
TT	0.35/0.488	6.00E-16	1.36E-13	2.01E-14	7.60E-13	6.60E-12	2.16E-11
SS	0.35/0.488	2.08E-16	1.22E-14	1.68E-15	2.73E-14	9.55E-13	2.57E-12
FF	0.35/0.488	1.37E-14	4.32E-12	3.37E-13	2.44E-11	4.99E-11	2.46E-10
SNFP	0.35/0.488	2.49E-15	7.35E-13	8.37E-14	4.21E-12	1.88E-11	7.05E-11
FNSP	0.35/0.488	2.46E-16	2.51E-14	4.95E-15	1.08E-13	2.34E-12	6.82E-12

*GMINDC=1.0E-15

5 Shallow Trench Isolation (STI) or LOD Stress Effect

5.1 Simulation Results for NMOS at SA=SB=0.324 um (SAREF=3.6um)

The simulation results for the NMOS characterization are summarized in the table below.
All parameters are given for T=25 °C, $V_{BS}=0$ V unless specified otherwise.

Parameter	Condition	Unit	Simulation Value
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Long channel device (W/L= 10 μ m / 10 μ m)

V_{TLIN}	$V_{DS} = 0.1$ V	V	0.563
I_{ON}	$V_{DS} = 5$ V	μ A/ μ m	70.42

Wide and short channel device (W/L= 10 μ m / 0.5 μ m)

V_{TLIN}	$V_{DS} = 0.1$ V	V	0.567
V_{TSAT}	$V_{DS} = 5$ V	V	0.456
Body Effect V_T shift	$V_{DS} = 5$ V $V_{BS} = 0 \sim -3$ V	V	0.520
DIBL	$V_{DS} = 0.1 \sim 5$ V	V	0.112
I_{ON}	$V_{DS}=V_{GS} = 5$ V	μ A/ μ m	570.95
I_{OFF}	$V_{DS} = 5$ V, $V_{GS}=0$ V	A/ μ m	6.0E-12

Narrow and long channel device (W/L= 0.35 μ m / 10 μ m)

V_{TLIN}	$V_{DS} = 0.1$ V	V	0.559
V_{TSAT}	$V_{DS} = 5$ V	V	0.541
I_{ON}	$V_{DS}=V_{GS} = 5$ V	μ A/ μ m	62.51

Narrow and short channel device (W/L= 0.35 μ m / 0.5 μ m)

V_{TLIN}	$V_{DS} = 0.1$ V	V	0.536
V_{TSAT}	$V_{DS} = 5$ V	V	0.438
I_{ON}	$V_{DS}=V_{GS} = 5$ V	μ A/ μ m	661.60
I_{OFF}	$V_{DS} = 5$ V, $V_{GS}=0$ V	A/ μ m	1.0E-11

5.1.1 Model verification plots for NMOS

The L and W in the plots refer to LDES and WDES respectively.

V_T vs. SA

The following figures show the SA dependence of linear and saturation threshold voltage extracted at $I_D=300\text{nA} \cdot W_{DES} \cdot 0.9 / (L_{DES} \cdot 0.9 + 12\text{nm})$ at 25°C .

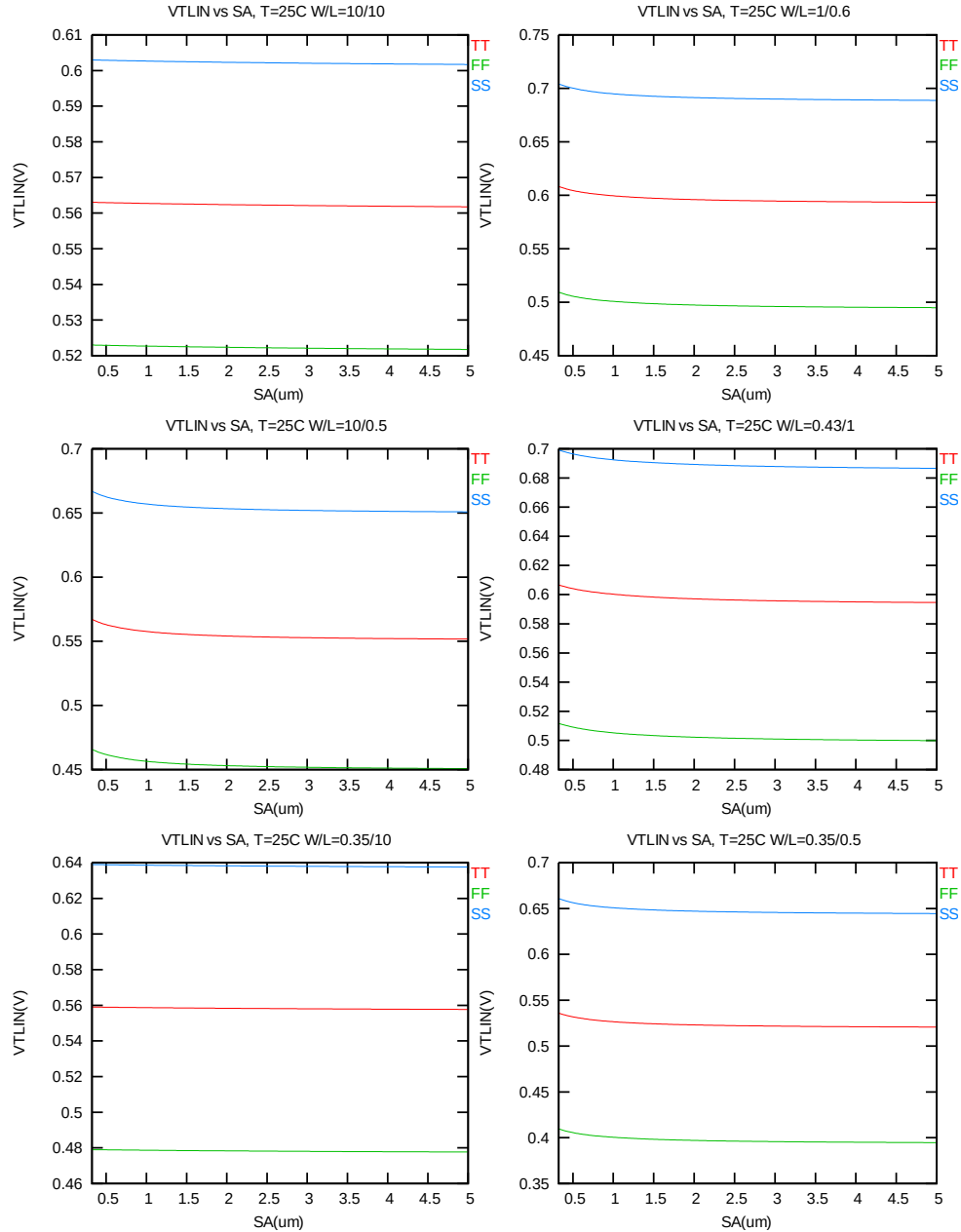


Figure 5-1 V_{TLIN} vs. SA ($V_{DS}=0.1$ V) for NMOS

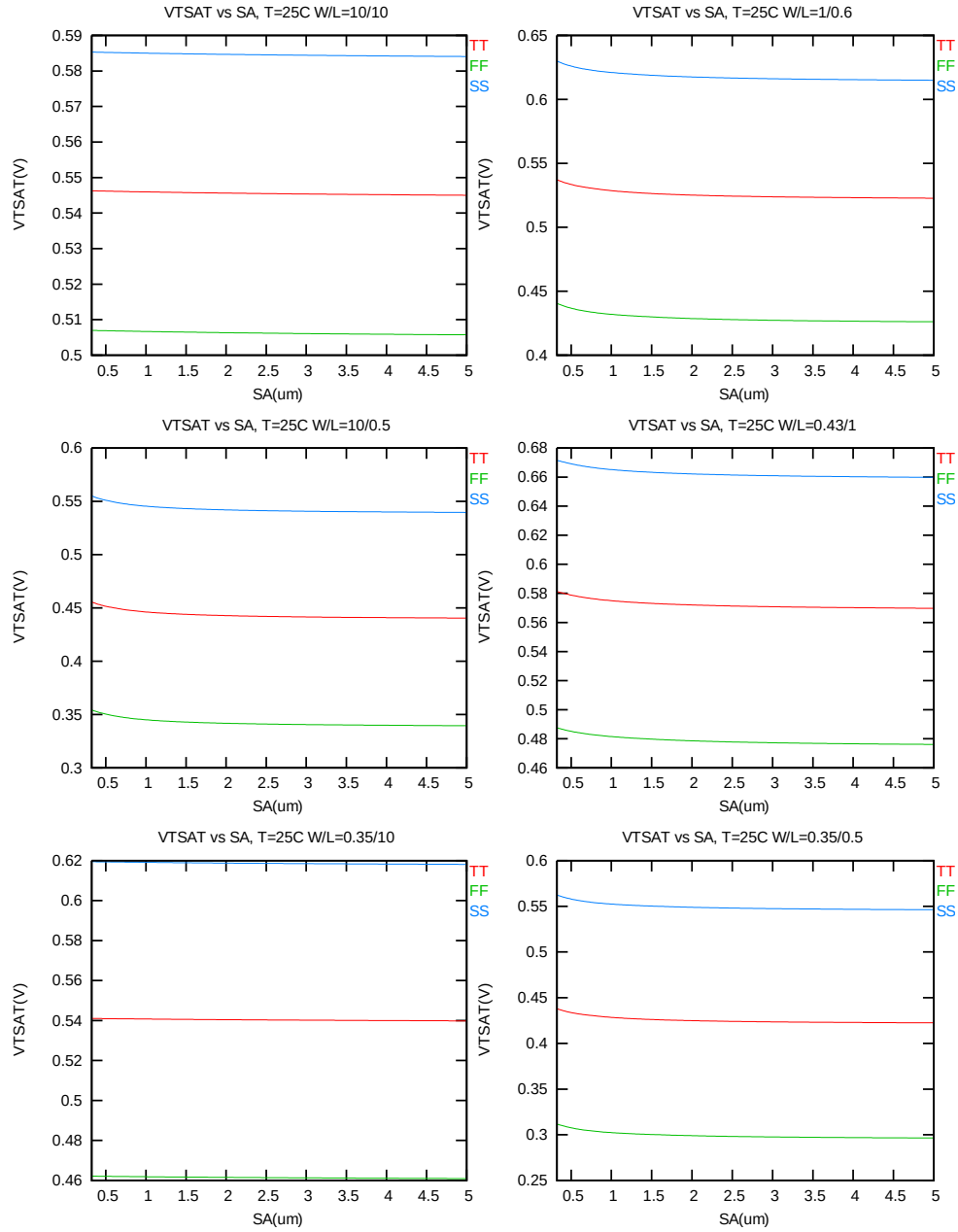


Figure 5-2 V_{TSAT} vs. SA ($V_{DS}=5$ V) for NMOS

I_{ON} vs. SA

The following plots show the SA dependence of I_{ON} extracted at $V_{DS}=V_{GS}=5$ V for $V_{BS}=0$ at 25C.

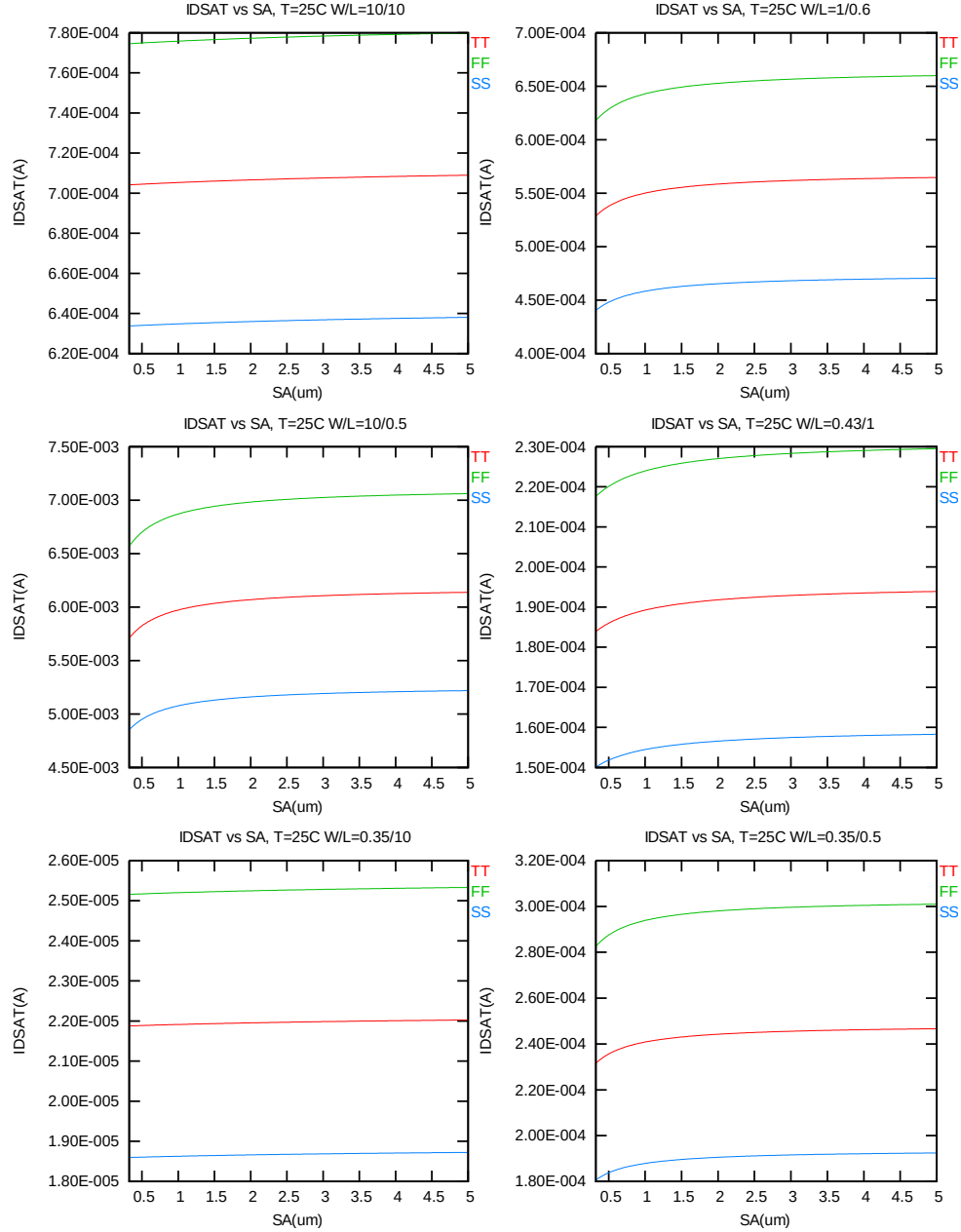


Figure 5-3 Saturation current I_{ON} vs. SA for NMOS

5.2 Simulation Results for PMOS at SA=0.324 um (SAREF=3.6um)

The simulation results for the PMOS characterization are summarized in the table below.

Table 5-1 Simulation results for PMOS

All parameters are given for T=25 °C, $V_{BS}=0$ V unless specified otherwise.

Parameter	Condition	Unit	Simulation Value
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Long channel device (W/L= 10 μ m / 10 μ m)

V_{TLIN}	$V_{DS} = -0.1$ V	V	-0.651
I_{ON}	$V_{DS} = -5$ V	μ A/ μ m	-16.12

Wide and short channel device (W/L= 10 μ m / 0.5 μ m)

V_{TLIN}	$V_{DS} = -0.1$ V	V	-0.603
V_{TSAT}	$V_{DS} = -5$ V	V	-0.548
Body Effect V_T shift	$V_{DS} = -5$ V $V_{BS} = 0 \sim 3$ V	V	-0.703
DIBL	$V_{DS} = -0.1 \sim -5$ V	V	-0.055
I_{ON}	$V_{DS}=V_{GS}= -5$ V	μ A/ μ m	-273.44
I_{OFF}	$V_{DS}= -5$ V, $V_{GS}=0$ V	A/ μ m	-2.6E-12

Narrow and long channel device (W/L= 0.35 μ m / 10 μ m)

V_{TLIN}	$V_{DS} = -0.1$ V	V	-0.674
V_{TSAT}	$V_{DS}= -5$ V	V	-0.669
I_{ON}	$V_{DS}=V_{GS}= -5$ V	μ A/ μ m	-16.17

Narrow and short channel device (W/L= 0.35 μ m / 0.5 μ m)

V_{TLIN}	$V_{DS} = -0.1$ V	V	-0.612
V_{TSAT}	$V_{DS}= -5$ V	V	-0.549
I_{ON}	$V_{DS}=V_{GS}= -5$ V	μ A/ μ m	-264.98
I_{OFF}	$V_{DS}= -5$ V, $V_{GS}=0$ V	A/ μ m	-2.2E-12

5.2.1 Model verification plots for PMOS

The L and W in the plots refer to LDES and WDES respectively

V_T vs. SA

The following figures show the SA dependence of linear and saturation threshold voltage extracted at $I_D = -70\text{nA} \cdot W_{DES} \cdot 0.9 / (L_{DES} \cdot 0.9 + 12\text{nm})$ at 25°C .

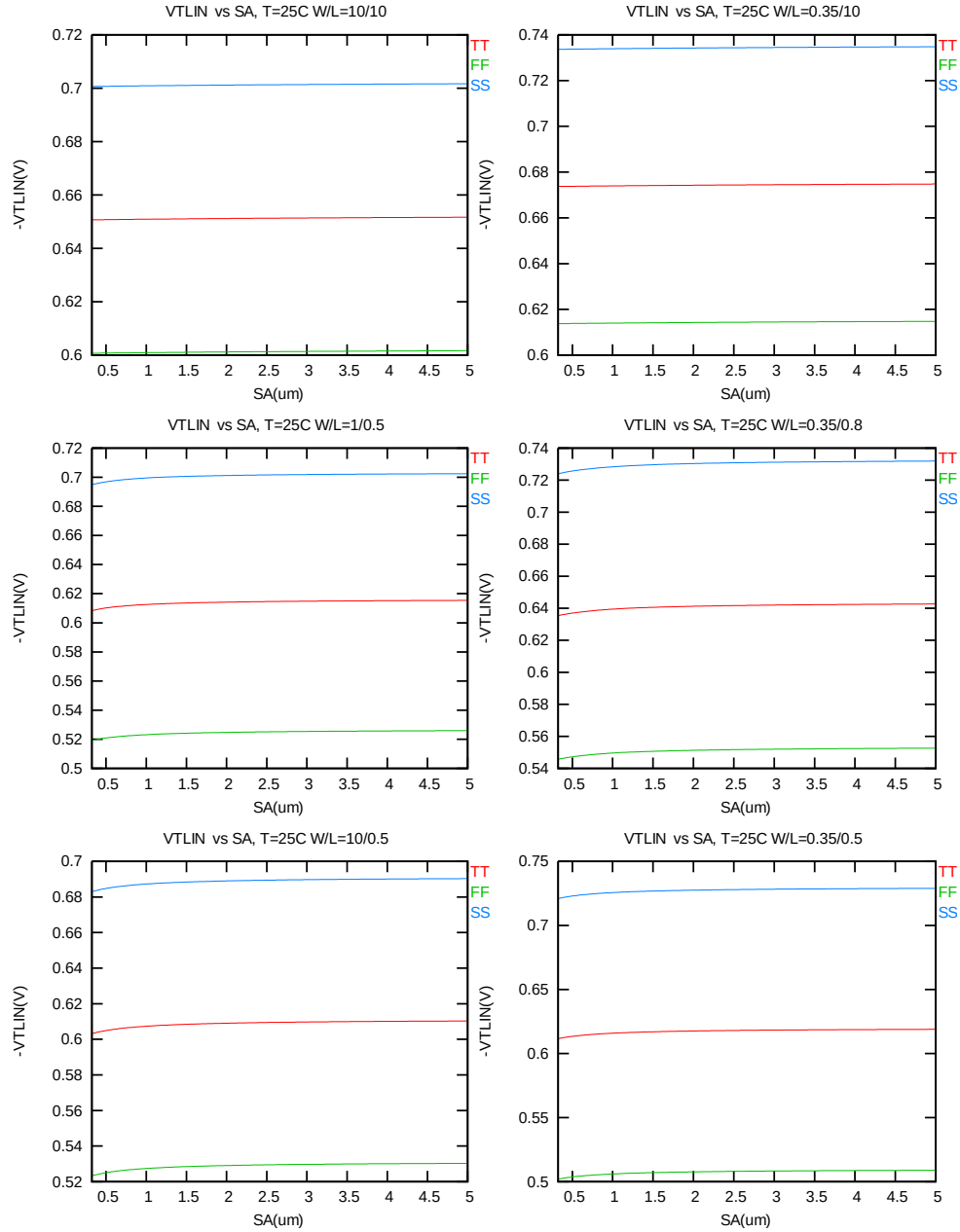


Figure 5-4 V_{TLIN} vs. SA ($V_{DS} = -0.1\text{ V}$) for PMOS

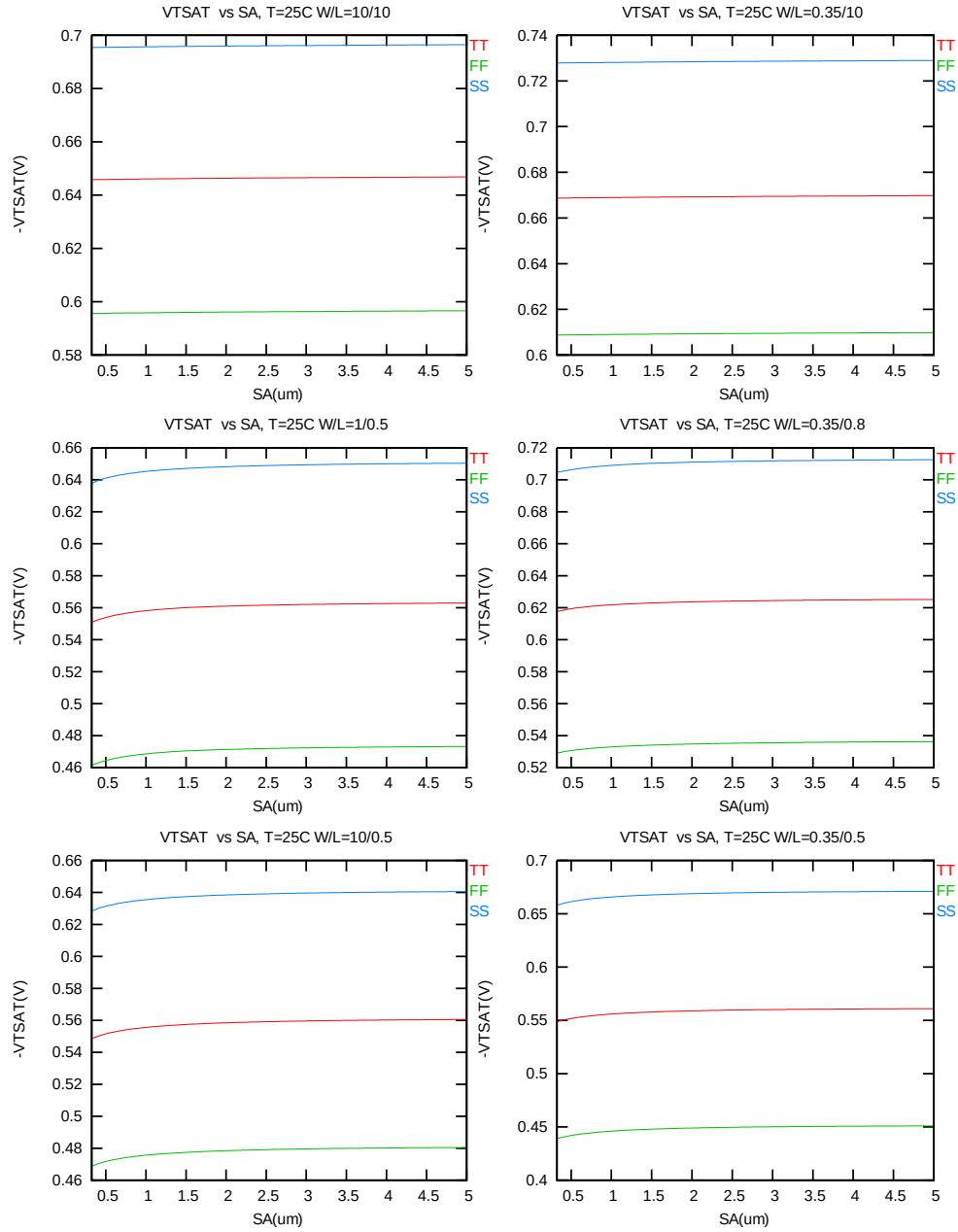


Figure 5-5 V_{TSAT} vs. S_A ($V_{DS} = -5 \text{ V}$) for PMOS

The following plots show the SA dependence of I_{ON} extracted at $V_{DS}=V_{GS}= -5$ V for $V_{BS}=0$ at 25C.

I_{ON} vs. SA

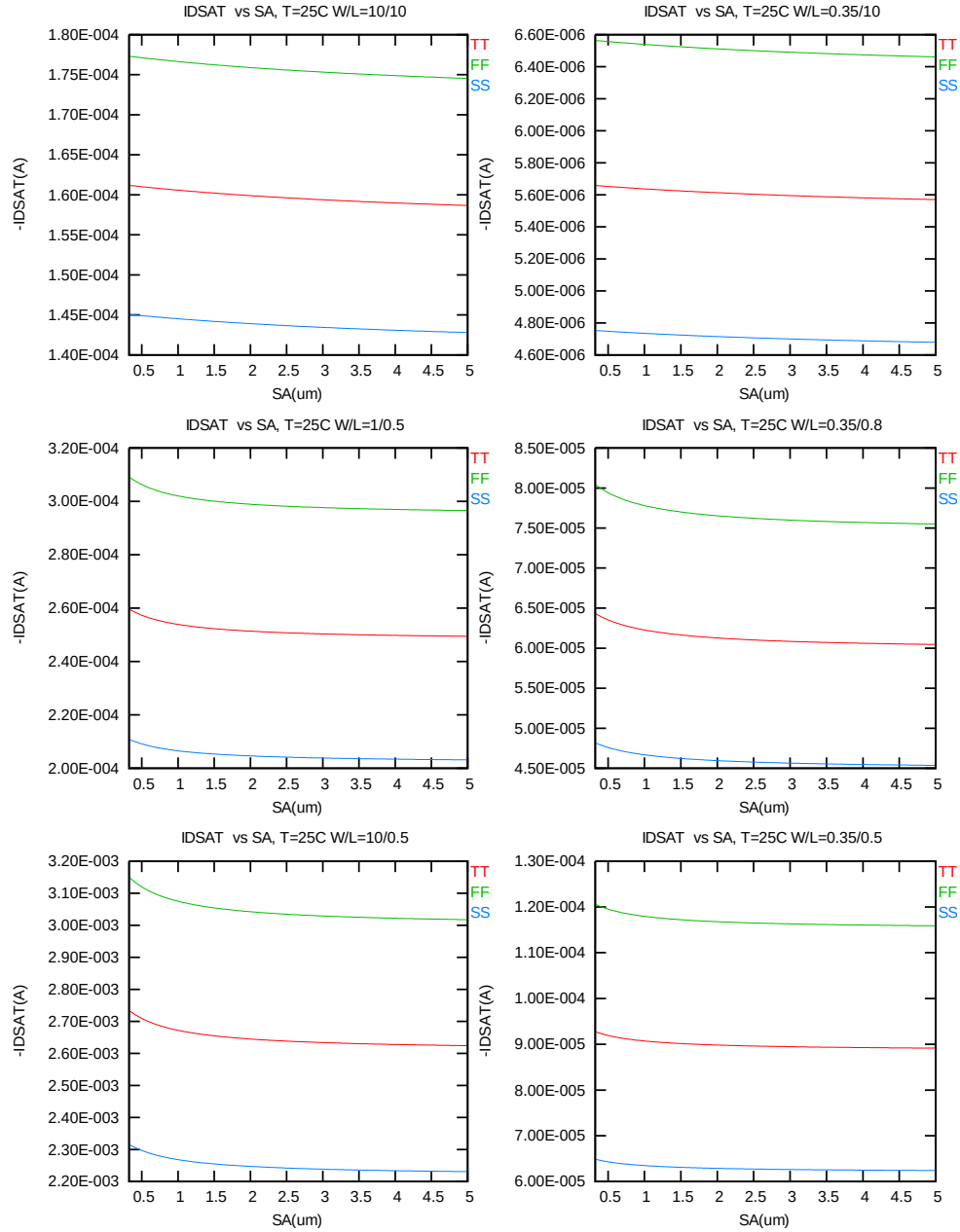


Figure 5-6 Saturation current I_{ON} vs. SA for PMOS

5.3 V_T and I_{ON} Variation v. SA for both N/PMOS

Delta V_T is defined as $V_T(SA) - V_T(3.6\mu m)$ for NMOS and $-[V_T(SA) - V_T(3.6\mu m)]$ for PMOS. Delta I_{ON} is defined as $[I_{ON}(SA) - I_{ON}(3.6\mu m)] / I_{ON}(3.6\mu m) * 100\%$.

Delta V_T and I_{ON} vs. SA

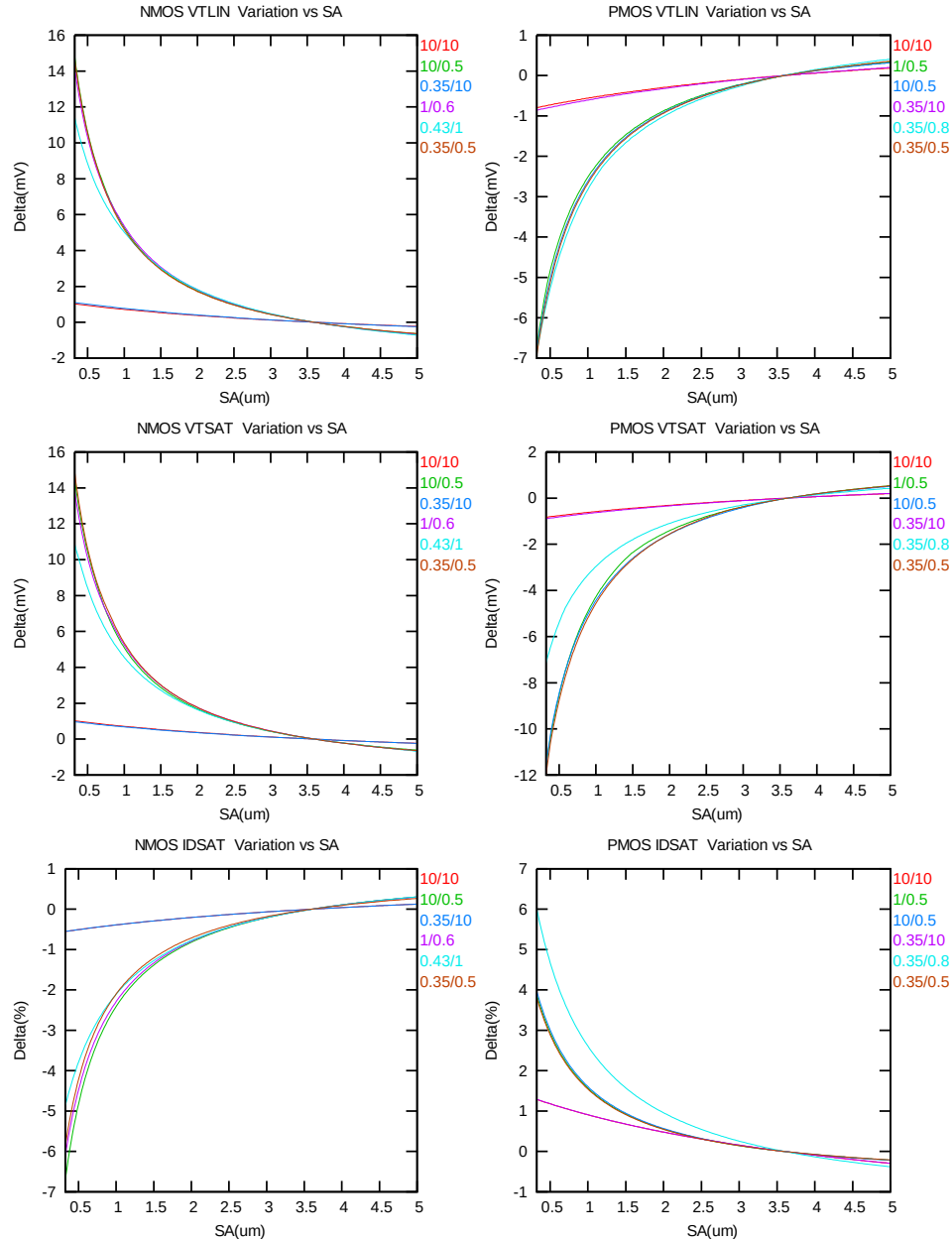


Figure 5-7 Delta V_T and I_{ON} vs. SA for both N/PMOS

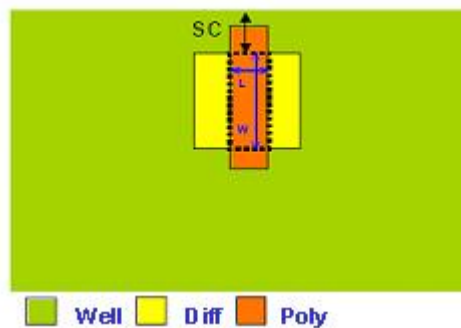
6 Well Proximity Effect (WPE)

6.1 Simulation Results for NMOS at SC=0.4um/0.75um for 1V/1P structure (SCREF=2um)

Note:

1. SA and SB are set to 0.324um in all the following plots and tables.
2. 1V structure: device is close to only one well edge, which is vertical relative to the poly of device (Fig. 6-1a); 1P structure: device is close to only one well edge, which is parallel to the poly of device (Fig. 6-1b)

(a) 1V structure



(b) 1P structure

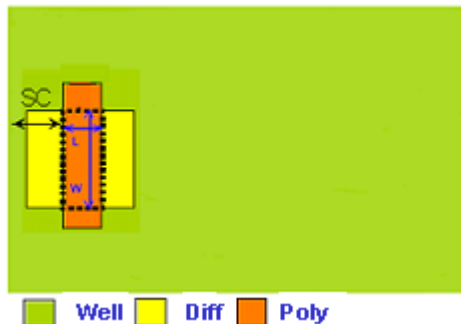


Figure 6-1 Layout example for WPE effect

The simulation results for the NMOS characterization are summarized in the table below.

Table 6-1 Simulation results for NMOS

All parameters are given for T=25°C, V_{BS}=0 V unless specified otherwise.

Parameter	Condition	Unit	Simulation Value	
			1V structure	1P structure

Long channel device (W/L= 10 μm / 10 μm)

V _{TLIN}	V _{DS} = 0.1 V	V	0.568	0.563
I _{ON}	V _{DS} = 5 V	μA/μm	70.22	70.41

Wide and short channel device (W/L= 10 μm / 0.5 μm)

V _{TLIN}	V _{DS} = 0.1 V	V	0.572	0.588
V _{TSAT}	V _{DS} = 5 V	V	0.461	0.476
Body Effect V _T shift	V _{DS} = 5 V V _{BS} = 0 ~ -3 V	V	0.520	0.520
DIBL	V _{DS} = 0.1~5 V	V	0.111	0.112
I _{ON}	V _{DS} =V _{GS} = 5 V	μA/μm	570.09	567.28
I _{OFF}	V _{DS} = 5 V, V _{GS} =0 V	A/μm	5.5E-12	4.2E-12

Narrow and long channel device (W/L= 0.35 μm / 10 μm)

V _{TLIN}	V _{DS} = 0.1 V	V	0.694	0.559
V _{TSAT}	V _{DS} = 5 V	V	0.676	0.541
I _{ON}	V _{DS} =V _{GS} = 5 V	μA/μm	57.64	62.51

Narrow and short channel device (W/L= 0.35 μm / 0.5 μm)

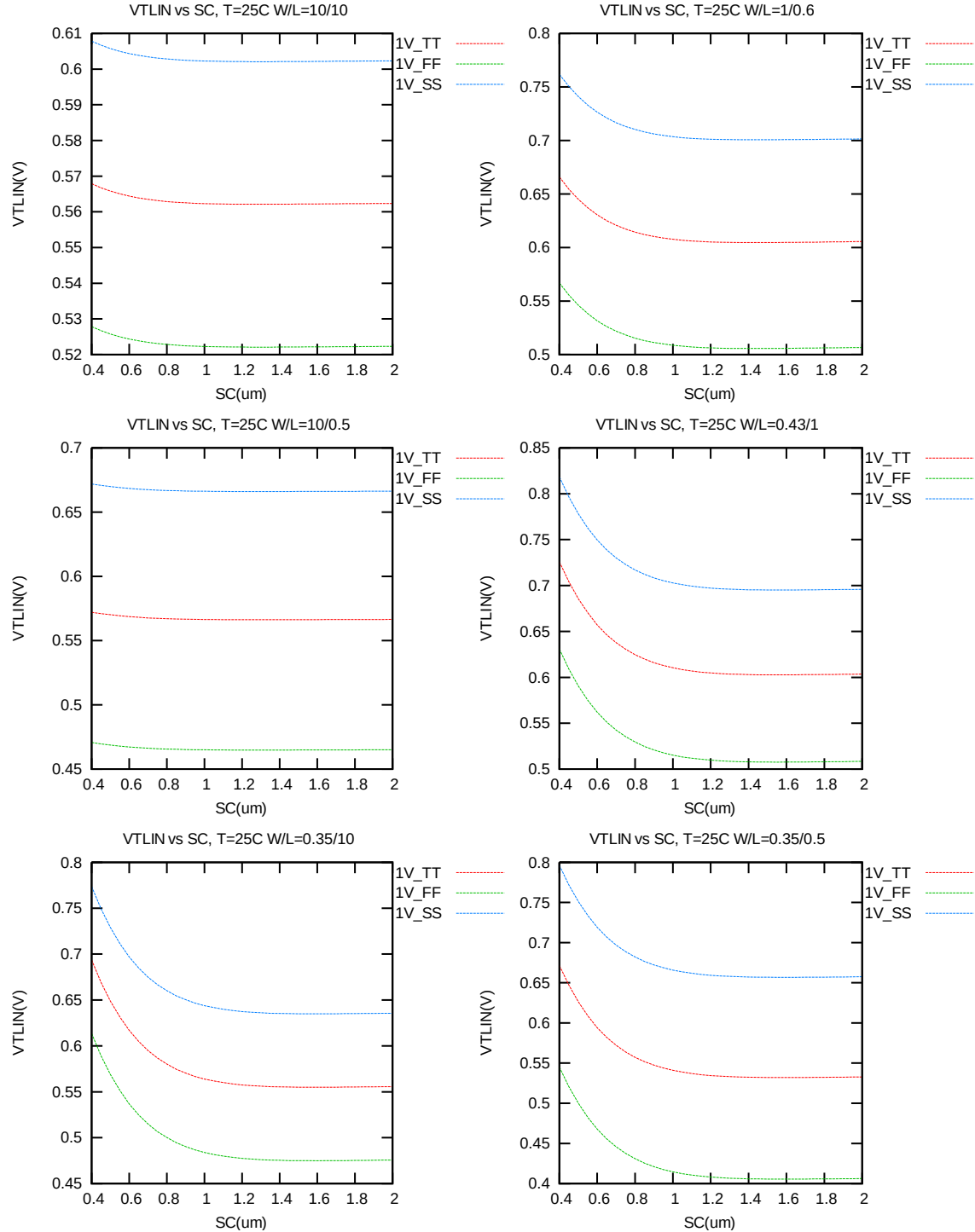
V _{TLIN}	V _{DS} = 0.1 V	V	0.672	0.557
V _{TSAT}	V _{DS} = 5 V	V	0.573	0.459
I _{ON}	V _{DS} =V _{GS} = 5 V	μA/μm	637.94	657.97
I _{OFF}	V _{DS} = 5 V, V _{GS} =0 V	A/μm	2.2E-12	6.7E-12

6.1.1 Model simulation plots for NMOS

The L and W in the plots refer to LDES and WDES respectively.

V_T vs. SC

The following figures show the SC dependence of V_{tlin} and V_{tsat} were extracted by $ID=300nA*WDES*0.9/(LDES*0.9+12nm)$ at 25°C.



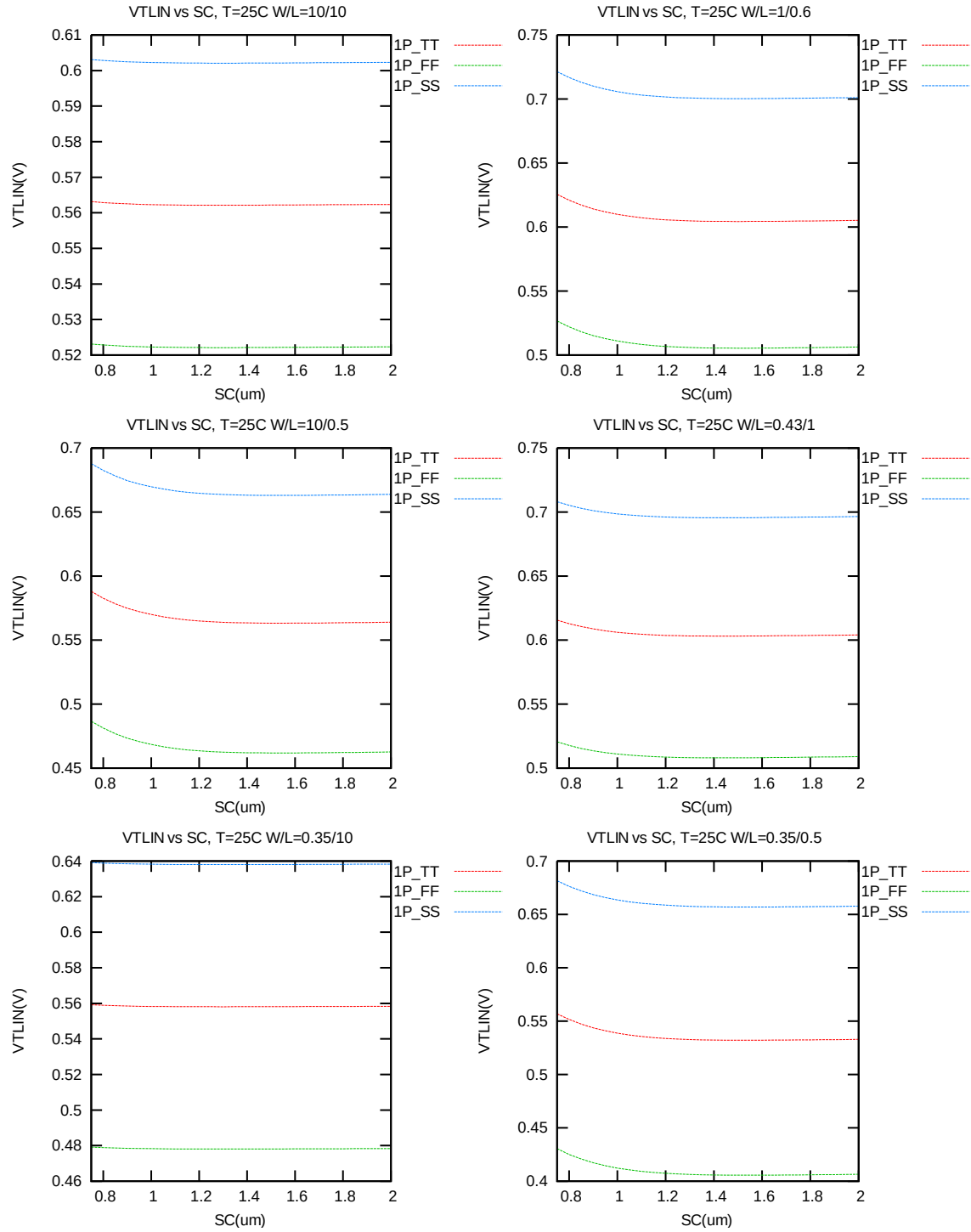
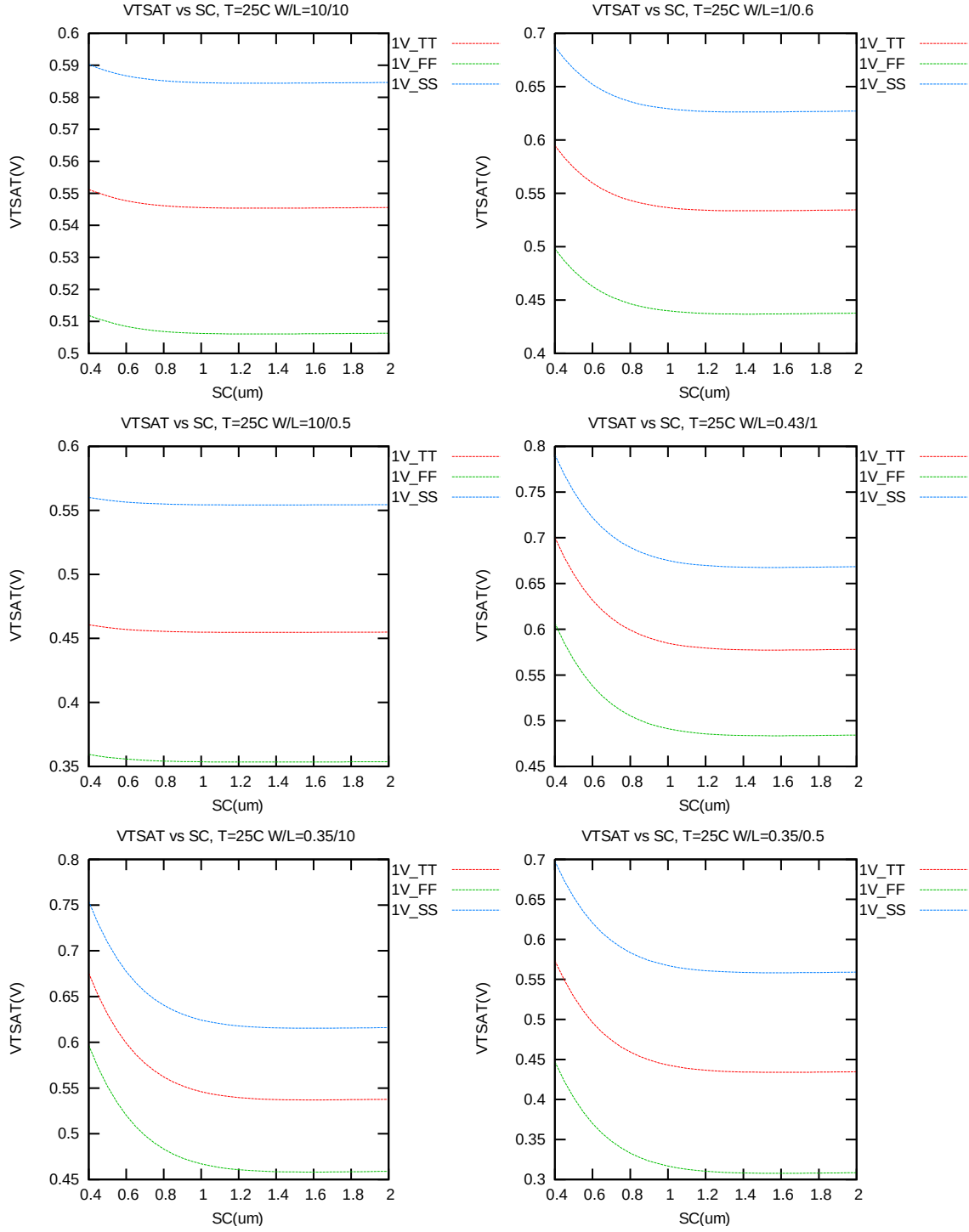


Figure 6-2 V_{TLIN} vs. SC ($V_{DS}=0.1\text{ V}$) for NMOS



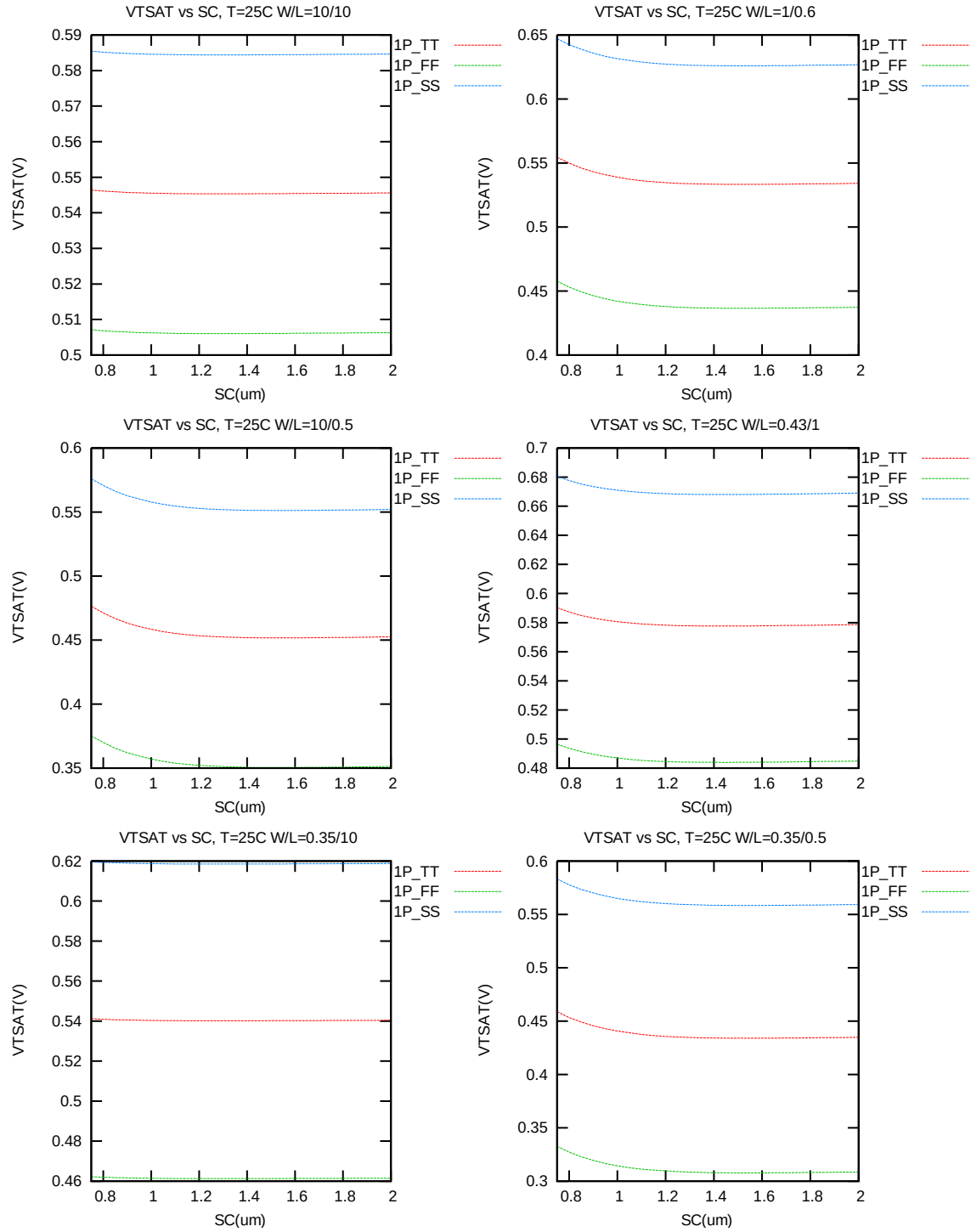
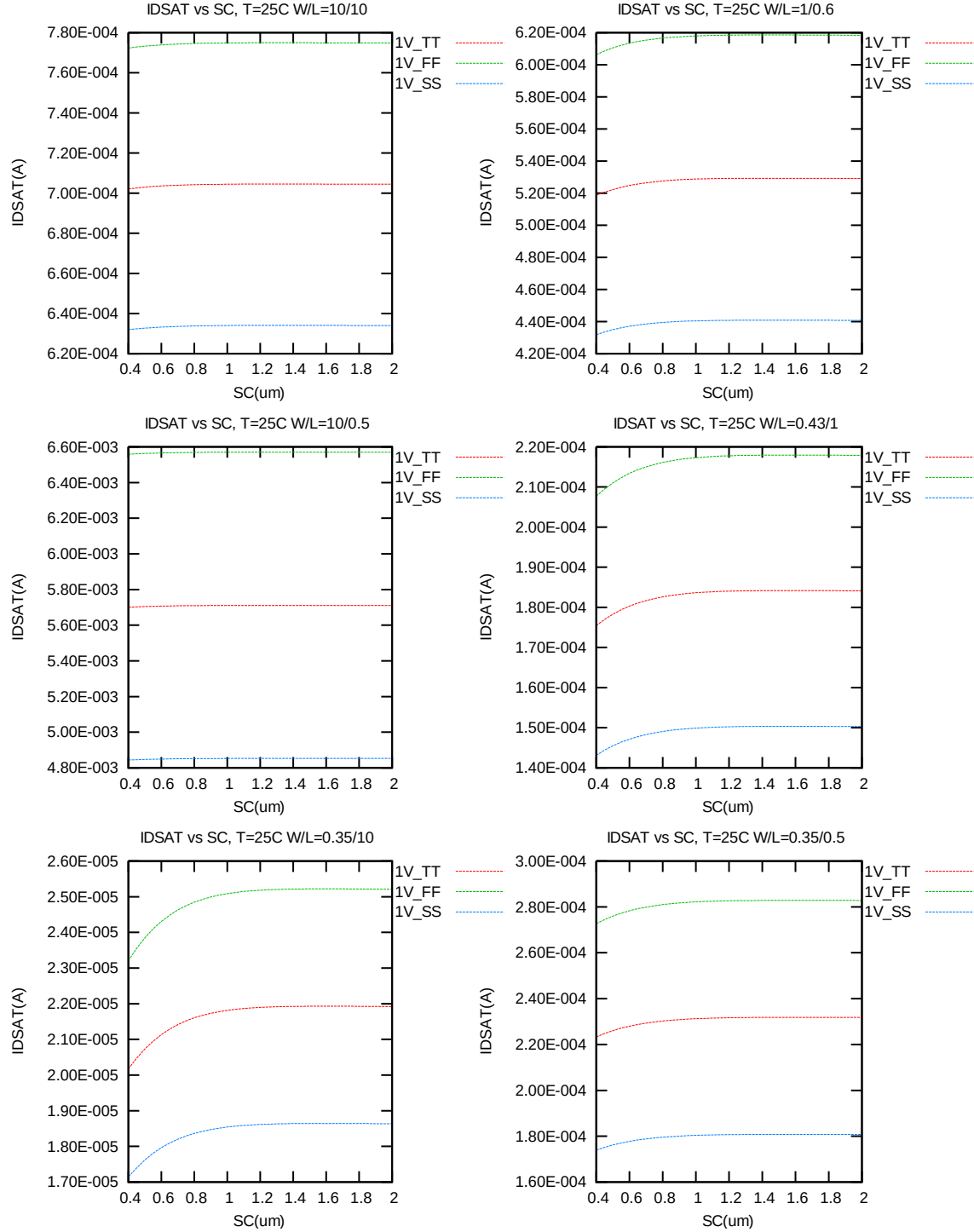


Figure 6-3 V_{TSAT} vs. SC ($V_{DS}=5$ V) for NMOS

I_{ON} vs. SC

The following plots show the SC dependence of I_{ON} extracted at $V_{DS}=V_{GS}=5$ V for $V_{BS}=0$ at 25C.



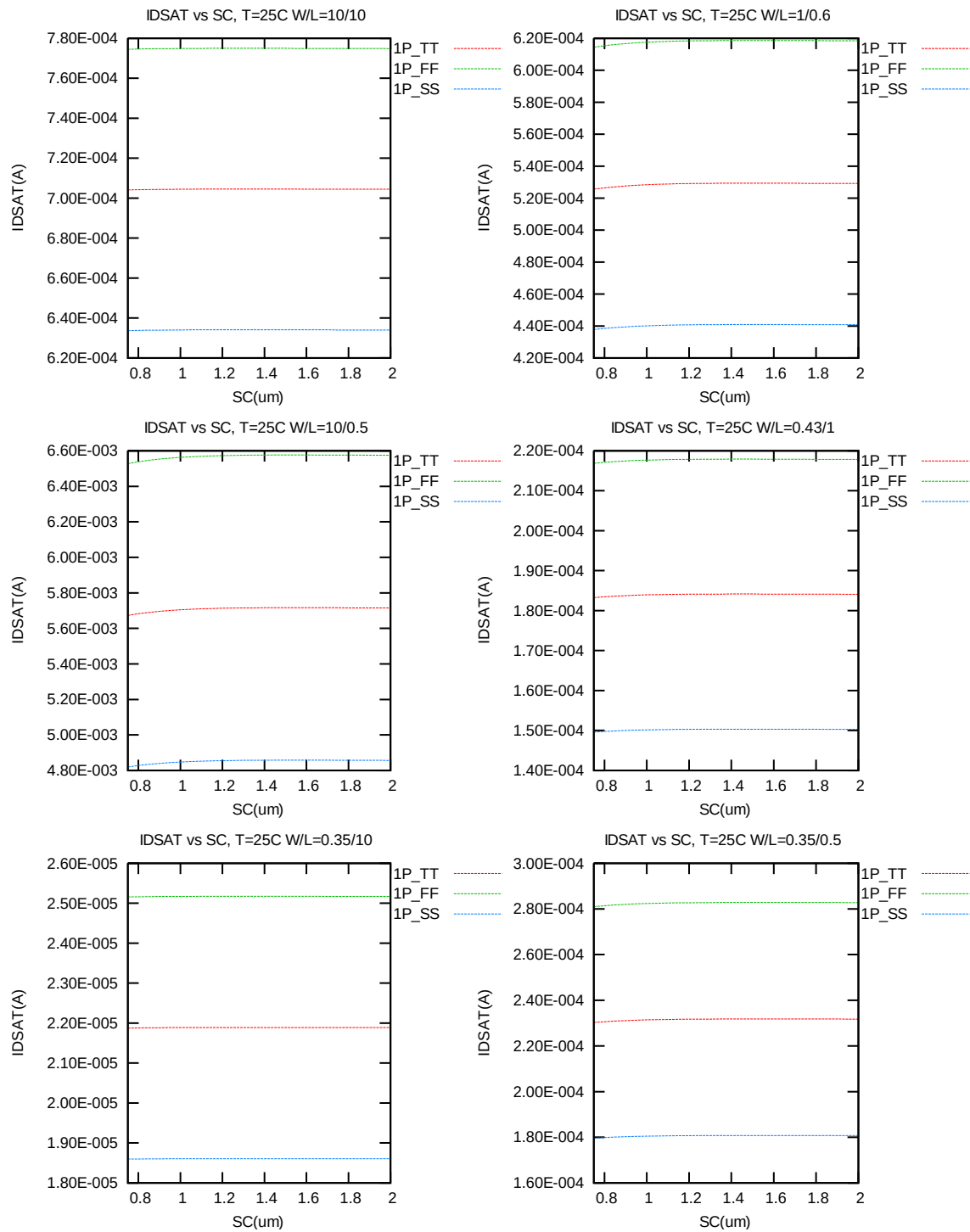


Figure 6-4 Saturation current I_{ON} vs. SC for NMOS

6.2 Simulation Results for PMOS at SC=0.4um/0.75um for 1V/1P structure (SCREF=2um)

The simulation results for the PMOS characterization are summarized in the table below.

Table 6-2 Simulation results for PMOS

All parameters are given for T=25 °C, V_{BS}=0 V unless specified otherwise.

Parameter	Condition	Unit	Simulation Value	
			1V structure	1H structure

Long channel device (W/L= 10 μm / 10 μm)

V _{TLIN}	V _{DS} = -0.1 V	V	-0.654	-0.650
I _{ON}	V _{DS} = -5 V	μA/μm	-16.09	-16.12

Wide and short channel device (W/L= 10 μm / 0.5 μm)

V _{TLIN}	V _{DS} = -0.1 V	V	-0.606	-0.622
V _{TSAT}	V _{DS} = -5 V	V	-0.551	-0.567
Body Effect V _T shift	V _{DS} = -5 V V _{BS} = 0 ~ 3 V	V	-0.702	-0.703
DIBL	V _{DS} = -0.1~-5 V	V	-0.055	-0.055
I _{ON}	V _{DS} =V _{GS} = -5 V	μA/μm	-273.10	-271.32
I _{OFF}	V _{DS} = -5 V, V _{GS} =0 V	A/μm	-2.6E-12	-2.5E-12

Narrow and long channel device (W/L= 0.35 μm / 10 μm)

V _{TLIN}	V _{DS} = -0.1 V	V	-0.774	-0.673
V _{TSAT}	V _{DS} = -5 V	V	-0.769	-0.668
I _{ON}	V _{DS} =V _{GS} = -5 V	μA/μm	-15.16	-16.17

Narrow and short channel device (W/L= 0.35 μm / 0.5 μm)

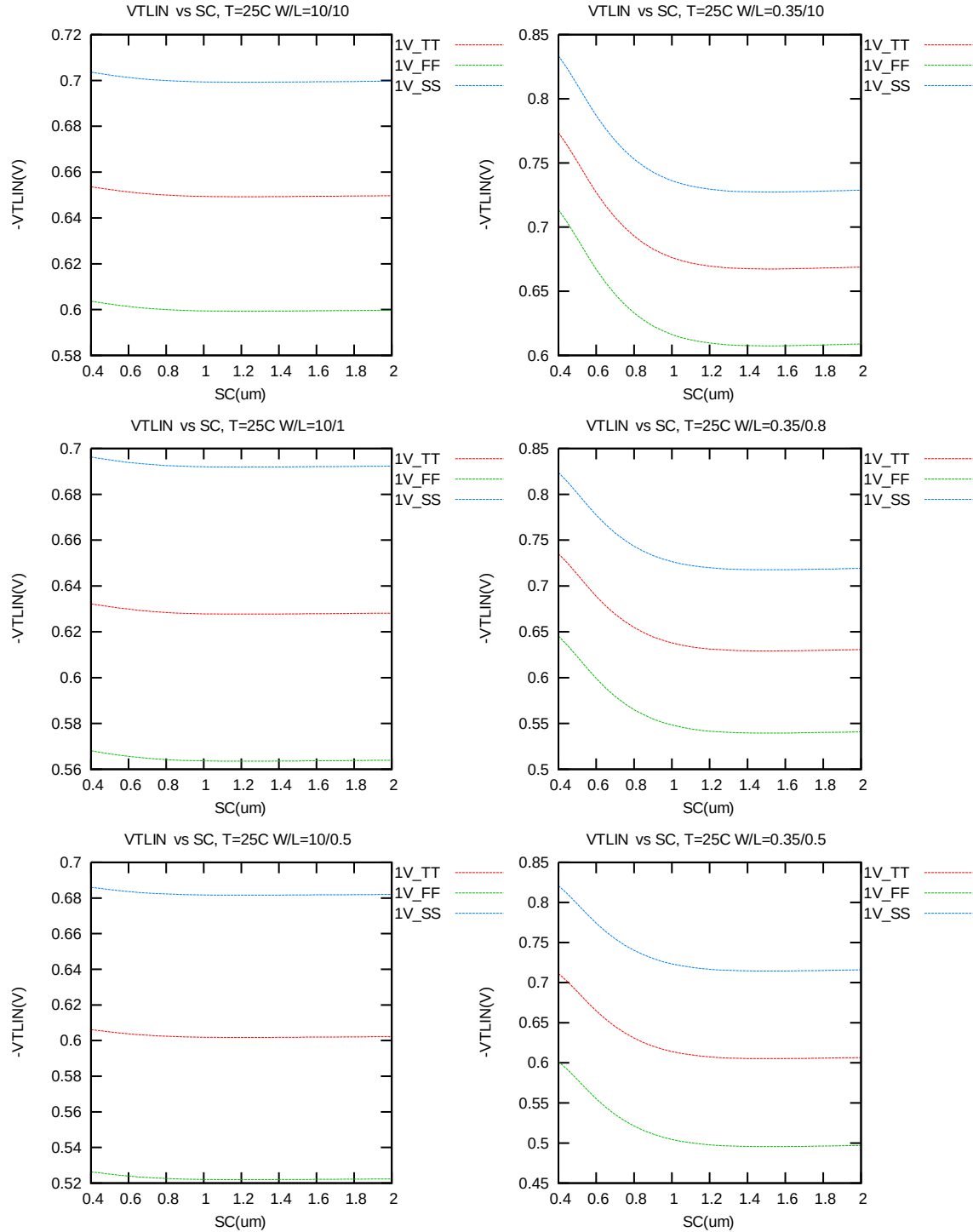
V _{TLIN}	V _{DS} = -0.1 V	V	-0.711	-0.630
V _{TSAT}	V _{DS} = -5 V	V	-0.648	-0.567
I _{ON}	V _{DS} =V _{GS} = -5 V	μA/μm	-254.19	-262.95
I _{OFF}	V _{DS} = -5 V, V _{GS} =0 V	A/μm	-2.0E-12	-2.1E-12

6.2.1 Model simulation plots for PMOS

The L and W in the plots refer to LDES and WDES respectively.

V_T vs. SC

The following figures show the SC dependence of Vtlin and Vtsat were extracted by $ID=-70nA*WDES*0.9/(LDES*0.9+12nm)$ at 25°C.



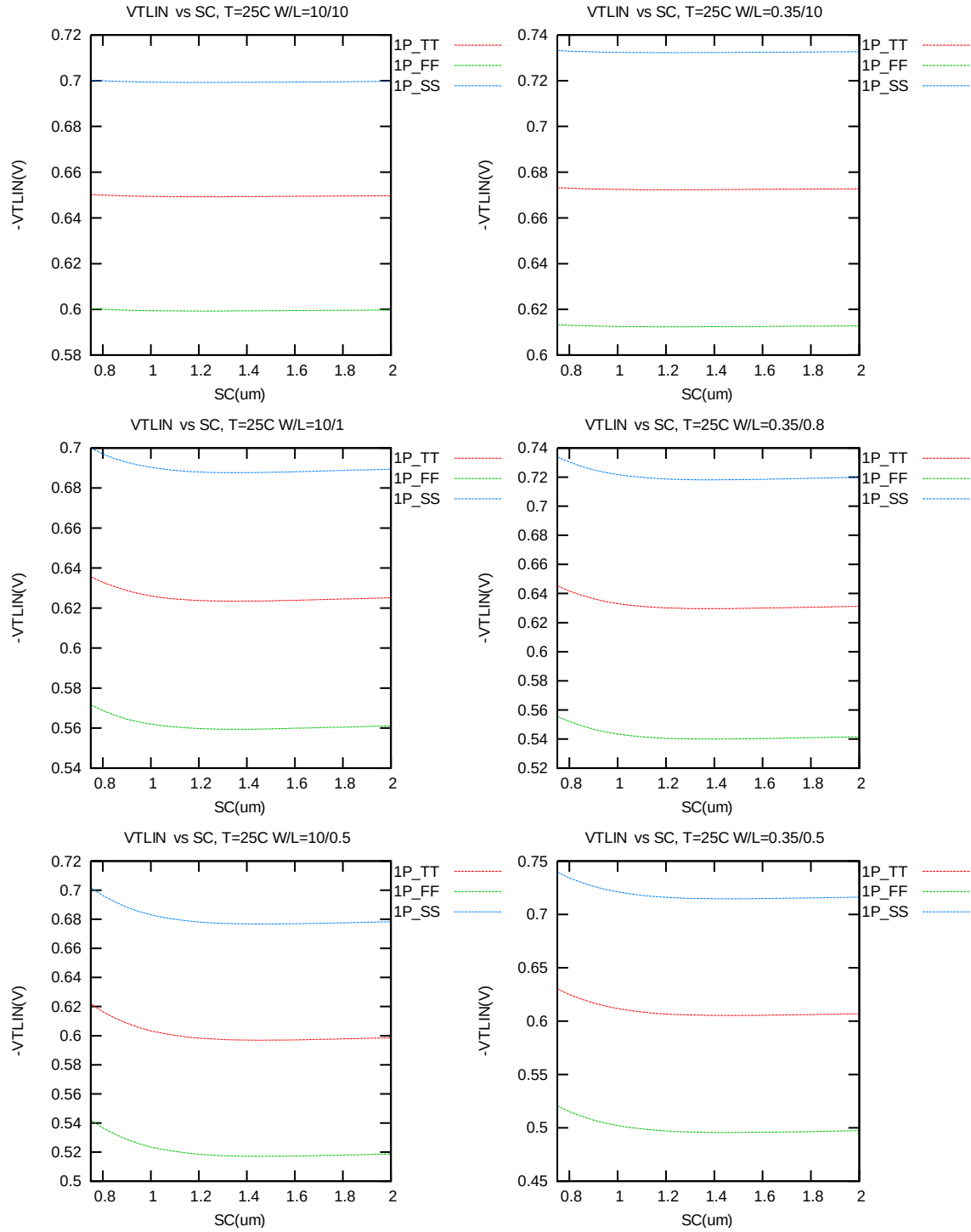
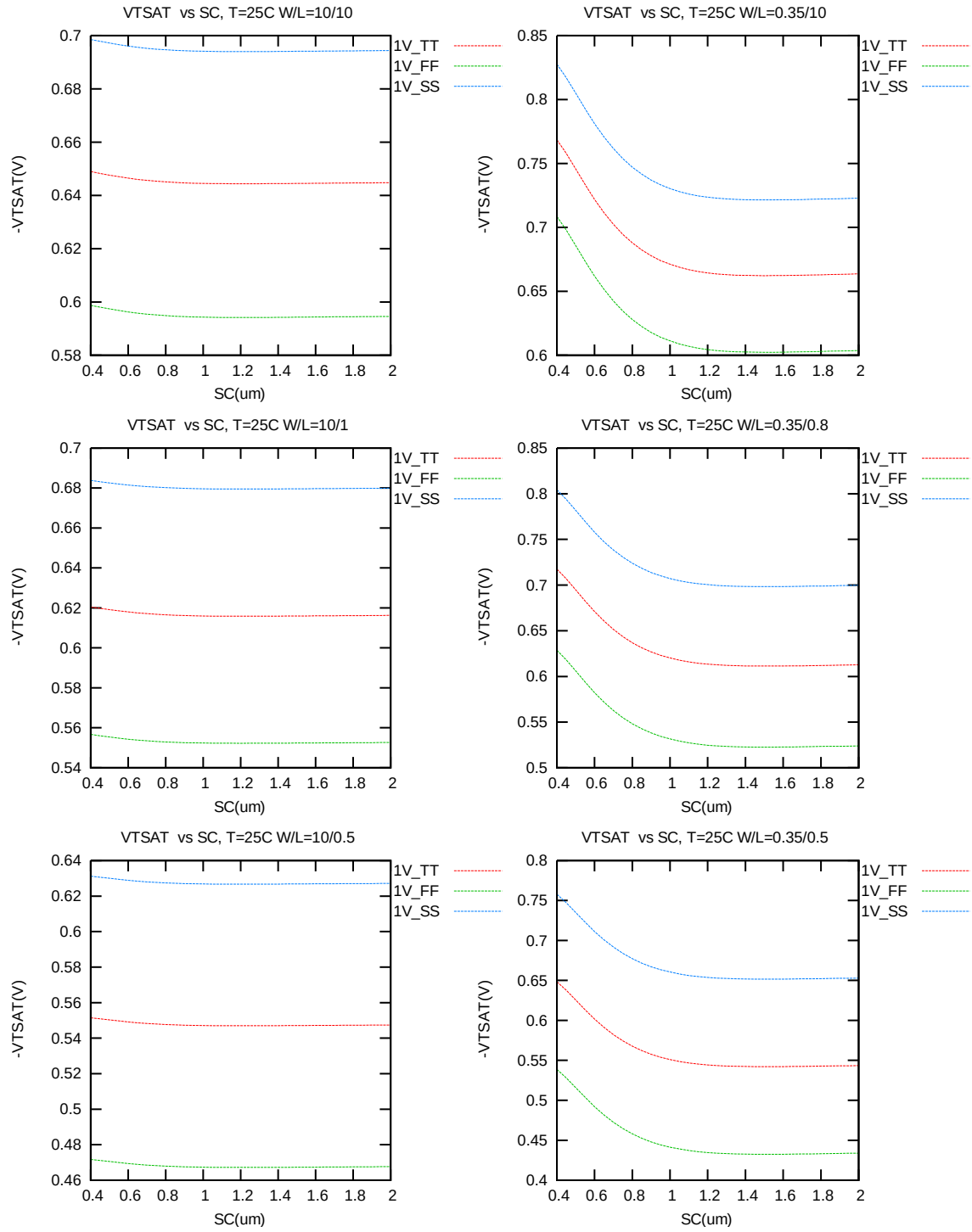


Figure 6-5 V_{TLIN} vs. SC ($V_{DS} = -0.1$ V) for PMOS



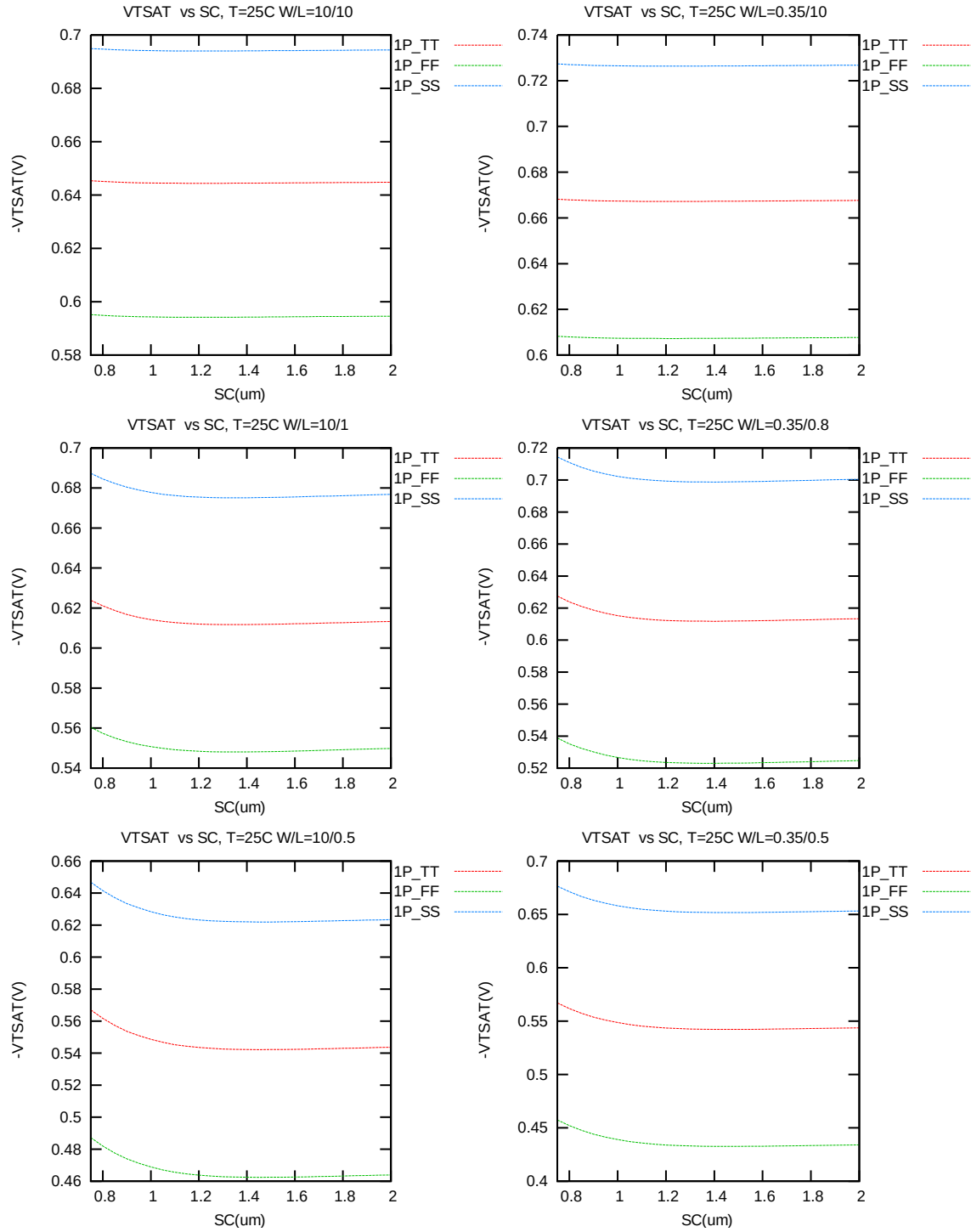
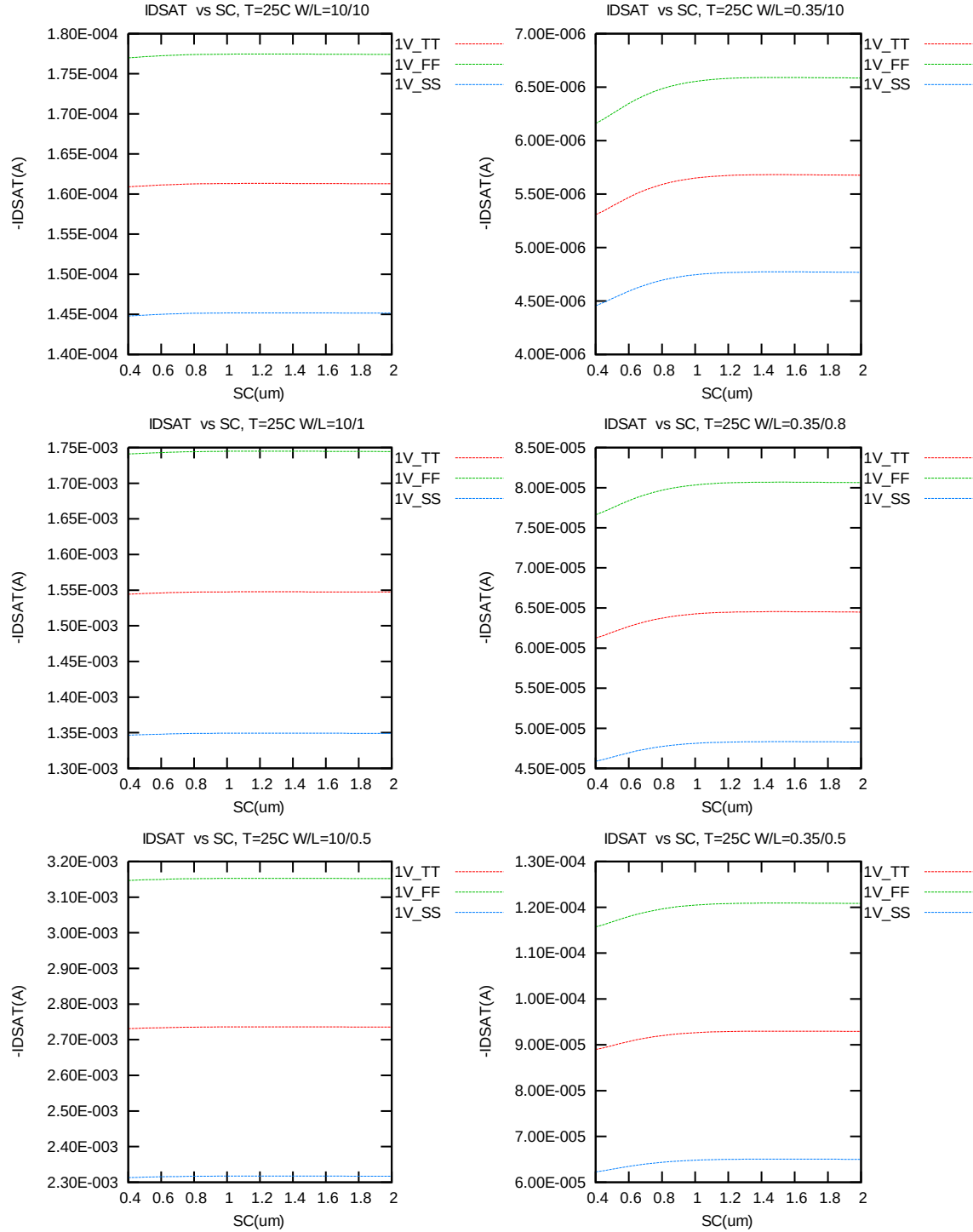


Figure 6-6 V_{TSAT} vs. SC ($V_{DS} = -5\text{ V}$) for PMOS

The following plots show the SC dependence of I_{ON} extracted at $V_{DS}=V_{GS}=-5$ V for $V_{BS}=0$ at 25C.



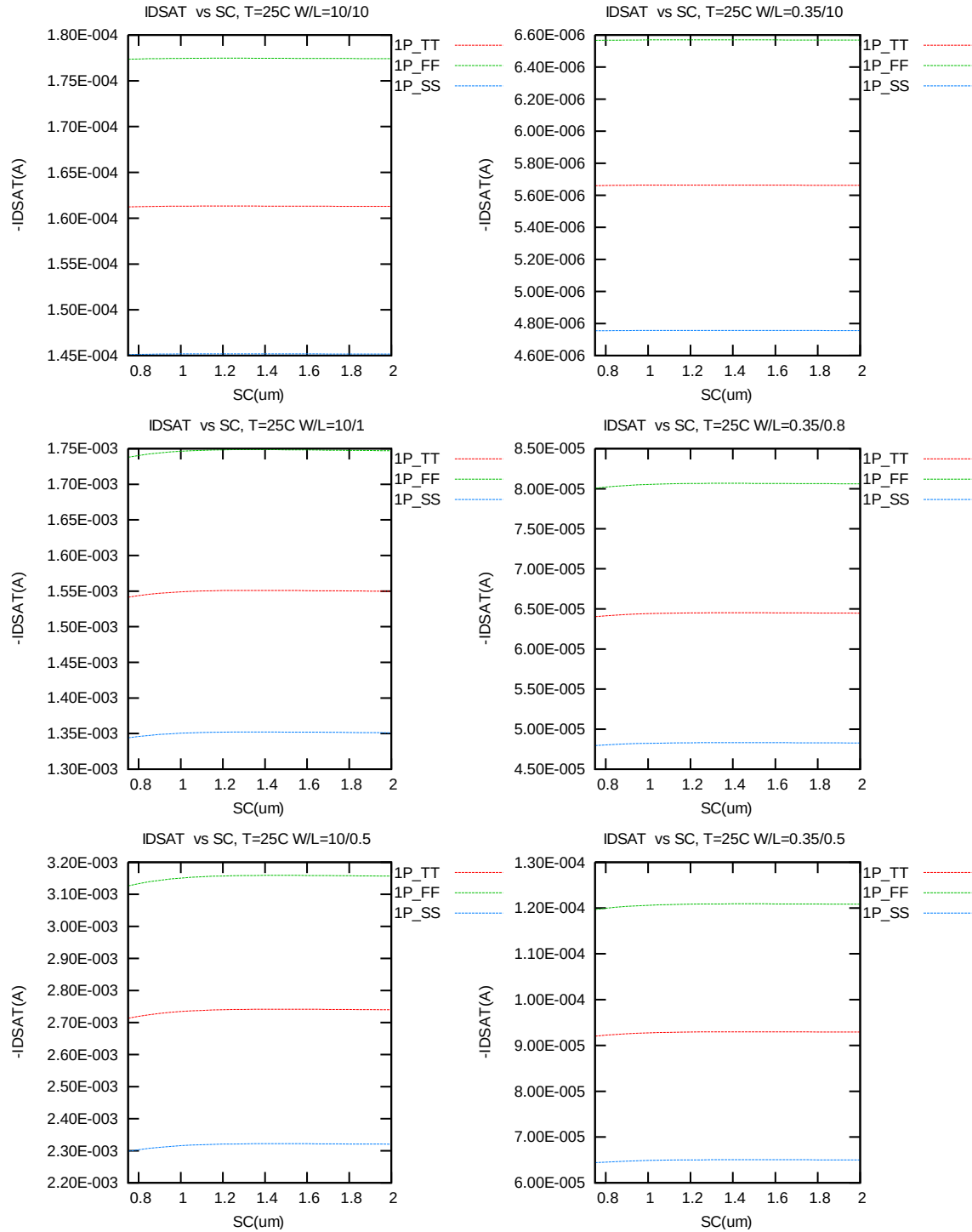


Figure 6-7 Saturation current I_{ON} vs. SC for PMOS

6.3 V_T and I_{ON} Variation v. SC for both N/PMOS

Delta V_T is defined as $V_T(SC) - V_T(2\mu m)$ for NMOS and $-[V_T(SC) - V_T(2\mu m)]$ for PMOS. Delta I_{ON} is defined as $[I_{ON}(SC) - I_{ON}(2\mu m)]/I_{ON}(2\mu m) * 100\%$.

Delta V_T and I_{ON} vs. SC

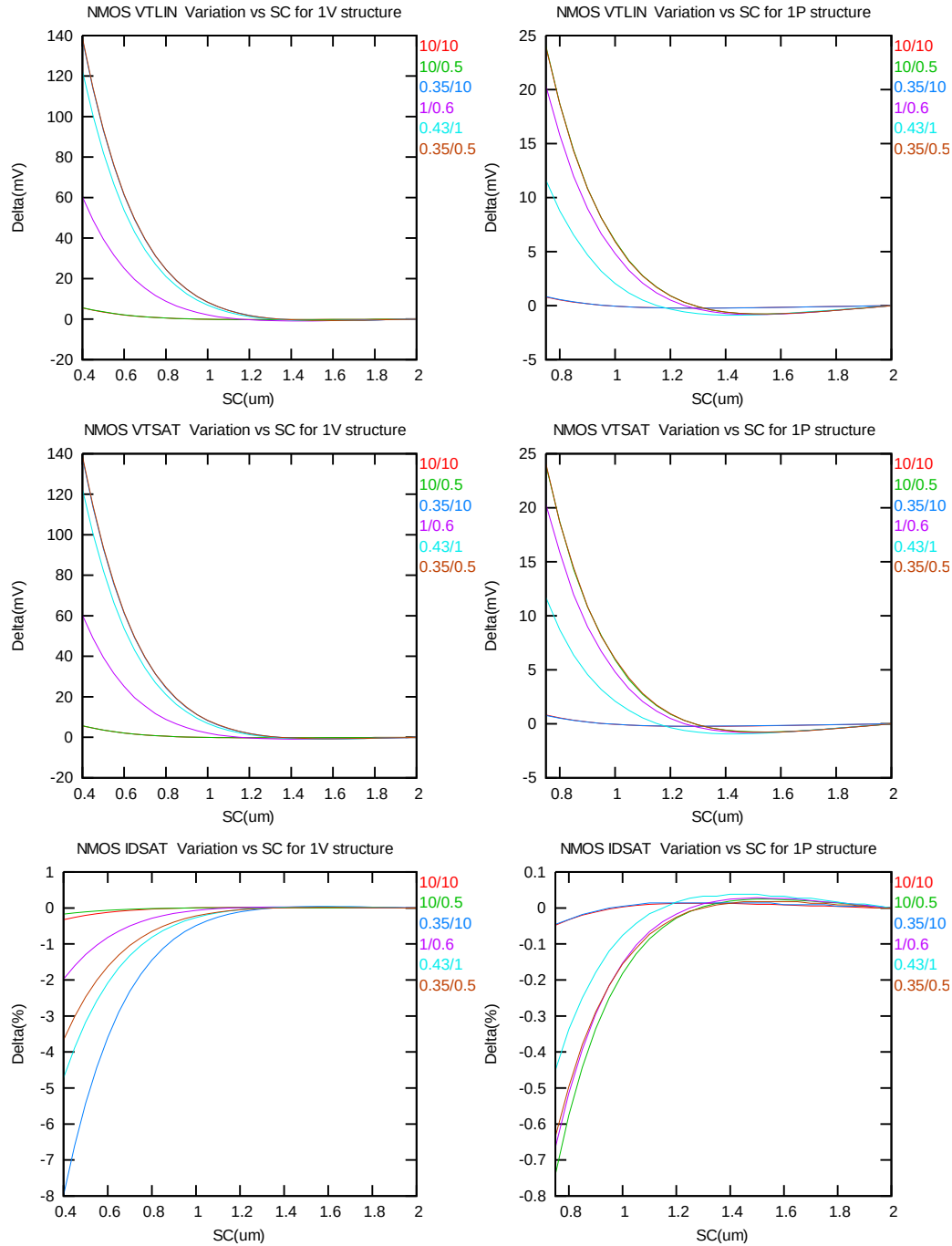


Figure 6-8 Delta V_T and I_{ON} vs. SC for NMOS

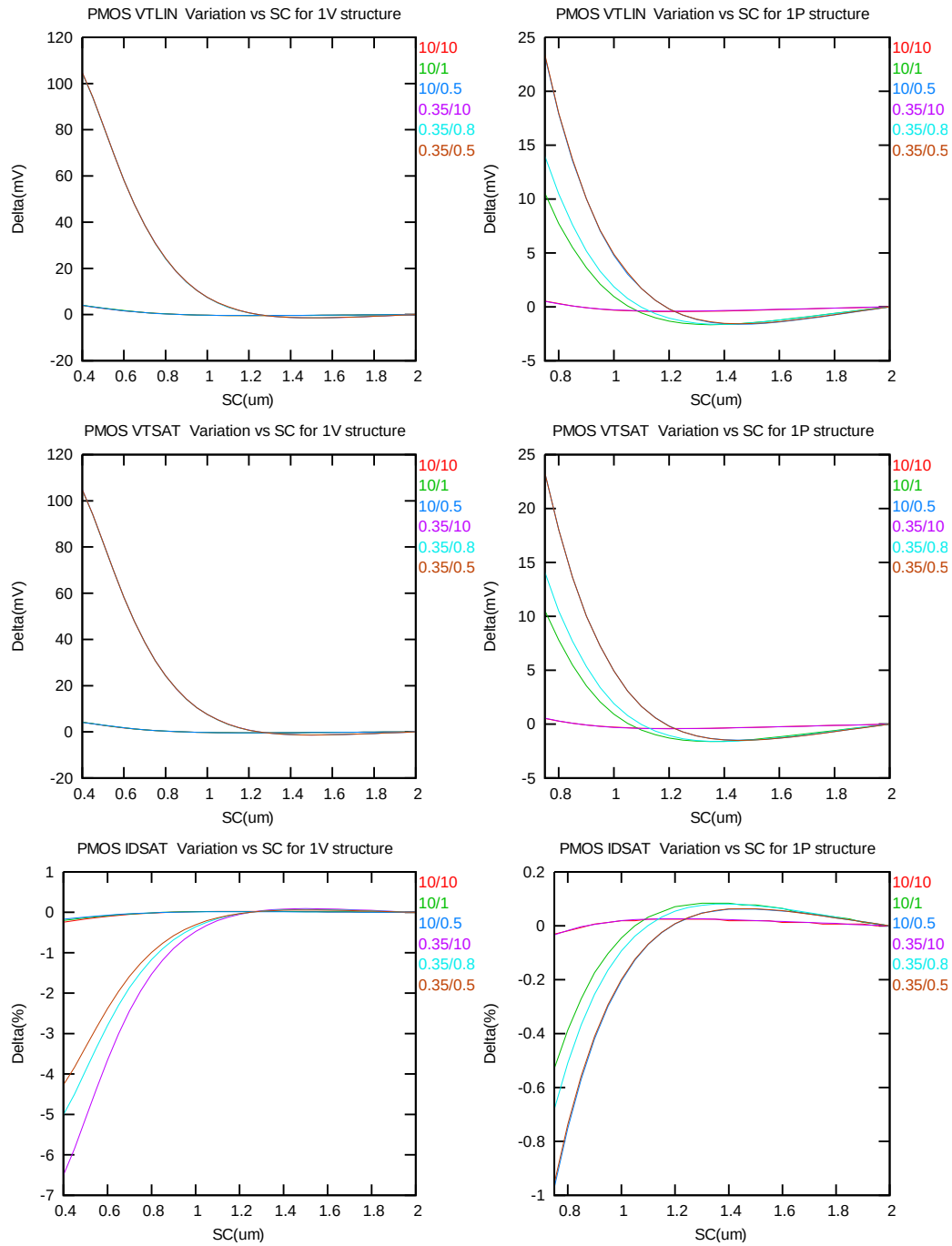


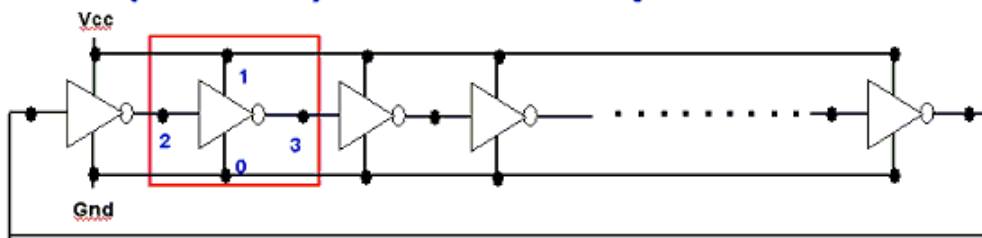
Figure 6-9 Delta V_T and I_{ON} vs. SC for PMOS

7 Dynamic Performance

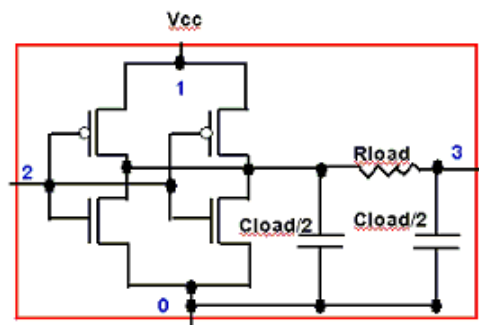
7.1 Test Circuit

To assess the dynamic performance of these devices, a ring oscillator circuit with the following configurations is adopted.

R.O. (Inverter) circuit description



Ring Oscillator circuit



single stage R.O. circuit

Figure 7-1 Ring oscillator netlist description

7.2 Dynamic Performance

Table 7-1 Ring oscillator (inverter) structure description

R.O. (Inverter) Structure description		
Contact size	0.144 um X 0.144 um	
Polysilicon gate to contact spacing	0.14 um	
Contact to diffusion edge spacing	0.06 um	
Without RC loading	R = 0 ohm	C = 0 fF
Inverter		
Logic gate type	Inverter	
Channel width for n-ch MOSFET	2.5 um X 2	
Channel length for n-ch MOSFET	0.5 um	
Channel width for p-ch MOSFET	3.75 um X 2	
Channel Length for p-ch MOSFET	0.5 um	

Table 7-2 Electrical parameters for ring oscillator (ps/stage)

The EDR target of Electrical parameters for ring oscillator is listed below. The condition of this EDR target is base on Vcc =5V and with RC loading. (Extracted by command file)

Table 7-3 Electrical parameters for ring oscillator of EDR target with RC loading

Parameter	Condition	Unit	Target	Centered Value
Tpd	Vcc = 5V	ps/stage	56.5	58.62

The simulation results at different temperatures and using different worst-case models are listed below.

Table 7-4 Ring oscillator simulation results (ps/stage)

Inverter, R=0, C=0f, Fanout=1, Vcc=5V				Inverter, R=0, C=0f, Fanout=1, T=25C			
Temp(C)	TT	FF	SS	Vcc(V)	TT	FF	SS
-40	44.415	39.654	51.236	3.5	64.721	56.819	75.878
0	48.734	43.595	56.096	4	58.786	52.042	68.354
25	51.478	46.101	59.181	4.5	54.588	48.639	63.083
85	58.192	52.233	66.722	5	51.478	46.101	59.181
125	62.762	56.410	71.850	5.5	49.066	44.115	56.160

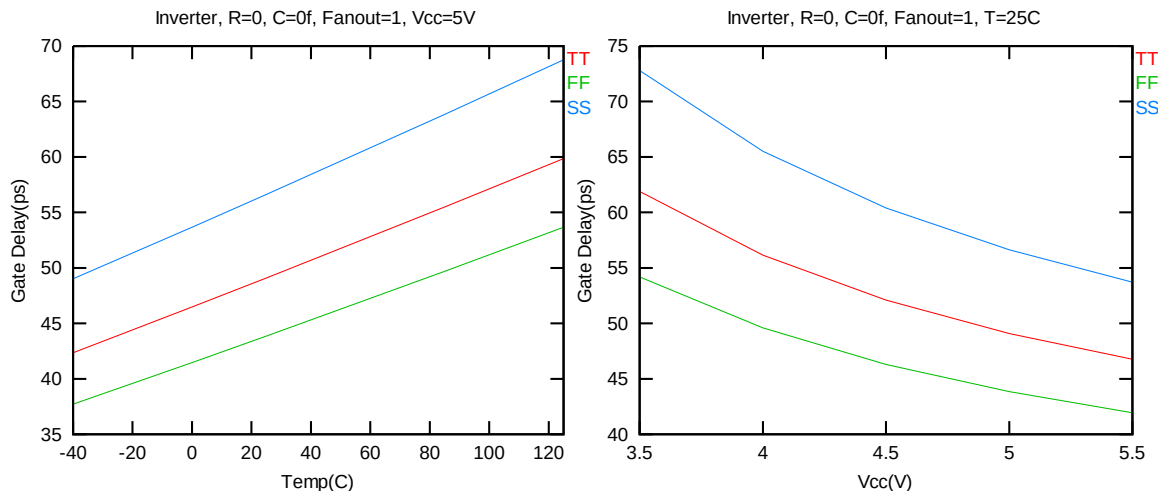


Figure 7-2 Ring oscillator (inverter) simulation

The (Inverter) ring oscillator gate delay simulation vs. width

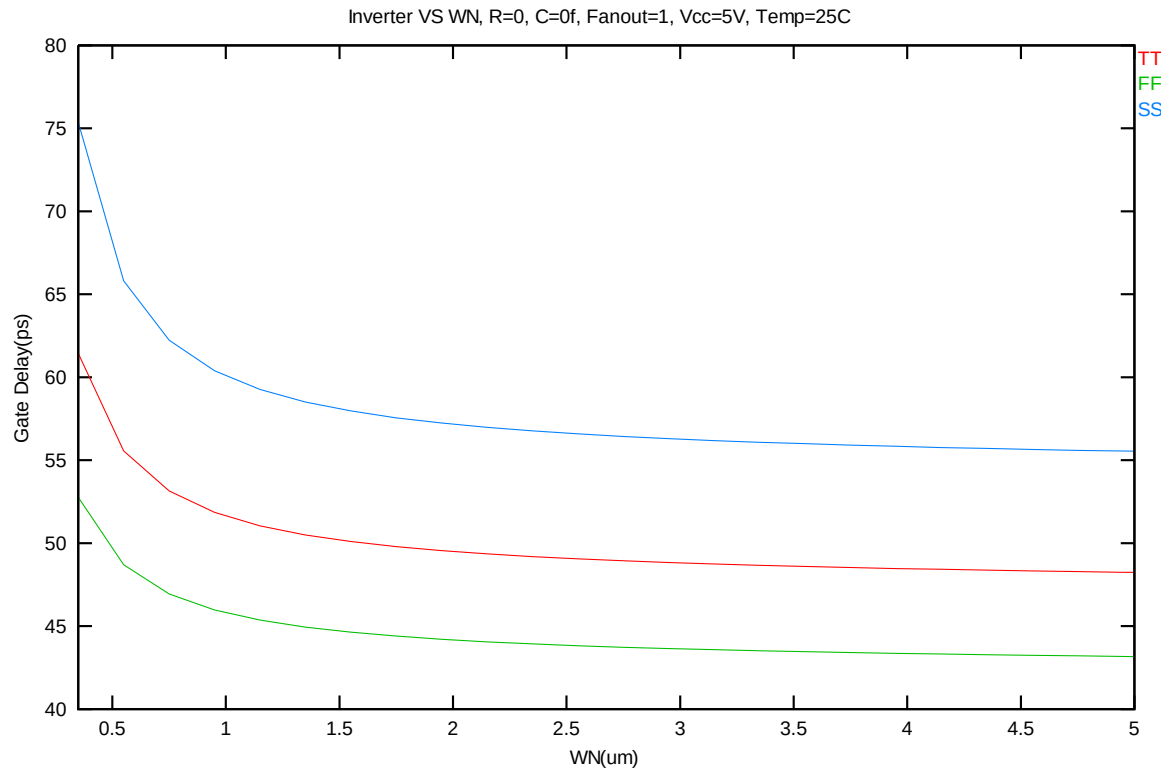


Figure 7-3 Ring oscillator (inverter) gate delay simulation vs. width

8 Noise Characteristics

8.1 Verification Plots

NMOS, W=10um L=10um

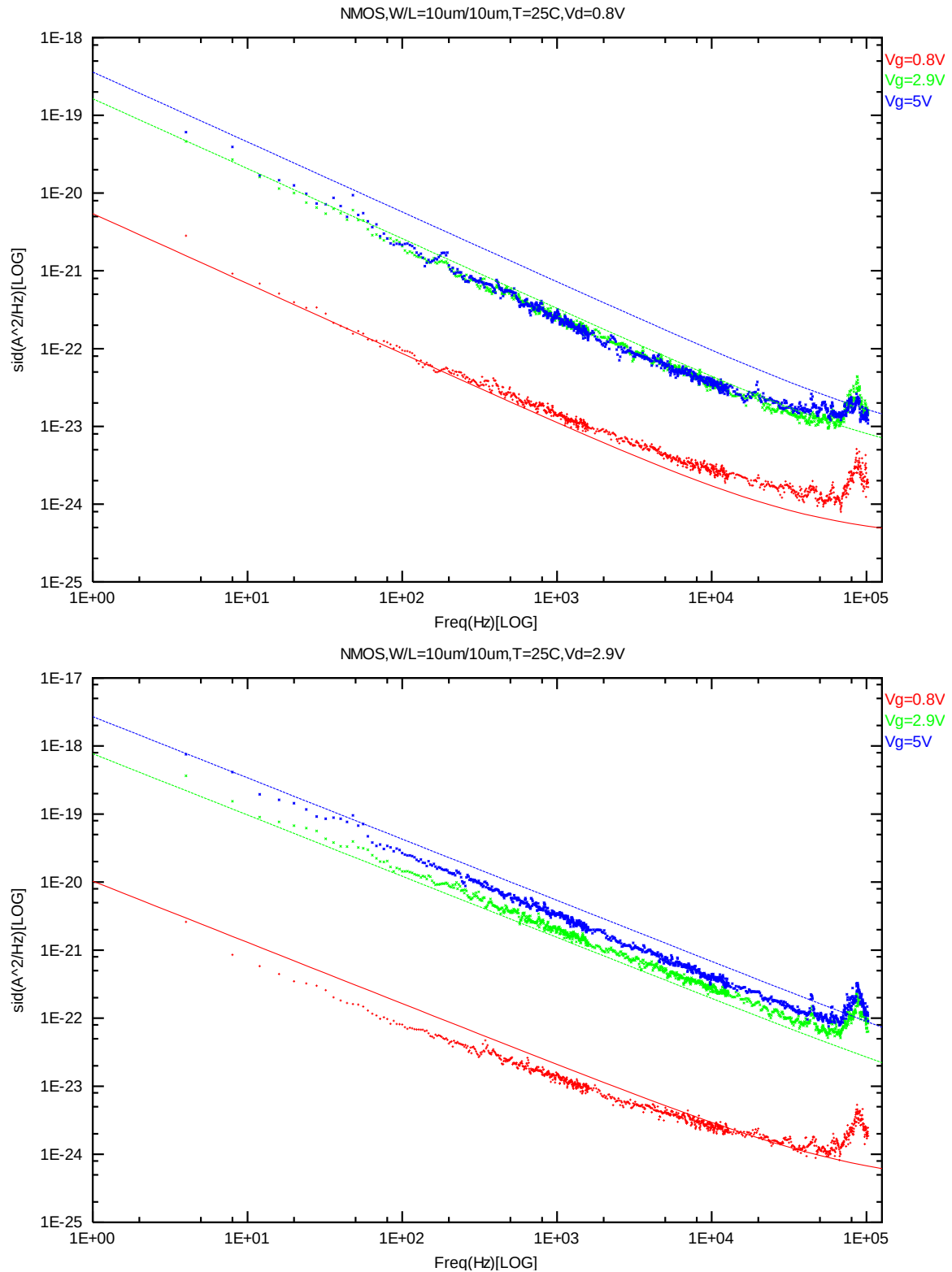


Figure 8-1 NMOS W/L=10um/10um @ Vd=0.8 V, Vd=2.9V

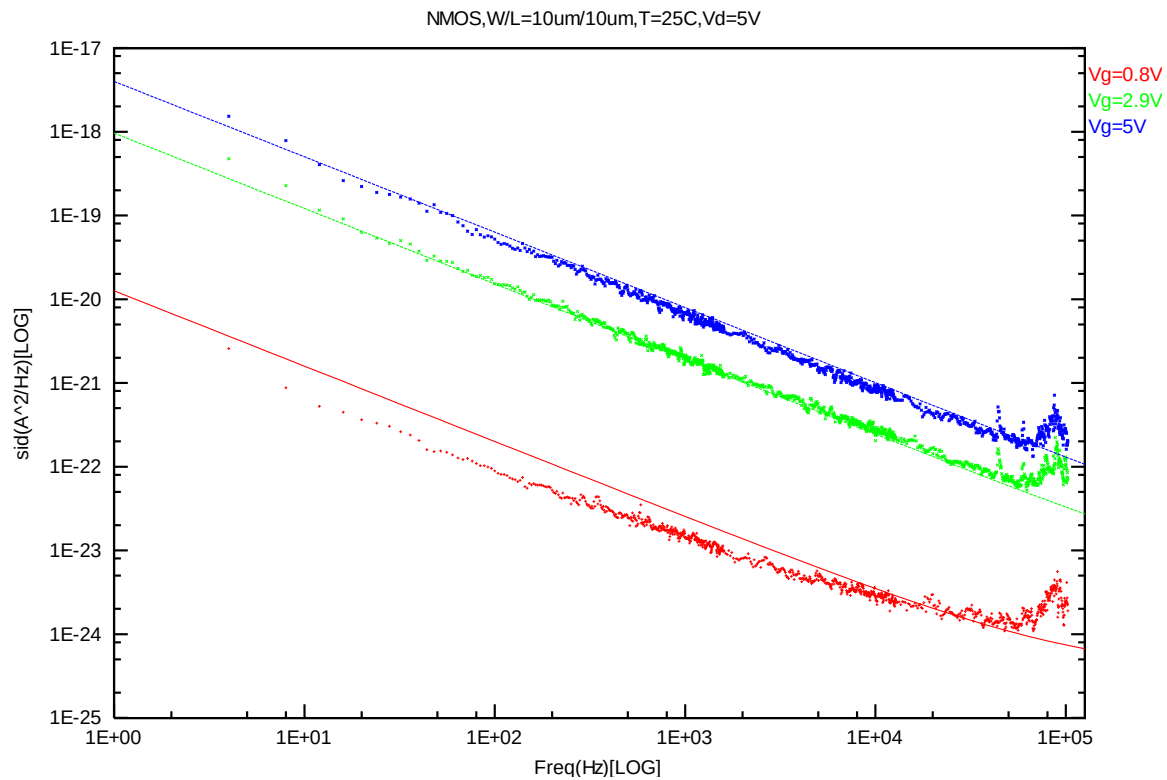


Figure 8-2 NMOS W/L=10um/10um @ Vd=5 V

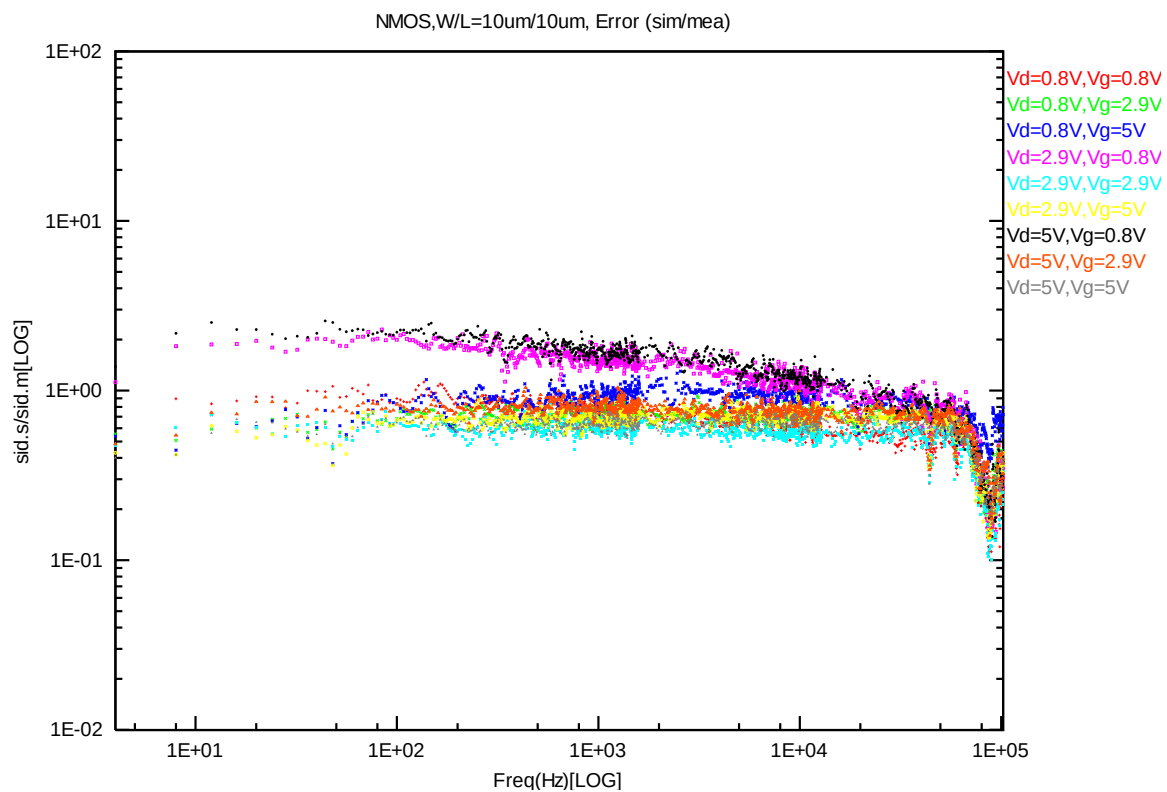


Figure 8-3 NMOS W/L=10um/10um error distribution plot

NMOS, W=10um L=1um

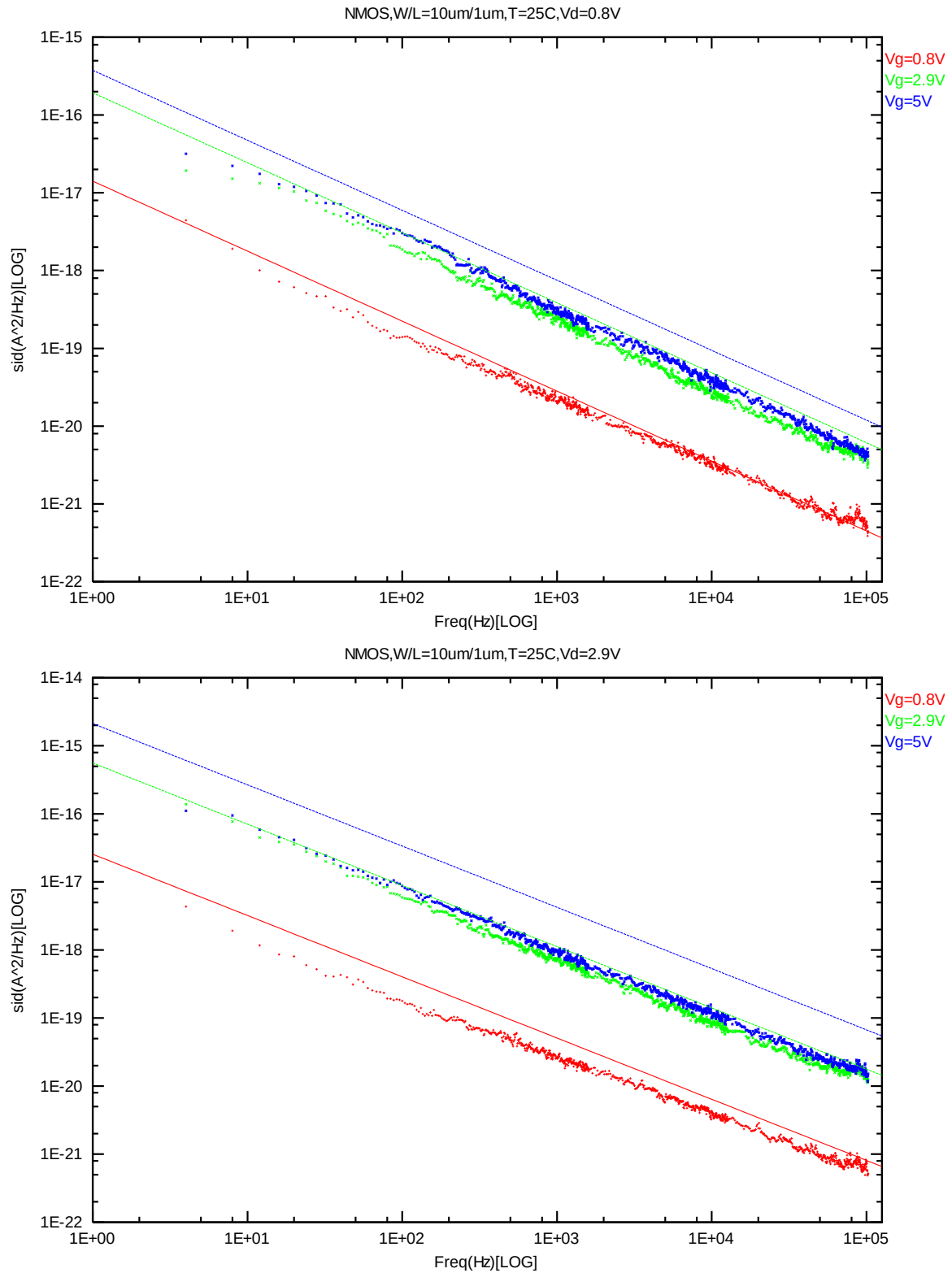
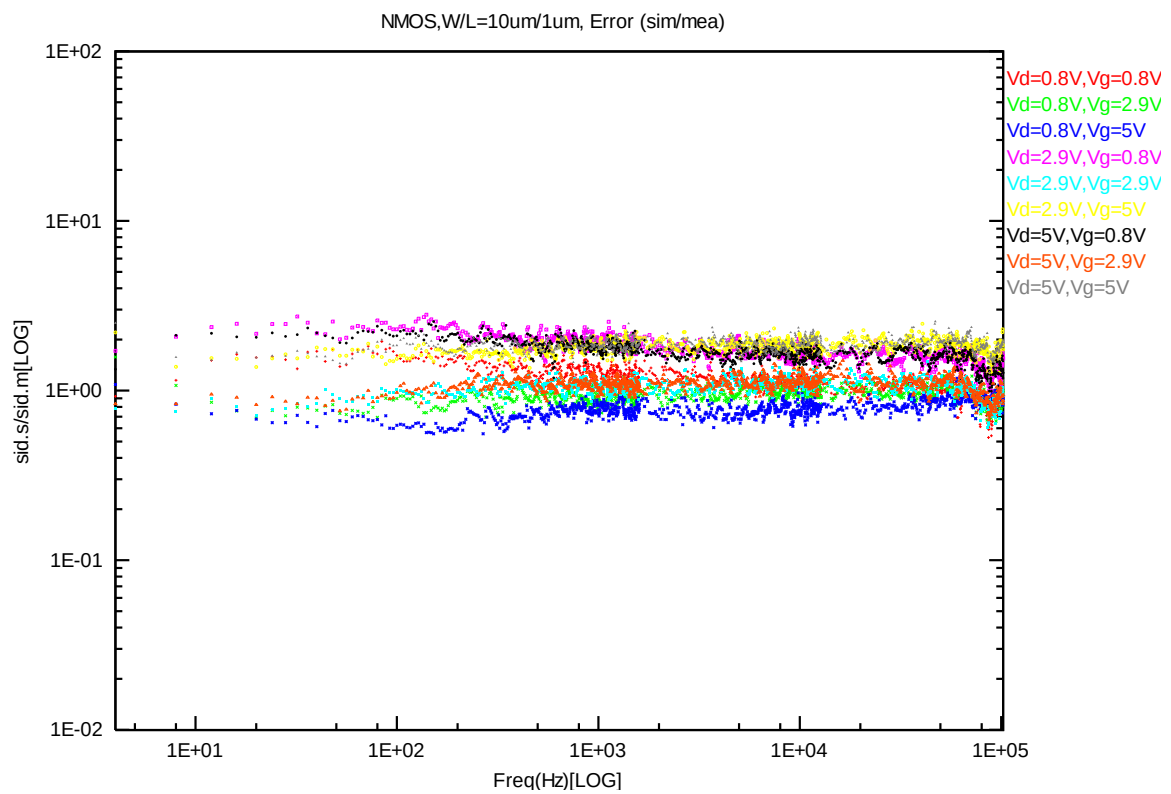
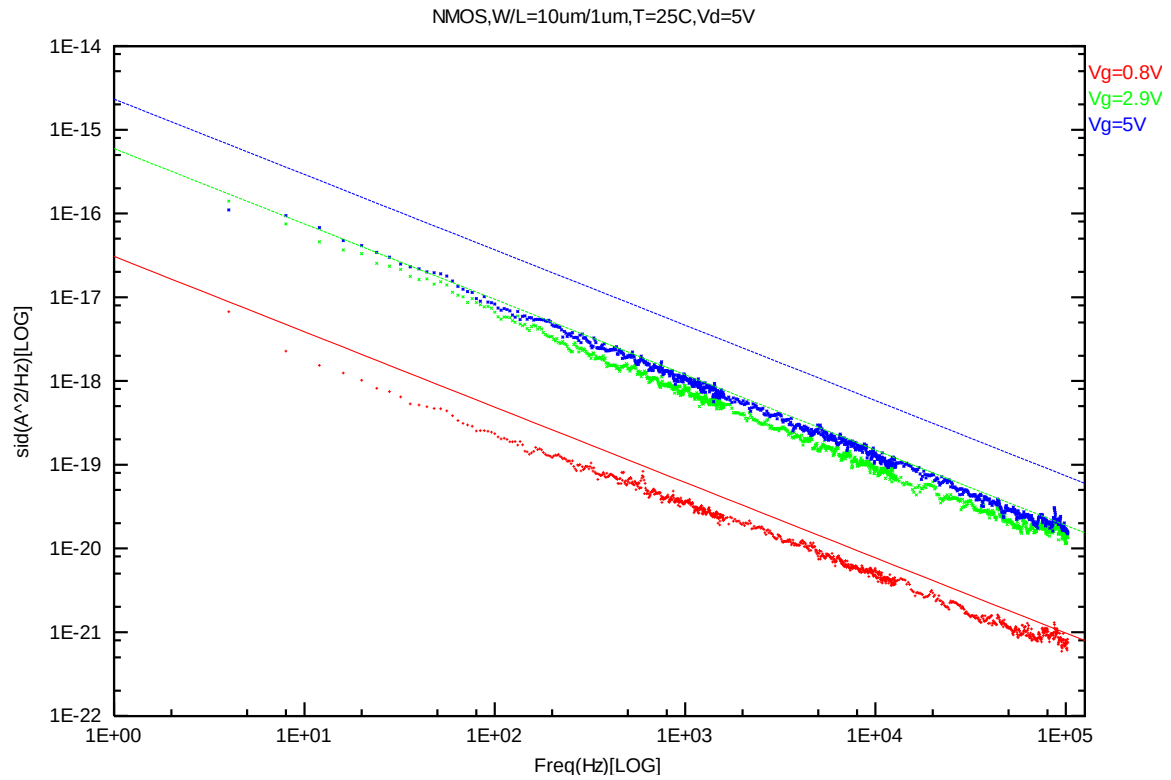


Figure 8-4 NMOS W/L=10um/1um @ Vd=0.8 V, Vd=2.9V



NMOS, W=10um L=0.5um

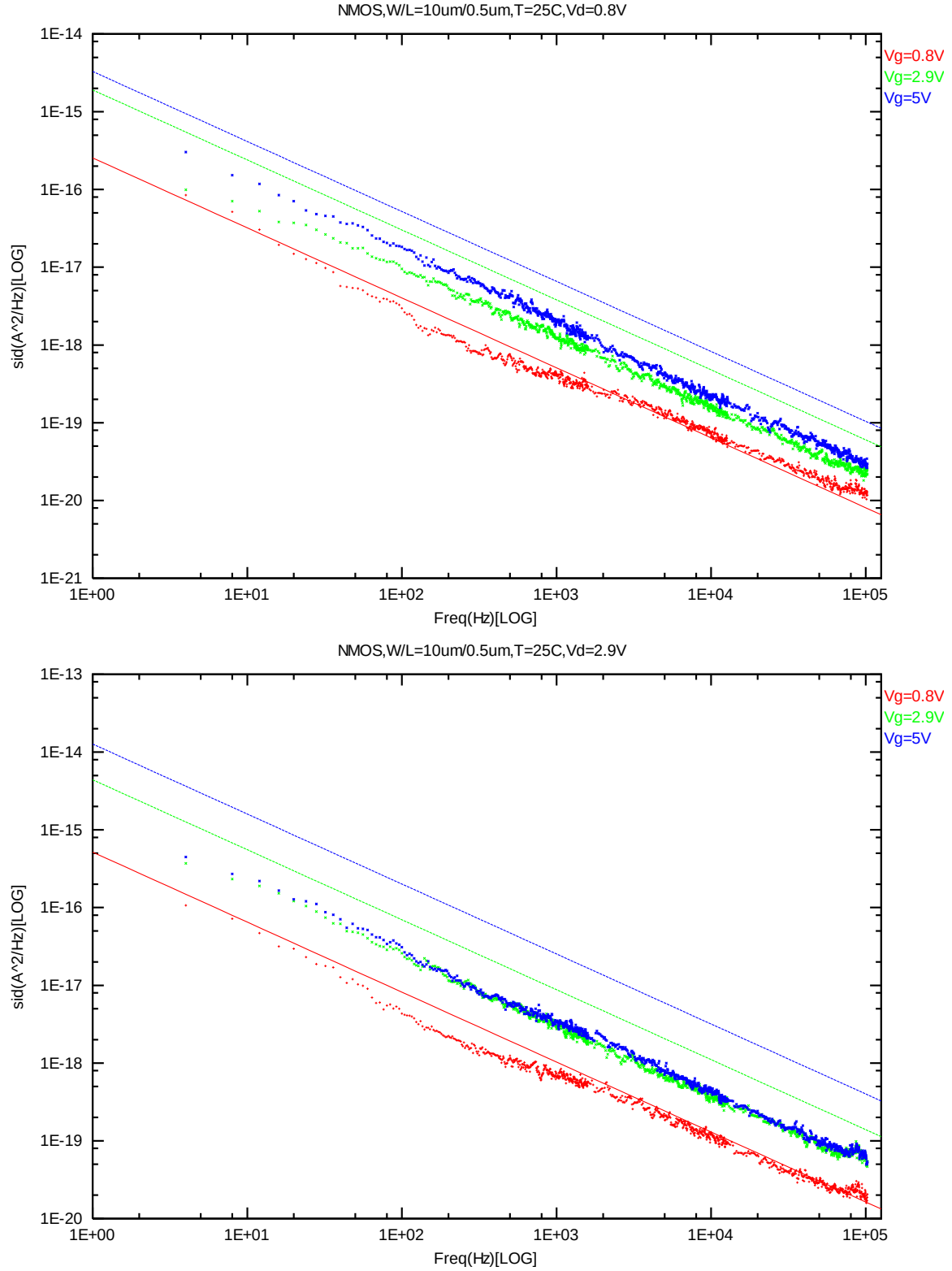


Figure 8-7 NMOS W/L=10um/0.5um @ Vd=0.8 V, Vd=2.9V

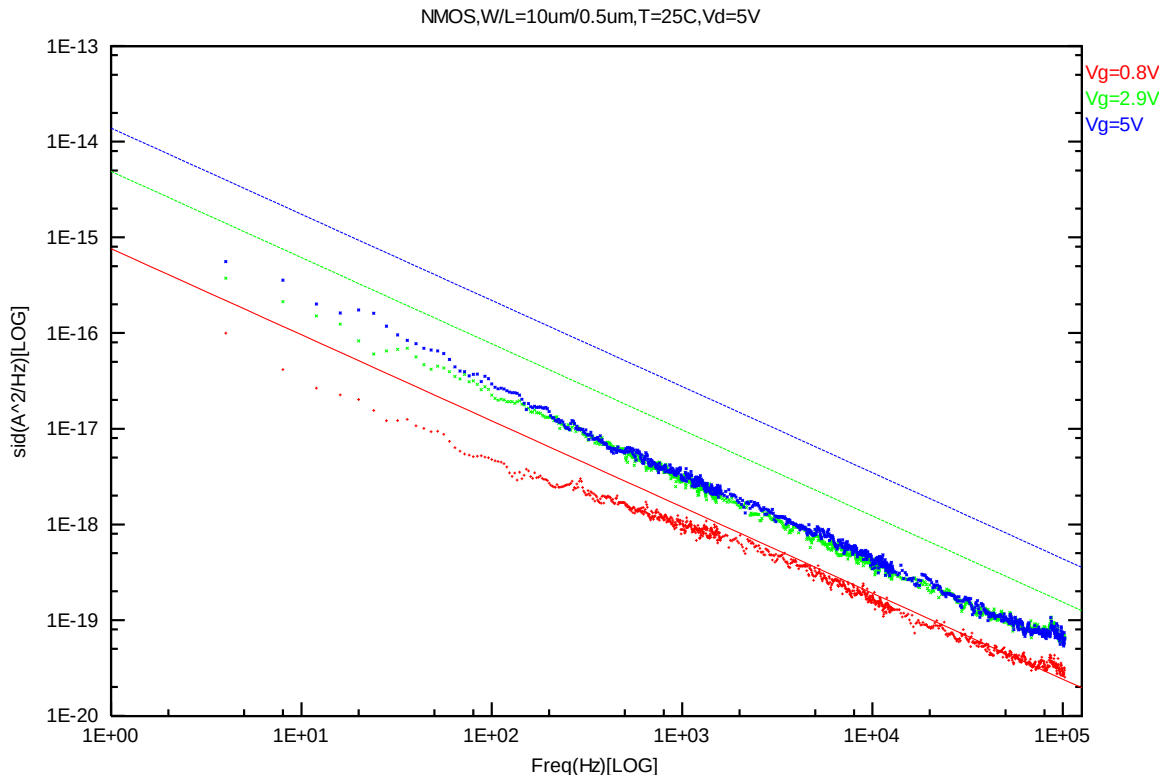


Figure 8-8 NMOS W/L=10um/0.5um @ Vd=5 V

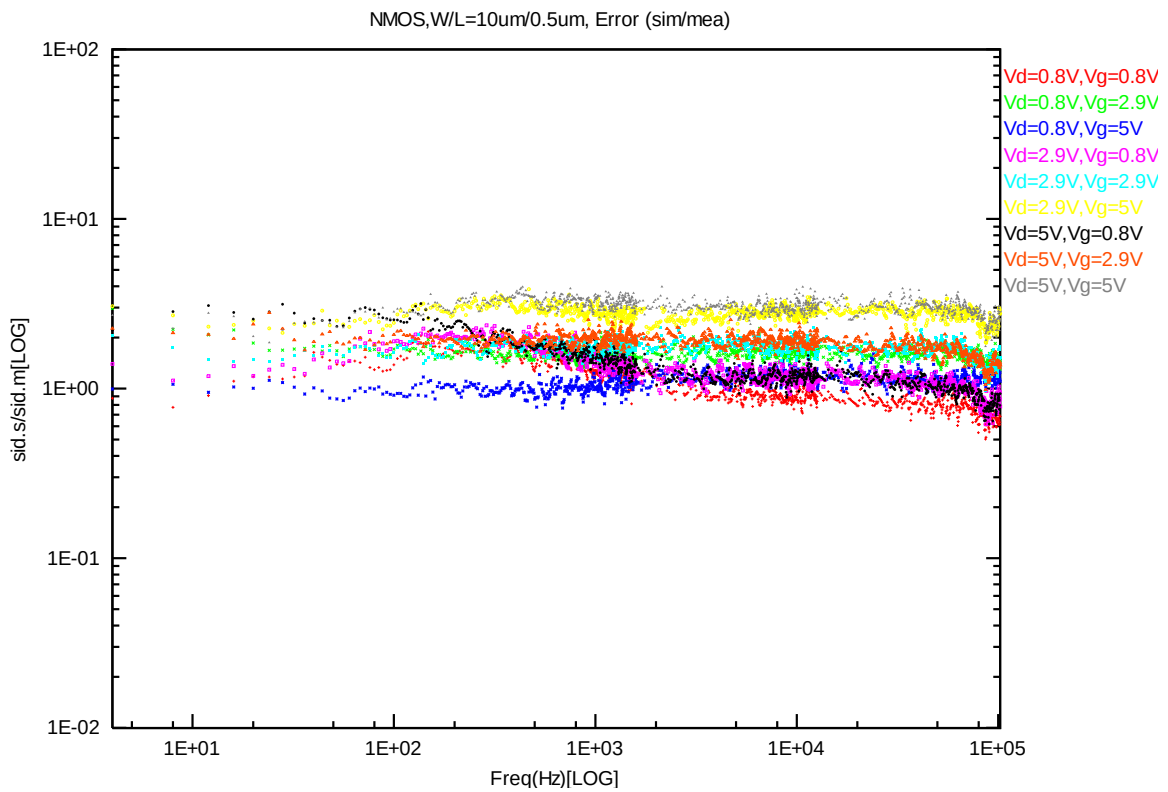


Figure 8-9 NMOS W/L=10um/0.5um error distribution plot

PMOS, W=10um L=10um

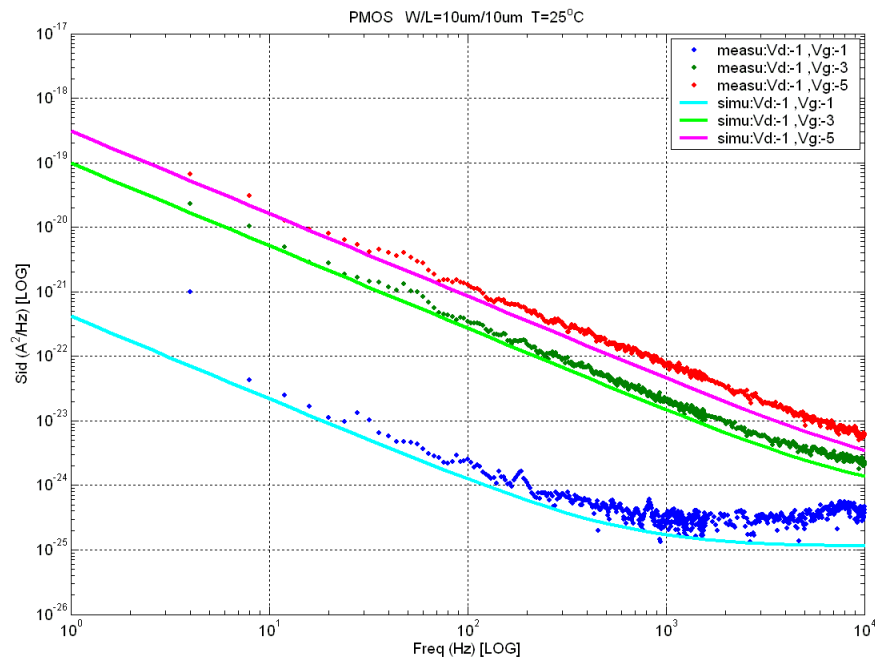


Figure 8-10 PMOS W/L=10um/10um @ Vd=-0.8 V

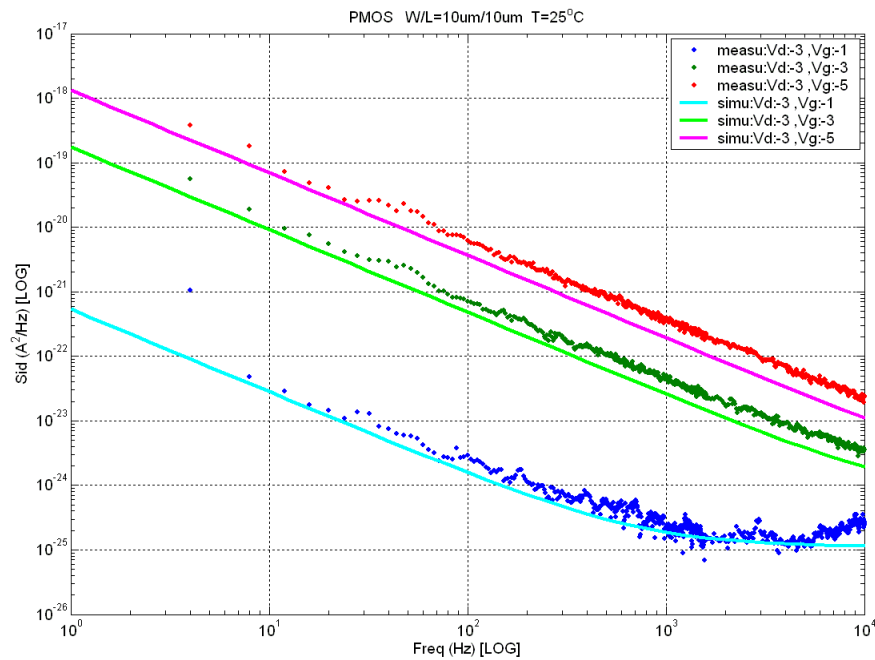


Figure 8-11 PMOS W/L=10um/10um @ Vd=-2.9 V

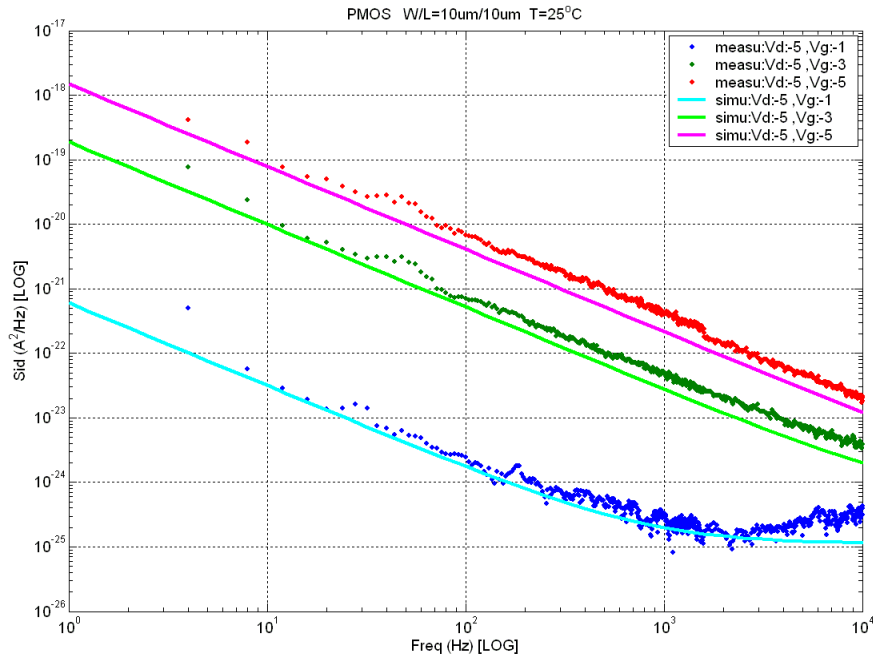


Figure 8-12 PMOS W/L=10um/10um @ Vd=-5 V

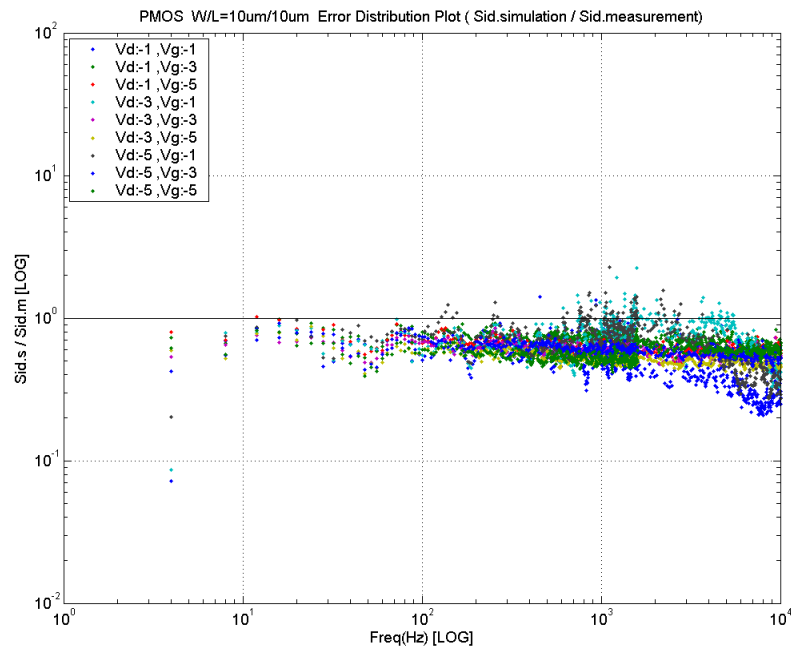


Figure 8-13 PMOS W/L=10um/10um error distribution plot

PMOS, W=10um L=1um

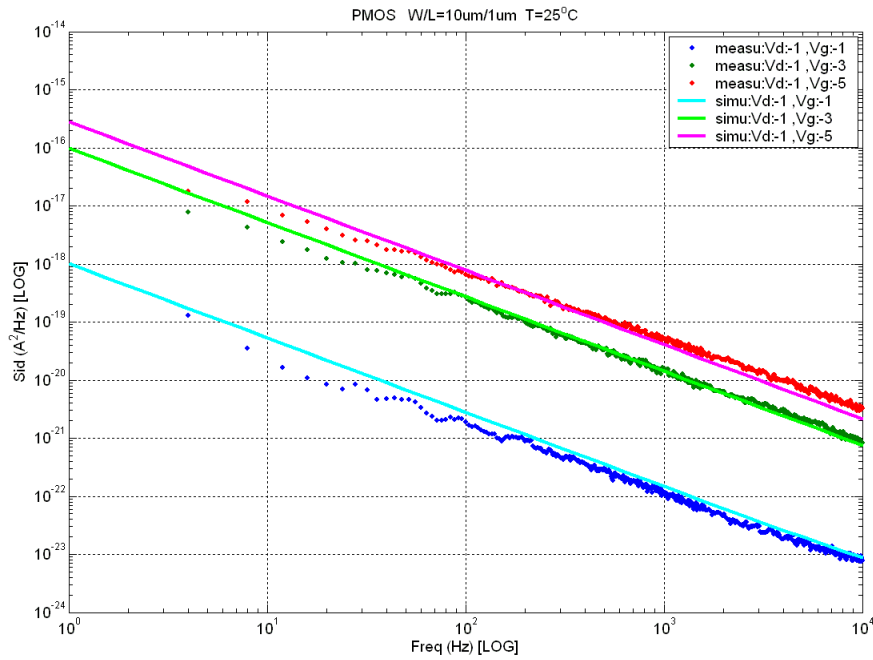


Figure 8-14 PMOS W/L=10um/1um @ Vd=-0.8 V

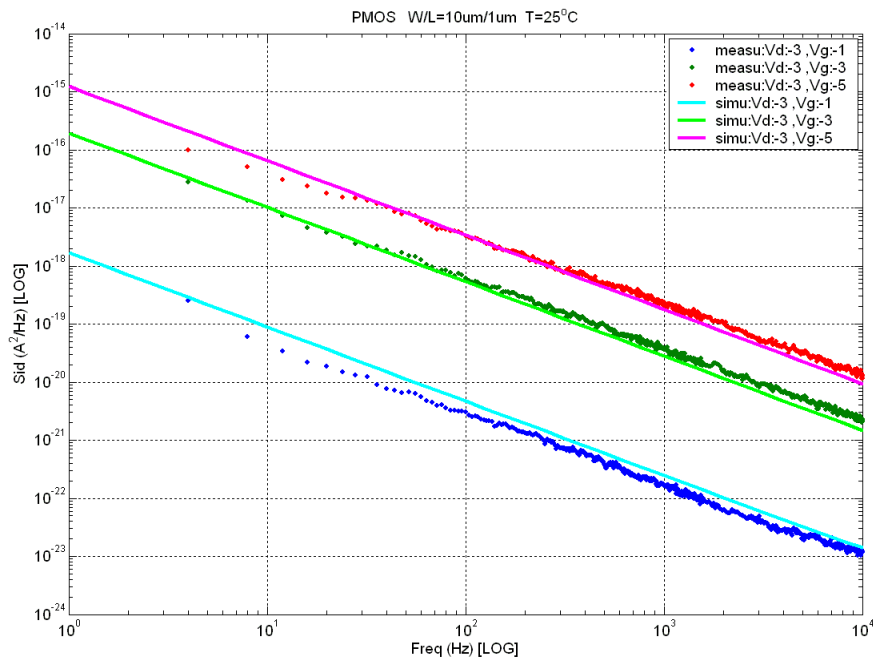


Figure 8-15 PMOS W/L=10um/1um @ Vd=-2.9 V

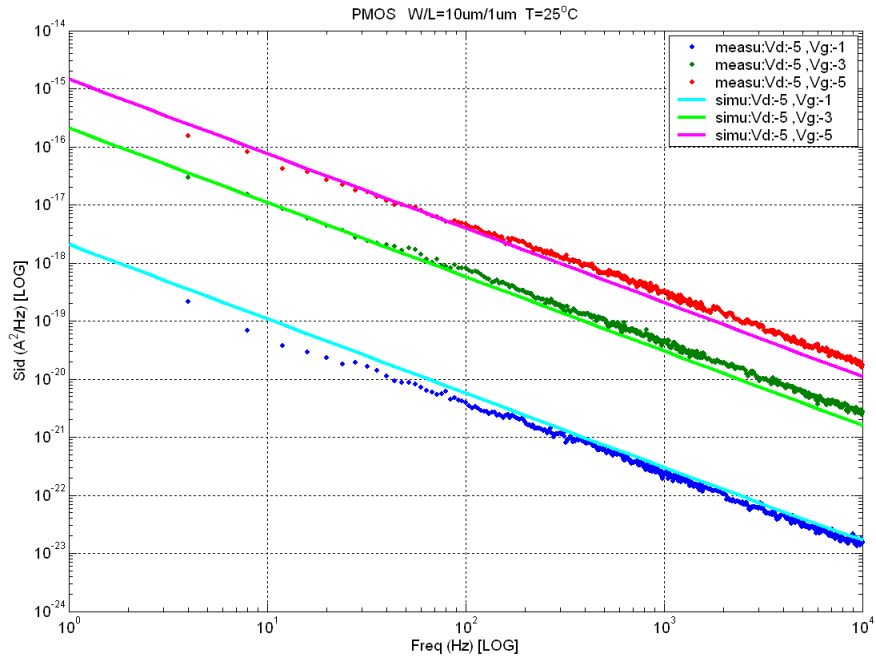


Figure 8-16 PMOS W/L=10um/1um @ Vd=-5 V

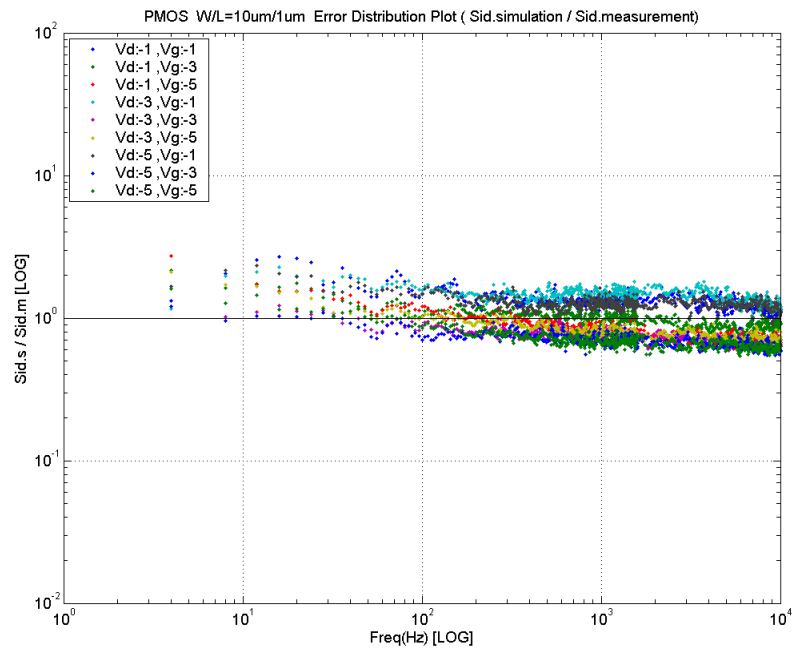


Figure 8-17 PMOS W/L=10um/1um error distribution plot

PMOS, W=10um L=0.5um

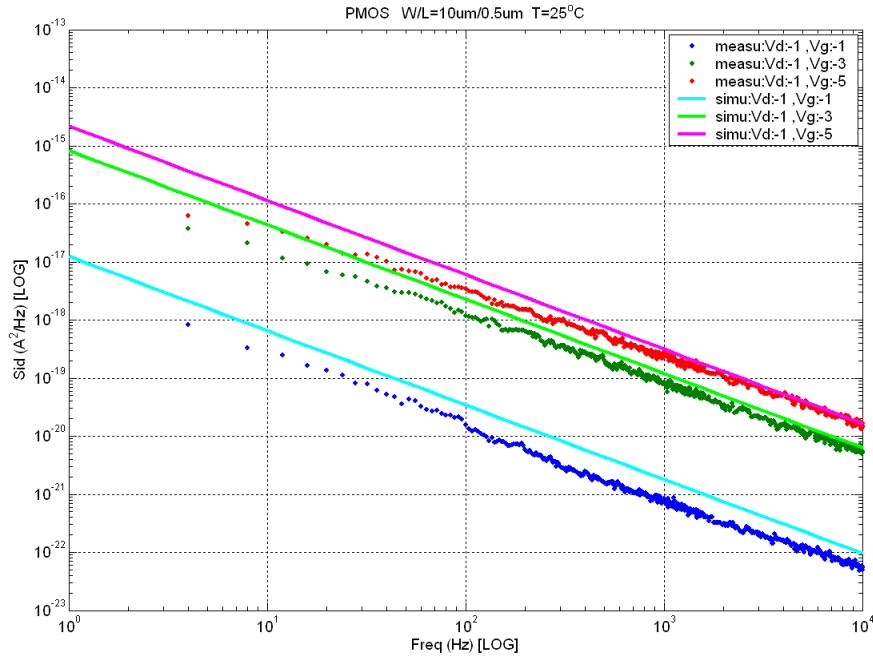


Figure 8-18 PMOS W/L=10um/0.5um @ Vd=-0.8 V

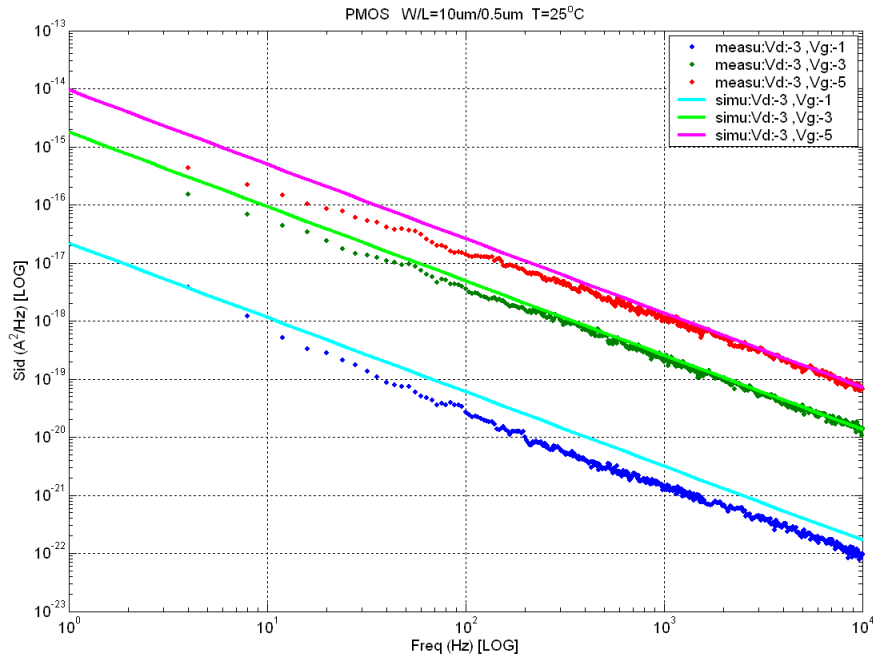


Figure 8-19 PMOS W/L=10um/0.5um @ Vd=-2.9 V

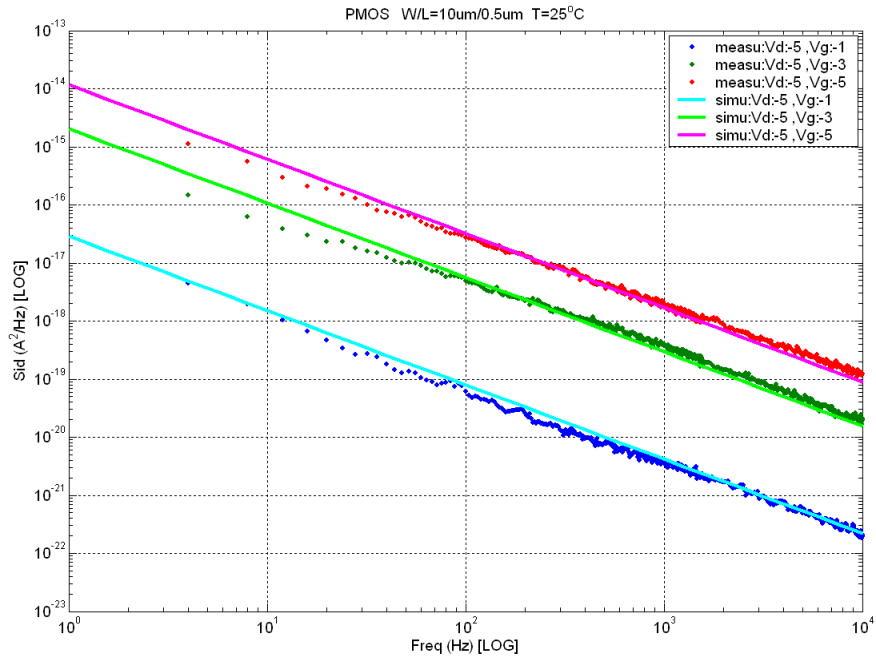


Figure 8-20 PMOS W/L=10um/0.5um @ Vd=-5 V

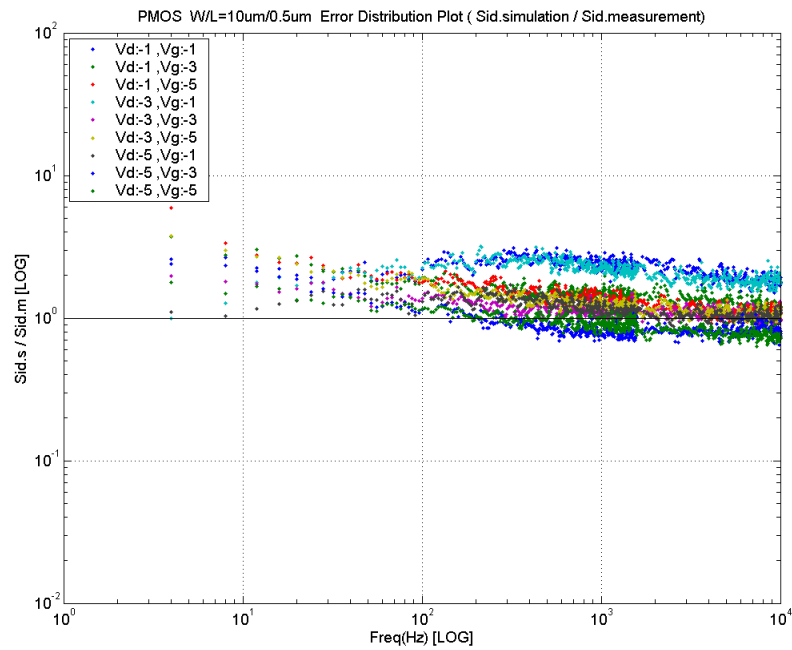


Figure 8-21 PMOS W/L=10um/0.5um error distribution plot

8.2 Trend Plots at Frequency=100Hz

A. NMOS

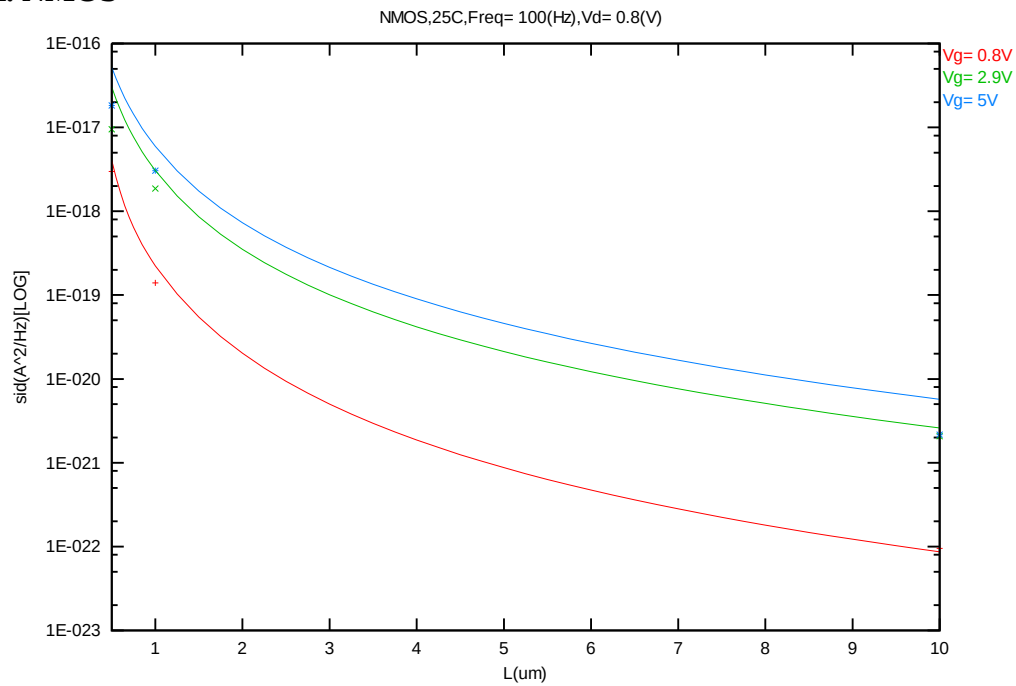


Figure 8-22 Noise vs. Length for NMOS @ Vd=0.8 V

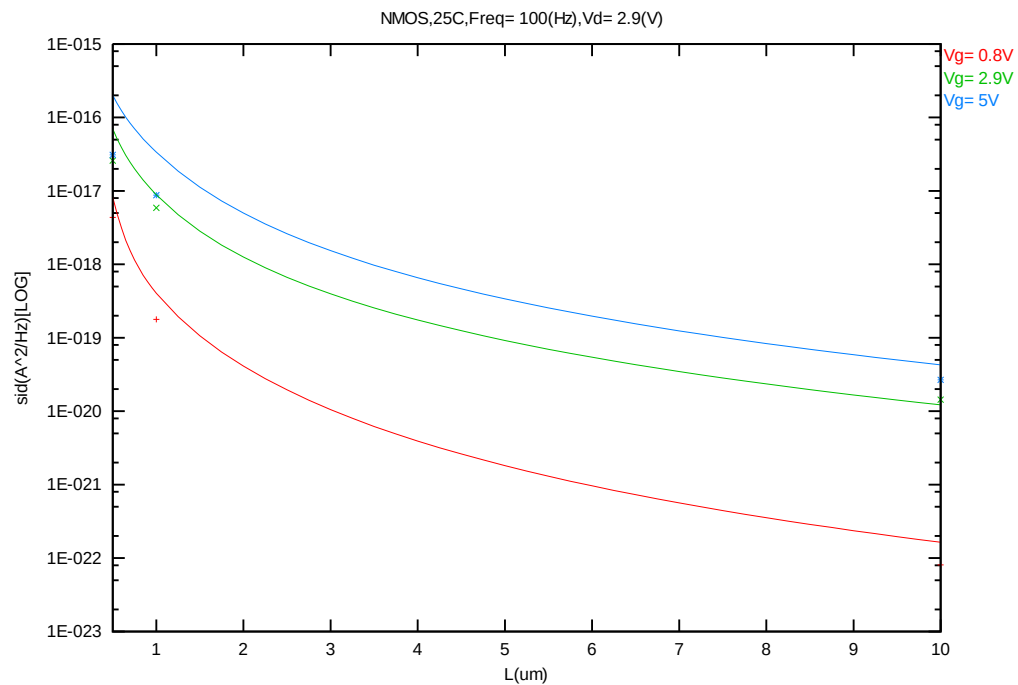


Figure 8-23 Noise vs. Length for NMOS @ Vd=2.9 V

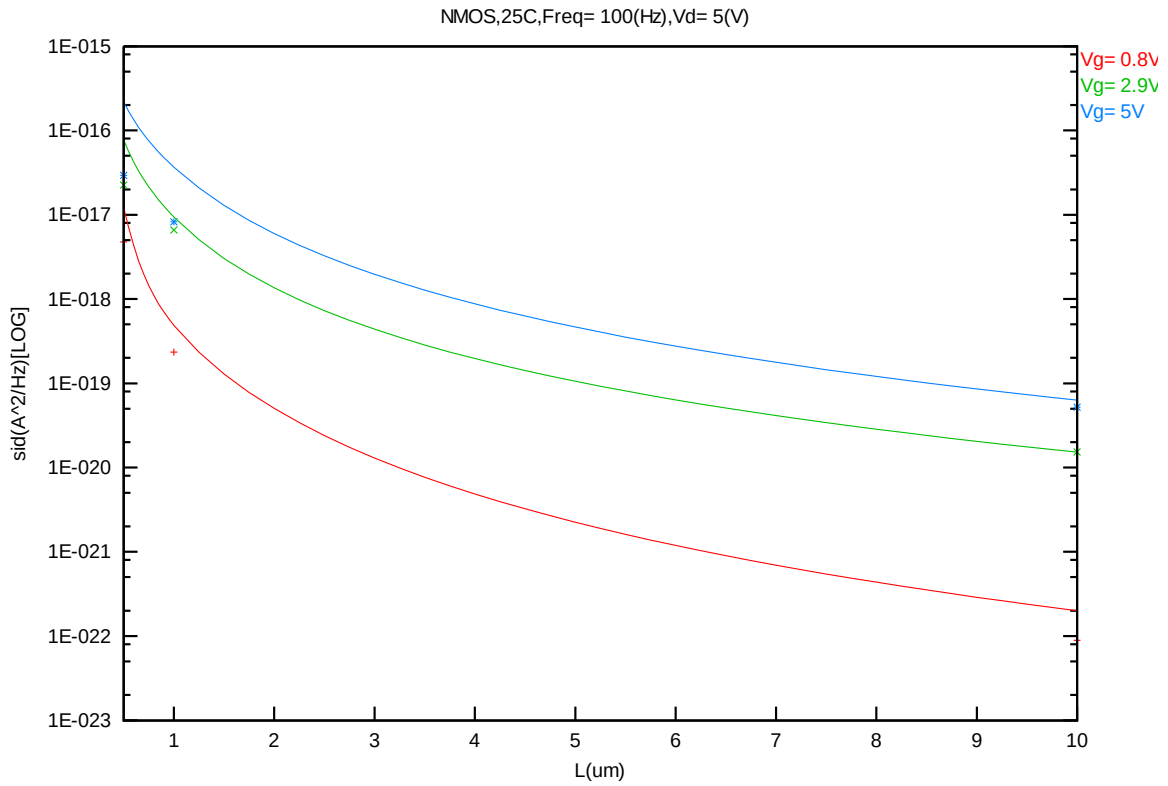


Figure 8-24 Noise vs. Length for NMOS @ Vd=5 V

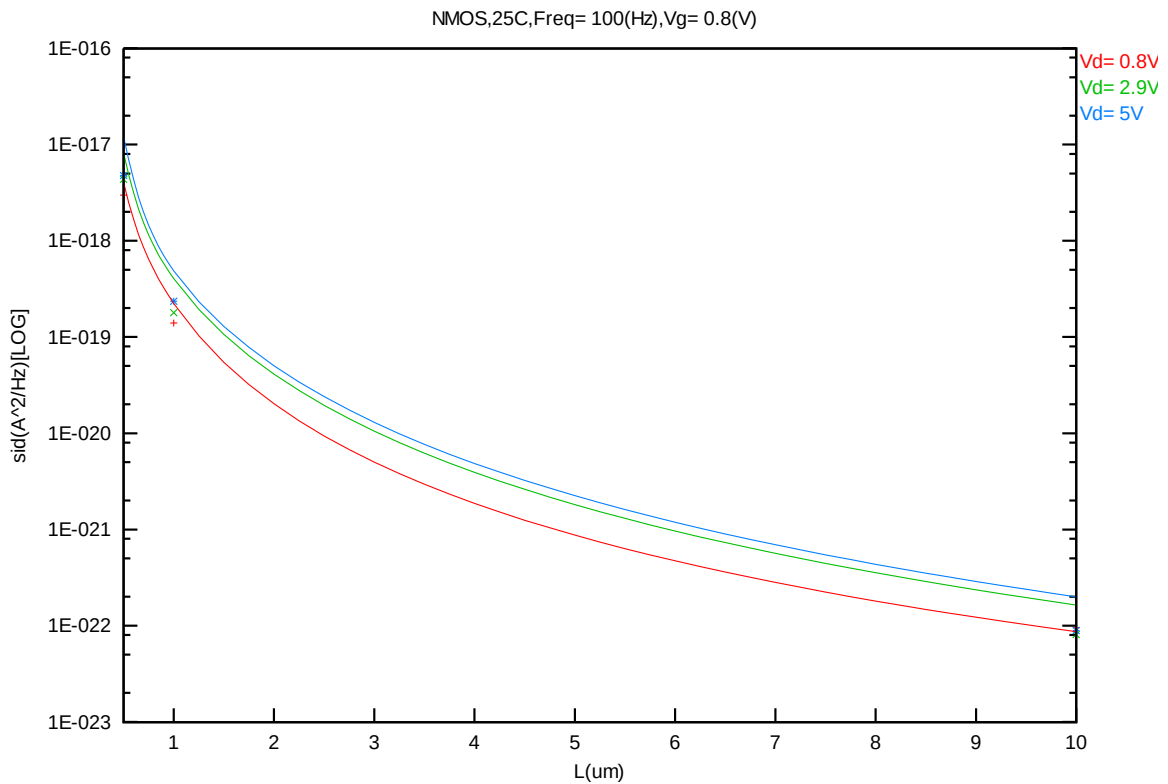


Figure 8-25 Noise vs. Length for NMOS @ Vg=0.8 V

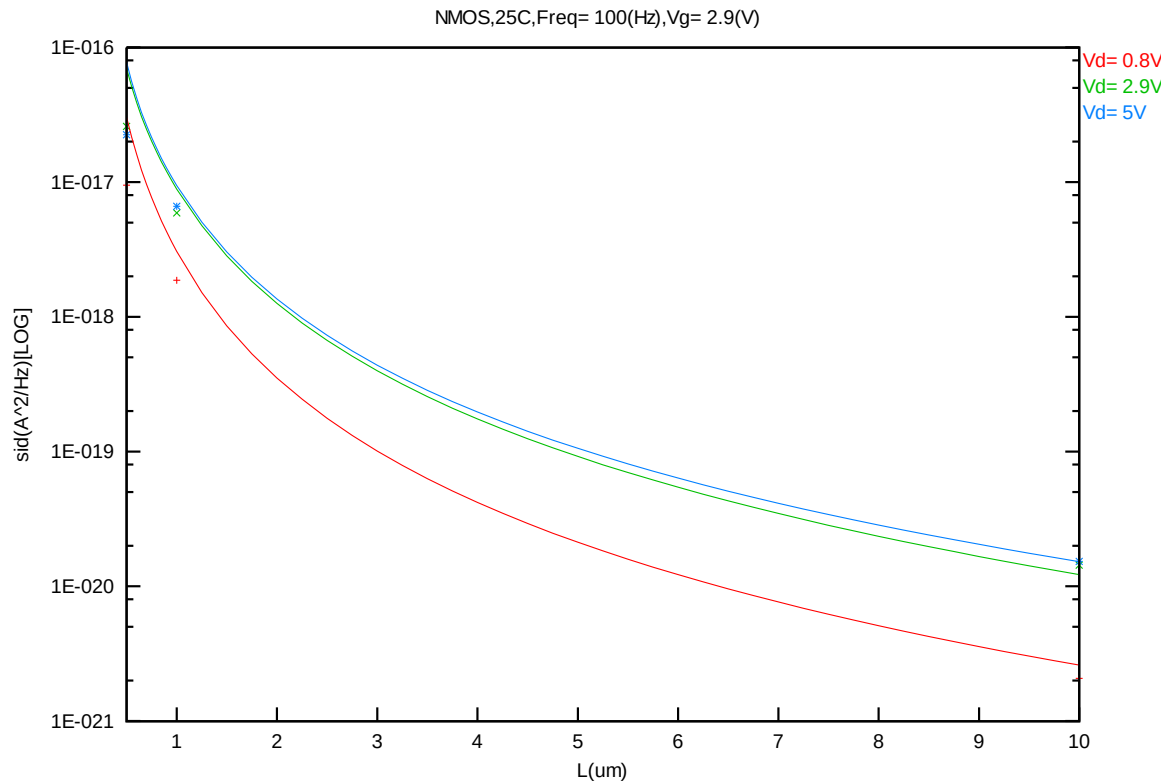


Figure 8-26 Noise vs. Length for NMOS @ Vg=2.9 V

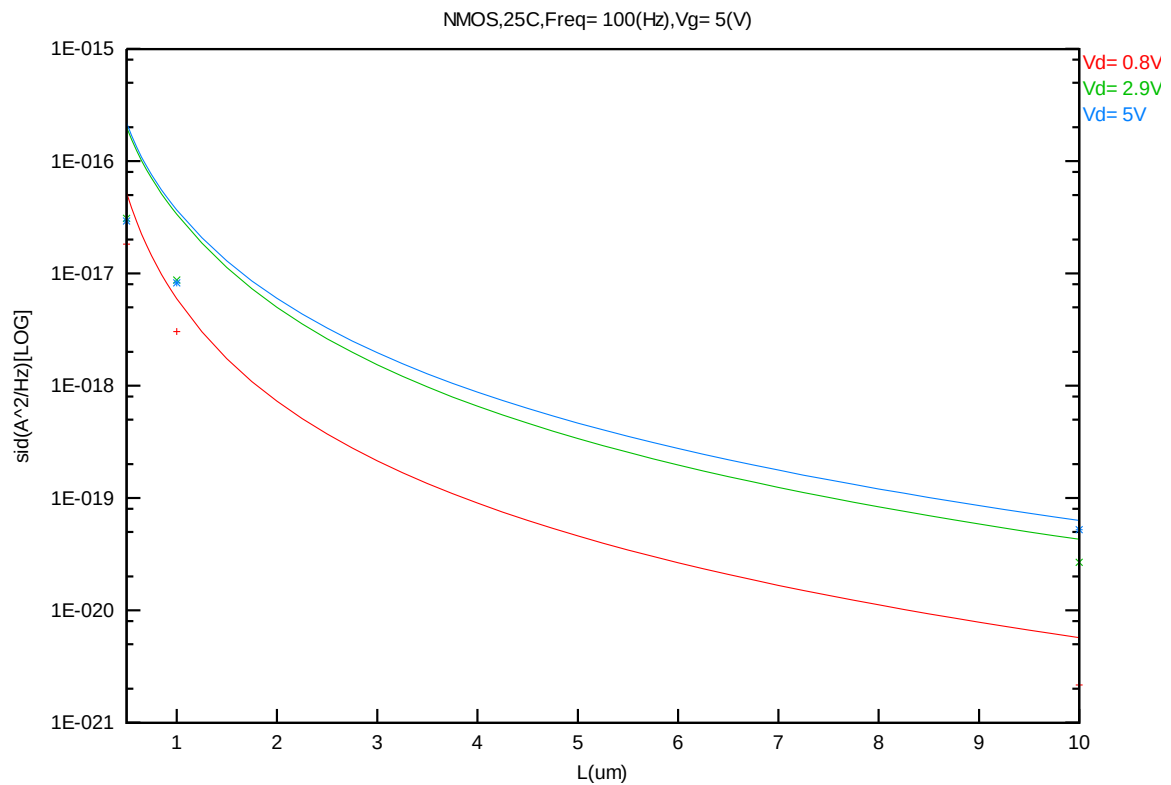


Figure 8-27 Noise vs. Length for NMOS @ Vg=5 V

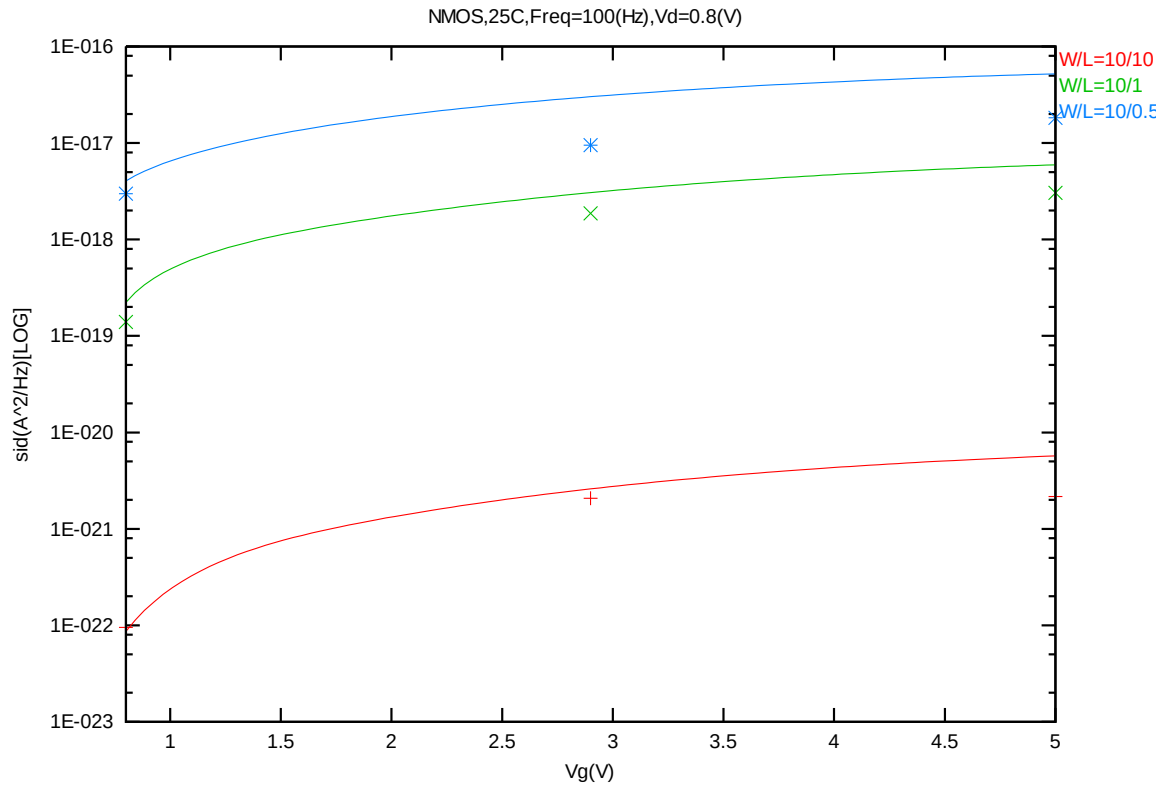


Figure 8-28 Noise vs. Vg for NMOS @ Vd=0.8 V

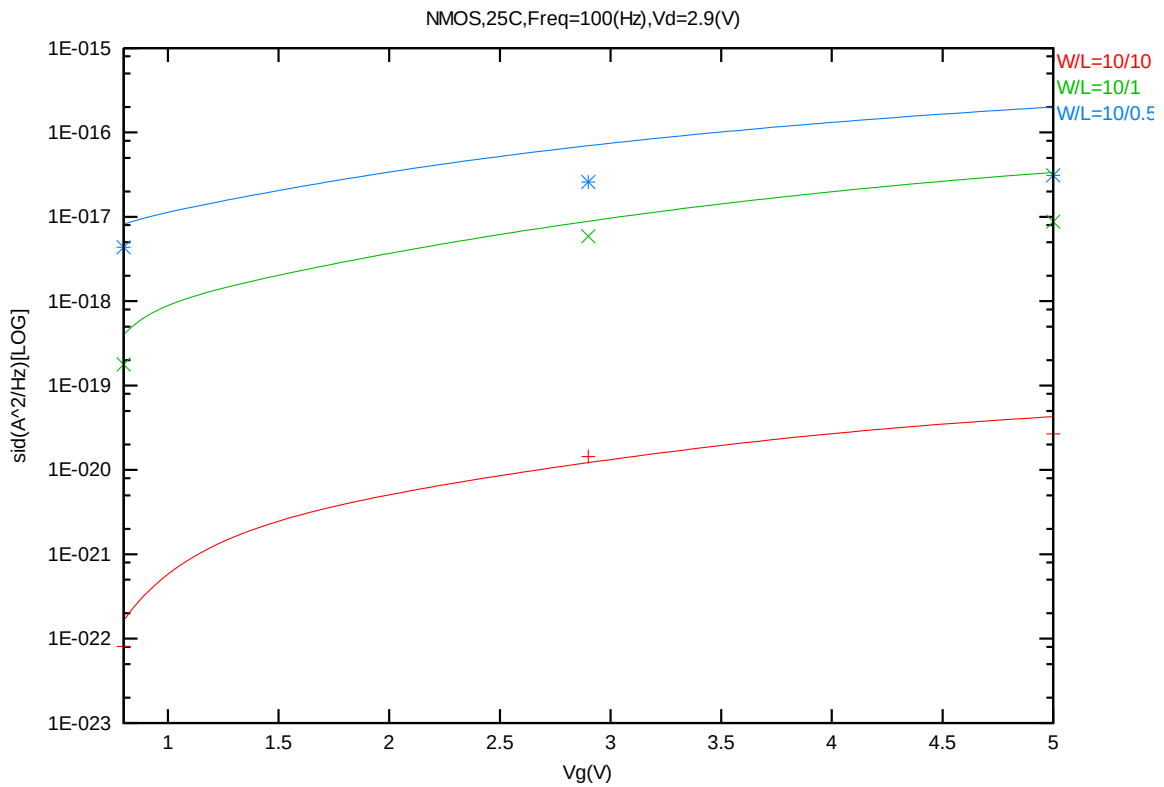


Figure 8-29 Noise vs. Vg for NMOS @ Vd=2.9 V

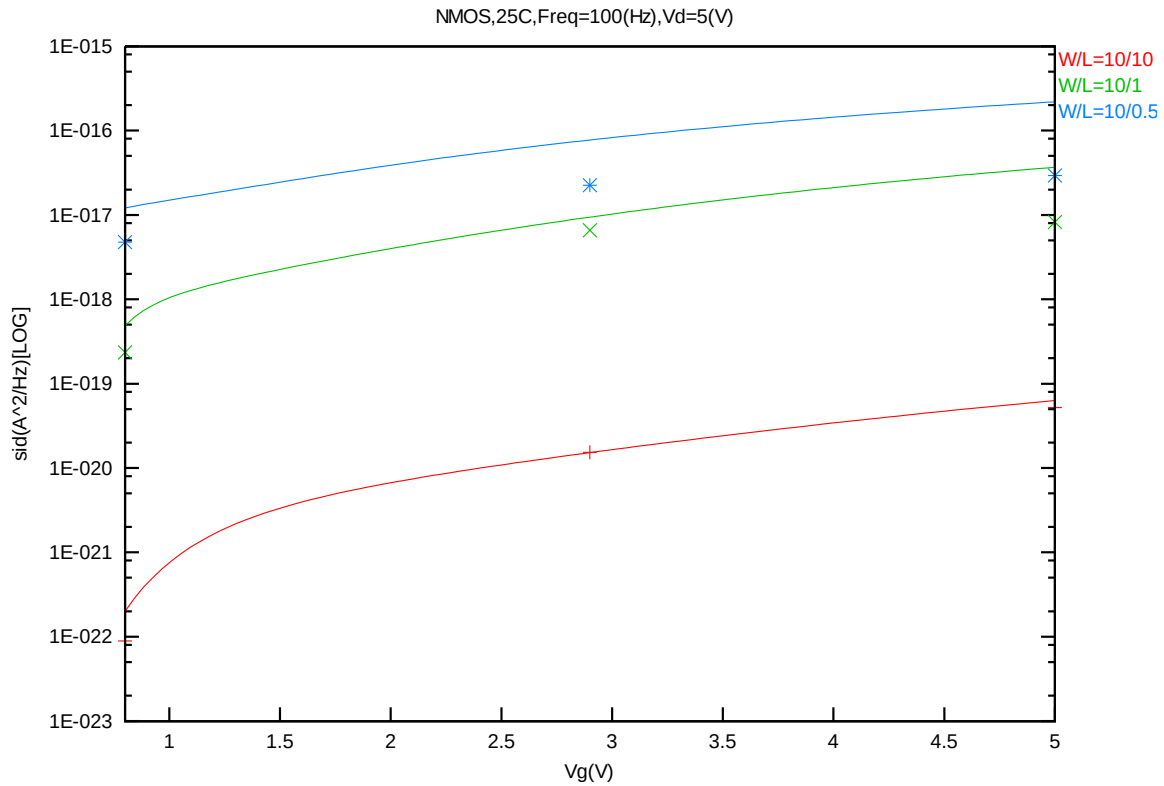


Figure 8-30 Noise vs. Vg for NMOS @ Vd=5 V

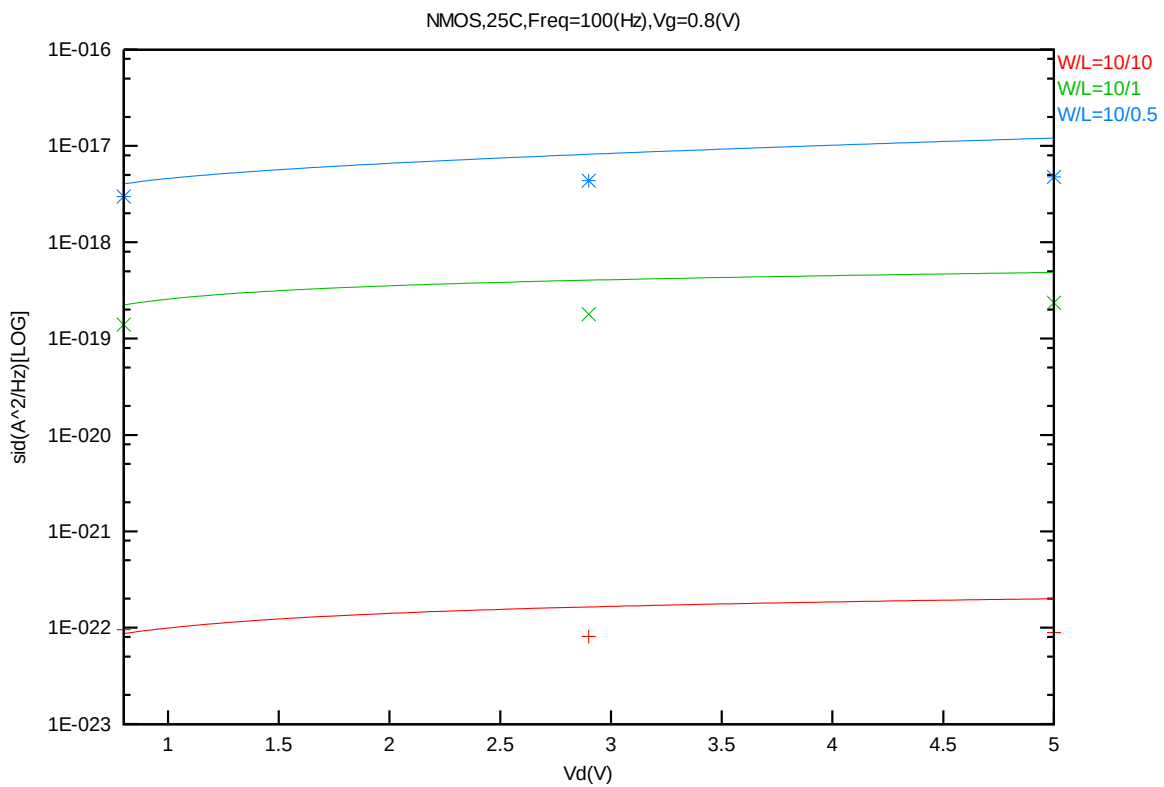


Figure 8-31 Noise vs. Vg for NMOS @ Vg=1 V

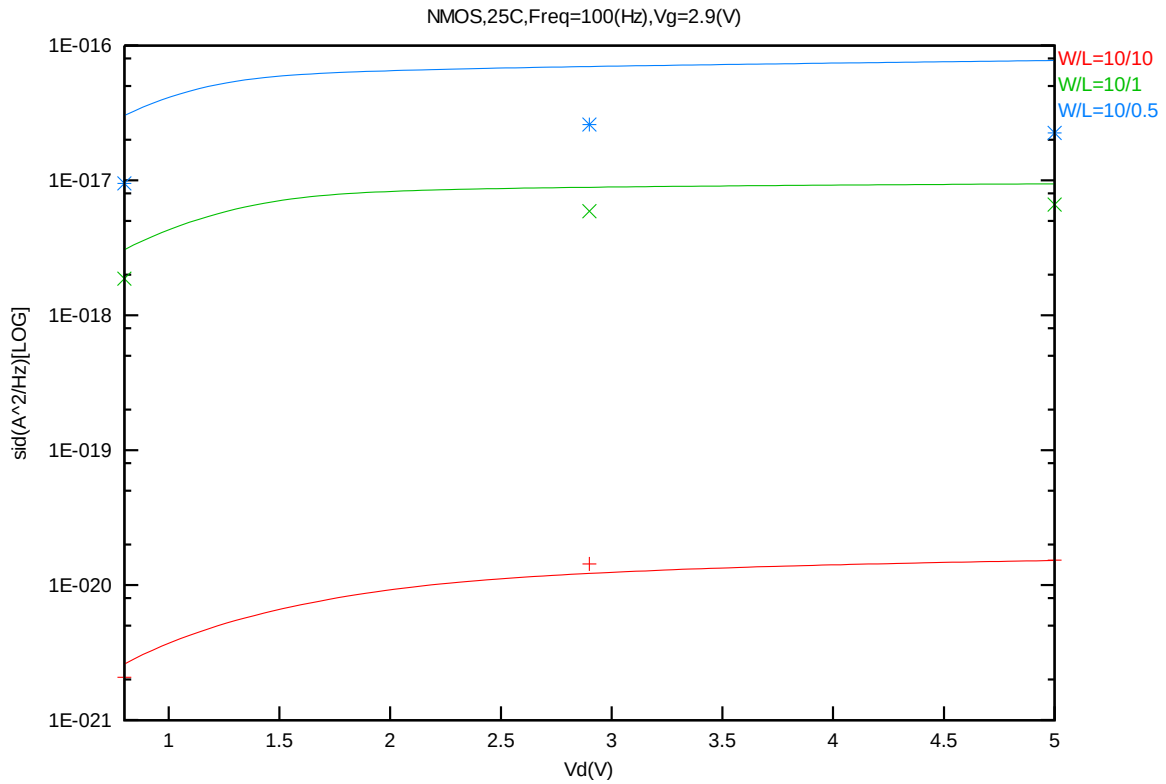


Figure 8-32 Noise vs. Vd for NMOS @ Vg=3 V

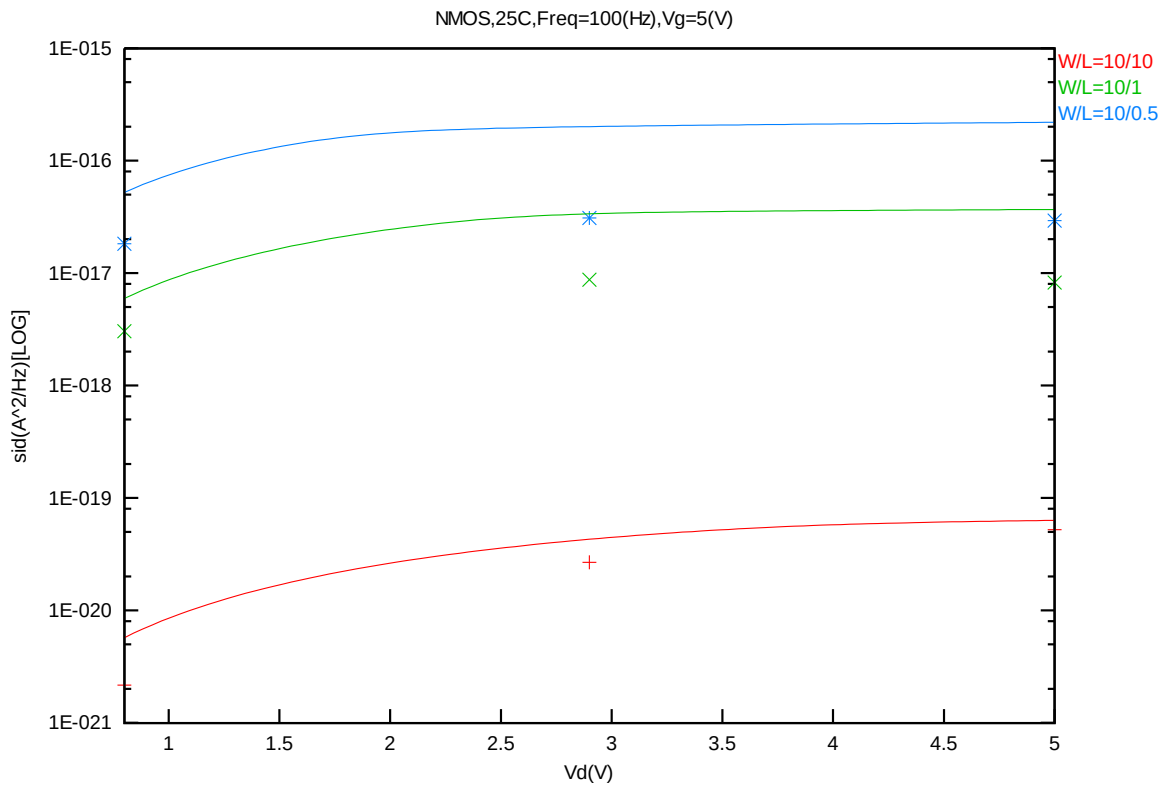


Figure 8-33 Noise vs. Vd for NMOS @ Vg=5 V

B. PMOS

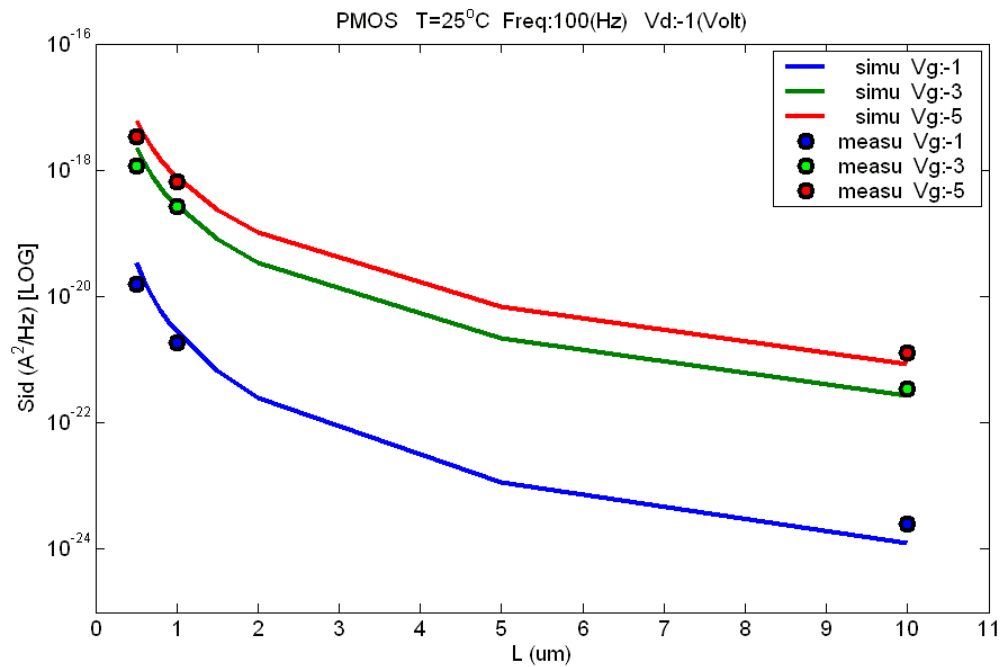


Figure 8-34 Noise vs. Length for PMOS @ Vd=-0.8 V

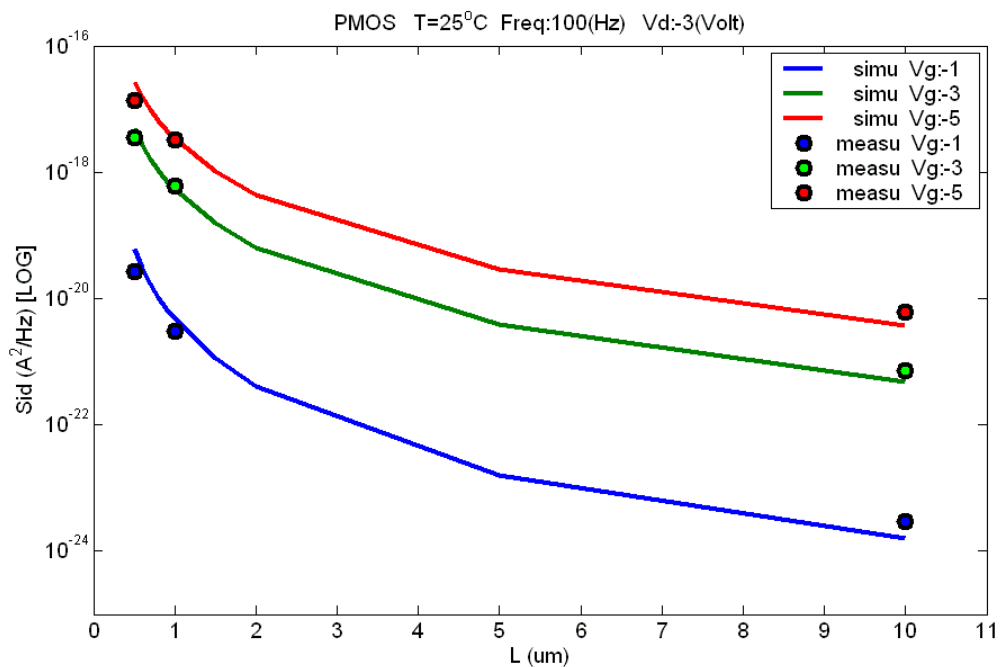


Figure 8-35 Noise vs. Length for PMOS @ Vd=-2.9 V

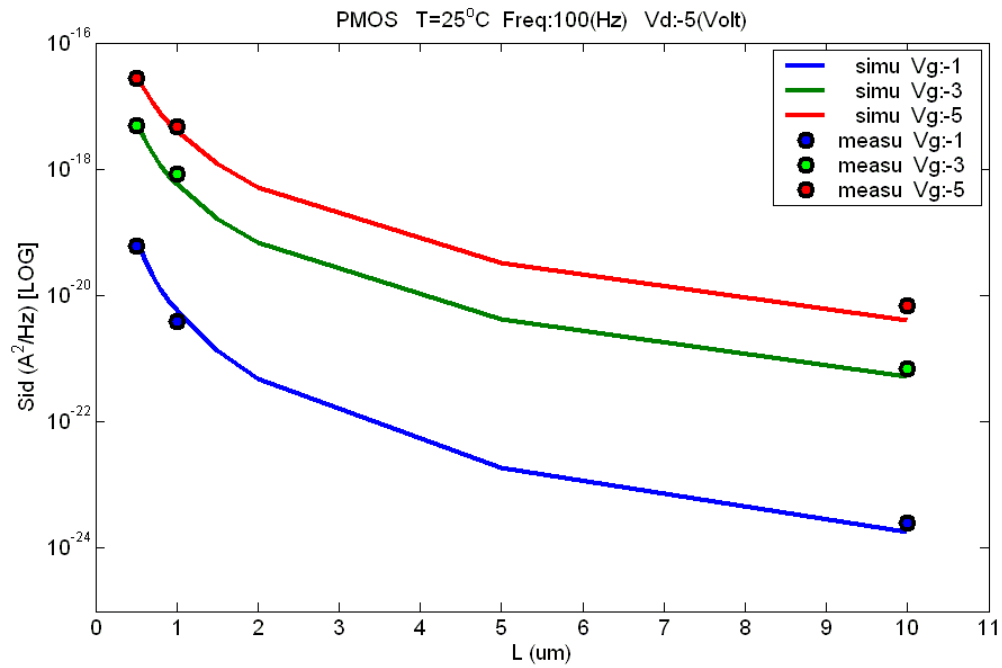


Figure 8-36 Noise vs. Length for PMOS @ Vd=-5 V

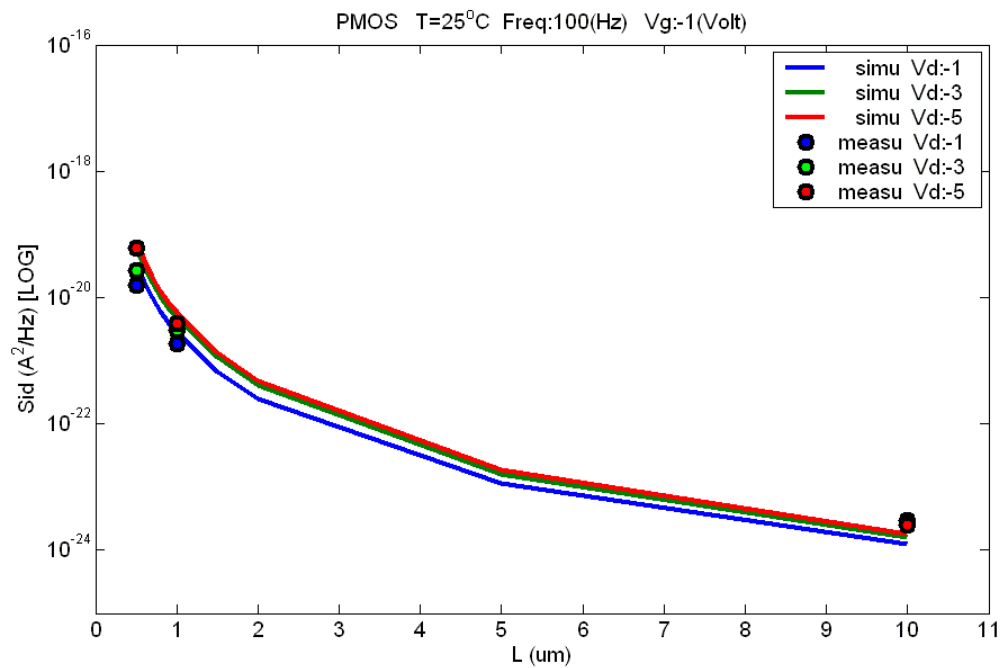


Figure 8-37 Noise vs. Length for PMOS @ Vg=-1 V

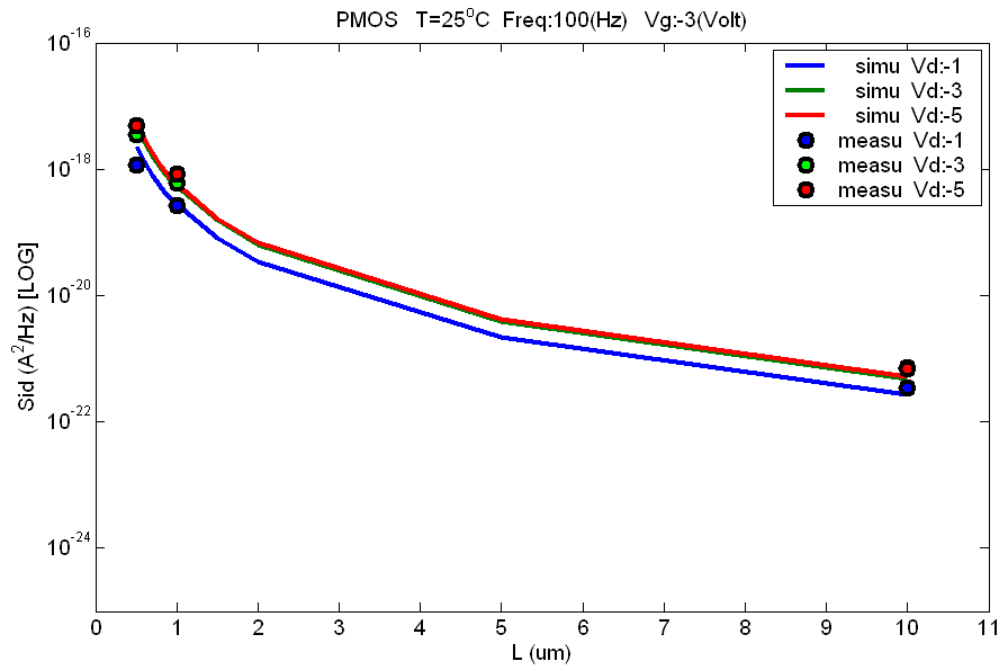


Figure 8-38 Noise vs. Length for PMOS @ Vg=-3 V

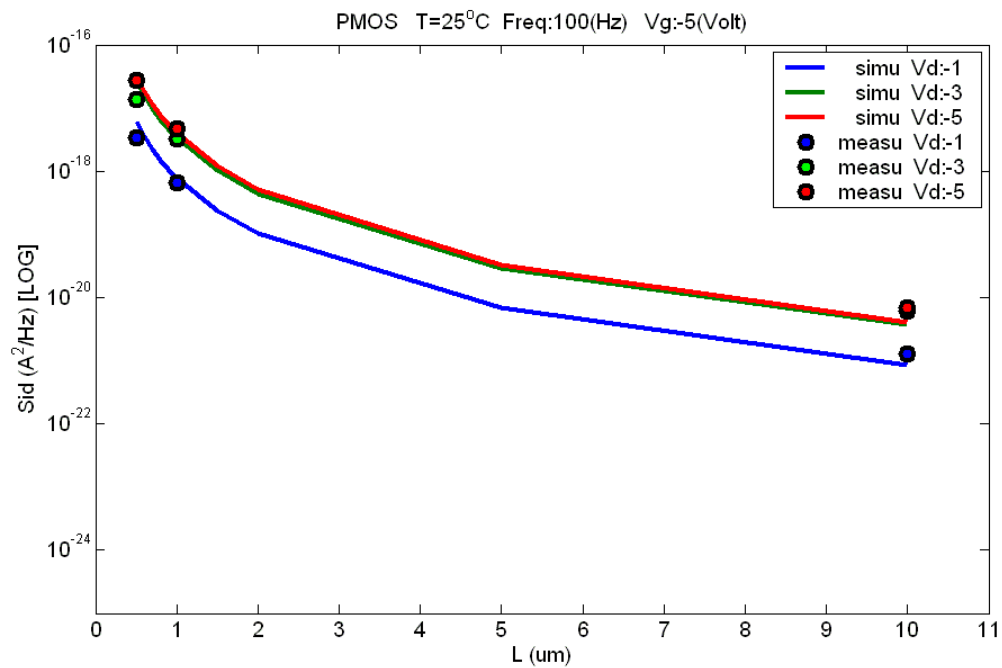


Figure 8-39 Noise vs. Length for PMOS @ Vg=-5 V

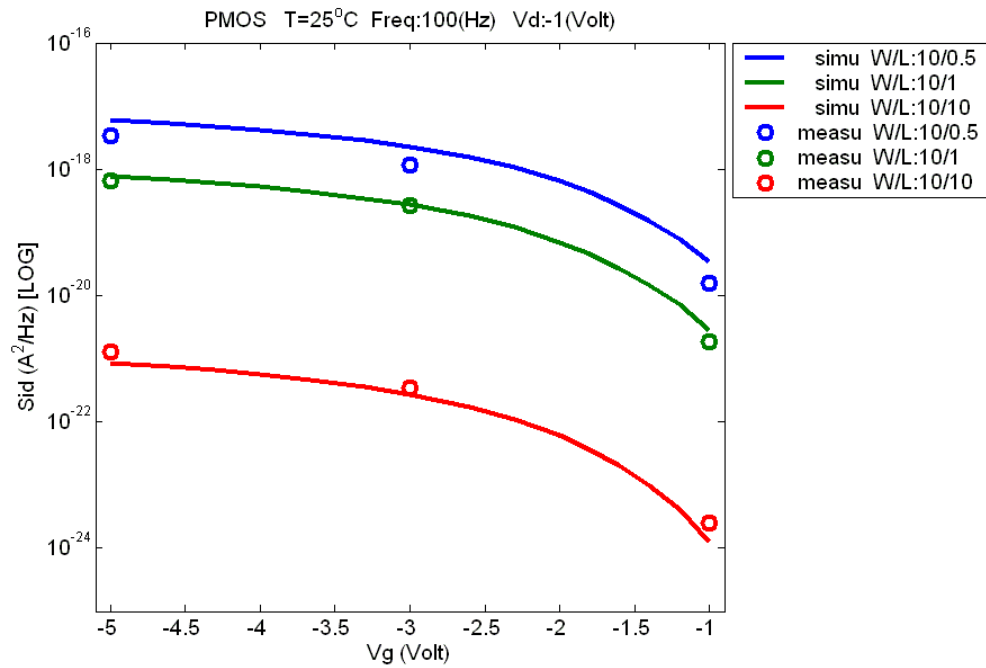


Figure 8-40 Noise vs. Vg for PMOS @ Vd=-0.8 V

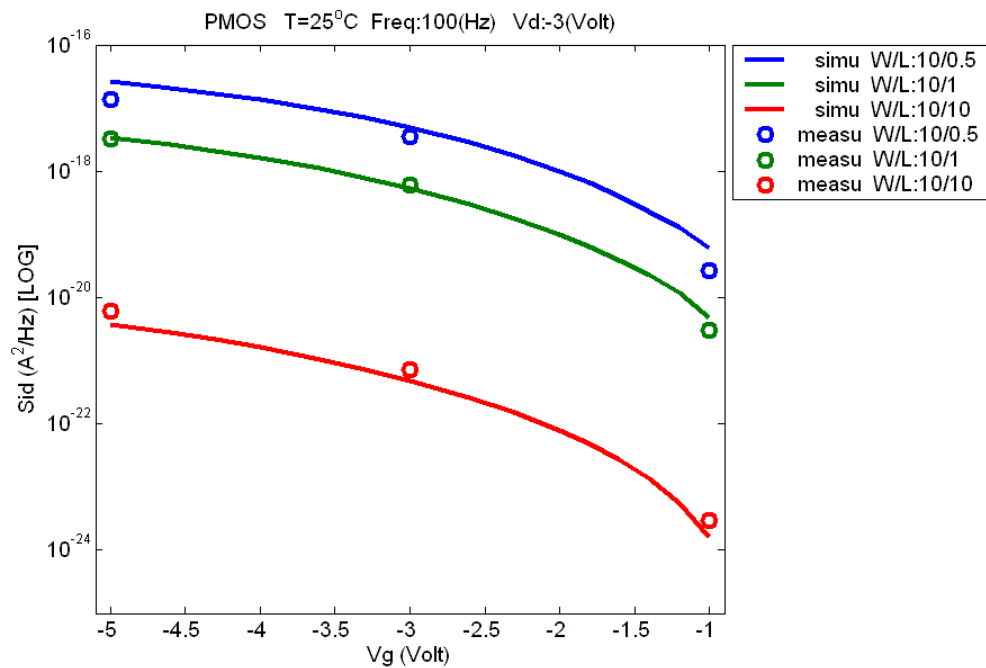


Figure 8-41 Noise vs. Vg for PMOS @ Vd=-2.9 V

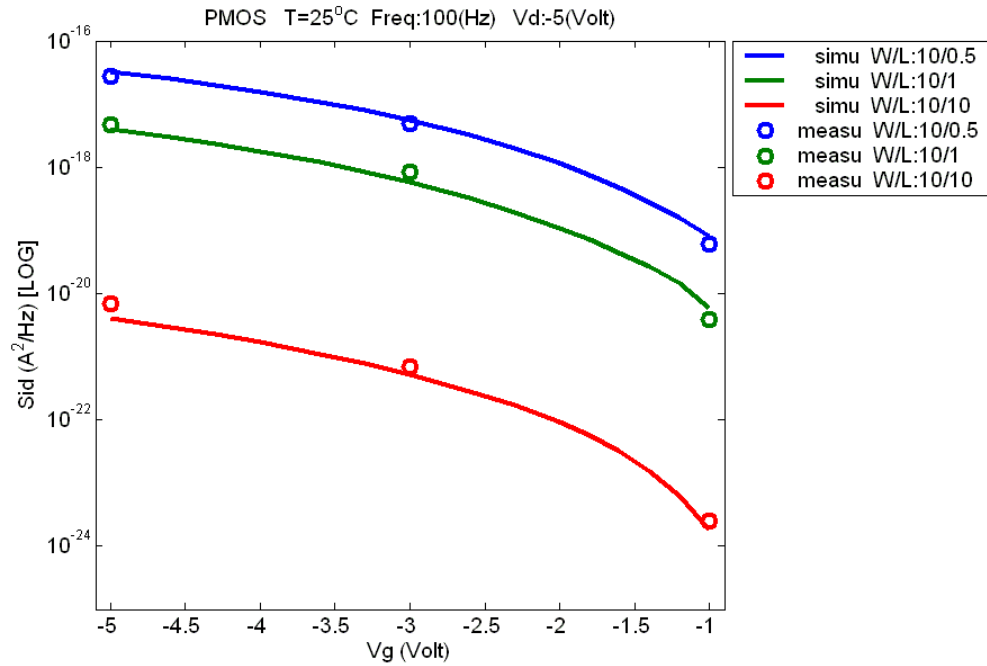


Figure 8-42 Noise vs. Vg for PMOS @ Vd=-5 V

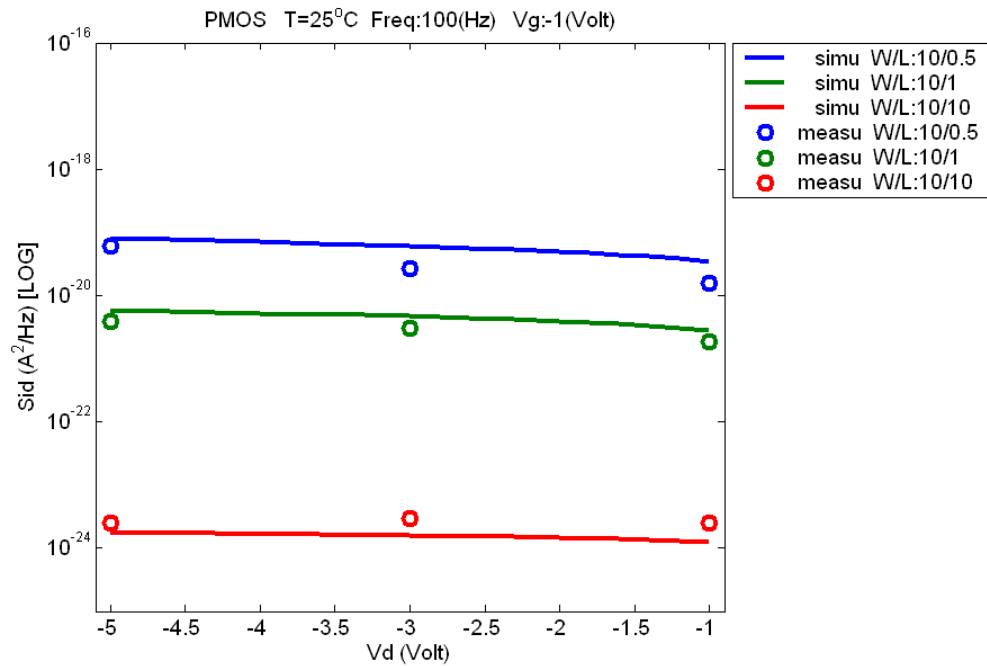


Figure 8-43 Noise vs. Vd for PMOS @ Vg=-1 V

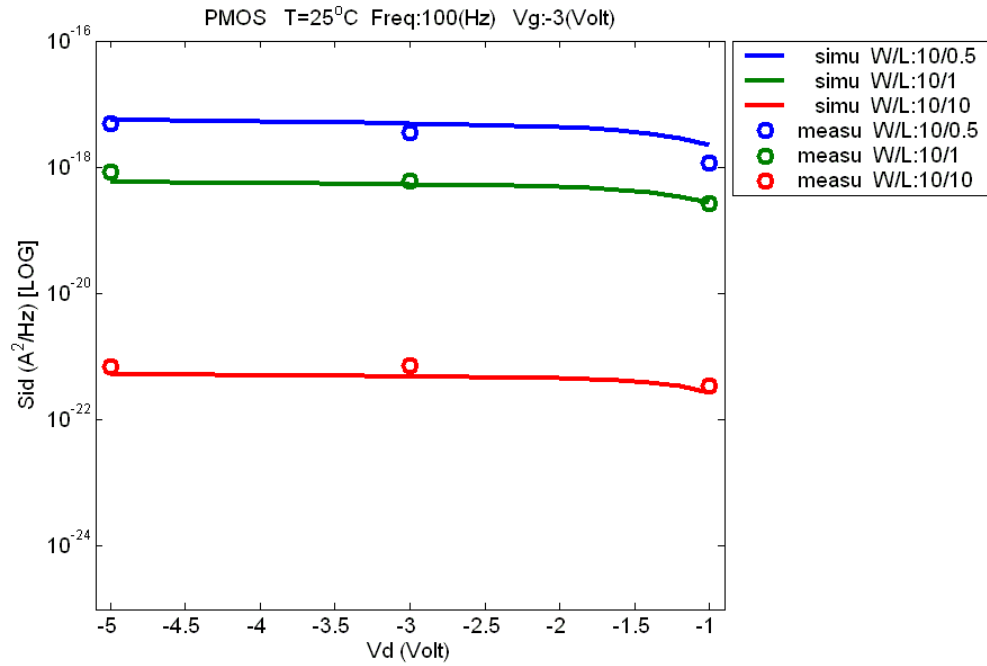


Figure 8-44 Noise vs. Vd for PMOS @ Vg=-3 V

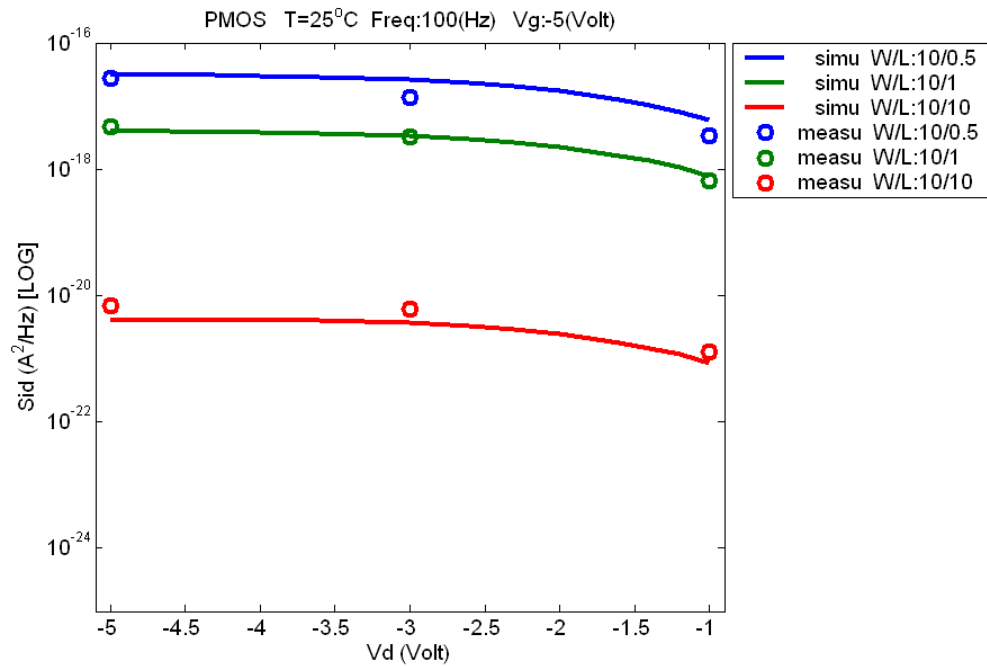


Figure 8-45 Noise vs. Vd for PMOS @ Vg=-5 V

9 Appendix

9.1 HSPICE Warning Message

Table 9-1 Hspice warning message

NO.	Warning Message	Explanation
1.	-	-

9.2 ELDO Warning Message

Table 9-2 Eldo warning message

NO.	Warning Message	Explanation
1.	-	-

9.3 SPECTRE Warning Message

Table 9-3 Spectre warning message

NO.	Warning Message	Explanation
1.	`Eta0' is negative.	A negative parameter ETA0 for NMOS is needed to get best fitting for threshold voltage in saturation region. But it will cause a harmless warning message.